Corte

x

Clou

d

Runti

me

Secur

ity
Docu

ment

ation
-

Detail

ed

1. Get

started

with

Cortex

Cloud



1.1. What is
Cortex
Cloud?

 1.1.1. Key
features



1.2. What is
Cortex
Cloud
Runtime
Security?



1.3. Agentic
AI in Cortex
Cloud



💡🔵

tip

The system
Help Center
agent
delivers fast,
context-
aware

assistance
to answer
your
questions.
You can ask
natural
language
questions,
such as
"How do I
create a
dashboard?"
or "Where
can I review
my data
retention
policies?"
and the
agent
retrieves
concise,
relevant
information
from the
documentati
on. If a
question
remains
unresolved,
the agent...

🧭➡️

navigation

Settings →
Configuratio
ns →

General →
Server
Settings →
Agentic
Assistant

1.3.1. Agentic
Assistant
use cases

 📝🟢

note

If a team
member in
the channel
sees the
summary
and
attempts to
ask the
agent, "Give
me more

details on
the first SLA
issue," the
team
member
receives an
access

denied
message
because the
active
session is
only
available to
you, the
initiator.

1.3.2. Agentic
Assistant
security



1.4.

Supported
web
browsers

 📝🟢

note

Cortex
Cloud is
optimized
for best
performance
using the
HTTPv2

protocol. If
you are
using a non-
supported
browser
version or a
network
middleware
that uses a
protocol
other than
HTTPv2, you
may
experience
slowness in
loading and
response
times
between the
browser and
the Cortex
Cloud
service.
Consult with
your
browser or
middleware
vendor to
see how...

1.5. Use the
interface



1.6. In-
product
support
case creation



📝🟢

note

When opening a support case through the
Customer Support Portal, you need to manually
select Cortex Cloud as the product. While
there may be discrepancies between the
categories in this wizard and the Customer
Support Portal process, that's because this
wizard is designed specifically to focus on
options relevant to Cortex Cloud.

📝🟢

note

Selecting an
endpoint
from the
endpoint
table and
retrieving
TSF requires
full Retrieve
Endpoint
Data
permissions
under
Endpoint
Administratio

n.

📝🟢

note

Console
recordings
cannot
exceed 10
minutes. The
current

recording
time is
displayed at
the top of
the window.

1.7.
Understand
your user
persona



1.8.
Understand
license plans



📝🟢

note

A Cloud Runtime license includes both Posture
Scanning and Runtime Protection on the same
asset. Usage, including any overflow, is tracked
automatically to ensure accurate reporting
across both licenses without duplicate counting.

1.8.1. Data
retention



📝🟢

note

Case data is
retained
according to
the Last
Updated
date.
Issue data is
retained
according to
the
Observation
Time. Data
collected
within these
dates is kept
and
displayed
for 186 days.
To ensure
the
accuracy of
issues,
Cortex
Cloud
provides a
grace period
of up to 31
days for
issues
displayed in
the Issues
View, Issues
table, and
Cases View.

📝🟢

note

Requires the
Forensics
add-on.

1.9. Fair
Usage policy
for Cortex
Cloud



2.

Onboard

and

configure
Cortex

Cloud



2.1. Plan and
prepare



2.2.
Deployment
steps and
checklist



2.2.1.
Activate
Cortex
Cloud



☑️🟣

prerequisite

The Cortex
Cloud
activation
email.
A Customer
Support
Portal (CSP)
account.
You need to
set up your
CSP
account. For
more

information,
see How to
Create Your
CSP User
Account.
When you
create a
CSP
account,
you can set
up two-
factor
authenticatio
n (2FA) to
log into the
CSP by
using an
Email, Okta
Verify, or
Google
Authenticato
r (non-
FedRAMP
accounts)....

📝🟢

note

Activation
can take
about an
hour and
does not
require you
to remain on
the
activation
page. Cortex
Cloud sends
a

notification
to your
email when
the process
is complete.

📝🟢

note

You can only
change the
subdomain
once, and it
cannot be
undone.
After
deleting the
subdomain,
you can
reuse it after
7 days.

2.2.1.1.
Cortex
Cloud
supported
regions



2.2.1.2.
Enable
access to

required
PANW
resources



📝🟢

note

<tenant-
name>

refers to the
selected
subdomain
of your
Cortex
Cloud
tenant, and
<region> is
the region in
which your
tenant is
deployed.
For more
information,
see Cortex
Cloud
supported
regions.

📝🟢

note

Refer to
Regional
Docker
registry
mapping for
your specific
tenant
location and
correspondin
g Docker
registry URL.

📝🟢

note

For agents
on

endpoints,
you must
allow the IP
address for
the closest
region to
ensure

connectivity.
Endpoints
use latency-
based
routing. An
agent that
belongs to a
US tenant,
for example,
but that is
physically
located in
Singapore,
routes to
Singapore to
get the
verdict.

🧭➡️

navigation

SSL
decryption:
If you use
SSL
decryption
and
experience
difficulty
connecting
the Cortex
XDR agent
to the
server, we
recommend
that you add
the FQDNs
required for
access to
your SSL
Decryption
Exclusion list
in Device →
Certificate
Management
→ SSL
Decryption
Exclusion

2.2.1.2.1.
Regional
egress
resources



2.2.1.2.2.
Engines IP
addresses
(outbound)



2.2.1.2.3.
Inbound
source
resources



2.2.1.2.4.
FedRamp
and the US
Federal
Government
required
resources



2.2.2.
Upgrade
from Prisma
Cloud to
Cortex
Cloud



2.2.2.1.
About the
Upgrade
Helper

 📝🟢

note

Your Prisma
Cloud
tenant
remains
available
and fully
operational
during the
agreed
upgrade
period.

2.2.2.2. Link
Cortex
Cloud to
Prisma Cloud

 🧭➡️

navigation

Settings →
Configuratio
ns →

Upgrade
Helper

2.2.2.3. Copy
content



2.2.2.3.1.
Copy Global
configuration
s



📝🟢

note

Keep the
following
caveats in
mind:
Scope-
Based
Access
Control
(SBAC)
configuration
s, such as
resources or

account lists
are not

copied. You
can

manually
assign
scope-
based
access to
the relevant
users or

groups.
When
migrating
permission
groups and
roles, the
total count
of items
successfully
copied may
be lower
than the
initial
number
selected.
This...

🧭➡️

navigation

Settings →
Configuratio
ns → Access
Management

2.2.2.3.2.
Copy CSPM
configuration
s

 🧭➡️

navigation

Investigation
& Response
→

Automation

→

Automation
Rules

Posture
Management
→

Compliance
→ Standards

2.2.2.3.3.
Copy CWP
configuration
s

 🧭➡️

navigation

Posture
Management
→ Rules &
Policies →
Cloud
Workload

2.2.2.3.4.
Copy Cortex
Cloud
Application
Security
configuration
s

 👀🟠

important

If labels are
imported
into Cortex
Cloud less
than three
hours after
they were
last added
or modified
in Prisma
Cloud, they
may not be
included in
the import.

2.2.2.4.
Migrate
Cortex CLI

 ☑️🟣

prerequisite

Before you begin, ensure you have the following:
Cortex Cloud API key: An active API key for
your Cortex Cloud tenant with associated CLI
role permissions. Refer to Manage API keys for
more information
Install the Cortex CLI. You can find the
installation instructions here

2.2.3. Set up
users and
roles



📝🟢

note

The user
who
activated
the Cortex
product is
assigned the
Account
Admin role.
You cannot
create
additional
Account
Admin roles
in the Cortex
Cloud
tenant. If you
do not want
the user to
have
Account
Admin
permission,
you must
remove the
Account
Admin role
in Cortex
Gateway.

📝🟢

note

Relevant for
Behavioral
Analytics &
Identity
Threat
Detection
and
Response
(ITDR).

📝🟢

note

Relevant for
Behavioral
Analytics &
Identity
Threat
Detection
and
Response
(ITDR).

📝🟢

note

Relevant for
Identity
Posture &
Cloud
Infrastructur

e

Entitlements
(CIEM).

📝🟢

note

Relevant for
Identity
Posture &
Cloud
Infrastructur

e

Entitlements
(CIEM).

💡🔵

tip

To quickly
see exactly
which pages
and actions
a role
allows, click
on the role
name, which
opens a

read-only
view of all
checked
permissions.

🧭➡️

navigation

Settings →
Configuratio
ns → Access
Management
→ Roles

2.2.3.1. User
group
management



📝🟢

note

If users have
been
created in
the CSP, but
you want
them to
access the
tenant

through SSO
only, skip
this field and
add only
SAML group
mapping
after SSO is
set up,
otherwise,
users can

access the
tenant

through
both the
CSP and
SSO.
If you have
not yet
created any
users, skip
this field and
add them
later. See
Set up
authenticatio

n .

📝🟢

note

Capitalization is vital. String evaluation enforces
absolute case mapping rules. The name must
match your IdP's string configuration exactly.
When using Microsoft Entra ID for SSO, the
SAML group mapping needs to be provided
using the group object ID (GUID) and not the
group name.

Relevant strategy: For functional context and
the session mechanics of...

📝🟢

note

Visibility of
Security
domain
Issues that
refer to
assets with
agents is
controlled
by the
Endpoints
scoping
configuration
.

📝🟢

note

To
automaticall
y

synchronize
group

membership
with your
organization'
s Active
Directory,
you can
import an
AD group.
When
someone

joins or
leaves a
team in AD,
their Cortex
permissions
update
automaticall
y.

The Import
AD Group
feature is
only enabled
when the
Cloud
Identity
Engine (CIE)
is connected
and
configured.

📝🟢

note

Only CSP
and SSO
users

already
existing in
Cortex will
be imported.

📝🟢

note

When using
Microsoft
Entra ID for
SSO, the
SAML group
mapping
needs to be
provided
using the
group

object ID
(GUID) and
not the
group name.

📝🟢

note

If an
imported
group is
later deleted
from your
Active
Directory,
Cortex
Cloud
automaticall
y deletes the
correspondin
g user group
at the next
sync cycle.

👀🟠

important

The Cloud Identity Engine (CIE) is used
exclusively for directory group management; it
is not utilized for individual user account
management, provisioning, or authentication
workflows. Consequently, disabling, suspending,
or removing user objects directly within CIE
does not automatically disable, restrict, or
delete those corresponding users inside...

👀🟠

important

By default,
Enable
Scope
Based
Access
Control is
disabled in
Settings →
Configuratio
ns →

General →
Server
Settings, and
granular
scoping is
not
enforced.
Before
enabling
SBAC, we
recommend
that an
administrator
or a user

with Access
Management
permissions
first ensures
that the
users, user

groups, and
API Keys
defined in
Cortex
Cloud are
granted...

2.2.3.2.
Assign user
roles and
groups



📝🟢

note

If an existing
user in the
Cortex
Gateway no
longer has a
role or a
user group

assigned,
the user is
revoked. Any
roles, user
groups, or
egress

configuration
s created by
that user are
shown as
created by
Revoked
user instead
of the user's
email
address.

📝🟢

note

Visibility of
Security
domain
Issues that
refer to
assets with
agents is
controlled
by the
Endpoints
scoping
configuration
.

👀🟠

important

Setting
Cortex
Query
Language
(XQL)
dataset
access

permissions
for a user
role can
only be
performed
from Cortex
Cloud
Access
Management.
For more
information,
see Manage
user roles.

👀🟠

important

Before
configuring,
ensure that
you review
Understand
scoping in
the Manage
user scope
section.

👀🟠

important

By default,
Enable
Scope
Based
Access
Control is
disabled in
Settings →
Configuratio
ns →

General →
Server
Settings, and
granular
scoping is
not
enforced.
Before
enabling
SBAC, we
recommend
that an
administrator
or a user

with Access
Management
permissions
first ensures
that the
users, user

groups, and
API Keys
defined in
Cortex
Cloud are
granted...

💡🔵

tip

To apply the same settings to multiple users,
select them, and then right-click and select Edit
Users Permissions.

2.2.4.
Manage API
keys



📝🟢

note

Visibility of
Security
domain
Issues that
refer to
assets with
agents is
controlled
by the
Endpoints
scoping
configuration
.

👀🟠

important

Before
configuring,
ensure that
you review
Understand
scoping in
the Manage
user scope
section.

👀🟠

important

By default,
Enable
Scope
Based
Access
Control is
disabled in
Settings →
Configuratio
ns →

General →
Server
Settings, and
granular
scoping is
not
enforced.
Before
enabling
SBAC, we
recommend
that an
administrator
or a user

with Access
Management
permissions
first ensures
that the
users, user

groups, and
API Keys
defined in
Cortex
Cloud are
granted...

👀🟠

important

You will not
be able to
view the API
key again
after you
complete
this step.
Ensure that
you copy

the API key
before
closing the
notification.

2.2.5. Set up
authenticatio

n



💡🔵

tip

You should
have at least
one user in
the
Customer
Support
Portal for
backup, in
case of any
authenticatio
n issues with
your IdP
provider.

2.2.5.1.
Authenticate
users

through the
Customer
Support
Portal



📝🟢

note

You must be assigned the Super User role in the
Customer Support Portal to add users in the
Customer Support Portal.
The first Super User who logs into Cortex
Gateway is automatically assigned the Account
Admin role and has access to the tenant. The
user who activates the Cortex Cloud tenant will
also be assigned the Account Admin role (if
there is no...

📝🟢

note

After users
are added to
the
Customer
Support
Portal and
they accept
the
invitation,
you can
manage
them in
Cortex
Gateway or
the Cortex
Cloud tenant.

2.2.5.2.
Authenticate
users using
SSO



📝🟢

note

To set up
SSO
authenticatio
n in the
tenant, you
must be
assigned an
Instance
Administrato
r or Account
Admin role.
SAML 2.0
users must

log in to
Cortex
Cloud using
the FQDN
(full URL) of
the tenant.
To allow
login
directly
from the IdP
to , you must
set the relay
state on the
IdP to the
FQDN of the
tenant.

If you have
multiple
tenants, you
must set up...

📝🟢

note

The first
SSO cannot
be deleted,
it can only
be
deactivated
by toggling
SSO Enabled
to off.
The Domain
parameter is
predefined
for the first
SSO.
If you add
additional
SSO
providers,
you must
provide the
email
Domain in
the SSO
Integration
settings for
all providers
except the
first. Cortex
Cloud uses
this domain
to determine
to which
identity
provider...

📝🟢

note

Cortex Cloud requires the IdP to send the
group membership as part of the SAML token.
Some IdPs send values in a format that include
a comma, which is not compatible with Cortex
Cloud. In that case, you must configure your IdP
to send a single value without a comma for
each group membership. For example, if your
IdP sends the Group DN (a...

👀🟠

important

PROGRAMM
ATIC
CONTRAINT:
There is no
public API
endpoint
available to
provision or
de-provision
users

programmati
cally within
Cortex
Cloud. All
target
accounts
must be
initialized or
explicitly
managed
using the
native
interactive
Single Sign-
On (SSO) or
Customer
Support
Portal (CSP)
interface
workflows
defined in
this guide.
To review
the list of...

2.2.5.3. Set
up Okta as
the Identity
Provider
Using SAML
2.0

 📝🟢

note

When using SAML 2.0, users are required to
authenticate by logging in directly at the tenant
URL. They cannot log in via Cortex Gateway.

2.2.5.4. Set
up Microsoft
Entra ID as
the Identity
Provider
Using SAML
2.0



📝🟢

note

The default
attributes
shown on
the main
single sign-
on page in
Microsoft
Entra ID are
not the
values you
need. You
must click
Edit next to
Attributes
and Claims
to view and
copy the
actual
values.

📝🟢

note

The
attribute
claim name
must exactly
match the
value sent
by your IdP.
In some
cases, this
may be the
full attribute
name/names
pace,

depending
on the
configuration
of our IdP

📝🟢

note

When using SAML 2.0, users are required to
authenticate by logging in directly at the tenant
URL. They cannot log in via Cortex Gateway.

👀🟠

important

When
copying the
Single Sign-
On URL
value,
remove

idp/saml and
leave the
trailing /.
For example,
if the Single
Sign-On URL
is
https://client
name.panpro

duct.region.p
aloaltonetwo
rks.com/idp/s
aml, just
copy

https://client
name.panpro

duct.region.p
aloaltonetwo
rks.com/.

2.2.6. Cloud
service
provider
(CSP)
onboarding



2.2.6.1. Cloud
service
provider
onboarding



2.2.6.2.
Amazon
Web
Services
cloud
onboarding



2.2.6.2.1.
Onboard
Amazon
Web
Services



2.2.6.2.2.
Prerequisites

 📝🟢

note

The
permissions
included in
Sid:
CortexBaseM
oduleDeploy
ment are

always
required for
the
discovery
module and
stack
lifecycle
management.
The
permissions
included in
Sid:
CortexAuditL
ogsModule
are required
only when
deploying
the audit log
collection
module. It
enables
creating S3,
SQS, SNS,
CloudTrail,
and KMS
resources.

2.2.6.2.3.
How to
onboard
Amazon
Web
Services



📝🟢

note

Scanning
with an
outpost may
require
additional
AWS
permissions
and may
incur
additional
CSP costs.

👀🟠

important

It is critical
to ensure

that your
KMS key
region and
SNS topic
region are
the exact
same as the
AWS region
where you
are

deploying
the
CloudFormat
ion stack.

2.2.6.2.4.
Deploy the
CloudFormat
ion template
in AWS



2.2.6.2.5.
AWS post-
deployment
verification



2.2.6.3.
Microsoft
Azure cloud
onboarding



2.2.6.3.1.
Onboard
Microsoft
Azure



2.2.6.3.2.
Prerequisites



2.2.6.3.3.
How to
onboard
Microsoft
Azure



📝🟢

note

Scanning
with an
outpost may
require
additional
GCP
permissions
and may
incur
additional
CSP costs.

📝🟢

note

Cortex
Cloud
performs a
live
verification
of each
tenant's
approval
status
against
Azure each
time the list
is displayed.
If a
previously
approved
tenant no

longer
shows a
green
checkmark,
the Cortex
Cloud
application
may have
been
removed
from the
Azure
tenant. Click
Approve in
Azure to re-
authorize.

👀🟠

important

It is critical
to ensure

that your
namespace
and Event
Hub belong
to the
specific
Azure
subscription
being
onboarded.
Cross-
subscription
or

centralized
logging is
not

currently
supported.

2.2.6.3.4.
Finalize
Microsoft
Azure
onboarding
by executing
the
authenticatio
n template



☑️🟣

prerequisite

Before you
begin,
ensure you
have:
An Azure
subscription.
A user with
the required
permissions
for the
relevant
scope

(subscription,

management
group,
tenant). We
recommend
you create a
dedicated
role.
Tenant ID
and
subscription
ID. You can
view these in
Microsoft
Azure Portal
in
Management
groups.
Installed
Terraform on
your local
machine.
You can...

☑️🟣

prerequisite

Before you
begin,
ensure you
have:
An Azure
subscription.
A user with
the required
permissions
for the
relevant
scope

(subscription,

management
group,
tenant). We
recommend
you create a
dedicated
role.
Tenant ID
and
subscription
ID. You can
view these in
Microsoft
Azure Portal
in
Management
groups.
Installed the
Azure CLI
tool.
Authorization
to create...

☑️🟣

prerequisite

Before you
begin,
ensure you
have:
An Azure
subscription.
A user with
the required
permissions
for the
relevant
scope

(subscription,

management
group,
tenant). We
recommend
you create a
dedicated
role.
Tenant ID
and
subscription
ID. You can
view these in
Microsoft
Azure Portal
in
Management
groups.
Installed the
Azure CLI
tool.
Authorization
to create...

👀🟠

important

Do not
delete or
move the
Terraform
files from
this folder. It
will prevent
you from
being able
to edit your
cloud
instance in
the future.

2.2.6.4. GCP
cloud
onboarding



2.2.6.4.1.
Onboard
Google
Cloud
Platform



2.2.6.4.2.
Prerequisites



2.2.6.4.3.
How to
onboard
Google
Cloud
Platform



📝🟢

note

Scanning
with an
outpost may
require
additional
GCP
permissions
and may
incur
additional
CSP costs.

📝🟢

note

If you want to connect to your GCP Workspace,
you must first complete onboarding with the
option disabled. Once the GCP cloud instance
is created, perform the steps detailed in
Connect Google Workspace with your GCP
cloud instance.

2.2.6.4.4.
Deploy the
Terraform
authenticatio
n template
in GCP



☑️🟣

prerequisite

Before you
begin,
ensure you
have:
A GCP
account.
Permission
to create the
required
resources in
Google
Cloud
Deployment
Manager.
Installed
Terraform on
your local
machine.
You can
download
Terraform
from the
official
Terraform
website and
follow the
installation
instructions
for your
operating
system.
Installed the
GCP gcloud
CLI tool.
Reviewed
the...

📝🟢

note

The
directory
you create
must be a
subdirectory
of the home
directory.

👀🟠

important

You must not delete or move the Terraform files
from this folder. It will prevent you from being
able to edit your cloud instance in the future.

2.2.6.4.5.
Connect
Google
Workspace
with your
GCP cloud
instance



☑️🟣

prerequisite

Ensure you
have the
Super Admin
role in
Google
Workspace.

☑️🟣

prerequisite

Ensure you
have the
organization
ID of the
Google
Workspace
you want to
connect:
Log in to
your Google
Admin
Console. and
navigate to
Account →
Account
settings →
Profile. Next
to Customer
ID is your
organization
ID.
Ensure you
have the
organization
ID of the
Google
Workspace
you want to
connect:
Log in to
your Google
Admin
Console. and
navigate...

2.2.6.4.6.
Monitor
GCP
resources

inside
service
perimeters

 📝🟢

note

Each GCP
cloud
instance is
assigned a
scope within
GCP. If the
scope,
whether it
be
organization,
folder, or
project,
includes any
projects
with a
service
perimeter,
this
procedure
must be
performed
for that
cloud
instance to
authorize
Cortex
Cloud to
scan the
resources in
the
perimeter.

2.2.6.5.
Oracle
Cloud
Infrastructur
e cloud
onboarding



2.2.6.5.1.
Onboard
Oracle
Cloud
Infrastructur

e



2.2.6.5.2.
Prerequisites



2.2.6.5.3.
How to
onboard
Oracle
Cloud
Infrastructur

e



2.2.6.5.4.
Deploy the
Terraform
authenticatio
n template
in OCI

 ☑️🟣

prerequisite

Before you
begin,
ensure you
have:
An Oracle
Cloud
Infrastructur
e account
and the
tenancy
OCID.
Permission
to deploy a
custom

template
and create
its resources
in OCI.
Installed
Terraform on
your local
machine.
You can
download
Terraform
from the
official
Terraform
website and
follow the
installation
instructions
for your
operating
system.
Installed...

2.2.6.6.
Alibaba
Cloud cloud
onboarding



2.2.6.6.1.
Security
capabilities
and
deployment
planning



2.2.6.6.2.
Resource
inventory



2.2.6.6.3.
Security
model and
authenticatio

n



2.2.6.6.4.
Onboard
Alibaba
Cloud



2.2.6.6.5.
Prerequisites



2.2.6.6.6.
How to
onboard
Alibaba
Cloud



☑️🟣

prerequisite

Before you
begin,
ensure you
have:
Installed
Terraform on
your local
machine.
You can
download
Terraform
from the
official
Terraform
website and
follow the
installation
instructions
for your
operating
system.
Installed the
Alibaba
Cloud CLI
tool.

📝🟢

note

If the
following
error

appears,
contact
support:
"Validation
failed: No
valid
managed
outpost
found for
cloud
provider
ALIBABA_CL
OUD".

👀🟠

important

Do not
delete or
move the
Terraform
files from
this folder. It
will prevent
you from
being able
to edit your
cloud
instance in
the future.

2.2.6.6.7.
Alibaba
Cloud post-
deployment
verification



2.2.6.7.
Manually
connect a
cloud
instance



2.2.6.8.
Manage
cloud
instances

 📝🟢

note

Currently,
automation
permission
errors or

missing
automation
permissions
do not
affect the
Automation
health
status. You
can view any
permission
errors or

missing
permissions
in the the
Permissions
health status.

2.2.6.9.
Pending
cloud
instances



2.2.6.10. Edit
your
onboarded
CSP
configuration

 👀🟠

important

When using Terraform authentication
templates, you must execute the updated
Terraform template from the same folder where
the original Terraform template was executed.

2.2.6.11.
Update
cloud
permissions
after Cortex
release
updates



☑️🟣

prerequisite

Ensure that
the user
account
used to
modify
permissions
has the
necessary

privileges
within both
the Cortex
platform
and your
cloud
environment,
for example,
AWS or
Azure.
You received
a

notification
regarding a
new version
available
that requires
permission
updates, or
viewed a
Needs
Update
status in the
Data
Sources &
Integrations
page.

📝🟢

note

Instances
requiring
updates will
not change
their
connection
status, for
example,
Connected,
Warning,
Error,
Disabled,
due to the
pending
permission
update.

2.2.6.12.
Troubleshoot
errors on

cloud
instances

 📝🟢

note

Errors
related to
collection of
audit logs in
the cloud
instance are
recorded in
the
collection_au
diting
dataset. For
more

information,
see Audit
logs fields
and query
examples.

2.2.6.13.
Outposts



👀🟠

important

Outpost scan is an alternative to the
recommended standard cloud scan. Cloud scan
is recommended because it is fully managed by
Palo Alto Networks and incurs no compute
costs for your organization. Outpost scan is an
advanced deployment model reserved for
specific data residency or architectural
requirements.

2.2.6.13.1.
Outpost
fundamental
s and
planning



📝🟢

note

Notes:

Before you
create your
outpost,
verify that
your internet
connection
is active. An
active
internet
connection
is necessary
for the
notification
to be sent to
Cortex
Cloud to
create the
new outpost.
(Azure) Due
to limitations
in Terraform,
the Azure
subscription
name cannot
contain
blanks. Take
this into
account
while
onboarding.

👀🟠

important

While
outposts
provide
maximum
control over
the scanning
environment,
cloud scan
mode is the
recommende
d default for
most
organizations
.

2.2.6.13.2.
Create an
outpost



☑️🟣

prerequisite

Before you begin, ensure you have:
An AWS account
Permission to create a stack and its resources in
AWS
Installed Terraform on your local machine. You
can download Terraform from the official
Terraform website and follow the installation
instructions for your operating system.
Installed the AWS CLI tool and configured your
profile with the aws...

☑️🟣

prerequisite

Before you
begin,
ensure you
have:
A GCP
account
Permission
to create the
required
resources in
Google
Cloud
Deployment
Manager
Installed
Terraform on
your local
machine.
You can
download
Terraform
from the
official
Terraform
website and
follow the
installation
instructions
for your
operating
system.
Installed the
GCP gcloud
CLI tool

☑️🟣

prerequisite

Before you
begin,
ensure you
have:
An active
Azure
subscription.
Installed the
Azure CLI
tool.
Permission
to deploy a
custom

template
and create
its resources
in Microsoft
Azure
("Owner" or
"Contributor"
on the
designated
outpost
subscription
scope, and
Active
Directory
"Cloud
Application
Administrato
r" or
"Application
Administrato
r" privileged...

📝🟢

note

Verify that
your internet
connection
is active. An
active
internet
connection
is necessary
for
notifications
to be sent to
Cortex
Cloud for
creating the
new outpost.
If you are
unable to
establish an
internet
connection,
contact
customer

support for
a manual
workaround.

📝🟢

note

When
creating an
outpost for
a specific
Azure
subscription,
the outpost
account
must be in
the same
Azure
organization
as the
monitored
subscriptions
.

📝🟢

note

Due to
limitations in
Terraform,
the Azure
subscription
name cannot
contain
blanks.

👀🟠

important

While
outposts
provide
maximum
control over
the scanning
environment,
cloud scan
mode is the
recommende
d default for
most
organizations
. For details,
see When to
choose
outpost scan.

2.2.6.14.
Introduction
to Terraform
for Cloud
service
provider
(CSP)
onboarding



👀🟠

important

Always run
these
commands
in the same
folder where
the original
.tf files and
.terraform
folder live—
this is where
the state is
stored.

👀🟠

important

This .tfstate
state file is
critical
because it is
needed by
the
terraform
destroy
command to
clean up
created
resources. It
is critical
that you
never delete
this file.

2.2.6.15.
Cloud
service
provider
permissions



⚠️🟡

caution

Reducing
permissions
below the
recommende
d level may
cause

specific
commands
to fail or
limit
functionality
in future
pack
updates.

2.2.6.15.1.
Amazon
Web
Services
provider
permissions



📝🟢

note

Unified Cortex platform cloud content packs
require a specific set of automation permissions
to enable full integration with your cloud
environment. Before configuring access for
these packs, review the automation permission
scope guidelines.

👀🟠

important

All
conditional
capabilities
documented
below
require the
mandatory
Base and
Discovery
Engine
permissions
to be
deployed
alongside
them. Base
provides the
foundational
CortexPlatfo
rmRole and
AWS-
managed
read-only
baseline.
Discovery
Engine
extends that
baseline
with the
asset-
inventory
coverage

that every
other
capability
assumes.

2.2.6.15.2.
Microsoft
Azure
provider
permissions

 📝🟢

note

Unified Cortex platform cloud content packs
require a specific set of automation permissions
to enable full integration with your cloud
environment. Before configuring access for
these packs, review the automation permission
scope guidelines.

2.2.6.15.3.
Google
Cloud
Platform
provider
permissions

 📝🟢

note

Unified Cortex platform cloud content packs
require a specific set of automation permissions
to enable full integration with your cloud
environment. Before configuring access for
these packs, review the automation permission
scope guidelines.

2.2.6.15.4.
Oracle
Cloud
Infrastructur
e provider
permissions



2.2.7.
Onboard the
Kubernetes
Connector



📝🟢

note

This option
is not
supported
for Fargate.
📝🟢

note

Enabling
Realtime
Protection
installs the
agent on
your
Kubernetes
clusters as a
DaemonSet.

📝🟢

note

For Fargate,
you must
provide the
cluster
resource

identifier.
The format
of the
identifier is
arn:aws:eks:
<region>:
<account-
id>:cluster/<
cluster-
name>.

📝🟢

note

Basic
authenticatio
n is only
supported in
Posture
Management.
If deploying
Realtime
Protection,
select None .

📝🟢

note

If you install
the latest
version but
select the N-
1 strategy,
this policy
will take
effect
starting
from the
next

upgrade
cycle (it will
not

immediately
downgrade
your current
installation).
If you
choose an
older version
and keep the
latest
strategy, the
latest
version will
be installed.

2.2.7.1.
What's new
in
Kubernetes
Connector?



2.2.7.2.
Supported
Kubernetes
distributions



📝🟢

note

Does not
include
Autopilot.

📝🟢

note

In Google
Container-
Optimized
OS release
100 and
earlier,
where the
FANOTIFY
EXEC flag is
not

supported,
the Kernel
configuration
may be
partial for
the user
mode agent
to properly
function. In
such cases,
the agent
will fallback
to

asynchronou
s mode.
In RHCOS
version 4.12
and earlier,
the Kernel
configuration
may be
partial for
the user
mode agent
to...

2.2.8.
FedRAMP
overview



2.2.8.1.
Cortex
Cloud
federal
compliance



2.2.8.2.
Onboarding
&
configuration

 📝🟢

note

If you
choose an
outpost
scan, you
must select
an outpost
that has the
environment

type defined
as

Government
to maintain
compliance
and
connectivity.

2.2.8.3.
Limitations
& supported
regions



2.3. Post-
deployment
steps



2.3.1. Set up
your
environment



2.3.1.1.
Configure
server

settings



📝🟢

note

Keyboard
shortcuts,
timezone,
and
timestamp
format are
not set

universally
and only
apply to the
user who
sets them.

📝🟢

note

The Cortex
Agentic
Assistant
and AI case
summarizatio
n are

currently
available for
users in
limited
regions. For
more

information,
see Agentic
AI in Cortex
Cloud.
For multi-
tenant/MSSP
environments
, the Cortex
Agentic
Assistant
and AI case
summarizatio
n are not
available in
the main
tenant.

📝🟢

note

If the
Password
Protection
(for
downloaded
files) setting
under
Settings →
Configuratio
n → General
→ Server
Settings is
enabled,
enter the
password
'suspicious'
to download
the file.

📝🟢

note

In MSSP
environments
, this option
is not
available on
the parent
tenant.

🧭➡️

navigation

Settings →
Configuratio
ns →

General →
Server
Settings

2.3.1.2.
Configure
security
settings



2.3.1.3. Data
and log
forwarding



2.3.1.3.1.
Forward
logs and
data from
Cortex
Cloud to
external
services



2.3.1.3.1.1.
Configure
external
applications
for
forwarding

 📝🟢

note

There are
two options
for
configuring
external
applications.
To configure
relevant
external
applications
before you
begin
creating
forwarding
notifications,
follow the
steps in the
following
topics using
the menu
path
Settings →
Configuratio
ns →

Integrations
→ External
Applications.
You can also
configure an
external
application
as part of
the...

2.3.1.3.1.2.
Configure
notification
forwarding



☑️🟣

prerequisite

Before you
can select
an external
service for
notification
forwarding,
you must
integrate the
external
service with
Cortex
Cloud. For
more

information,
see

Configure
external
applications
for
forwarding.
No prior
configuration
is required
to send data
to an email
distribution
list.

📝🟢

note

Forwarding
destinations:
Only issues
and cases
can be
forwarded
to Slack,
Splunk,
Amazon
SQS,
Amazon S3,
or Webhook.
Notification
forwarding
by domain:
To configure
notification
forwarding
for issues by
domain,
select Issues
and filter the
Issues table
by Issue
Domain.
Alert vs.
issue
format:By
default, new
configuration
s use the
issue format,
but you...

📝🟢

note

Not all data
and log
types can be
sent to all
external
services. For
more

information,
see Forward
logs and
data from
Cortex
Cloud to
external
services.Forw
ard logs and
data from
Cortex
Cloud to
external
services

📝🟢

note

The
Grouping
Timeframe
defines the
time frame,
in minutes,
of how often
Cortex
Cloud sends
notifications.
Every 20
issues or 20
events

aggregated
within this
time frame
are sent

together in
one

notification,
sorted
according to
severity. To
send a
notification
when one
issue or
event is
generated,
set the time
frame to 0.

2.3.1.3.1.3.
Monitor
administrativ
e activity



2.3.1.3.2.
Data and log
notification
formats



📝🟢

note

Issues can
be
forwarded
to email,
syslog
servers, and
Slack in the
alert format,
if you prefer.
The alert
format can
be selected
when you
configure
your

forwarding
notification.

2.3.1.3.2.1.
Management
audit log
messages



2.3.1.3.2.2.
Issue
notification
format

 📝🟢

note

For issues
with relevant
assets, issue
notifications
sent to
Amazon S3,
Amazon
SQS,
Webhook,
Splunk, and
email
provide
asset and
remediation
information
including the
asset name,
cloud
resource

name, asset
tags,
account
name,

region, and
evidence.

2.3.1.3.2.3.
Agent Audit
log
notification
format



2.3.1.3.2.4.
Management
Audit log
notification
format



2.3.1.3.2.5.
Log format
for IOC and
BIOC issues



2.3.1.3.2.6.
Analytics
log format



2.3.2. Cortex
MCP server

 2.3.2.1.
Cortex MCP
server

overview



📝🟢

note

The Cortex
MCP Server
empowers
you to
integrate AI
into your
security
workflows
using natural
language.
When using
LLM-based
suggestions,
always
review and
approve
actions
suggested
by the AI
before
they're
executed.
We
recommend
deploying
the Cortex
MCP server
in a secure
environment
where
access is
limited to
authorized
users.

2.3.2.1.1.
Install the
Cortex MCP
server



☑️🟣

prerequisite

If you are
running the
Cortex MCP
server in a
Poetry
virtual
environment,
you must
have Python
3.13 or higher.
If you plan
to run the
Cortex MCP
server in a
Docker
container,
you must
have Docker
installed.

📝🟢

note

The MCP
Server uses
public APIs
to
communicat
e and is
limited by
the license
quotas
available in
your tenant.
This is
particularly
relevant
when
running XQL
queries. For
more

information
on running
XQL query
APIs, see
Run XQL
query APIs.

📝🟢

note

It is critical
to avoid
assigning
excessive
permissions
when
creating an
API key for
the Cortex
MCP Server.
Since the
key has both
read and
write
capabilities,
overly broad
permissions
can lead to
unintended
actions and
potentially
compromise
your
environment.
Ensure the
key follows
the principle
of least
privilege and
is granted
only the
minimum...

📝🟢

note

When using Docker, we recommend using an
.env file to set the Cortex API credentials as
environment variables. While the credentials
can be provided in the MCP client configuration
settings, the .env file provides safer handling of
API credentials and makes your configuration
easily reproducible.

📝🟢

note

By default, stdio (standard input/output) is
used. You can also configure Streamable HTTP,
to send requests directly to the tenant instead
of through the MCP client. Streamable HTTP
can be useful for testing in the browser without
a MCP client and to bypass limits that may be in
place for your MCP client. For Docker, you can
include the Streamable HTTP...

👀🟠

important

To configure
the Cortex
MCP Server,
you need the
Cortex API
URL, Cortex
API key, and
Cortex API
key ID. You
will not be
able to view
the API key
again after
you

complete
this step.
Ensure that
you copy

the API key
before
closing the
notification.

2.3.2.1.2.
Configure
the MCP
client

 🧭➡️

navigation

Settings →
Developer
→ Edit
Config
2.3.2.1.3. Use
the Cortex
MCP server

 📝🟢

note

The built-in
tools
retrieve
information,
but do not
write to the
tenant. You
can create
your own
tools that
include write
actions. The
examples
below
include both.

2.3.2.1.4.
Create
custom
Cortex MCP
server tools

 📝🟢

note

Any new or
updated
components
provided by
Cortex are
automaticall
y
downloaded
into the
remote_com
ponents
folder.
During each
update, the
folder is
fully
replaced
and all
existing
contents are
recreated.
Do not add
custom
tools to this
directory, as
it is
managed
entirely by
Cortex and
is
overwritten
at every
update.

2.3.3.
Manage user
roles and
access

management



☑️🟣

prerequisite

Managing
users, roles,
scopes, user
groups,
authenticatio
n settings in
Cortex
Cloud
Access
Management
requires
View/Edit
RBAC
permissions
for Access
Management
(under
Configuratio
ns). Account
Admin and
Instance
Administrato
r roles are
granted this
permission
by default.
For more
information,
see

Predefined
user roles in
Set up users
and roles.

📝🟢

note

To remove
users added
to your CSP
account,
you must do
so in the
CSP, not in
Cortex
Gateway.

📝🟢

note

For more
information
on assigning
user roles
when
generating
an API key,
see Manage
API keys.

📝🟢

note

Some
features are
license-
dependent.
Accordingly,
users may
not see a

specific
feature if
the feature
is not
supported
by the
license type
or if they do
not have
access

based on
their
assigned
role or
scope.

📝🟢

note

For features
where
scoping is
not

applicable,
Role-Based
Access
Control
(RBAC) is
used and
can be
configured
when
managing
user roles.
For more
information,
see Manage
user roles.

2.3.3.1.
Manage user
roles

 ☑️🟣

prerequisite

Managing
user roles in
Cortex
Cloud
Access
Management
requires
View/Edit
RBAC
permissions
for Access
Management
(under
Configuratio
ns). Account
Admin and
Instance
Administrato
r roles are
granted this
permission
by default.
For more
information,
see

Predefined
user roles in
Set up users
and roles.

2.3.3.2.
Manage user
access



☑️🟣

prerequisite

Managing
users, roles,
scopes, user
groups,
authenticatio
n settings in
Cortex
Cloud
Access
Management
requires
View/Edit
RBAC
permissions
for Access
Management
(under
Configuratio
ns). Account
Admin and
Instance
Administrato
r roles are
granted this
permission
by default.
For more
information,
see

Predefined
user roles in
Set up users
and roles.

📝🟢

note

You can only
reduce the
permissions
of an
Account
Admin user
via Cortex
Gateway.

📝🟢

note

User
permissions
for
components
and datasets
are based on
the access
permissions
set in the
user role.
For more
information
on editing
these user
role
permissions,
see Manage
user roles.

📝🟢

note

Visibility of
Security
domain
Issues that
refer to
assets with
agents is
controlled
by the
Endpoints
scoping
configuration
.

👀🟠

important

Before
configuring,
ensure that
you review
Understand
scoping in
the Manage
user scope
section.

👀🟠

important

By default,
Enable
Scope
Based
Access
Control is
disabled in
Settings →
Configuratio
ns →

General →
Server
Settings, and
granular
scoping is
not
enforced.
Before
enabling
SBAC, we
recommend
that an
administrator
or a user

with Access
Management
permissions
first ensures
that the
users, user

groups, and
API Keys
defined in
Cortex
Cloud are
granted...

💡🔵

tip

To apply the same settings to multiple users,
select them, and then right-click and select Edit
User Permissions.

💡🔵

tip

To apply the same settings to multiple users,
select them, and then right-click and select Edit
User Permissions.

💡🔵

tip

To apply the same settings to multiple users,
select them, and then right-click and select Edit
User Permissions.

🧭➡️

navigation

Settings →
Configuratio
ns → Access
Management
2.3.3.2.1. User → Users
access

reference
information



2.3.3.3.
Manage user
scope



☑️🟣

prerequisite

Configuring
user scopes
in Cortex
Cloud
Access
Management
requires
View/Edit
RBAC
permissions
for Access
Management
(under
Configuratio
ns). Account
Admin and
Instance
Administrato
r roles are
granted this
permission
by default.
For more
information,
see

Predefined
user roles in
Set up users
and roles.
By default,
Enable
Scope
Based
Access
Control is...

📝🟢

note

Use the
existing
Realm
attribute
whenever
you need to
scope based
on the
Account ID.

📝🟢

note

This
configuration
can impact
the visibility
of the
related
Security
domain in
the Cases
and Issues
scope area,
but with not
affect asset
visibility.

📝🟢

note

XQL-based
dashboard
widgets may
require a
few hours to
initially
reflect
changes to
the list or
definitions
of asset
groups used
for scoping.
To view the
most current
data
immediately,
refresh the
dashboard
or its XQL
widgets.

📝🟢

note

For Cases
and Issues
domains, a
workaround
is to create
a Dataset
View for
each
required
combination
of domains,
and allow
the relevant
entity
access only
to this
Dataset
View, not to
the
underlying
cases and
issues
datasets.

📝🟢

note

Make sure to
assign the
required
default
granular
scoping for
users. This
depends on
the
structure
and divisions
within your
organization
and the
particular
purpose of
each
organization
al unit to
which
scoped
users belong.

📝🟢

note

Visibility of
Security
domain
Issues that
refer to
assets with
agents is
controlled
by the
Endpoints
scoping
configuration
.

👀🟠

important

Some areas
and features
in Cortex
Cloud do
not respect
SBAC. In
these cases,
use RBAC
permissions
to restrict
access.

👀🟠

important

Before
configuring,
ensure that
you review
the
Understand
scoping
section.

2.3.3.4.
Manage
access to

objects



📝🟢

note

Only the
Owner or an
Administrato
r can delete
a custom

object.

👀🟠

important

While Per-
object
access

controls the
visibility of
the object
(such as a
dashboard
or saved
query), the
underlying
data remains
governed by
Scope-
Based
Access
Control
(SBAC). A
user must
have the
appropriate
SBAC
permissions
to view the
data
available
through an
object.

2.3.3.4.1.
Manage
access to
custom
dashboards



☑️🟣

prerequisite

Configure
tenant-level
settings: An
administrator
must first
establish the
sharing
framework
under
Settings →
Configuratio
ns → Access
Management
→ Objects.
The
configuration
of these
settings
defines the
authorized
sharing
workflows
for for all
custom

objects,
including
dashboards:
Enable
"Owners can
Share
objects they
created":
Grants
owners...

📝🟢

note

Only Account Admins and Instance
Administrators have the authority to change the
owner of an object.

📝🟢

note

If you're
designated
as an Editor,
you can

always
duplicate
the widget
and make
your

changes on
the copy.

2.3.3.4.2.
Manage
access to
report
templates



☑️🟣

prerequisite

Configure
tenant-level
settings: An
administrator
must first
establish the
sharing
framework
under
Settings →
Configuratio
ns → Access
Management
→ Objects.
The
configuration
of these
settings
defines the
authorized
sharing
workflows
for all
custom

objects,
including
report
templates:
Enable
"Owners can
Share
objects they
created":
Grants...

📝🟢

note

Only Account Admins and Instance
Administrators have the authority to change the
owner of an object.

2.3.3.4.3.
Manage
access to

playbooks
and scripts



☑️🟣

prerequisite

Configure
tenant-level
settings: An
administrator
must first
establish the
sharing
framework
under
Settings →
Configuratio
ns → Access
Management
→ Objects.
The
configuration
of these
settings
defines the
authorized
sharing
workflows
for all
custom

objects,
including
report
templates:
Enable
"Owners can
Share
objects they
created":
Grants...

📝🟢

note

Playbooks
can only be
Enabled
when Scripts
are Enabled
first.

📝🟢

note

To detach a
playbook or
script, a user
must have
both the
Create and
the Edit
Public sub-
permissions
assigned to
their role.
Reattaching
an existing
automation
to a

Marketplace
content

pack can be
performed
by the
playbook or
script Owner
or an

Administrato

r.

2.3.3.4.4.
Manage
access to
saved
queries



☑️🟣

prerequisite

Configure
tenant-level
settings: An
administrator
must first
establish the
sharing
framework
under
Settings →
Configuratio
ns → Access
Management
→ Objects.
The
configuration
of these
settings
defines the
authorized
sharing
workflows
for saved
queries in
the Query
Library,
including the
options that
appear to
users when
clicking the
three dot,...

📝🟢

note

If the role of
a user is set
to Edit
Public
Queries but
not Create
Queries,
they can
update
existing
public
queries, but
the Save as
drop-down
menu will be
hidden,
preventing
them from
creating new
Query
Library
entries.

📝🟢

note

When the
tenant-level
setting
Owners and
editors can
change the
general
access is
unselected,
the drop-
down is
disabled and
only an
administrator
can

configure
this option.

2.3.4.
Configure
the Cortex
Agentic
Assistant



2.3.4.1.
Agentic
Assistant
components
and
concepts

 📝🟢

note

Agents are
bound by
the same
rules and
robust
permissions
as a human
user. In
addition, you
can mark
actions that
make real-
world
changes in
production
systems as
sensitive,
requiring a
quick
manual
review and
confirmation,
ensuring
peace of
mind before
critical
system
changes are
made.

2.3.4.2.
Agents Hub



📝🟢

note

To manage
agents in the
Agents Hub,
you must
have the
proper
permissions.
For more
information,
see Agentic
Assistant
role-based
access

control.

2.3.4.2.1.
Manage
actions



📝🟢

note

To manage
actions in
the Agents
Hub, you
must have
the correct
permissions.
For more
information,
see Agentic
Assistant
role-based
access

control.

💡🔵

tip

System
actions may
rely on
content

packs that
need to be
installed and
configured.

2.3.4.2.2.
Register
actions



📝🟢

note

To register
actions,
ensure you
have the
correct
permissions.
For more
information,
see Agentic
Assistant
role-based
access

control.

📝🟢

note

Mandatory
arguments
cannot be
deselected.

📝🟢

note

The
execution of
system or
custom
actions that
are based on
integration
commands
can be
restricted
using
integration
permissions.

👀🟠

important

A single
content item
can be
registered as
different
actions, with
each action
using
different
inputs and
outputs
from the
same script.
Only register
the same
script,
command,
or AI prompt
as a new

action if it is
required for
your use
case, as

providing an
agent with
many
actions with
overlapping
abilities can
reduce the
ability of the
agent to
choose...

2.3.4.2.3.
Manage
agents



📝🟢

note

In some
cases, an

agent may
include
actions with
content
items that
are not
relevant for
all licenses.
If that
occurs, the
grey circle
appears, but
you are not
able to
install the
related
content.

💡🔵

tip

System
agents can
include
actions that
require
additional
content

packs to be
installed and
configured.
To view all
actions
assigned to
a system
agent,
including
actions not
available
due to
missing
content,
click on the
system
agent in the
Agents Hub.
There may
be actions
assigned to
a system
agent that
are not
relevant to
your

organization.
For...

2.3.4.2.4.
Build agents



📝🟢

note

To build
agents in the
Agents Hub,
you must
have
view/edit
permissions.
For more
information,
see Agentic
Assistant
role-based
access

control.

📝🟢

note

If you clear the checkbox for a role, all actions
associated with that role are also cleared. The
exception is if another role is also selected,
which is associated with the same actions.
If you clear the checkbox for an action, all roles
associated with that action are cleared. For
example, if you select the Investigator role, and
Send Mail and Tavily...

2.3.4.2.5.
Expand
agent
capabilities
with MCP
integrations



📝🟢

note

For agentic
actions to
be created
from tools
on an MCP
server, the
tools must
have input
and output
descriptions
on the MCP
server. These
descriptions
are required
for Cortex
Cloud to
understand
the tools'
capabilities
and the
correct use
cases.

📝🟢

note

If you are
using OAuth-
based
authenticatio
n, the Test
button
returns an
error

containing
the
command to
run in the
playground
in order to
test the
connection.

📝🟢

note

If an MCP
tool is
removed
from the
MCP server,
the action
will be
unavailable
due to
missing
content. If
the tool is
restored on
the MCP
server, the
action is
automaticall
y reenabled.

2.3.4.3.
Agentic
Assistant
role-based
access

control

 📝🟢

note

Agents are
limited by
the
individual
permissions
of the user.
For example,
if users do
not have
sufficient
permissions
to isolate an
endpoint,
they cannot
use an agent
to isolate an
endpoint.
The
execution of
system or
custom
actions that
are based on
integration
commands
can be
restricted to
specific
roles using
integration
permissions.

2.3.5. XQL
query
management



☑️🟣

prerequisite

Setting
query limits
requires
View/Edit
permissions
for
Configuratio
ns → Query
Management.

📝🟢

note

A scheduled
query or
report is run
on behalf of
the user that
originally
created it,
even if it is
edited and
run by
another user.

📝🟢

note

Queries run
by
correlation
rules are not
restricted by
the query
limit.
Very short
queries do
not count
towards
concurrent
queries.

📝🟢

note

To ensure
optimal
system
performance,
all queries
(user-
generated
and
otherwise)
adhere to a
default
timeout limit
of 60
minutes that
is defined by
Palo Alto
that takes
priority over
the
administrator
defined
value.
Therefore,
regardless
of the value
specified in
this field,
queries will
be stopped
after 60
minutes.
You can
override the
default
timeout...

📝🟢

note

Query
visibility is
subject to
Role-Based
Access
Control
(RBAC);
users can't
see queries
for datasets
they do not
have
permission
to access.

🧭➡️

navigation

You can find
Query
Management
options
under
Settings →
Configuratio
ns →

General →
Query
Management

2.3.6.
Customize
cases and
issues

 2.3.6.1.
Customize
cases and
issues



2.3.6.1.1. Set
up case
scoring



📝🟢

note

For scoped
users, a
small lock
icon
indicates
that you
don't have
permissions
to edit a rule.

💡🔵

tip

Right-click
an issue field
to add it as
match
2.3.6.1.2. criteria.
Create a
starring
configuration

 💡🔵

tip

Right-click
an issue field
to add it as
match
2.3.6.1.3. criteria.
Service
Level
Agreements
(SLAs) for
issue
resolution

 📝🟢

note

These filters
support
filtering of
whole days
only, for
example
Resolution
SLA > 1
filters for
issues that
have a
Resolution
SLA of
greater than
one day.

2.3.6.1.4.
Create issue
exceptions



☑️🟣

prerequisite

You must
have
Exception
Approver
Admin
permission
to add or
delete issue
exception
approvers
and to turn
off and on
approver

functionality.
You must
have
Exception
Management
Admin
permission
to create
issue
exception
rules, delete
or disable
exception
rules, and
view the All
Exception
Rules page.

📝🟢

note

Issue
exceptions
apply
specifically
to issues,
not
individual
findings.
Issue
exceptions
do not
automaticall
y update
standard
dashboards,
reporting
widgets, or
compliance
reporting
profiles. To
view the
aggregate
impact of
exceptions,
you must
manually
filter the
issues table.
Exceptions
do not alter
Attack Path
policies.
Exceptions
do not
bypass any...

2.3.6.1.4.1.
Configure
the issue
exception
approval
workflow



☑️🟣

prerequisite

You must
have
Exception
Approver
Admin
permission
to add or
delete
approvers.

☑️🟣

prerequisite

You must
have
Exception
Approver
Admin
permission
to add or
delete
approvers.

☑️🟣

prerequisite

You must
have the
Exception
Approver
Admin
permission
to disable or
re-enable
the issue
exception
approval
workflow.

2.3.6.1.4.2.
Create an
issue
exception
rule

 📝🟢

note

You can
create a
maximum of
10 issue
exceptions
per day.

2.3.6.1.4.3.
Create an
exception
rule from an
issue

 📝🟢

note

You can
create a
maximum of
10 issue
exception
rules per day.

2.3.6.1.4.4.
View issue
Exception
Rules



2.3.6.1.4.5.
Disable issue
exception
rules

 📝🟢

note

You cannot
re-enable an
issue
exception
rule after it
has been
disabled.

2.3.6.1.4.6.
View
excepted
issues



2.3.6.1.5.
Optimize
case

grouping in
correlations

 📝🟢

note

Field
mapping
does not
guarantee
grouping.
Cortex
Cloud
grouping
uses

machine
learning and
platform
logic that
evaluates
confidence,
context,
relationships,
and
additional
case signals.
Field
mapping
helps
influence
grouping by
contributing
relevant
artifacts, but
final
grouping
decisions are
determined
by the
platform's
overall
grouping
logic.
For...

2.3.7.
Dashboards
and reports



3.

Endpoint

security



3.1. Endpoint
protection



3.1.1.
Malware
protection



3.1.2. Exploit
protection



3.1.3. File
analysis and
protection
flow

 📝🟢

note

To enhance
performance
and
efficiency,
hash verdict
requests
from the
Cortex XDR
agent will be
routed to
the WildFire
service with
the lowest
latency. File
uploads for
analysis will
strictly
adhere to
the
designated
Cortex
Cloud and
WildFire
regions,
ensuring
data remains
within the
appropriate
geographical
boundaries.

3.1.4.
Endpoint
protection
capabilities



3.1.5.
Endpoint
protection
modules



3.1.6.
Processes
protected
by exploit
security
policy



3.1.7. File
Integrity
Monitoring
(FIM)



🔒🟣

content

File Integrity
Monitoring
usage
requires a
dedicated
FIM add-on
or the Cloud
Runtime
Security
add-on, with
XDR Pro
capabilities
enabled.

📝🟢

note

It is
recommende
d to create a
policy that
targets only
the
necessary
files and
folders.

📝🟢

note

Platform
cannot be
changed
once a rule
group has
been created

📝🟢

note

A rule group
can contain
up to 100
rules.

📝🟢

note

When
importing a
policy,
select
whether to
enable the
associated
policy
targets.
Rules within
the imported
policy are
managed as
follows:
New rules
are added to
the top of
the list.
Default rules
override the
default rule
in the target
tenant.
Rules
without a
defined
target are
disabled
until the
target is
specified.

📝🟢

note

It is possible
to ingest up
to 15,000
events per
day (24
hours) for
each
host/contain
er.

🧭➡️

navigation

Creating,
modifying
and viewing
FIM rule
groups and
rules is be
done in the
Rule Groups
page,
located at
the Inventory
→ Endpoints
→ File
Integrity
Monitoring
menu

In Inventory
→ Endpoints
→ Policy
management
→

Extensions
→ Profiles,
select +Add
Profile and
then select
either
Create New
or Import
from File

3.1.8. CaaS
Workloads



🔒🟣

content

Requires the
Cortex
Cloud
Runtime
Security
add-on.
Every 10
container-
embedded
agents will
consume a

single
Cortex
Runtime
Security
license.

🧭➡️

navigation

Inventory →
Endpoints →
Installations,
click Create
3.1.9.
WildFire
analysis
concepts

 📝🟢

note

In cases
when the
Cortex
Cloud agent
gets a failed
status from
the WF
service due
to a general
error or

unsupported
file type,
and the
Local
Analysis is
set to
disabled or
not

applicable,
Cortex
Cloud will
not generate
an alert on
the file.

3.1.10.
Guidelines
for keeping
Cortex XDR
agents and
content

updated



📝🟢

note

The Cortex
XDR agent
keeps the
name of the
original
installation
package
after every
upgrade.

📝🟢

note

Custom
dashboards
that include
upgrade
status

widgets, and
the All
Endpoints
page display
upgrade
status.

During the
upgrade
process, the
endpoint
operating
system
might
request a
reboot.
However,
you do not
have to
perform the
reboot for
the Cortex
XDR agent
upgrade
process to
complete it
successfully.
After you
upgrade on
an endpoint
with Cortex...

📝🟢

note

These
profiles can
be
configured
on one or
more

endpoints,
static/dynami
c groups,
tags, IP
ranges,

endpoint
names, or
other
parameters
that allow
the creation
of logical
endpoint
groups. See
how to
define
endpoint
group.

📝🟢

note

These
profiles can
be
configured
on one or
more

endpoints,
static/dynami
c groups,
tags, IP
ranges,

endpoint
names, or
other
parameters
that allow
the creation
of logical
endpoint
groups.

👀🟠

important

When
planning
Cortex XDR
agent
upgrades
and content
updates,
consult with
the
appropriate
stakeholders
and teams
and follow
the change
management
strategy in
your
organization.

⚠️🟡

caution

Before
enabling
Auto-

Update for
Cortex XDR
agents, make
sure to
consult with
all relevant
stakeholders
in your
organization.

3.1.11. About
content

updates

 📝🟢

note

Cortex
Cloud
delivers the
content

update to
the agent in
parts and
not as a

single file,
allowing the
agent to
retrieve only
the updates
and
additions it
needs.

3.1.12.
Endpoint
data
collection



📝🟢

note

For specific
files only
and only if
the file was
written.

👀🟠

important

Registry key is collected as a real key name, and
not as a symbolic link.
Instead of
HKEY_LOCAL_MACHINE\System\CurrentControl
Set, which is a symbolic link,
KEY_LOCAL_MACHINE\System\ControlSet001
will be collected.
Instead of HKEY_CURRENT_USER, HKEY_USERS\
<SID> will be collected, where SID is a SID of
the current user.

3.2. Install
and manage
endpoints



3.2.1. Set up
endpoint
protection



3.2.1.1. Set up
endpoint
profiles and
exception
rules



📝🟢

note

Profiles
associated
with one or
more targets
that are
beyond the
scope of
your defined
user

permissions
are locked,
and cannot
be edited.

3.2.1.1.1. Set
up malware
prevention
profiles



🔒🟣

content

Requires the
ITDR add-on.

📝🟢

note

In this
profile, the
Report
options
configure
the
endpoints to
report the
correspondin
g suspicious
files,
actions,
processes,
or behaviors
to Cortex
Cloud,
without
blocking
them. The
Disabled
options
configure
the
endpoints to
neither
analyze nor
report the
correspondin
g malware
or behavior.

📝🟢

note

New profiles
based on
imported
profiles are
added, and
do not
replace
existing ones.

📝🟢

note

As part of
the anti-
malware
security
flow, the
Cortex XDR
agent
leverages
the
operating
system's
capability to
identify
revoked
certificates
for
executables,
and DLL files
that attempt
to run on the
endpoint by
accessing
the Windows
Certificate
Revocation
List (CRL).
To allow the
Cortex XDR
agent
access the
CRL, you
must enable
internet
access over...

📝🟢

note

The
Quarantine
Malicious
Executables
feature is
not available
for malware
identified on
network
drives.

📝🟢

note

The
Quarantine
Malicious
Script Files
feature is
not available
for malware
identified on
network
drives.

📝🟢

note

If on-write
actions were

configured
in earlier
versions of
Cortex
Cloud, the
same

configuration
has been
preserved
and applied
globally for
all file types.
On-write file
protection
may have an
impact on
the
resources

required by
the Cortex
XDR agent.

📝🟢

note

On-demand
file
protection
may have an
impact on
the
resources

required by
the Cortex
XDR agent.

📝🟢

note

We
recommend
that you
disable
scheduled
scanning.
VDI machine
scans are

based on the
golden
image and
additional
files will be
examined
upon
execution.

📝🟢

note

When
periodic
scanning is
enabled in
your profile,
the Cortex
XDR agent
initiates an
initial scan
when it is
first
installed on
the
endpoint,
regardless
of the
periodic
scanning
scheduling
time.

📝🟢

note

This feature
only comes
into effect
after a

📝🟢 restart.

note

When you
are actively
evaluating
with BAS
tools, it is
recommende
d to enable
the BAS
mode
setting only
for the
duration of
your
evaluation,
and for a
limited
number of
agents.

📝🟢

note

BAS tools
mode with
content
older than
version 1850
cannot be
configured,
the agent
will be
treated as
Enabled.

📝🟢

note

New profiles
based on
imported
profiles are
added, and
do not
replace
existing ones.

📝🟢

note

We
recommend
that you
disable
scheduled
scanning.
VDI machine
scans are

based on the
golden
image and
additional
files will be
examined
upon
execution.

📝🟢

note

When
periodic
scanning is
enabled in
your profile,
the Cortex
XDR agent
initiates an
initial scan
when it is
first
installed on
the
endpoint,
regardless
of the
periodic
scanning
scheduling
time.

📝🟢

note

The
Quarantine
Malicious
Mach-O
Files feature
is not
available for
malware
identified on
network
drives.

📝🟢

note

This module
is supported
by Cortex
XDR agent
8.1.0 and
later
releases.

📝🟢

note

The
Quarantine
Malicious
Executables
feature is
not available
for malware
identified on
network
drives.

📝🟢

note

When you
are actively
evaluating
with BAS
tools, it is
recommende
d to enable
the BAS
mode
setting only
for the
duration of
your
evaluation,
and for a
limited
number of
agents.

📝🟢

note

BAS tools
mode with
content
older than
version 1850
cannot be
configured,
the agent
will be
treated as
Enabled.

📝🟢

note

New profiles
based on
imported
profiles are
added, and
do not
replace
existing ones.

📝🟢

note

The
Quarantine
Malicious
ELF Files
feature is
not available
for malware
identified on
network
drives.

📝🟢

note

This module
is supported
by Cortex
XDR agent
8.1.0 and
later
releases.

📝🟢

note

Enabling on-
demand
scanning will
automaticall
y scan these
core system
directories:
/etc, /tmp,
/home, /usr,
/bin, /sbin,
/lib, /var,
/opt, /dev,
/root, /boot.

📝🟢

note

We
recommend
that you
disable
scheduled
scanning.
VDI machine
scans are

based on the
golden
image and
additional
files will be
examined
upon
execution.

📝🟢

note

When
periodic
scanning is
enabled in
your profile,
the Cortex
XDR agent
initiates an
initial scan
when it is
first
installed on
the
endpoint,
regardless
of the
periodic
scanning
scheduling
time.

📝🟢

note

When you
are actively
evaluating
with BAS
tools, it is
recommende
d to enable
the BAS
mode
setting only
for the
duration of
your
evaluation,
and for a
limited
number of
agents.

📝🟢

note

BAS tools
mode with
content
older than
version 1850
cannot be
configured,
the agent
will be
treated as
Enabled.

⛔🔴

warning

For optimal
user

experience,
we

recommend
that you set
the action
mode to
either Allow
or Run Local
Analysis.

⛔🔴

warning

For optimal
user

experience,
we

recommend
that you set
the action
mode to
either Allow
or Run Local
Analysis.

⛔🔴

warning

For optimal
user

experience,
we

recommend
that you set
the action
mode to
either Allow
or Run Local
Analysis.

⛔🔴

warning

For optimal
user

experience,
we

recommend
that you set
the action
mode to
either Allow
or Run Local
Analysis.

3.2.1.1.2. Set
up exploit
prevention
profiles



📝🟢

note

In this
profile, the
Report
options
configure
the
endpoints to
report the
correspondin
g exploit
attempts to
Cortex
Cloud,
without
blocking
them. The
Disabled
options
configure
the
endpoints to
neither
analyze nor
report the
correspondin
g malware
or behavior.

📝🟢

note

New profiles
based on
imported
profiles are
added, and
do not
replace
existing ones.

📝🟢

note

In exploit
prevention
profiles, if
you change
the action
mode for
processes,
you must
restart the
protected
processes
for the
following
security
modules to
take effect
on the
process and
its forked
processes:
Brute Force
Protection
Java
Deserializatio

n

SO Hijacking

📝🟢

note

This step
provides a
temporary
workaround
for the
following
publicly
known
information-
security
vulnerabilitie
s and
exposures:
CVE-2021-
24074, CVE-
2021-24086
and CVE-
2021-24094.

📝🟢

note

If you have
Windows
endpoints in
your
network that
are

unpatched
and exposed
to a known
vulnerability,
we strongly
recommend
that you
upgrade to
the latest
Windows
Update that
has a fix for
that
vulnerability.

📝🟢

note

New profiles
based on
imported
profiles are
added, and
do not
replace
existing ones.

📝🟢

note

In exploit
prevention
profiles, if
you change
the action
mode for
processes,
you must
restart the
protected
processes
for the
following
security
modules to
take effect
on the
process and
its forked
processes:
Brute Force
Protection
Java
Deserializatio

n

SO Hijacking

📝🟢

note

New profiles
based on
imported
profiles are
added, and
do not
replace
existing ones.

📝🟢

note

In exploit
prevention
profiles, if
you change
the action
mode for
processes,
you must
restart the
protected
processes
for the
following
security
modules to
take effect
on the
process and
its forked
processes:
Brute Force
Protection
Java
Deserializatio

n

SO Hijacking

⛔🔴

warning

This
workaround
applies only
to the
specific
Windows
versions
listed as
exposed to
these CVEs,
and requires
a Cortex
XDR agent
release 7.1 or
later and
content 167-
51646 or
later. This
workaround
is not
recommende
d for non-
persistent,
stateless, or
linked-clone
environments
. In some
cases,

enabling this
workaround
can affect
the network
functionality..
.

3.2.1.1.3. Set
up agent
settings
profiles



🔒🟣

content

Requires a
Cortex XDR
Pro per
Endpoint
license.
When you
enable this
feature, a
Cortex XDR
Pro per
Endpoint
license is
consumed.

🔒🟣

content

Requires
Host Insights
add-on.

🔒🟣

content

Requires
Forensics
Add-on.

🔒🟣

content

Requires a
Cortex XDR
Pro per
Endpoint
license.
When you
enable this
feature, a
Cortex XDR
Pro per
Endpoint
license is
consumed.

🔒🟣

content

Requires
Host Insights
add-on.

🔒🟣

content

Requires
Forensics
Add-on.

🔒🟣

content

Requires a
Cortex XDR
Pro per
Endpoint
license.
When you
enable this
feature, a
Cortex XDR
Pro per
Endpoint
license is
consumed.

🔒🟣

content

Requires
Host Insights
add-on

☑️🟣

prerequisite

User Space
Agent mode
requires
Linux kernel
5.0.0 or
higher.

📝🟢

note

New profiles
based on
imported
profiles are
added, and
do not
replace
existing ones.

📝🟢

note

Disabling
Device
Control
Violation
notifications
is only
supported
on endpoints
running
Cortex XDR
agent
version 8.6
and above.

📝🟢

note

You can
enable the
option to
maintain a
persistent
notification
regarding
the
disconnectio
n of the
endpoint
from the
network. The
settings
Persistent
Isolation
Notification
and Blocked
Connectivity
Notification
must be
enabled.
Until the
threat on the
endpoint has
been
removed, the
endpoint
remains
disconnected
from the
network.

📝🟢

note

In Traps 5.0.6
and later
releases,
when
protection is
enabled,
access will
be read-
only. In
earlier Traps
releases,
enabling
protection
disables all
access to
services,
processes,

files, and
registry
values.

📝🟢

note

If you
choose the
Enable
option, you
must also
enable XDR
Agent
Tampering
Protection in
the malware
profile and
set it to
Block.
Ensure that
both profiles
are assigned
to the same
endpoints.

📝🟢

note

When you
enable
Cortex XDR
agent
registration
with the
Windows
Security
Center,
Windows
automaticall
y shuts
down
Microsoft
Defender on
Windows-
based
workstation
endpoints. If
you still
want to
allow
Microsoft
Defender to

run on a

workstation
endpoint
where
Cortex
Cloud is
installed,
you must
use the
Disable
option.
However,
Palo Alto
Networks
does not...

📝🟢

note

To enable
access to
these
options,
scroll down
to Network
Location

Configuratio
n, and set
Action Mode
to Enabled.

📝🟢

note

When using
Nmap, the
Cortex XDR
agent
downloads
an Nmap
driver for
the duration
of the scan
and removes
the driver
upon

completion.
If an Nmap
scan is in
process,
Cortex
Cloud
identifies the
Nmap driver
and places
any
additional
scans in a
queue.

📝🟢

note

If you
disable
content

updates for
a newly
installed
agent, the
agent
retrieves the
content for
the first time
from Cortex
Cloud, and
then
disables
content

updates on
the endpoint.
When you
add a
Cortex XDR
agent to an
endpoint
group with a
disabled
content
auto-

upgrades
policy, the
policy is
applied to
the added
agent as well.

📝🟢

note

Automatic
upgrades
are not

supported
with non-
persistent
VDI and
temporary
sessions.

📝🟢

note

The delay
timer starts
from the
date of the
target agent
version's
availability
in the tenant.

📝🟢

note

Limitations
in the
content
download
process:

When you
install the
Cortex XDR
agent, the
agent
retrieves the
latest
content

update
version
available. A
freshly
installed
agent can
take
between five
to ten
minutes
(depending
on your
network and
content

update
settings) to
retrieve the
content for
the first
time. During
this time,
your

endpoint is
not...

📝🟢

note

If the Cortex
XDR agent is
initially
unable to
communicat
e without
the local
store,
enforcement
is not
enabled and
the agent
will show as
partially
protected.

📝🟢

note

New profiles
based on
imported
profiles are
added, and
do not
replace
existing ones.

📝🟢

note

Disabling
Device
Control
Violation
notifications
is only
supported
on endpoints
running
Cortex XDR
agent
version 8.6
and above.

📝🟢

note

You can
enable the
option to
maintain a
persistent
notification
regarding
the
disconnectio
n of the
endpoint
from the
network. The
settings
Persistent
Isolation
Notification
and Blocked
Connectivity
Notification
must be
enabled.
Until the
threat on the
endpoint has
been
removed, the
endpoint
remains
disconnected
from the
network.

📝🟢

note

If you
choose the
Enabled
option, you
must also
set Anti
Tampering
Protection in
the malware
security
profile to
Block, and
ensure that
both profiles
are assigned
to the same
endpoints.

📝🟢

note

When
protection is
enabled,
access to
services,
processes,

files, and
registry
values will
be read-only.

📝🟢

note

If you
disable
content

updates for
a newly
installed
agent, the
agent
retrieves the
content for
the first time
from Cortex
Cloud, and
then
disables
content

updates on
the endpoint.

📝🟢

note

Automatic
upgrades
are not

supported
with non-
persistent
VDI and
temporary
sessions.
When a
Cortex XDR
agent is
added to an
endpoint
group, it
inherits the
group's
policy,
including the
disabled
content
auto-

upgrades
setting.

📝🟢

note

The delay
timer starts
from the
date of the
target agent
version's
availability
in the tenant.

📝🟢

note

Limitations
in the
content
download
process:

When you
install the
Cortex XDR
agent, the
agent
retrieves the
latest
content

update
version
available. A
freshly
installed
agent can
take
between five
to ten
minutes
(depending
on your
network and
content

update
settings) to
retrieve the
content for
the first
time. During
this time,
your

endpoint is
not...

📝🟢

note

If the Cortex
XDR agent is
initially
unable to
communicat
e without
the local
store,
enforcement
is not
enabled and
the agent
will show as
partially
protected.

📝🟢

note

New profiles
based on
imported
profiles are
added, and
do not
replace
existing ones.

📝🟢

note

If you
disable
content

updates for
a newly
installed
agent, the
agent
retrieves the
content for
the first time
from Cortex
Cloud, and
then
disables
content

updates on
the endpoint.
When you
add a
Cortex XDR
agent to an
endpoint
group with a
disabled
content
auto-

upgrades
policy, the
policy is
applied to
the added
agent as well.

📝🟢

note

Automatic
upgrades
are not

supported
with non-
persistent
VDI and
temporary
sessions.

📝🟢

note

Automatic upgrades are not supported for XDR
agents running on K8s.

📝🟢

note

The delay
timer starts
from the
date of the
target agent
version's
availability
in the tenant.

📝🟢

note

Limitations
in the
content
download
process:

When you
install the
Cortex XDR
agent, the
agent
retrieves the
latest
content

update
version
available. A
freshly
installed
agent can
take
between five
to ten
minutes
(depending
on your
network and
content

update
settings) to
retrieve the
content for
the first
time. During
this time,
your

endpoint is
not...

📝🟢

note

User Space
Agent mode
requires
Linux kernel
5.0.0 or
higher.
Endpoints
running an
older Linux
kernel
version with
this fallback
enabled, will
not start
using User
Space Agent
mode, and
will operate
asynchronou
sly.

⛔🔴

warning

If you
disable or
delay
automatic-
content

updates
provided by
Palo Alto
Networks, it
may affect
the security
level in your
organization.

⛔🔴

warning

If you
disable or
delay
automatic-
content

updates
provided by
Palo Alto
Networks, it
may affect
the security
level in your
organization.

⛔🔴

warning

If you
disable or
delay
automatic-
content

updates
provided by
Palo Alto
Networks, it
may affect
the security
level in your
organization.

3.2.1.1.4. Set
up
restrictions
prevention
profiles



📝🟢

note

New profiles
based on
imported
profiles are
added, and
do not
replace
existing ones.

📝🟢

note

Configure
custom
BIOC
prevention
📝🟢 rules here:

note

New profiles
based on
imported
profiles are
added, and
do not
replace
existing ones.

📝🟢

note

Configure
custom
BIOC
prevention
📝🟢 rules here:

note

New profiles
based on
imported
profiles are
added, and
do not
replace
existing ones.

📝🟢

note

Configure
custom
BIOC
prevention
📝🟢 rules here:

note

New profiles
based on
imported
profiles are
added, and
do not
replace
existing ones.

📝🟢

note

Wildcards
are

supported.

3.2.1.1.5. Set
up
exception
profiles and
rules



3.2.1.1.5.1.
Exception
configuration



👀🟠

important

If you don't
migrate the
legacy
exceptions,
you can
continue to
create
exceptions
through the
profiles.
Add a new
exceptions
security
profile
Add a
global
endpoint
policy
exception
Set up
exploit
prevention
profiles
Set up
malware
prevention
profilesSet
up malware
prevention
profiles
Set up
restrictions
prevention
profiles

🧭➡️

navigation

Settings →
Exception
Configuratio
n → Legacy
Exceptions
and click
Start
Simulation

3.2.1.1.5.2.
Issue
exclusions



📝🟢

note

The agent
continues to
generate
excluded
issues on the
endpoint,
but they are
not saved or
displayed in
Cortex
Cloud.
Configuratio
n of an issue
exclusion
does not

remove or

delete any
of the logs
that would
have
triggered
the issue
notification.
You can also
set up issue
exceptions
by creating
global
endpoint
policy
exceptions.
For more
information,...

🧭➡️

navigation

The Settings
→ Exception
Configuratio
n → Issue
Exclusions
page

displays the
issue
exclusion
rules in
Cortex
Cloud

3.2.1.1.5.3.
Add an IOC
or BIOC rule
exception



📝🟢

note

Cortex Cloud only supports exceptions with
one attribute. See Add an alert exclusion rule to
create advanced exceptions based on your
filtered criteria.Add an alert exclusion rule

🧭➡️

navigation

Settings →
Exceptions
Configuratio
n →

IOC/BIOC
Suppression
3.2.1.1.5.4. Rules
Add a
disable
prevention
rule for
endpoints



👀🟠

important

All
applicable
prevention
actions are
skipped for
the files and
process that
match the
properties
defined in
the rule.
Consider the
consequence
s of
disabling a
prevention
rule before
you add the
exception,
and monitor
it over time.
You can only
apply a
Disable
Prevention
Rule to
endpoints
running
Cortex XDR
agents
version 7.9
and later.

🧭➡️

navigation

Settings →
Exception
Configuratio
n → Disable
Prevention
Rules
3.2.1.1.5.5.
Add a
disable
injection and
prevention
rule



👀🟠

important

Exceptions
are limited
up to 48
hours by
default and
configurable
up to one
week.
Consider the
consequence
s of
disabling a
prevention
rule before
you add the
exception
and monitor
it over time.
You can only
apply a
Disable
Prevention
Rule to
agents
version 7.9
and later.

🧭➡️

navigation

Settings →
Exception
Configuratio
n → Disable
Injection and
Prevention

3.2.1.1.5.6.
Add a
support
exception
rule for
endpoints



👀🟠

important

If you don't
migrate the
legacy
exceptions,
you can
continue to
create
exceptions
through the
profiles.
Add a new
exceptions
security
profile
Add a
global
endpoint
policy
exception
Set up
exploit
prevention
profiles
Set up
malware
prevention
profilesSet
up malware
prevention
profiles
Set up
restrictions
prevention
profiles

🧭➡️

navigation

Your
migrated
rules are
displayed on
the Settings
→ Exception
Configuratio
ns →

Support
Exception
Rules page

From
Settings →
Exception
Configuratio
n → Support
Exception
Rules, click
+ Import
from file

3.2.1.1.5.7.
Add a
legacy
exception
rule for
endpoints



📝🟢

note

This
exception
prevents the
agent from
examining
the specified
file. Use with
caution, as it
may
unintentionall
y allow
unwanted or
malicious
behavior to
go
undetected.

👀🟠

important

If you don't
migrate the
legacy
exceptions,
you can
continue to
create
exceptions
through the
profiles.
Add a new
exceptions
security
profile
Add a
global
endpoint
policy
exception
Set up
exploit
prevention
profiles
Set up
malware
prevention
profilesSet
up malware
prevention
profiles
Set up
restrictions
prevention
profiles

🧭➡️

navigation

Your
migrated
rules are
displayed on
the Settings
→ Exception
Configuratio
ns → Legacy
Agent
Exceptions
3.2.1.1.6. Set page
up Identity
profiles



🔒🟣

content

Requires the
ITDR add-on.

📝🟢

note

The identity
profile is
available on
Cortex
XSIAM 3.6,
Cortex XDR
5.1, and
Cortex
Cloud
Runtime 2.1
or later, and
requires
Cortex XDR
agent
version 9.1 or
later. This
feature is
not available
on Cortex
XSIAM 2.x or
Cortex XDR
3.x tenants.
In
environments
with mixed
agent
versions, you
can assign
policies
containing
an identity
profile;
however,
any...

📝🟢

note

New profiles
based on
imported
profiles are
added, and
do not
replace
existing ones.

📝🟢

note

This feature
only comes
into effect
after a
restart.

3.2.1.2.
Define
endpoint
groups



📝🟢

note

Configuratio
n based on
user

granular
policy is
optimized
for VDI and
session-
persistent
environments
; it is not
recommende
d for
decentralize
d or
traditional
endpoint
architectures

.

📝🟢

note

Disconnectin
g Cloud
Identity
Engine in
your Cortex
Cloud
deployment
can affect
existing
endpoint
groups and
policy rules
based on
Active
Directory
properties.

3.2.1.3.
Configure
global agent
settings



📝🟢

note

Disabling
this
capability
takes
immediate
effect on
new hashes,
fresh agent
installations,
and existing
security
policies. It
could take
up to a week
to take
effect on
existing
agents in
your
environment
pending
agent
caching.

📝🟢

note

If you have
any Cortex
Cloud
filters,
starring
policies,
exclusion
policies,
scoring
rules, log
forwarding
queries, or
automation
rules
configured
for
XSOAR/3rd
party SIEM,
we advise
you to
update
those to
support the
changes
before
activating
the feature.
For example,
change the
query to
include the
previous
description
that is still
available in
the...

3.2.1.4. Apply
profiles to
endpoints



📝🟢

note

When
importing a
policy,
select
whether to
enable the
associated
policy
targets.
Rules within
the imported
policy are
managed as
follows:
New rules
are added to
the top of
the list.
Default rules
override the
default rule
in the target
tenant.
Rules
without a
defined
target are
disabled
until the
target is
specified.

📝🟢

note

The
exported file
is encoded
in Base64
and cannot
be edited.

3.2.1.5.
Create an
agent
installation
package



🔒🟣

content

Requires the
Cortex
Cloud
Runtime
Security
add-on.
Every 10
container-
embedded
agents will
consume a

single
Cortex
Runtime
Security
license.

📝🟢

note

The profile
will be
applied if
the security
policy
cannot be
retrieved in
real-time.

📝🟢

note

The Cortex
serverless
library must
be imported
after other
libraries to
activate the
hooks that
enable
auditing.

3.2.1.5.1.
Manage an
agent
installation
package

 📝🟢

note

Since Cortex Cloud relies on the installation
package ID to approve agent registration during
the installation, we recommend that you don't
delete the installation package of active
endpoints. If you install the Cortex XDR agent
from a package after you delete it, Cortex
Cloud denies the registration request leaving
the agent in an unprotected state....

3.2.1.6.
Harden
endpoint
security



🧭➡️

navigation

From
Inventory →
Endpoints →
Policy
Management
→

Extensions
→ Profiles,
you can
create

profiles for
the
following
hardened
endpoint
security
capabilities

To apply the
profiles,
from
Inventory →
Endpoints →
Policy
Management
→

Extensions
→ Policy
Rules, you
can view all
the policy
rules per
operating
system

3.2.1.6.1.
Device
control



📝🟢

note

Operating
systems
report on
devices in
different
ways.
Sometimes,
the same
BLE device
will report
different
services and
interfaces,
depending
on the host's
operating
system. This
may have an
effect on
the specific
BLE services
that are
blocked for
each
operating
system.
Depending
on your
defined user
scope
permissions,
creating
device
profiles,...

📝🟢

note

For network
printer print
jobs, ensure
that you
also
configure
the Agent
Settings
profile,
Network
Location

Configuratio
n option.
This setting
must be set
to Enabled,
and
configured.
If you do not
enable and
configure
this setting,
all network
printer
operations
will be
treated as
internal
network
print jobs.
For network
printer print
jobs, ensure
that...

📝🟢

note

Currently,
the default
is set to Use
Default
(Allow),
however,
Palo Alto
Networks
may change
the default
definition at
any time.

📝🟢

note

You cannot
edit or
delete the
default
profiles pre-
defined in
Cortex
Cloud.

📝🟢

note

You cannot
edit or
delete the
predefined
profiles in
Cortex
Cloud.

📝🟢

note

When
importing a
policy,
select
whether to
enable the
associated
policy
targets.
Rules within
the imported
policy are
managed as
follows:
New rules
are added to
the top of
the list.
Default rules
override the
default rule
in the target
tenant.
Rules
without a
defined
target are
disabled
until the
target is
specified.

📝🟢

note

Permanent
exceptions
apply across
platforms,
allowing the
devices on
all operating
systems.

📝🟢

note

Disabling
Device
Control
Violation
notifications
is only
supported
on endpoints
running
Cortex XDR
agent
version 8.6
and above.

🧭➡️

navigation

Endpoints →
Policy
Management
→ Settings
→ Device
Management

3.2.1.6.2. Host
firewall



3.2.1.6.2.1.
Host firewall
for Windows



📝🟢

note

A rule is
always part
of a rules
group. It
cannot
stand on its
own.

A rule can
belong to
one rules
group only
and cannot
be reused in
different
groups.

📝🟢

note

When
selecting
ICMP
protocol,
you must
enter a the
ICMP Type
and Code.
Without
these values
the ICMP
protocol is
ignored by
the Windows
and macOS
Cortex XDR
agents.

📝🟢

note

When
importing a
policy,
select
whether to
enable the
associated
policy
targets.
Rules within
the imported
policy are
managed as
follows.
New rules
are added to
the top of
the list.
Default rules
override the
default rule
in the target
tenant.
Rules
without a
defined
target are
disabled
until target
is specified.

📝🟢

note

The data is
aggregated
and
reported
periodically
every 60
minutes
since the
first time the
host firewall
policy was
enforced on
the
endpoint,
not every
round hour.
The table
lists
enforcement
events only
for rules set
to Report
Matching
Traffic.

🧭➡️

navigation

From
Inventory →
Endpoints →
Policy
Management
→

Extensions
and select +
Add Profile
or Import
from File

3.2.1.6.2.2.
Host firewall
for macOS



📝🟢

note

When
importing a
policy,
select
whether to
enable the
associated
policy
targets.
Rules within
the imported
policy are
managed as
follows:
New rules
are added to
the top of
the list.
Default rules
override the
default rule
in the target
tenant.
Rules
without a
defined
target are
disabled
until the
target is
specified.

🧭➡️

navigation

From
Inventory →
Endpoints →
Policy
Management
→

Extensions
Profiles →
Profiles,
select +
New Profile
or Import
from File

Additionally,
to monitor
the
communicati
on on your
macOS
endpoint,
you can use
the
following
operating
system
utilities:
From the
endpoint
System
Preferences
→ Security
and Privacy
→ Firewall
→ Firewall
options, you
can view the
list of
blocked and
allowed
applications
in the firewall

3.2.1.6.3. Disk
encryption



📝🟢

note

You must
select the
same

encryption
method
configured
by the
Microsoft
Windows
Group
Policy in
your

organization
for the
target
endpoints.
Otherwise, if
you select a
different
encryption
method than
the one
already
applied
through the
Windows
Group
Policy,
Cortex
Cloud
displays
errors.

📝🟢

note

When
importing a
policy,
select
whether to
enable the
associated
policy
targets.
Rules within
the imported
policy are
managed as
follows:
New rules
are added to
the top of
the list.
Default rules
override the
default rule
in the target
tenant.
Rules
without a
defined
target are
disabled
until the
target is
specified.

🧭➡️

navigation

You can
monitor the
Encryption
Status of an
endpoint in
the Inventory
→ Endpoints
→ Disk
Encryption
Visibility
table

3.2.1.6.4.
Host
Inventory



📝🟢

note

If you have
Cloud
Posture
Security,
Cloud
Runtime
Security, or
Cortex
XSIAM
Premium
licenses, go
to Inventory
→ Host
Insights →
Host
Inventory.

📝🟢

note

Cortex
Cloud
includes
only the first
10,000
results per
endpoint.
Cortex
Cloud lists
only users
that belong
to each
group
directly, and
does not
include
users who
belong to a
group within
the main
group.
If a local
users group
includes a
domain user
(whose
credentials
are stored
on the
Domain
Controller
server and
not on the
endpoint),
Cortex
Cloud...

🧭➡️

navigation

Inventory →
Endpoints →
Host
Inventory

3.2.1.7. Set a
Cortex XDR
agent
Critical
Environment
version



📝🟢

note

To ensure
the stability
of the line,
CE versions
maintenance
release
cadence is
longer than
in the
standard
line, we
recommend
that
deployment
is adjusted
accordingly.

🧭➡️

navigation

Settings →
Configuratio
ns →

General →
Agent
Configuratio
ns

Inventory →
Endpoints →
All Endpoints
3.2.1.8. Set an
application
proxy for
Cortex XDR
agents

 ⛔🔴

warning

If adding a
proxy to the
allow list be
aware of the
following:
This allows
any

application
on the
isolated
endpoint to
communicat
e through
that proxy,
potentially
allowing a
threat actor
or malware
to bypass
the isolation
and reach
external
resources.

Ensure that
no other
non-security
applications
are utilizing
this proxy if
you choose
to allow it.

3.2.2.
Manage
endpoint
protection



3.2.2.1. Move
agents
between
managing
servers



☑️🟣

prerequisite

Consider the
following
before
making
changes:
Endpoint
type is not
Kubernetes
Node.
Installation
type is not
VDI.
Ensure you
have
administrator
privileges
for Cortex
Cloud in the
hub.

📝🟢

note

Cortex
Cloud does
not support
moving
agents
between
FedRamp
and
commercial
tenants.

🧭➡️

navigation

Inventory →
Endpoints →
Agent
Installations

3.2.2.2.
Manage
endpoint
tags



📝🟢

note

The
following
explanations
use

Windows
operating
system
installation
parameters
and Cytool
argument
examples.

📝🟢

note

Tag names
are case

sensitive.
In Windows
and Mac, a
tag name
can contain
spaces.
Linux does
not support
tag names
with spaces
as command
line
arguments
to the shell
installer.
Instead, tags
can be set in
the
/etc/panw/co
rtex.conf
configuration
file, which
supports all
Linux
installers.

📝🟢

note

Tag names
are case

sensitive and
can contain

📝🟢 spaces.

note

Tag names
are case

sensitive and
can contain

📝🟢 spaces.

note

When SBAC
is enabled,
assigning
tags may
impact user
permissions.

📝🟢

note

If you
remove the
tag and
there are
assigned
users or user

groups with
scope

settings, this
can impact
user

permissions
in the system.

3.2.2.3. Set
an alias for
an endpoint

 💡🔵

tip

If you
change your
mind in the
future, you
can select
Endpoint
Control →
Change
Endpoint
Alias again,
and delete
the required
aliases.

3.2.2.4.
Manage
endpoint
prevention
profiles



📝🟢

note

The filter is
stored within
the policy
definition
and
assessed
during
runtime to
extract the
functions
that match
the filter
criteria.

🧭➡️

navigation

From
Inventory →
Endpoints →
Policy
Management
→

Prevention
→ Profiles,
locate the
profile that
you want to
remove

Inventory →
Endpoints →
Policy
Management
→

Prevention
→ Profiles

3.2.2.5.
Upgrade
Cortex XDR
agents



📝🟢

note

The Cortex
XDR agent
keeps the
name of the
original
installation
package
after every
upgrade.

📝🟢

note

It is possible
to cancel an
upgrade if
the Status is
Pending.
Once the
Status is In
Progress,
the action is
already
received by
the agent
locally and
cannot be
canceled
from the
management
console.
Custom
dashboards
that include
upgrade
status

widgets, and
the All
Endpoints
page display
upgrade
status.

During the
upgrade
process, the
endpoint
operating...

⛔🔴

warning

You must ensure that the System Extensions
were approved on the endpoint. Otherwise, if
the extensions were not approved, after the
upgrade the extensions remain on the endpoint
without any option to remove them which could
cause the agent to display unexpected
behavior. To check whether the extensions were
approved, you can either verify that the...

👀🟠

important

The
following list
includes
important
points to
take into
account
when
upgrading
the Cortex
XDR agent:
Review the
recommende
d guidelines
for keeping
Cortex XDR
agents and
content

updated.
Read more
here.
You cannot
upgrade the
Cortex XDR
agent on VDI
endpoints or
a Golden
Image.
You must
reinstall
(uninstall
and install
again) the
relevant
agent...

3.2.2.6.
Restart agent



3.2.2.7.
Uninstall the
Cortex XDR
agent



📝🟢

note

Before
upgrading a
Cortex XDR
agent
running on
macOS
10.15.4 or
later, you
must ensure
that the
System
Extensions
were

approved on
the
endpoint.
Otherwise, if
the
extensions
were not

approved,
after the
upgrade the
extensions
remain on
the endpoint
without any
option to
remove them
which could
cause the
agent to
display
unexpected
behavior. To
check...

💡🔵

tip

If needed,
use the filter
to filter the
list of
endpoints by
attribute or
3.2.2.8. Clear group name.
agent
database



📝🟢

note

Clearing the
agent
database is
supported
on all
platforms
with Cortex
XDR agent
version 7.9 or
later, and is
available
only when
using the
debugging
mode.

📝🟢

note

You can only
right-click
to cancel
the clear
agent
database for
actions with
a pending
status.

3.2.2.9.
Delete
Cortex XDR
agents

 📝🟢

note

To reinstate
an endpoint,
you have to
uninstall and
reinstall the
agent.
3.2.2.10.
Manage
agent tokens



📝🟢

note

Agent
tokens are
only
supported
for Windows
and Mac.

📝🟢

note

You can
select a one
or multiple
agents to
add a
temporary
token.

3.2.2.11.
Retrieve
support file
password



3.2.2.12.
Monitor
agent
operational
status



📝🟢

note

Any of the
listed items
could lead
to a partially
protected
state. Refer
to the
Cortex
Cloud
management
console for
specific
reasons for
the state.

⚠️🟡

caution

Status can
have the
following
implications
on the
endpoint:
*(Status):
The exploit
protection
module is
not running.
**(Status):
XDR data
collection is
not running
Behavioral
threat
protection is
not running
Anti-
malware
flow is
asynchronou
s

Local
privilege
escalation
protection is
asynchronou
s

**(Status):
XDR data
collection is
not...

🧭➡️

navigation

You can
monitor the
Cortex XDR
agent
Operational
Status in
Inventory →
Endpoints →
All Endpoints

3.2.2.13.
Monitor
agent
activity



3.2.2.14.
Monitor
agent
upgrade
status

 🧭➡️

navigation

From
Inventory →
Endpoints →
All
Endpoints,
you can view
the upgrade
status of
any Cortex
XDR agent
that you
manage

3.3. Endpoint
DLP



3.3.1. Cortex
Data Loss
Prevention
(DLP)
module
overview



☑️🟣

prerequisite

Endpoint
DLP add-on
license
Cortex
agent 9.1 and
above for
Windows
and macOS

🧭➡️

navigation

Settings →
Configuratio
n → Access
Management
3.3.2. → Roles
Personas
workflow for
DLP



3.3.3. Best
Practices



📝🟢

note

A good
example
would be to
add the
description
from the
Raised Issue
Name of the
specific
Data-in-
Motion rule.

📝🟢

note

When adding a source, it becomes a mandatory
requirement that must be met for the rule to
trigger. You should only apply this setting if you
want the rule to take action exclusively when
files originate from those specific locations.

🧭➡️

navigation

Refer to
Modules →
Data
Security →
Data
Security
Issues →
Threats

Modules →
Data
Security →
Endpoint
Data-in-
Motion Rules
→ Endpoint
DLP Settings
to configure
the tenant
settings for
DLP

3.3.4.
Configure
DLP end-to-
end

 3.3.4.1.
Onboarding
checklist for
DLP



3.3.4.1.1.
Install DLP
browser
extension on
your

endpoint



📝🟢

note

The
extensions
are not
enabled for
Incognito or
InPrivate
modes in
Chrome and
Edge. It is
recommende
d to disable
these modes
in the
organization.

📝🟢

note

It is highly
recommende
d to refer to
Install the
Cortex XDR
Agent using
JAMF for a
comprehensi
ve overview

of setting up
a JAMF
profile.Install
the Cortex
XDR Agent
Using JAMF

👀🟠

important

Endpoints
should be
linked to the
domain.

🧭➡️

navigation

Modules →
Data
Security →
Endpoint
Data-in-
Motion Rules
→ Endpoint
DLP Settings

3.3.4.1.2.
Create
endpoint
applications

 🧭➡️

navigation

Modules →
Endpoint
Data-in-
Motion Rules
→ Endpoint
Applications

3.3.4.1.3.
Create
endpoint
application
groups

 📝🟢

note

The catalog
contains a
list of all
supported
and tested
applications.
If an
application
you need is
not on the
list, contact
support for
assistance.

3.3.4.1.4.
Create data-
in-motion
rules



📝🟢

note

Third-party
application
behaviour:
When a file
upload is
blocked, the
local
application
may display
its own
generic error
message in
response to
the DLP
restriction. In
similar
cases, some

third-party
applications
might still
proceed by
sending a
dummy file
or an error

placeholder
instead of
the actual
data.

📝🟢

note

To maximize
data
security, if
you use
Microsoft
Purview for
extensive
manual and
automated
file
classification
tagging and
are looking
to integrate
those labels
directly with
the DLP
policies to
trigger
protective
actions
based on a
file's
sensitivity,
refer to How
to use
information
protection
labels in
Cortex
Cloud Data
Security.

📝🟢

note

Distribution
of the
Endpoint
DLP
package is
restricted to
agents
assigned to
the data-in-
motion
policy. This
ensures that
only
endpoints
requiring
DLP

functionality
are affected,
rather than
all eligible
endpoints in
the tenant.

3.3.4.1.5.

Configure
endpoint
DLP settings



📝🟢

note

When a fail-
close action
occurs, the
system
creates a
Data
movement

blocked by
Endpoint
DLP fail-
close action
issue.

📝🟢

note

The agent
does not
force-install
the
extension if
it is already
managed by
the MDM on
the endpoint.
The XDR
agent
ensures full
coverage by
force-
installing the
extension on
both
managed
and
unmanaged
browsers.
But if a
browser
becomes
officially
managed by
the
organization
later, the
extension
must be
redeployed
through the
central
management
console...

📝🟢

note

In the case
of MDM, the
extension is
user-

managed, so
Cortex
doesn't
remove an

installed
MDM
extension, it
only
disables
communicati
on with the
DLP
extension.

📝🟢

note

You can
specify the
end-user
message per
rule.

3.3.5. DLP
status in all
endpoints



📝🟢

note

It is
recommende
d to work
with the
latest
security
content.

📝🟢

note

Close and
reopen your
browser to
activate the
DLP
extension.

3.3.6. Cortex
DLP threat
detection
and issues



📝🟢

note

Access to
this page is
restricted to
users with
the roles:
Data
Security
Admin,
Instance
Administrato
r, and
Account
Admin.

📝🟢

note

If the default action configured in Endpoint DLP
Settings is set to Block file movement (fail-
close), an issue is raised where the assigned
severity is set to low, and includes the Name
Data movement blocked by Endpoint DLP
default action

🧭➡️

navigation

Data
Security →
Data
Security
Issues →
Threats

From Data
Security →
Data
Security
Issues →
Threats, you
can view the
DLP issues

4. Threat

managem
ent



4.1.
Detection
rules

 4.1.1. What
are

detection
rules?



4.1.1.1.
What's an
IOC?



📝🟢

note

Cortex
Cloud
supports a
maximum of
4,000,000
IOCs.
4.1.1.1.1. IOC
rule details



📝🟢

note

Field cannot
exceed 36
characters.

🧭➡️

navigation

In the Threat
Management
→ Detection
Rules → IOC
page, you
can view all
configured
or uploaded
indicators of
compromise
(IOCs

4.1.1.1.2.
Create an
IOC rule



📝🟢

note

To ensure
your IOC
rules
generate
issues
efficiently
and do not
overcrowd
your Issues
table,
Cortex
Cloud
automaticall
y does the
following:
Disables any
IOC rules
that reach
5000 or
more hits
over 24
hours.
Creates a
rule
exception
based on the
PROCESS
SHA256
field for IOC
rules that hit
more than
100
endpoints
over 72
hours.

🧭➡️

navigation

In Threat
Management
→ Detection
Rules →
IOC, select +
Add IOC

4.1.1.2.
What's a
BIOC?



4.1.1.2.1. BIOC
rule details

 🧭➡️

navigation

Manage
your
behavioral
indicator of
compromise
(BIOC) rules
in Threat
Management
→ Detection
Rules →
BIOC

4.1.1.2.2.
Create a
BIOC rule



🔒🟣

content

Custom
prevention
rules are
supported
on Cortex
Cloud agent
7.2 and later
versions and
enable you
to configure
and apply
user-defined
BIOC rules
to
Restriction
profiles
deployed on
your
Windows,
Mac, and
Linux
endpoints.

📝🟢

note

To ensure
your BIOC
rules
generate
issues
efficiently
and do not
overcrowd
your Issues
table,
Cortex
Cloud
automaticall
y does the
following:
Disables
BIOC rules
that reach
5000 or
more hits
over a 24-
hour period.
Creates a
rule
exception
based on the
PROCESS
SHA256
field for
BIOC rules
that hit more
than 100
endpoints
over a 72
hour period.

📝🟢

note

A
cloud_audit_l
og dataset
requires a
Cortex XDR
Pro per GB
license.
Currently,
you cannot
create a
BIOC rule on
customized
datasets and
only the
filter stage,
alter stage,
and
functions
without any
aggregations
are

supported
for XQL
queries that
define a
BIOC.
For BIOC
rules, the
field values
in XQL are
evaluated as
case

insensitive

(config...

📝🟢

note

To
demonstrate
the
expected
behavior of
the rule
before you
save it,
Cortex
Cloud tests
the BIOC on
historical
logs. After
you save a
BIOC rule, it
will operate
both on
historical
logs (up to
10,000 hits)
and on new
data
received
from your
log sensors.

📝🟢

note

To
demonstrate
the
expected
behavior of
the rule
before you
save it,
Cortex
Cloud tests
the BIOC on
historical
logs. After
you save a
BIOC rule, it
will operate
on both
historical
logs (up to
10,000 hits)
and new
data
received
from your
log sensors.

📝🟢

note

You can only
import files
that were
exported
from Cortex
Cloud. You
can not edit
an exported
file.

📝🟢

note

You can only
add to
existing
profiles you
created,
Cortex
Cloud
Default
profiles will
not appear
as an option.
When you
want to add
to
restrictions
profile, you
can only use
fields or
options that
exist in pre-
built
process-
type BIOCs

📝🟢

note

Auto-
disable will
turn off
both the
BIOC rule
detection
and the
BIOC
prevention
rule.

4.1.1.2.3.
Manage
Global BIOC
Rules



4.1.1.3.
What's a
correlation
rule?



🧭➡️

navigation

After you
configure
your
correlation
rules, you
can manage
them in
Threat
Management
→ Detection
Rules →
Correlations,
and view
and analyze
the
generated
issues in
Cases and
the Issues
Table

4.1.1.3.1.
Correlation
rule details



📝🟢

note

Only an
administrator
can create
and view
queries built
with an
unknown
dataset that
currently
does not
exist in
Cortex
Cloud .

📝🟢

note

For more
information

on

troubleshooti
ng server
errors in
scheduled
correlation
rules,
Troubleshoot
server

errors in
scheduled
correlation
rules.

📝🟢

note

Certain
fields are
exposed and
hidden by
default. An
asterisk (*) is
beside every
field that is
exposed by
default.

📝🟢

note

Only an
administrator
can create
and view
queries built
with an
unknown
dataset that
currently
does not
exist in
Cortex
Cloud.

🧭➡️

navigation

If you are assigned a role that enables
Investigation → Rules privileges, you can
manage all user-defined Correlation Rules from
Threat Management → Detection Rules →
Correlations

4.1.1.3.2.
Create a
correlation
rule



📝🟢

note

When you open the New Correlation Rule
editor from XQL Search, this XQL Search field is
already populated with the XQL query that you
defined.
An administrator can create and view queries
built with an unknown dataset that currently
does not exist in Cortex Cloud . All other users
can only create and view queries built with an
existing dataset.

📝🟢

note

The XQL features for call, top, and wildcards in
datasets (dataset in (<dataset prefix>_*)) are
currently not supported in Correlation Rules. If
you add them to the XQL definition, you will not
be able to Create or Save the Correlation Rule.
The XQL features for transaction in datasets
(dataset in (<dataset prefix>_*)) are currently
not supported in...

📝🟢

note

If a
generated
issue has
severity
Medium or
above, a
case is
automaticall
y opened.
Low severity
issues
generated
by
correlation
rules are not
grouped into
cases.

📝🟢

note

There is no
validation or
auto

complete
for these
parameters
and the
values can
be null or
empty. In
these
scenarios,
Cortex
Cloud does
not display
the null or
empty
values, but
adds the
text NULL or
EMPTY in
the
descriptions.

📝🟢

note

Cortex
Cloud
groups into
cases based
on a

grouping
logic that
evaluates
relationships,
context,
and shared
artifacts

across

issues.
However, not
all fields
influence

case

grouping.
For
information
about the
specific
fields that
influence
grouping,
see Optimize
case

grouping in
correlations.

📝🟢

note

When more
than one
XDM field is
listed,
Cortex
Cloud looks
for the field
value
according to
the order of
the fields
listed.

📝🟢

note

When Use
the Cortex
Cloud
default
incident
enrichment
is not
selected and
you have not
mapped any
issue fields,
the issue is
dispatched
into a new
incident.

👀🟠

important

The maximum size of a lookup dataset is 50 MB.
If the data exceeds this limit, the add to lookup
action fails.

💡🔵

tip

When
creating
your
correlation
rule, you can
use

predefined
values for
different
fields in the
editor, such
as Alert
Name, Alert
Description,
and Drill-
Down Query.
For more
information,
see Field
replacement
syntax in
correlation
rules.

🧭➡️

navigation

You can
create a new
correlation
rule from
either the
Threat
Management
→ Detection
Rules →
Correlation
Rules page
or when
building a
query in XQL
Search

4.1.1.3.3. Field
replacement
syntax in
correlation
rules



4.1.1.3.4.
Manage
correlation
rules



4.1.1.3.5.
Monitor
correlation
rules

 📝🟢

note

In the
dataset, the
Query start
time and
Query end
time indicate
the time
frame of the
data that
was queried.
The actual
start and
end times of
the
correlation
rule
execution
are recorded
in the _time
field for the
Initiated and
Completed
entries.

4.1.1.3.6. Troubleshoot server errors in scheduled

correlation rules 

4.1.1.4.
Manage IOC
and BIOC
rules



📝🟢

note

For
Analytics
BIOC rules,
you can only
disable and
enable rules.

📝🟢

note

You cannot
delete
global
BIOCs
delivered
with content
updates.

📝🟢

note

For BIOC
rules that
are applied
to
prevention
profiles:
If you
disable a
rule only on
the agent,
detection on
the server
works as
usual.
If you
disable a
rule only on
the server,
prevention
on the agent
works as
usual.
If you
disable a
rule only on
the agent,
detection on
the server
works as
usual.
If you
disable a
rule only on
the server,
prevention...

📝🟢

note

When a
BIOC rule is
disabled
automaticall
y by Cortex
Cloud, for
example due
to the server
anti flooding
mechanism,
prevention
on the agent
works as
before.

🧭➡️

navigation

Threat
Management
→ Detection
Rules →
BIOC

4.2. Analytics  4.2.1.
Analytics
overview



4.2.1.1.
Analytics
engine



4.2.1.2.
Analytics
sensors

 📝🟢

note

Enhanced application logs (EAL) are now non-
billable and are excluded from the dashboard
to remove them from your Data Collection (GB)
license. Data is still ingested and available in the
relevant datasets.

4.2.1.3.
Coverage of
MITRE
Attack
tactics



4.2.1.4.
Review
MITRE
ATT&CK
framework
coverage

 📝🟢

note

When a
contributing
data source
type is
active, it
does not
imply that
all the rules
and
detectors
associated
with the
data source
type are
active. Rule
applicability
is dependent
on the data
source's
context and
configuration
. To enable
an active
status, data
source types
require the
following
setup:
Endpoint:
Installed
Cortex XDR
agent.
Network: A...

4.2.1.5.
Analytics
detection
time intervals

 📝🟢

note

When establishing a baseline, detectors
compute limits beyond which network activity
will require an issue. In some cases, detectors
do not compute baseline limits; instead they are
predetermined by Cortex Cloud engineers. The
engineers determine the values used for
predetermined limits using statistical analysis of
malicious activity recorded...

4.2.1.6.
Analytics
issues and
Analytics
BIOCs



4.2.1.7. View
and manage
Analytics
rules



📝🟢

note

The default
severity is
updated by
content

updates. If a
content

update
determines a
new default
severity for
the rule
that's the
same as the
value you
had
previously
determined,
you won't
have the
option to
reset to
default. For
example, if
the default

was

Informational
, and you
changed the
severity to
Medium, and
after a
content

update
Cortex
Cloud...

📝🟢

note

When you
select
multiple
rows, you

can only
enable or
disable the
selected
rules.

⛔🔴

warning

Changing
the default
severity may
result in
issues not
being
triggered or
too many
issues being
triggered.
Please
consider this
carefully
before you
change the
severity
recommende
d by Cortex
Cloud. Any
responsibility
for not
getting
issues
triggered as
a result of
changing the
severity will
be yours.

4.2.1.8.
Identity
Analytics

 🧭➡️

navigation

After
configuring
your Cloud
Identity
Engine
instance and
Cortex
Cloud
Analytics,
select
Settings () →
Configuratio
ns → Cortex
Cloud -
Analytics,
and in the
Featured in
Analytics
section,
Enable
Identity
Analytics

4.2.1.9.
Identity
Threat
Module
(ITDR)

 🧭➡️

navigation

Modules →
Identity
Security →
Unified
Human
Identities,
click the
number of
unified
human
identities

4.2.1.10. AI
Detection &
Response in
Cortex
Cloud (Beta)



☑️🟣

prerequisite

To enable AIDR, you must have one of the
following user roles: Instance Admin or Account
Admin.

🧭➡️

navigation

Settings →
Configuratio
ns → Cortex
- Analytics
4.2.1.10.1.
Data
sources and
supported
services



4.2.1.10.2. AI
Detection &
Response
Dashboard



4.2.1.10.3.
Collect
prompt logs



4.2.1.10.3.1.
Prompt log
collection in
AWS



4.2.1.10.3.2.
Enable
prompt log
collection in
Azure



5. Web

and API

Security

(WAAS)



5.1. Overview  🔒🟣

content

Requires the
Cortex
Cloud
Runtime
Security
add-on.

5.2. Personas
workflow



5.3. Secure
your API
landscape



5.3.1. Gain
visibility and
assess risk
of API
endpoints



📝🟢

note

Data
classification
findings are
only
available for
enabled

profiles. 📝🟢
note

Indicates that a JSON Web Token was identified,
but it is unknown how to determine its type
from the given string. It could be a non-
standard JSON Web Token creation algorithm.

📝🟢

note

To link APIs
to business
applications,
the two
prerequisites
must be met:
The VM
must have
XDR agent
with WAAS
enabled.
Applications
must be
defined in
Application
Security
Posture
Management
(ASPM).Appli
cation
Security
Posture
Management
(ASPM)

5.3.2.
Monitor and
investigate
API threats



5.3.3.

Configure
API security
from end to
end



5.3.3.1. Third-
party
integrations



5.3.3.1.1.
Ingest AWS
API Gateway



📝🟢

note

You must
generate a
new key if
you did not
📝🟢 save.

note

If no logs are
showing,
send some
API requests
on Postman
or CURL.

👀🟠

important

Leave
'–'filter-
pattern
empty as
displayed
above.

5.3.3.1.2.
Ingest Azure
APIM



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

Microsoft
Azure APIM
service must
be running
before
starting to
configure
the
integration.

📝🟢

note

From the
URL, save
the UUID
and the
resource

group -
/resource/su
bscriptions/<
UUID>/resour
ceGroups/<R
esourceGrou
p>.
The UUID is
the Azure
account/sub
scription ID
and the
resource

group, which
is the group
where the
APIM
Service is
defined.

📝🟢

note

Use the
search to
navigate to
the specific
section.

📝🟢

note

131072 bytes = 128 KB. This value determines
the size (in bytes) of request and response
bodies to send to Cortex. Any bytes beyond this
limit are truncated.

📝🟢

note

If any other
inbound
policies
should be
added, they
must be
added after
these
elements.

👀🟠

important

If you want
to add
additional
data to the
<outbound>,
add it at the
start of the
<outbound>
code.

5.3.3.1.3.
Ingest
Apigee Proxy



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

The steps
include the
prerequisites
that run the
automated
script that
deploys files
and
configuration
s to Apigee.
For manual
configuration
, refer to the
section
Manual
deployment.

📝🟢

note

If you have
an existing
hookand
would like to
integrate
with the
shared flow,
run the
deploy.sh
script, and
select n' and
exit at the
prompt to
create a new
hook. Refer
to the
section
Connect to
existing hook.

📝🟢

note

Start with
the hook in
pre-proxy.

📝🟢

note

Archive
deployments
are currently
in preview
and are
subject to
change.

👀🟠

important

This should
only be
done when
there are
already
existing
hooks, and
API security
shared flows
can't be
hooked as a
standalone.
Run the
deployment
script, but
skip step 9
by passing n.
This step
publishes
API security
shared flows
to the
desired
Apigee
environment
without
setting them
to flow
hooks.

5.3.3.1.4.
Ingest Kong



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

Contact
support to
obtain the
custom

📝🟢 plugin file.

note

The queue
parameters
can be
updated to
change
when the
plugin
mirrors data
to Cortex.

📝🟢

note

Use commas
to separate
tags.

👀🟠

important

The size of
the buffer
must be
equal or
larger than
the max
body size
setting in the
plugin
configuration
, on every

data plane
node.

5.3.3.1.5.
Ingest-F5



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

Extract the
files from
the F5
plugin to a
folder
before
selecting to
upload to F5.

📝🟢

note

The provider
for
"context_pro
vider"
should
always be
uppercase.

Supported
providers:
AWS, GCP,
Azure, On-
prem.

📝🟢

note

Select
panw_apisec
_set_ssl_data
only if your
client SSL
profile is
enabled.

5.3.3.2.
Agent-
based
protection



📝🟢

note

Web and API
Security
(WAAS)
profiles and
policies are
currently a
Beta feature.

5.3.3.2.1. Set
up Web and
API Security
profiles



📝🟢

note

Web and API
Security
profiles and
policies are
currently a
Beta feature.

📝🟢

note

In this profile, the Report options configure the
workload to report the corresponding malicious
applications or APIs to Cortex Cloud, without
blocking them. The Disabled options configure
the workloads to neither analyze nor report the
corresponding malware or behavior.

5.3.3.2.2.
Apply Web
and API
Security
profiles to
workloads



📝🟢

note

Web and API
Security
profiles and
policies are
currently a
Beta feature.

📝🟢

note

When
importing a
policy,
select
whether to
enable the
associated
policy
targets.
Rules within
the imported
policy are
managed as
follows:
New rules
are added to
the top of
the list.
Default rules
override the
default rule
in the target
tenant.
Rules
without a
defined
target are
disabled
until the
target is
specified.

📝🟢

note

The
exported file
is encoded
in Base64
and cannot
be edited.

5.3.3.2.3.
Manage
Web and API
Security
prevention
profiles



5.3.3.2.4. Add
a disable
prevention
rule for
cloud
workloads



👀🟠

important

All
applicable
prevention
actions are
skipped for
the files and
process that
match the
properties
defined in
the rule.
Consider the
consequence
s of
disabling a
prevention
rule before
you add the
exception,
and monitor
it over time.

🧭➡️

navigation

Settings →
Exceptions
Configuratio
n → Disable
Prevention
Rules

5.3.3.2.5. Add a support exception rule for cloud
workloads  🧭➡️
navigation

From
Settings →
Exceptions
Configuratio
n → Support
Exception
Rules, click
+ Import
from file

5.3.3.2.6. Add
a legacy
exception
rule for
cloud
workloads

 🧭➡️

navigation

Your rule is
created, and
can be
viewed at
the
following
location:
Settings →
Exceptions
Configuratio
n → Legacy
Agent
Exceptions

5.3.3.2.7.
Additional
workload
management
tasks

 🧭➡️

navigation

Several
management
activities
that can be
performed
on

workloads
are

accessible
from
Inventory →
Endpoints →
All
Endpoints
and from
Inventory →
Endpoints →
Groups

5.3.4. API
specification
inventory



📝🟢

note

Even if you
have already
imported the
specification,
you can edit
the API
specification
in Cortex
Cloud and
add or
update the
server list.

5.3.4.1.
Import API
specification

 📝🟢

note

Even if you
already
imported the
file, you can
edit the API
asset and
add or
update the
server list.

6. Review

inventory
and

explore

your
cloud

environm

ent

 6.1. Asset
management



6.1.1. Asset
inventory
overview



6.1.2. All
assets

 📝🟢

note

To maintain
an accurate
and clutter-
free asset
inventory, an
automated
background
cleanup
process

periodically
removes

outdated
assets.

6.1.3. All
cloud assets

 6.1.3.1.
Discovery
Engine



6.1.4. Asset
classes



🔒🟣

content

This feature
is included
with the
AppSec add-
🔒🟣 on.

content

This feature
is included
with the
Attack
Surface
Management
(ASM) add-
on.

6.1.4.1. AI
assets

 🧭➡️

navigation

Inventory →
All Assets →
AI

6.1.4.2. API
assets



6.1.4.3.
Application
assets



📝🟢

note

The
topology
graph is
available
only when
all
application
components
(code,
pipeline,
build and
deploy), are
configured.

🧭➡️

navigation

Inventory →
All Assets →
Business
Applications

6.1.4.4. Code
and CI/CD
assets



6.1.4.4.1. IaC
resources

assets

 📝🟢

note

Scope: The
IaC asset
represents
individual
infrastructur

e resources

defined in
Terraform,
CloudFormat
ion, or
Kubernetes
manifests.
The IaC
asset does
not
represent
the physical
cloud
resource in
the runtime
environment;
those are
managed
under the
Cloud asset
class.

6.1.4.4.2.
Repository
assets



📝🟢

note

Scope: The
repository
asset
represents a
VCS
repository
onboarded
into Cortex
Cloud. The
repository
asset does
not
represent
container
image
repositories,
artifact
registries, or
cloud
resource

inventories;
those asset
categories
are

managed
under the
Compute
and Cloud
asset
classes
respectively.

📝🟢

note

For detailed
information

on

investigating
and
remediating
issues, refer
to Code
Security
scanners.

🧭➡️

navigation

Inventory,
select All
Assets →
Code →
Repositories

6.1.4.4.3. VCS
organization
assets



📝🟢

note

Scope: The
VCS
organization
asset
represents a
VCS
organization
discovered
through an
active VCS
integration.
It captures
the
organization
al identity,
provider
type, and
aggregated
security
posture
across all
child
entities. It
does not
represent
individual
repositories,
CI/CD
pipelines, or
CI/CD
instances,
nor does it
represent
business
applications.

🧭➡️

navigation

Inventory,
select All
Assets →
Code →
VCS
Organization
s

6.1.4.4.4.
CI/CD
pipelines
assets



📝🟢

note

Scope: The
CI/CD
pipeline
asset
represents a
CI/CD
pipeline
definition
associated
with an
onboarded
repository or
CI/CD
integration.
The CI/CD
pipeline
asset

captures the
pipeline
configuration
and build
activitypipeli
ne

configuration
and build
activity as
discovered
through
CI/CD
scanning.
The CI/CD
pipeline
asset does
not
represent
individual...

📝🟢

note

This requires
active CI/CD
integrations
and
successful
build log
analysis.
Pipelines
without
successful
build log
analysis
display only
the
repository
and pipeline
nodes

📝🟢

note

Navigate to
the
dedicated
Application
Security →
Issues →
CI/CD Risks
page to
manage the
remediation
lifecycle at
scale
through bulk
status

updates,
team
assignments,
and SLA
tracking for
compliance
monitoring.

🧭➡️

navigation

Inventory,
select All
Assets →
Code →
CI/CD
Pipelines

6.1.4.4.5.
CI/CD
instances
assets



📝🟢

note

Scope: The
CI/CD
instance
asset
represents a
CI/CD
platform
instance
discovered
through an
active CI/CD
integration.
The CI/CD
instance
asset

captures the
platform
identity,
version, and
aggregated
security
postureplatf
orm identity,
version, and
aggregated
security
posture
across all
pipelines
hosted on
the instance.
The CI/CD
instance
asset does
not...

📝🟢

note

Navigate to
the
dedicated
Application
Security →
Issues →
CI/CD Risks
page to
manage the
CI/CD risks
remediation
lifecycle at
scale
through bulk
status

updates,
team
assignments,
and SLA
tracking for
compliance
monitoring.

🧭➡️

navigation

Inventory,
select All
Assets →
Code →
CI/CD
Instances
6.1.4.4.6.
Software
packages
assets



📝🟢

note

Scope: The
software
package
asset
represents
an open-
source or

third-party
dependency
discovered
through SCA
scanning of
onboarded
repositories.
The
software
package
asset does
not
represent
first-party
application
code,
container
image layers,
or cloud
runtime
packages;
those asset
categories
are

managed
under their
respective
asset...

🧭➡️

navigation

Inventory,
select All
Assets →
Code →
Software
Packages

6.1.4.5.
Compute
assets



📝🟢

note

The Vulnerabilities tab is only available if the
cluster you wish to analyze has a K8s connector.

🧭➡️

navigation

Inventory →
All Assets →
Compute

6.1.4.5.1.
Container
images
assets

 🧭➡️

navigation

Inventory,
select All
Assets →
Compute →
Container
Images

6.1.4.5.2.
Serverless
functions
assets

 📝🟢

note

For details
of all
serverless
function
issues
generated
by Cortex
Cloud from
vulnerability
findings,
refer to
Serverless
function
usage.

6.1.4.5.3. VM
images
assets



📝🟢

note

The data in
the Scanners
column is
accurate

only for
Core Image
assets.

Ignore the
Scanners
value for
assets

categorized
as Registry,
Build, or
Runtime
images, as it
may not
reflect an
accurate
scan status.

📝🟢

note

The Last
Scan value is
only
accurate for
Core Image
assets;
ignore the
Last Scan
values for
Registry,
Build, and
Runtime
images, as
they may be
incorrect.

📝🟢

note

If the VM
image is not
used to
create any
VM instance,
the graph
section will
show no
results. This
feature
enables you
to precisely
identify the
registry and
repository
source of
any running
image,
directly
linking
runtime
security
findings to
their origin.
As a result,
you can
answer audit
and security
questions,
such as
determining
which
registry...

🧭➡️

navigation

Inventory,
select All
Assets →
Compute →
VM Images

6.1.4.6. Data
assets



📝🟢

note

The data
assets
inventory
requires a
Cortex
XSIAM
Premium
license, or a
base Cortex
XSIAM
license that
includes the
Cloud
Posture
Security or
Cloud
Runtime
Security
add-on.

🧭➡️

navigation

Inventory →
All Assets →
Data → All
Data Assets

Inventory →
All Assets →
Data →
Backups

6.1.4.7.
Device assets



📝🟢

note

The device assets inventory requires deployed
Cortex XDR agents, which are included with
Cortex XSIAM Enterprise and Premium licenses,
or available as an add-on for Cortex XSIAM NG-
SIEM.

🧭➡️

navigation

Inventory →
All Assets →
Device →
General
6.1.4.8. Devices
External
Surface
assets

 🔒🟣

content

Requires the
Attack
Surface
Management
(ASM) add-
on

6.1.4.9.
Identity
assets



6.1.4.10.
Network
Assets



6.1.4.11.
Security
services
assets



6.1.5. Asset
groups



📝🟢

note

If an asset
group uses
fields
outside of
this
supported
list, it cannot
be used for
scoping in
Access
Management.

📝🟢

note

Note: You
cannot
create SBAC
based on
static
groups.

When using
dynamic
asset groups
for SBAC,
you can limit
access

based only
on the
following
attributes:
Asset Class,
Category,
Provider,
Region,
Organization,
Realm,
Business
Application
Names,
Kubernetes
Cluster,
Kubernetes
Namespace,
Code
Repository,
and Asset
Tags.

📝🟢

note

When you
create or
edit an Asset
Group, the
changes are
applied
immediately
to new
assets and
to existing
assets that
have been
updated.
However, it
may take a
few hours
for the
changes to
appear on
existing
assets that
have not
been
updated.

6.1.6.
Network
configuration



🔒🟣

content

This feature
is included
with the
Attack
Surface
Management
(ASM) add-
on.

📝🟢

note

You can add
a new range

that is fully
contained
within an
existing
range, but
you cannot
add a new
range that
partially
intersects
with another.

🧭➡️

navigation

Inventory →
Assets →
Network
Configuratio
n to define
the
boundaries
of your
organization'
s network

6.1.6.1.

Configure
your
network
parameters



📝🟢

note

You can add
a range that
is fully
contained in
an existing
range,

however, you
cannot add
a new range

that partially
intersects
with another
range.

🧭➡️

navigation

In Cortex
Cloud ,
select
Assets →
Network
Configuratio
n → Internal
Domain
Suffixes

In Cortex
Cloud ,
select
Assets →
Network
Configuratio
n → Trusted
Networks

6.1.7. Asset
Roles



📝🟢

note

Asset Roles
are available
only if the
Identity
Threat
Module add-
on is
enabled.

6.1.7.1.
Manage
Asset Roles
for Endpoints

 📝🟢

note

The Identity Security Module add-on is required
in order have the ability to explicitly edit the
host lists assigned to asset roles.

6.1.7.2.
Manage
Asset Roles
for Users



📝🟢

note

User Role
Management
is available
only if the
Identity
Threat
Module add-
on is

📝🟢 enabled.

note

Deleting a manually added user removes the
user from the Included Users list. If this user is
detected automatically as having this asset role
in the future, it will appear in the Included Users
list again.
Excluding a manually added user ensures that
even if in the future the user is detected as
having this asset role, this detection is
overridden and...

6.1.7.2.1.
Honey user

 ☑️🟣

prerequisite

The honey
user role is
available
only if the
Identity
Threat
Module add-
on is

6.1.8. Manage enabled.
Asset Scores



📝🟢

note

The Hosts
tab is
available if
the Identity
Threat
Module add-
on is
enabled.

📝🟢

note

As new
alerts are
associated
with
incidents,
the User and
Host Scores
are

recalculated.
You can
view the
latest User
and Host
Scores on
the Asset
Scores page,
or track the
Score trend
on the User
Risk View
and Host
Risk View.

📝🟢

note

Some User
Associated
Insights may
not appear
as part of
the User
Associated
Incidents
due to the
insight
generation
mechanism.
For example,
when an
insight
related to
one of the
assets in an
incident is
generated a
few days
after the
associated
incident, the
insight may
not be
associated
with the
incident.

7. Review

and

prioritize

posture
issues



7.1. Cases
and issues



7.1.1.
Overview of

cases



☑️🟣

prerequisite

To work with
cases, an
administrator
must

configure
your user
role with
specific
RBAC
permissions.
Permissions
must be
enabled in
the
following
order:
Playbooks:
This
component
(under
Investigation
& Response
→

Automations)
must be set
to Enabled
first. Role-
level
permissions
determine
your ability
to create
new

playbooks or
edit those
marked as...

7.1.1.1. What
are cases?



7.1.1.2.
Resolving
cases with AI



7.1.1.3. Case
lifecycle



7.1.1.4. Case
thresholds



7.1.1.5. Case
scope and
impact



7.1.1.6. Case
and issue
domains



7.1.2. Case
concepts



7.1.2.1. Issues,
findings, and
events



7.1.2.1.1. Issues 

7.1.2.1.2.
Findings and
events



7.1.2.2. Case
grouping



7.1.2.3. Case
scoring



7.1.2.4. Case
starring

 🧭➡️

navigation

You can
manage all
starring
configuration
s under Case
& Issues →
Case
Configuratio
n → Starred
Issues

7.1.2.5. What
is Causality?

 📝🟢

note

There are no
CGOs in the
Cloud
Causality
View, when
investigating
cloud
Cortex
Cloud alerts
and Cloud
Audit Logs,
or SaaS
Causality
View, when
investigating
SaaS-
related
alerts for
501 audit
events, such
as Office
365 audit
logs and
normalized
logs.

7.1.3. Analyze
and resolve
cases



7.1.3.1. Review
all cases

 📝🟢

note

Administrato
rs can

enable or
disable
legacy
mode. Go to
Configuratio
ns →
General →
Server
Settings →
Case display
modes.

7.1.3.2. Start
case analysis



📝🟢

note

This section
describes
the product
in default
mode and
using the
Split view. If
you are

using legacy
mode, see
Detailed
View.

7.1.3.2.1.
Agentic
Assistant-
Case
Investigation
agent



7.1.3.3.
Establish
case context



7.1.3.3.1. AI-
generated
case

summaries



📝🟢

note

The AI-
generated
title and
description
is a
calculated
value that is
regenerated
each time
you open a
case.

This value is
not a saved
static
description,
therefore it
is not
reflected in
the saved
case names

in the list of
cases in the
Split view ,
or in the
Case Name
and Case
Description
columns in
the Table
view.

🧭➡️

navigation

Configuratio
ns →

General →
Server
Settings →
AI

Configuratio
n and enable
the
following
settings

7.1.3.3.2.
Assess case
severity and
score



7.1.3.3.3.
Update case
attributes

 📝🟢

note

The defined values are shown in the Case Name
and Case Description columns in the Table view,
and saved to the cases dataset. The case title is
also shown the list of cases in the Split view.
These values do not replace the AI-generated
case title and summary. If the Summarize with
AI toggle is enabled, AI-generated case
summaries are automatically...

7.1.3.4.
Analyze
case details



📝🟢

note

If you prefer
a tabular or
legacy
layout,
switch the
case card to
the Detailed
view.
This view
preserves

the legacy
tab based
format and
custom

layouts,
ensuring full
backward
compatibility.
You can
switch
between the
new case

experience
and the
legacy view
based on
personal
workflow
preferences.
For more
information,
see Detailed
View.

7.1.3.4.1.
Grouping
graph

 📝🟢

note

Similar issues are displayed as individual entities
rather than in a parent/child hierarchy.

7.1.3.4.2.
Evidence



7.1.3.4.3. Issue
feed



7.1.3.4.4.
Associated
assets and
artifacts

 📝🟢

note

If you do not
have
permissions
to access an
asset of a
case (which
is shown as
grayed out
and locked),
check your
scoping
permissions
in Manage
Users or
Manage
User Groups.

7.1.3.4.5.
MITRE
ATT&CK
tactics and
techniques

 📝🟢

note

This
component
is available
for cases
associated
with the
Security
domain or
custom
domains.

7.1.3.4.6.
Case timeline

 📝🟢

note

If you
unmark a
record as
evidence,
the evidence
flag is
removed but
the
underlying
timeline
record and
its
attachments
are not
deleted.

7.1.3.4.7.
Detailed
View

 📝🟢

note

When an
issue is
resolved, it
remains
linked to a
case. Once
all of the
issues in a
case are

resolved, the
case is
automaticall
y closed.

7.1.3.5.
Resolve the
case



7.1.3.5.1.
Resolution
Center

 📝🟢

note

Tasks that
are already
In Progress
but require
your input
will appear
in both the
Pending and
In Progress
tabs.

7.1.3.5.2.
Collaborativ
e notes and
comments



7.1.3.5.3.
Resolve a
case

 📝🟢

note

If a case is
resolved
with the
status
Resolved -
Auto
Resolved,
Cortex
Cloud can
reopen the
case for up-
to six hours
if a new
issue is
triggered
that
matches the
case. The
six-hour
period is
defined by
the
timestamp
of the last
issue that
was grouped
into the
case. After
the six-hour
period, any
new issues
are linked to
a new case

for a new...

7.1.3.5.4.
Resolution
reasons for
cases and
issues



📝🟢

note

Cases and
issues
resolved as
True Positive
and False
Positive help
Cortex
Cloud to
identify real
threats in
your
environment
by
comparing
future cases
and
associated
issues to the
resolved
cases.

Therefore,
the handling
and scoring
of future
cases is
affected by
these
resolutions.

📝🟢

note

Cases and
issues
resolved as
True Positive
and False
Positive help
Cortex
Cloud to
identify real
threats in
your
environment
by
comparing
future cases
and
associated
issues to the
resolved
cases.

Therefore,
the handling
and scoring
of future
cases is
affected by
these
resolutions.

7.1.3.6.
Additional
case actions



7.1.3.6.1.
Create a
case



📝🟢

note

To create a
case

manually,
you must
have
View/Edit
permission
for Cases
and Issues
selected
under
Settings →
Configuratio
ns → Access
Management
→ Roles →
Components
→ Cases &
Issues.

📝🟢

note

This option
is only
relevant for
certain

📝🟢 domains.

note

This option
is only
relevant for
certain

📝🟢 domains.

note

You can't
attach files
to manually
created

💡🔵 cases.

tip

The issues
that you link
to a case
can be
linked to
multiple
cases, and
the issue
domains do
not need to
match the
case domain.

7.1.3.6.2.
Merge a case



7.2.
Investigation
and response



7.2.1.
Investigate
issues



☑️🟣

prerequisite

To work with
issues, an
administrator
must

configure
your user
role with
specific
RBAC
permissions.
Permissions
must be
enabled in
the
following
order:
Playbooks:
This
component
(under
Investigation
& Response
→

Automations)
must be set
to Enabled
first. Role-
level
permissions
determine
your ability
to create
new

playbooks or
edit those
marked as...

7.2.1.1.
Overview of
the Issues
page

 📝🟢

note

Every 12
hours, the
system
enforces a
cleanup
policy to
remove the
oldest issues
once the
maximum
limit is
exceeded.
The default
issue
retention
period in
Cortex
Cloud is 186
days.

7.2.1.2. Issue
card

 📝🟢

note

This section
is context-

specific and
shows data
according to
the issue
context.

7.2.1.3.
Resolution
actions



7.2.1.4. Link
or unlink
issues from

a case



7.2.1.5. Run
an

automation
on an issue



📝🟢

note

In addition
to
automation,
some

playbooks
contain
manual tasks
that prompt
the analyst
for input.
This enables
you to
enhance an
automation
workflow
with analyst
input.

📝🟢

note

You can also
manually
select a
playbook to
run from the
Issue Work
Plan tab.

7.2.1.6. Use
the War
Room in an
investigation



📝🟢

note

The case
War Room is
usually used
for
communicati
on

capabilities,
but unlike
the issue
War Room, it
does not
include
playbook
specific
entries. The
case War
Room
enables you
to
investigate
an entire
case, not
just an issue.

📝🟢

note

Cortex
Cloud does
not index
notes and

💡🔵 chats.

tip

In the
Playground,
you can
clear the
context
data, if
needed,
which
deletes
everything in
the
Playground
context
data, but
does not
affect the
actual issue
or case. To
clear the
context, run
!DeleteConte
xt all=yes'
from the CLI
or click
Clear
Context
Data while
viewing the
context data.

💡🔵

tip

You can use
the up/down
arrow

buttons in
the CLI to
do a reverse
history
search for
previous
commands
with the
same prefix.

🧭➡️

navigation

Investigation
& Response
→

Automation

→

Playground

7.2.1.7. Use
the Work
Plan in an
investigation

 👀🟠

important

A playbook
will not
continue its
execution

path if a
prior task
has failed;
you must
resolve the
failed task
before
subsequent
tasks can run.

7.2.1.8. Issue
syncing



☑️🟣

prerequisite

You must set
up an

integration
before you
can sync
issues. For
more

information,
see Set up
an

integration
for mirroring
issues.

☑️🟣

prerequisite

You must set
up an
integration
before you
can sync
issues. For
more

information,
see Set up
an

integration
for mirroring
issues.

☑️🟣

prerequisite

You must set
up an

integration
before you
can sync
issues. For
more

information,
see Set up
an

integration
for mirroring
issues.

📝🟢

note

Multiple
tickets can
be linked to
an issue with
outbound
syncing.
Issues with
inbound
syncing can
be linked to
a single
ticket only.

📝🟢

note

Using issue
fields as
variables is
not

currently
supported.

📝🟢

note

You can only
define a
single
inbound
profile. If
you change
the inbound
sync profile
the current
profile is
overwritten.
You can
define
multiple
outbound
profiles; one
issue can
update
multiple
tickets.

📝🟢

note

Using issue
fields as
variables is
not

currently
supported.

📝🟢

note

You can only
define a
single
inbound
profile. If
you change
the inbound
sync profile
the current
profile is
overwritten.
You can
define
multiple
outbound
profiles; one
issue can
update
multiple
tickets.

📝🟢

note

The inbound
syncing flow
runs every
two minutes,
and the
outbound
syncing flow
runs every
five minutes.
In a bi-
directional
set-up, if the
same field is
updated in
both tickets,
the most
recently
updated
value is used.
In the
external
ticket, the
logged
history
shows
updates to
the ticket.
The user
name that is
logged with
the history
reflects the

user...

📝🟢

note

After an
issue is
resolved,
ticket
syncing
remains
active for
up-to seven
days.
Therefore,
you still
update,
change, or
reopen the
issue or
external
ticket and
the tickets
will continue
to sync.

📝🟢

note

If you
change the
selected
inbound
sync profile,
the original
sync profile
is
immediately
overwritten.

📝🟢

note

You can only
define a
single
inbound
profile. If
you change
the inbound
sync profile
the current
profile is
overwritten.
You can
define
multiple
outbound
profiles; one
issue can
update
multiple
tickets.

⛔🔴

warning

If you leave
this field
blank, all
configured
instances
will be used.

⛔🔴

warning

If you leave
this field
blank, all
configured
instances
will be used.

💡🔵

tip

You can find
a sync

profile ID
under
Settings →
Configuratio
ns → Object
Setup →
Issues →
Sync
Profiles. By
default the
ID field is
not

displayed in
the table.
Click the
three dot
menu and
add it to the
table layout.

7.2.1.9. Issue
deduplicatio
n



📝🟢

note

Issues are
automaticall
y excluded
from
deduplicatio
n if the
agent_id is
missing, the
hash_id is
missing, or
the hash_id
is an all-zero
SHA256
string.

🧭➡️

navigation

File
Identification
(Hash): Use
the SHA256
hierarchy
(Action File
→ Action
Process →
Actor
Process

7.2.1.10.
Causality
view



💡🔵

tip

To view
statistics for
files on
VirusTotal,
you can

pivot from
the Initiator
MD5 or
SHA256
value of the
file on the
Files tab.

7.2.1.10.1.
Cloud
causality
view



7.2.1.10.2.
SaaS
causality
view



7.2.1.10.3.
Timeline

 📝🟢

note

The Timeline
view is not
available
when
investigating
cloud
Cortex
Cloud issues
and cloud
audit logs or
SaaS-
related
issues for
501 audit
events, such
as Office
365 audit
logs and
normalized
logs. Only
the
applicable
cloud
causality
view and
SaaS
causality
view is
available for
this data.

7.2.1.10.4.
Network
causality
view



📝🟢

note

The network
causality
view
displays only
the
information
it collects
from the
detectors. It
is possible
that the CI
may not
show some
of the
firewall or
agent
processes.

💡🔵

tip

To view
statistics for
files on
VirusTotal,
you can

pivot from
the Initiator
MD5 or
SHA256
value of the
file on the
Files tab.

7.2.1.10.5.
Causality
icons key



7.2.1.11. Issue
investigation
actions



7.2.1.11.1.
Copy issues



7.2.1.11.2.
Update issue
fields



📝🟢

note

You must
create a
custom
status

before you
can select it.

💡🔵

tip

You can
create
custom issue
statuses and
resolution
reasons, and
use the
setIssueStatu
s command
to set these
custom
statuses for
issues.
For example,
when a user
starts
investigating
an issue, the
issue status
is
automaticall
y changed
from New to
Under
Investigation.
In some
cases, it is
useful to
create an
interim
status, such
as Triage.
After you...

7.2.1.11.3.
Create
profile
exceptions

 🧭➡️

navigation

Inventory+En
dpoints →
Policy
Management
7.2.1.11.4. → Profiles
Investigate
contributing
events



7.2.1.11.5.
Create a
featured
field



📝🟢

note

Featured
Active
Directory
values are
displayed in
the User and
Host fields
accordingly.

🧭➡️

navigation

Cases &
Issues →
Case
Configuratio
n →

Featured
Fields and
select a
type of
featured
field

7.2.1.11.6.
View
generating
BIOC or IOC
rule



7.2.1.11.7. Add
a file path to
a malware
profile allow
list

 🧭➡️

navigation

Inventory+En
dpoints →
Policy
Management
→

Prevention
→ Profiles
and locate
the malware
profile you
selected

7.2.1.11.8.
Retrieve
additional
issue details

 💡🔵

tip

If you
require
assistance
from Palo
Alto
Networks
support to
investigate
the issue,
make sure to
provide the
downloaded
ZIP file.

7.2.1.11.9.
Export issue
details to a
file



7.2.1.11.10.
Exclude an
issue

 📝🟢

note

Cortex
Cloud
supports
exclusion of
up to
100,000
issues.

7.2.1.11.11.
Query case
and issue
data



7.2.2. Review
findings

 7.2.2.1.
Findings card

 📝🟢

note

The information in this card is context specific,
therefore some sections are not available for all
findings.

7.2.3.
Investigate
artifacts and
assets



7.2.3.1.
Investigate
an IP address

 📝🟢

note

Requires a
license key.
Select
Settings →
Configuratio
ns →

Integrations
→ Threat
Intelligence.

7.2.3.2.
Investigate
an asset

 📝🟢

note

The Asset
view is
available for
hosts with a
Cortex XDR
agent
installed.

7.2.3.3.
Investigate a
file and
process hash

 📝🟢

note

Requires a
license key.
Go to
Settings →
Configuratio
ns →

Integrations
→ Threat
Intelligence.

7.2.3.4.
Investigate a
user



🔒🟣

content

If the
Identity
Threat
module is
enabled, you
can open
the User Risk
View. This
view
displays
insights and
profiling
information
to help you
investigate
issues and
cases.

Viewing
anomalies in
the context
of baseline
behavior
facilitates
risk
assessment
and shortens
the time you
require for
making
verdicts.
If the
Identity
Threat
module is
not enabled...

🔒🟣

content

The User
Risk view is
available
with the
ITDR add-
on.

Depending
on your
permissions,
some

information
may be
limited by
your scope.

📝🟢

note

Cortex
Cloud
normalizes
and displays
case and
issue times
in your time
zone. If
you're in a
half-hour
time zone,
the activity
in the Issues
& Insights
Heatmap is
displayed in
the whole-
hour time
slot
preceding it.
For example,
if you're in a
UTC +4.5
time zone,
the time
displayed
for the
activity will
be UTC +4.5,
however, the
visualization
in the...

💡🔵

tip

You can also
see a list of
all users
under
Inventory →
Assets →
Asset Scores.

7.2.4.
Investigate
endpoints



7.2.4.1.
Overview of
the Action
Center



📝🟢

note

Blocking
files on
endpoints is
enforced by
the endpoint
malware
profile. To
block a hash
value,
ensure the
hash value is
configured
in the
Malware
security
profile.
Select
Override
Report
mode to
allow the
agent to
block
hashes, even
if the
Malware
Profile is set
to Report.

🧭➡️

navigation

Investigation & Response → Response → Action
Center

7.2.4.1.1.
Initiate and
monitor
endpoint
actions



7.2.4.1.2.
Action
Center
reference
information



7.2.4.2.
Manage
endpoints



📝🟢

note

For the
Include
endpoints
from auto
upgrade
action, you
cannot
enable auto
upgrade for
Mobile, VDI,
and TS
installations.

📝🟢

note

If an
endpoint is
excluded,
the auto
upgrade
profile
configuration
is not
available.
If you
exclude the
endpoint
from auto
upgrade
while the
auto

upgrade
action is In
progress,

the ongoing
upgrade will
still take
place.

📝🟢

note

Content
Status is
calculated
every 30
minutes.
Therefore,
there might
be a delay
of up to 30
minutes in
displaying
the data.

📝🟢

note

Only
supported
for Windows
and macOS.

📝🟢

note

Changes to
the agent
status can

take up to
ten minutes
to display on
Cortex
Cloud .

🧭➡️

navigation

Settings →
Configuratio
ns →

General →
Agent
Configuratio
ns →

Endpoint
Administratio
n Cleanup

7.2.4.3.
Retrieve files
from an
endpoint



📝🟢

note

You cannot
define a
path using
environment
variables on
Mac and
Linux
endpoints.

📝🟢

note

If the
Password
Protection
(for
downloaded
files) setting
under
Settings →
Configuratio
n → General
→ Server
Settings is
enabled,
enter the
password
'suspicious'
to download
the file.

📝🟢

note

Disabling
File Retrieval
does not
take effect
on file
retrieval
actions that
are in
progress.

7.2.4.4.
Retrieve
support logs
from an
endpoint

 🧭➡️

navigation

Investigation & Response → Response → Action
Center → + New Action

7.2.4.5. Scan
an endpoint
for malware

 🧭➡️

navigation

Investigation & Response → Response → Action
Center → New Action

7.2.5.
Investigate
files



7.2.5.1.
Manage file
execution



7.2.5.2.
Manage
quarantined
files

 📝🟢

note

This will
restore all
files with the
same hash
on all of
your

endpoints.

7.2.5.3.
Review
WildFire
analysis
details

 📝🟢

note

Cortex
Cloud
displays the
preview of
WildFire
reports that
were

generated
within the
last couple
of years. To
view a
report that
was

generated
more than
two years
ago, you can
download
the report.

7.2.5.4.
Import file
hash
exceptions



7.2.6. Cortex
Assistant



📝🟢

note

If you are in an eligible region and have enabled
the Cortex Agentic Assistant, the Cortex
Agentic Assistant replaces the Cortex Assistant.
The Cortex Assistant is available if you do not
have access to the Cortex Agentic Assistant
based on the tenant region or you have not
enabled it. For more information, see Cortex
Agentic Assistant.Cortex Agentic...

🧭➡️

navigation

To increase
usability,
you can
create a

personalized
keyboard
shortcut:
Settings →
Configuratio
ns → Server
Settings →
Keyboard
Shortcuts
and choose
the shortcut
you want to
use

7.2.6.1.
Cortex
Assistant
layout



7.2.6.2.
Cortex
Assistant
capabilities



📝🟢

note

In some cases, if the prompt does not include at
least one recognizable entity such as an IP,
hash, user, asset, domain, case, or XQL query,
no response is returned.

📝🟢

note

When you
choose an
option from
the Respond
column,
Cortex
Assistant
will always
prompt you
to approve
the action
before
executing.

7.2.7.
Automation



7.2.7.1.
Automation
in Cortex
Cloud

 📝🟢

note

You can
build end-to-
end
automation
workflows
from within
the
playbook
editor,
including
creating
automation
rules,
configuring
integration
instances,
and creating
and editing
tasks. For
more

information,
see

Playbooks.

7.2.7.2. Quick
Actions

 📝🟢

note

Quick
Actions
appear as
War Room
entries, but
do not
appear in
the Work
Plan.

7.2.7.3.
Automation
Exclusion
Center



📝🟢

note

By default,
all users
have read
and edit
permissions
to lists.
When
creating a
list of
critical
assets, we
recommend
limiting the
read and
edit
permissions
to specific
roles.

7.2.7.3.1.
Manage
automation
exclusion
policies



📝🟢

note

For the IAM
User Hard
Remediation
and User
Soft
Remediation
policies, we
recommend
including
username,

email, and ID
for each
user you
want to
exclude.
Example:
username1,
user@exampl
e.com,
userID112.

📝🟢

note

You can also
right click on
a policy
from the
main
Automation
Exclusion
Center page
to disable or
enable the
policy.
If you click
on a list
name in the
Exclude
column, that
list opens in
the Lists
page.

7.2.7.4.
Playbooks



☑️🟣

prerequisite

To work with
playbooks,
an

administrator
must

configure
your user
role with
specific
RBAC
permissions.
Permissions
must be
enabled in
the
following
order:
Scripts: This
component
(under
Investigation
& Response
→

Automations)
must be set
to Enabled
first. It is the
foundational
permission
for all
automation;
if Scripts are
not enabled,
you cannot...

7.2.7.4.1.
Playbooks
overview



7.2.7.4.2.
Access to
playbooks



7.2.7.4.3.
Playbook
development
checklist



7.2.7.4.4. Plan
your

playbook



7.2.7.4.5.
Manage
playbooks

 📝🟢

note

The library
by default
shows only
playbooks
that are not
adopted.
Click the
Show
Adopted
checkbox to
show the
adopted
playbooks,
indicated by
an Adopted
mark.
The library
shows the
most

updated
playbook
version.
Adopting an
older version
than shown
should be
done
through
Marketplace.
Adopting a
playbook
does not
make it run.
Some
content

packs
include...

7.2.7.4.6.
Build your
playbook



👀🟠

important

The ability
to create,
edit, or
share
custom

playbooks is
governed by
access

management.
If certain
options are
unavailable,
contact your
administrator
. For more
information,
see Manage
access to

playbooks
and scripts.

7.2.7.4.6.1.
Task 1.
Choose
from existing
playbooks or
create your
own



📝🟢

note

When a task
inside of a
sub-
playbook is
not

configured,
the alert is
propagated
to the main
playbook. If
multiple sub-
playbooks
are nested,
and any of
the sub-
playbooks
have non-
configured
components,
the alert
appears in
the main
playbook as
well as in the
sub-
playbooks.
Alerts also
appear for
the
individual

non-

configured
tasks within
the...

📝🟢

note

You must
have the
required
permissions
to create

playbooks
to view this
button. For
more

information,
see Manage
access to

playbooks
and scripts.

💡🔵

tip

If there are additional relevant playbooks in
Marketplace that are not in your Org repository,
you can click Explore them now to see them in
the Playbook Catalog and choose to adopt.

💡🔵

tip

To open
multiple
playbooks at
the same
time, edit
the first
playbook
and then
click New
next to the
playbook
name to
create a new
tab. You can
either create
a new

playbook, or
add an
existing one.
You can
view
recently
modified or
deleted
playbooks
by clicking
version
history for
all
playbooks .

🧭➡️

navigation

Open the
Investigation
& Response
→

Automation

→

Playbooks
page to find
an existing
playbook,
customize it,
or create a

playbook

7.2.7.4.6.2.
Task 2.
Configure
playbook
settings



📝🟢

note

If the
playbook
has inputs
and outputs,
the
Playbook
Starts task
will show
back and
forth arrows.
Clicking
them opens
the
Playbook
Settings
pane

Inputs/Outpu
ts tab.

📝🟢

note

For more
information

on

Automation
Rules, see
Create an
automation
rule.

📝🟢

note

This rule will
trigger the
playbook to
run if no
other
Automation
Rule triggers
the
playbook
first. You can
view and
edit the
order the
rules run in
the
Automation
Rules page.

📝🟢

note

If you do not
add any
fields, the
group will
be deleted
when you
click Save.

7.2.7.4.6.3.
Task 3. Add
objects from
the Task
Library



7.2.7.4.6.4.
Task 4. Add
custom

playbook
features



7.2.7.4.6.5.
Task 5. Test
and debug
the playbook



7.2.7.4.6.6.
Task 6.
Manage
playbook
content



7.2.7.4.7.
Customize
your

playbook



7.2.7.4.7.1.
Configure a
sub-
playbook
loop



📝🟢

note

Consider the
following
when adding
a loop:
The
maximum
number of
loops
(default is
100). A high
number of
loops or a
high wait
time
combined
with a large
number of
issues may
affect
performance.
Periodically
check
looping
conditions
to ensure

they are still
valid for the
data set.
If you want a
sub
playbook
task to loop
over an

array passed
into its...

💡🔵

tip

Balance
between the
number of
iterations
and the
interval so
you do not
overload the
server.

7.2.7.4.7.2.
Filter and
transform
data



⚠️🟡

caution

You can
change the
context data
root to filter,
but it is not
recommende
d to select a
different
root, as it
affects the
filter results.
The drop-
down list
displays the
filter root
for
backward
compatibility.

🧭➡️

navigation

Investigation
& Response
→

Automation
→ Scripts →
New Script

7.2.7.4.7.3.
Extend
context



7.2.7.4.7.4.
Extract
indicators



📝🟢

note

This
configuration
may delay
playbook
execution
(issue
creation).
While
indicator
creation is
asynchronou
s, indicator
extraction
and
enrichment
are run

synchronousl
y. Data is
placed into
the issue
context and
is available
via the
context for
subsequent
tasks.

📝🟢

note

When using
out of band,
the
extracted
indicators
do not
appear in
the context.
If you want
the
extracted
indicators to
appear
select inline.

7.2.7.4.7.5.
Update issue
fields with
playbook
tasks

 📝🟢

note

The setIssue
script
includes all
available
fields; use
the scroll
bar to see
all the fields.
The name
field has a
limit of 600
characters. If
there are
more than
600
characters,
you can
shorten the
name field
to under 600
characters
and then
include the
full
information
in a long
text field
such as the
description
field.

7.2.7.4.8. Test
your

playbook



📝🟢

note

The
debugger
does not
support
using
parentIncide
ntFields.

📝🟢

note

Using an existing issue in the debugger does not
affect the original issue or change the original
context data.

7.2.7.4.8.1.
Troubleshoot
playbook
performance



7.2.7.4.9.
Manage
playbook
content



📝🟢

note

Opening a
playbook
does not
immediately
lock it. A
playbook is
considered
to be in edit
mode only
when a user
makes a
modification
that causes
the Save
button to
become
available.

👀🟠

important

Manually
unlocking a
playbook
may cause
version
conflicts.
This turns
off
concurrent

editing
protection,
and if the
original user
is still
editing, their
changes
might be
lost or
overwritten.

🧭➡️

navigation

Investigation
& Response
→

Automations
→ Playbooks

7.2.7.4.10.
Best
practices



📝🟢

note

If you
reattach a
detached
playbook,
any
customizatio
ns you have
made to the
playbook
will be
overwritten
when the
playbook
updates to
the current
version.

📝🟢

note

Retries are not supported for data collection
tasks that have errors sending emails (indicated
by a server timeout). This is because retries only
work on automation execution failures, not on
email delivery issues.

7.2.7.5. AI
Prompts



📝🟢

note

In MSSP
environments
, this option
is not
available on
the main
tenant.

🧭➡️

navigation

On the AI
Prompts
page
(Investigation
→ Response
→

Automation
→ AI
Prompts),
you can
view, edit,
and
manually
create
prompts

7.2.7.5.1. AI
prompts
role-based
access

control



7.2.7.5.2. Use
existing
prompts

 🧭➡️

navigation

Investigation
& Response
→

Automation
→ AI

Prompts and
in the
Prompts
Library
search for
the prompt
you want to
use

7.2.7.5.3.
Create a
prompt

 🧭➡️

navigation

Investigation
& Response
→

Automation
→ AI
Prompts and
click + New
Prompt

7.2.7.5.4.
Write
effective
prompts



📝🟢

note

An AI
prompt
cannot
create new
context

keys; it can
only refer to
existing
ones. The
only key an
AI prompt
can create is
the output
key.
Therefore, if
a variable
like
${PostKey} is
not already
mapped to
existing
context
data, it
cannot be
used in the
prompt.

📝🟢

note

The LLM's
response is
always
returned in a
text format,
even if you
explicitly ask
the LLM to
return a
structured
format such
as JSON.
Therefore,
the
subsequent
task in your
playbook
will have to
parse the
response to
extract the
relevant data.

7.2.7.6.
Create an
automation
rule



📝🟢

note

You can also
define the
conditions
that trigger
a specific
playbook in
the
playbook
editor. For
more

information,
see Task 2.
Configure
playbook
settings

📝🟢

note

Quick Actions, by default, run using all
available integration instances that contain the
command. When selecting a Quick Action for
an automation rule, you can instead choose one
specific integration instance to use.

7.2.7.7. Scripts 

7.2.7.7.1. Use
existing
scripts

 🧭➡️

navigation

Investigation
& Response
→

Automation
→ Scripts
and in the
Scripts
Library
search for
the script
you want to
use

7.2.7.7.2.
Create a
script



📝🟢

note

You can
enable/disabl
e a script in
the Settings
without
having to
duplicate
the script.
You can
view
recently
modified or
deleted
scripts by
clicking the
version
history for
all scripts .

👀🟠

important

If you
choose
Python, from
the Agentic
Assistant
you can use
the
Automation
Engineer
agent.

💡🔵

tip

If there is an
error, you
can copy
the error
message
from the
test result
into the
Agentic
Assistant
prompt and
ask the
Automation
Engineer
agent to
correct the
error.

7.2.7.7.3. Use
the
Automation
Engineer
agent to
accelerate
script
development
and
deployment



📝🟢

note

The
Automation
Engineer
agent is
available
with the
Cortex
Agentic
Assistant,
for users
with script
editing
permissions.
For more
information,
see Cortex
Agentic
Assistant
and Agentic
Assistant
role-based
access

control.Corte
x Agentic
Assistant

📝🟢

note

If you make
manual edits
in the script
and don't
save the
changes and
you then
modify the
script with
the
Automation
Engineer
agent, you
are

prompted to
confirm
overwriting
the manual
edits.

💡🔵

tip

Use detailed
prompts, for
example,
Get failed
logins from
last 24
hours and
return as a
table.
Clearly
define
argument
names,

types, and
default
values.
Mention if
you expect
the output in
a specific
format, such
as a table,
JSON, or
plain text.

💡🔵

tip

If there is an
error, you
can copy
the error
message
from the
test result
into the
Agentic
Assistant
prompt and
ask the
Automation
Engineer
agent to
correct the
error.

7.2.7.7.4.
Change the
Docker
image in a
script

 7.2.7.7.4.1.
Connect an
engine to an
image
registry

 📝🟢

note

This
procedure
uses the --
username

and --
password
command
line options
to pass the
username

and
password
directly. For
environments
where
command
history or
logs are
visible to
others,
consider
more secure

methods like
Docker
configuration
files for
handling
authenticatio
n in

production
or CI/CD
environments
. For more
details, see
docker login
or...

7.2.7.8.
Context data



7.2.7.8.1. Issue
context data



7.2.7.8.2.
Case
context data



7.2.7.8.3.
Search
context data



7.2.7.8.4. Add
context data
to an issue



7.2.7.8.5. Add
context data
to a case

 📝🟢

note

If you run
the
command in
the issue
War Room,
the data is
added to the
following
places:
The case
context data.
The issue
context data
under the
case tab.
If you run
the
command in
the Case
War Room,
the data is
added to the
case context

data only.

7.2.7.8.6.
Delete
context data
from a case



7.2.7.8.7. Use
context data
in a playbook

 ⚠️🟡

caution

Users with
Trigger
Playbook
permissions
on a given
issue may
still be able
to modify
the parent
case via
commands
and scripts,
even without
full access
to the case.

7.2.7.9. Lists 

📝🟢

note

The
maximum
size of a list
is 209715
characters.

7.2.7.9.1.
Create a list



📝🟢

note

If you intend to use the list as part of an
Automation Exclusion Policy, we recommend
choosing specific roles for read and edit
access. For more information on using lists in
exclusion policies, see Manage automation
exclusion policies.Manage automation exclusion
policies

📝🟢

note

If you want
to edit a list
from a
content

pack, you
need to
duplicate or
detach a list.
Detached
lists do not
receive
updated
content in
subsequent
Cortex
Cloud
content
releases. To
retain an
updated list,
reattach it.

7.2.7.9.2. List
commands

 📝🟢

note

The list must
already exist
before you
use the
getList,
addToList,
setList, or
removeFomLi
st
commands.

7.2.7.9.3. Use
cases: JSON
lists

 🧭➡️

navigation

Settings →
Configuratio
ns → Object
Setup →
Lists → Add
a List

Investigation
& Response
→

Automation

→

Playbooks
→ New
Playbook

7.2.7.9.4.
Transform a
list into an
array

 🧭➡️

navigation

Investigation
& Response
→

Automation

→

Playbooks
and create
or edit a
playbook

7.2.7.10.
Integrations



7.2.7.10.1.
Integrations
in Cortex
Cloud



7.2.7.10.2.
Add an
integration
instance



7.2.7.10.3. Use
integration
commands
in the CLI

 💡🔵

tip

In the
Playground,
you can
clear the
context
data, if
needed,
which
deletes
everything in
the
Playground
context
data, but
does not
affect the
actual issue
or case. To
clear the
context, run
!DeleteConte
xt all=yes'
from the CLI
or click
Clear
Context
Data while
viewing the
context data.

7.2.7.10.4.
Troubleshoot
Integrations

 📝🟢

note

If an
integration
instance
consistently
encounters
API rate
limits when
running on
the tenant,
consider
configuring
the instance
to run on an
engine to
change the
source of
the
outbound
traffic.

7.2.7.10.5.
Manage
credentials

 🧭➡️

navigation

Settings →
Configuratio
ns →

Integrations
→

Credentials
→ New
Credential

7.2.7.11.
Engines



7.2.7.11.1.
What is an
engine?



📝🟢

note

You cannot
share a
multiple-
engine
installation
with a single-
engine
installation.

📝🟢

note

When you
add an
engine to a
load-
balancing
group, you
cannot use

that engine
separately.
The engine
does not
appear in
the engines
menu when
configuring
an

integration
instance, but
you can
choose the
load-
balancing
group.

7.2.7.11.2.
Engine
requirements



📝🟢

note

The Cron
package is
required to
install
engines on a
Linux
machine.

📝🟢

note

CentOS 8.x
reached End
of Life (EOL)
on

December
31, 2021, and
is no longer
supported
as an

operating
system.
CentOS 7.x
reached End
of Life (EOL)
on June 30,
2024, and is
no longer
supported
as an

operating
system.

📝🟢

note

If you have
configured a
range of
Approved IP
Ranges
under
Allowed
Sessions on
the Security
Settings
page, the
engine must
communicat
e through
one of the
approved IPs.

7.2.7.11.3.
Install an
engine



📝🟢

note

If you are
using DEB,
RPM, or Zip
installation,
install
Docker or
Podman.
Natively
running
Python or
PowerShell
integrations/s
cripts on
Windows or
Linux is not
supported
on Cortex
Cloud
engines.

📝🟢

note

When upgrading an engine that was installed
using the Shell installation, you can use the
Upgrade Engine feature in the Engines page. For
Amazon Linux 2 type engines, you need to
upgrade these engine types using a zip-type
engine and not use the Upgrade Engine feature.
If you use the shell installer, Docker/Podman is
automatically installed. We...

📝🟢

note

Use DEB and
RPM
installation
when the
shell
installation
is not
available.
You need to
manually
install
Docker or
Podman
and any
dependencie
s.

📝🟢

note

If you receive a permissions denied error, it is
likely that you do not have permission to access
the /tmp directory.
If the installer fails to start due to a permissions
issue, even if running as root, add one of the
following two arguments when running the
installer:
--target <path> - Extracts the installer files
into the specified custom...

👀🟠

important

For DEB/RPM engines, Python (including 3.x)
and the containerization platform
(Docker/Podman) must be installed and
configured. For Docker or Podman to work
correctly on an engine, IPv4 forwarding must be
enabled.

💡🔵

tip

For Linux
systems, we
recommend
using the
shell
installer. If
using
Amazon
Linux 2, use
the zip
installer (see
step 4).

7.2.7.11.3.1.
Docker



📝🟢

note

This section is relevant when installing an engine.

📝🟢

note

Docker
images can
be
downloaded
together
with their
relevant
content

packs for
offline
installation.

7.2.7.11.3.2.
Podman

 📝🟢

note

When
upgrading
an engine,
the engine
keeps the
previously
used
container
management
type
(regardless
of
distribution
version).

7.2.7.11.4.
Manage
engines



7.2.7.11.5.
Upgrade an
engine



7.2.7.11.6.
Remove an
engine



7.2.7.11.7.

Configure
engines



📝🟢

note

A day is
defined as
24 hours
and may not
exactly
correspond
to calendar
days due to
daylight
savings, leap
seconds, etc.

📝🟢

note

In addition,
to support
engine
upgrades
from the UI,
edit the
/usr/local/de
misto/upgrad
e.conf file
on the
engine to
include the
SERVER_URL
S setting
with the new
tenant's
address.
Include only
the host,
without
https:// or
any
additional
path at the
end of the
host name.
For example:
SERVER_URL
S="api-
example.us.p
aloaltonetwo
rks.com"

📝🟢

note

If the engine
loses
communicati
on for
longer than
this time, it
will
disconnect
and you
need to
restart the
service.

7.2.7.11.7.1.

Configure
the engine
to use a web
proxy



📝🟢

note

You need to
configure
Docker to
use a proxy.

When using
a BlueCoat
proxy,
ensure you
encode the
values
correctly.

📝🟢

note

In an environment with a single engine, go to
/usr/local/demisto/upgrade.conf. In an
environment with multiple engines, go to
/usr/local/demisto/<engine-
name>/upgrade.conf, replacing <engine-name>
with the name of the engine.
Note that the key is in the upgrade.conf file and
must be https_proxy, even if your proxy address
starts with http://.

7.2.7.11.7.2.

Configure
the engine
to call the
server

without
using a proxy



7.2.7.11.7.3.
Configure
an engine to
use custom
certificates



7.2.7.11.8. Use
an engine in
an

integration

 📝🟢

note

Long-
running
integrations
should not
run on load-
balancing
groups.

7.2.7.11.9. Run
a script
using an
engine

 🧭➡️

navigation

Investigation
& Response
→

Automation
→ Scripts

7.2.7.11.10.
Troubleshoot
engines



📝🟢

note

Ensure that
pop-ups are
not blocked
by your
📝🟢 browser.

note

If the
installer fails
to start due
to a
permissions
issue, even if
running as
root, add
one of the
following
two
arguments
when
running the
installer:
--target
<path> -
Extracts the
installer files
into the
specified
custom path.
--keep -
Extracts the
installer files
into the
current

working
directory
(without
cleaning at
the end).
If using
installer...

📝🟢

note

If the Allow
running
multiple
engines on
the same
machine
option is
selected, run
the
command:
sudo
systemctl
status

d1_<Engine
_name>

📝🟢

note

Any changes
made to
variables in
the
upgrade_eng
ine.sh file are
reset after
each
upgrade. We
recommend
instead
using the
upgrade.conf
file to set
variables.

📝🟢

note

You can
ignore the
following
error:

Cannot
create
folder
'/var/lib/demi
sto'

📝🟢

note

If the Allow
running
multiple
engines on
the same
machine
option is
selected, run
the
command:
sudo
systemctl
status

d1_<Engine
_name>

📝🟢

note

If the
installer fails
to start due
to a
permissions
issue, even if
running as
root, add
one of the
following
two
arguments
when
running the
installer:
--target
<path> -
Extracts the
installer files
into the
specified
custom path.
--keep -
Extracts the
installer files
into the
current

working
directory
(without
cleaning at
the end).
If using
installer...

📝🟢

note

If the Allow
running
multiple
engines on
the same
machine
option is
selected, run
the
command:
sudo
systemctl
status

d1_<Engine
_name>

📝🟢

note

If the Allow
running
multiple
engines on
the same
machine
option is
selected, run
the
command:
sudo
systemctl
status

d1_<Engine
_name>

7.2.7.11.11.
Troubleshoot
integrations
running on
engines



📝🟢

note

If the Allow
running
multiple
engines on
the same
machine
option is
selected, run
the
command:
sudo
systemctl
restart

d1_<Engine
_name>

📝🟢

note

Docker CE
installations
typically run
Docker,
while Docker
EE
installations
typically run
dockerroot.

📝🟢

note

If the Allow
running
multiple
engines on
the same
machine
option is
selected, run
the
command:
sudo
systemctl
restart

d1_<Engine
_name>

📝🟢

note

If the Allow
running
multiple
engines on
the same
machine
option is
selected, run
the
command:
sudo
systemctl
restart

d1_<Engine
_name>

7.2.8.
Response
actions



7.2.8.1.
Initiate a Live
Terminal
session



📝🟢

note

You can run
PowerShell
5.0 or a later
release on
Live Terminal
of Windows.

📝🟢

note

Network
drives are
not

supported.
📝🟢

note

On Windows
endpoints,
you cannot
run GUI-
based cmd
commands
like winver
or

appwiz.cpl.

📝🟢

note

Disabling
Live Terminal
does not
take effect
on sessions
that are in
progress.

💡🔵

tip

You can fine-
tune the Live
Terminal
session
visibility on
the endpoint
by adjusting
the User
Interface
options in
your Agent
Settings
Profile.

7.2.8.2.
Isolate an
endpoint



📝🟢

note

IP-based file
storage
protocol
traffic will
also be
blocked.
This might
affect
endpoint
functionality
if the
endpoint
uses such
mounts.

📝🟢

note

If file system
operations
become
unresponsive
during
isolation,
such as
being unable
to list folder
content,
unmount the
mounted
network
shares.

💡🔵

tip

If needed,
Filter the list
of endpoints.

7.2.8.3. Pause
endpoint
protection



📝🟢

note

Pausing your
endpoint
protection
modules
leaves your
machines
exposed to
risks.

🧭➡️

navigation

Investigation & Response → Response → Action
Center and locate Action Type Pause Endpoint
Protection or Resume Endpoint Protection

7.2.8.4.
Remediate
changes
from
malicious
activity



☑️🟣

prerequisite

To initiate
remediation
suggestions,
you must
have the
following
system
requirements:
An App
Administrato
r, Privileged
Responder,
or Privileged
Security
Admin role
permissions
which
include the
remediation
permissions.
EDR data
collection
enabled.
Agent
version 7.2 or
above on
Windows
endpoints.

📝🟢

note

Endpoints
that are part
of the Case
view and do
not meet the
required
criteria are
excluded
from the
remediation
analysis.

7.2.8.5.
Search and
destroy
malicious
files



☑️🟣

prerequisite

The
following
are

prerequisites
to enable
Cortex
Cloud to
search and
destroy files
on your

endpoints:
Supported
platforms:
Windows:
Cortex XDR
agent
version 7.2 or
a later. If you
plan to
enable
Search and
Destroy on
VDI sessions,
you must
perform the
initial scan
on the
Golden
Image.
Mac: Cortex
XDR agent
version 7.3 or
a later
release
running on...

📝🟢

note

The Cortex
Cloud agent
does not
include the
following
information
in the local
files
inventory:
Information
about files
that existed
on the
endpoint
and were
deleted
before the
Cortex
Cloud agent
was installed.
Information
about files
where the
file size
exceeds the
maximum
file size for
hash
calculations
that are pre-
configured
in Cortex
Cloud .
If...

7.2.8.6.
Manage
external
dynamic lists



🔒🟣

content

This
functionality
is not
supported
for XDR
Prevent.

📝🟢

note

Ensure EDL
sizes don't
exceed your
firewall

💡🔵 model limit.

tip

You can also
use the IP
list and URL
lists as part
of a URL
Filtering
policy, or the
domain list
as part of a
custom Anti-
Spyware
profile.

7.2.8.7.
Collect a
memory
image

 🔒🟣

content

This
functionality
has the
following
license
requirements:
Forensics
add-on
license.

7.2.9.
Forensics



🔒🟣

content

Requires the
Forensics
add-on

7.2.9.1.
Forensic
investigations



7.2.9.1.1.
Manage an
investigation



7.2.9.1.1.1.
Create a
new

investigation

 📝🟢

note

To set up
user

permissions,
you must
have Scope-
Based
Access
Control
(SBAC)
enabled.

7.2.9.1.1.2.
Edit an
investigation



7.2.9.1.1.3.
Close an
investigation

 📝🟢

note

When you close an investigation, Palo Alto
Networks has a grace period of 24 hours before
deleting any collections associated with the
investigation. During this timeframe, you have
the option to cancel the close investigation
action.

7.2.9.1.1.4.
User
permissions

 📝🟢

note

Even if a
user does
not have
access to
view an
investigation
via the
Forensics
Investigations
page, they
can still
query the
results of
the
collections
using an
XQL query.

7.2.9.1.2. Data
collection



7.2.9.1.2.1.
Hunting



7.2.9.1.2.2.
Triage



7.2.9.1.2.3.
Configure
collection



📝🟢

note

When
search fields

are

specified,
the search is
limited
based on
those filters.
If more than
one entry is
in a search
filter field,
the search
returns
entries that
match any
of them. For
example: A
File Search
with two
specified
paths
("C:\Test*"
and
"C:\Windows
*") will
return
results from
both the
Test and
Windows
folders.
If you
specify
multiple...

📝🟢

note

Data
collected
during a
Triage
Collection is
categorized
into
Artifacts,
Volatiles,
and File
Collection

📝🟢

note

Supported
only for the
UFW tool
(Firewall
managment
tool for
some Linux
distributions
such as
Ubuntu)

7.2.9.1.3.
Analysis and
documentati
on



7.2.9.1.3.1.
Review alerts



7.2.9.1.3.2.
Investigation
timeline



7.2.9.1.3.3. Key
assets &
artifacts



7.2.9.1.4.
Export

 📝🟢

note

You can
export a
collection
more than
once.

7.2.10. Build
XQL queries



📝🟢

note

If you have
the Cortex
Agentic
Assistant,
you can use
natural
language
prompts to
create and
run XQL
queries
within the
chat
interface.
For more
information,
see Create
and run XQL
queries with
Agentic
Assistant
chat.Cortex
Agentic
Assistant

7.2.10.1.
About the
Query
Builder



📝🟢

note

Schema
changes to
datasets
may not be
reflected in
the
autocomplet
e

suggestions
and
definitions
as you type
in real time
the XQL
query, and
can appear

with a slight
delay.

💡🔵

tip

When
creating
XQL queries,
you can:

Use the up
and down
arrow keys
to navigate
through the
auto-
suggestion
commands
and
definitions.
Select an
auto-
suggestion
command by
pressing
either the
Enter or Tab
key.
Press
Shift+Enter
to add a
new line, and
easily ignore
the auto-
suggestion
output.
Close the
auto-
suggestion
output by
pressing the
Esc key.

7.2.10.2. How
to build XQL
queries



📝🟢

note

Users with different dataset permissions can
receive different results for the same XQL query.
An administrator or a user with a predefined
user role can create and view queries built with
an unknown dataset that currently does not
exist in Cortex Cloud. All other users can only
create and view queries built with an existing
dataset.
When you have more...

📝🟢

note

You can
build a
query that
investigates
data in both
a cold
dataset and
a hot
dataset in
the same
query. In
addition, as
the hot
storage
dataset
format is the
default
option and
represents
the fully
searchable
storage, this
format is
used
throughout
this guide
for
investigation
and threat
hunting. For
more

information
on hot and
cold
storage, see...

👀🟠

important

Forensic
datasets are
not inlcuded
by default in
XQL query
results,
unless the
dataset
query is
explicitly
defined to

use a

forensic
dataset.

7.2.10.2.1. Get
started with
XQL queries



7.2.10.2.2.
Useful XQL

user

interface
features



📝🟢

note

Schema
changes to
datasets
may not be
reflected in
the
autocomplet
e

suggestions
and
definitions
as you type
in real time
the XQL
query and
can appear

with a slight
delay.

💡🔵

tip

When
creating
XQL queries,
you can:

Use the up
and down
arrow keys
to navigate
through the
auto-
suggestion
command
suggestions
and
definitions.
Select an
auto-
suggestion
command by
pressing
either the
Enter or Tab
key.
Press
Shift+Enter
to add a
new line, and
easily ignore
the auto-
suggestion
output.
Close the
auto-
suggestion
output by
pressing the
Esc...

7.2.10.2.3.
XQL Query
best
practices



7.2.10.2.4.
Expected
results when
querying
fields



7.2.10.2.5.
Create XQL
query



📝🟢

note

Whenever
the time
period is
changed in
the query
window, the
config
timeframe is
automaticall
y set to the
time period
defined for
the entire
query,

including
queries that
are part of
the join
stage. Yet,
this won't be
visible as
part of the
query. Only
if you
manually
type in the
config
timeframe
will this be
seen in the
query.
These time
picker...

💡🔵

tip

When
creating
XQL queries,
you can:

Use the up
and down
arrow keys
to navigate
through the
auto-
suggestion
command
suggestions
and
definitions.
Select an
auto-
suggestion
command by
pressing
either the
Enter or Tab
key.
Press
Shift+Enter
to add a
new line, and
easily ignore
the auto-
suggestion
output.
Close the
auto-
suggestion
output by
pressing the
Esc...

💡🔵

tip

While the
query is
running, you
can navigate
away from
the page. A
notification
is sent when
the query
has finished.
You can also
Cancel the
query or run
a new query,

where you
have the
option to
Run only
new query

(cancel
previous) or
Run both
queries.

7.2.10.2.6.
Review XQL
query results



📝🟢

note

It's also
possible to
graph the
results
displayed.
For more
information,
see Graph
query results.

📝🟢

note

Real time
query
results are
available
only in the
Query
Builder and
in free text
query.
Real time
results are
displayed
only for
queries run
on hot
datasets.
The Sort
option is
available
only after all
the data is
retrieved.
When you
formulate
complex
queries, the
results will
be displayed
when the
query has
finished
running
completely,
and not in
real...

📝🟢

note

Results are
received
incrementally
for the first
100K
records, or
up to 100MB
worth of
records,
whichever
comes first.
After that,
the next
update is
when the
query has
finished
running
completely.

📝🟢

note

In order for
Cortex
Cloud to
provide a
histogram
for a field,
the field
must not
contain an
array or a

JSON object.

7.2.10.2.7.
Translate to
XQL

 👀🟠

important

This feature is still in a Beta state and you will
find that not all Splunk queries can be
converted to XQL. This feature will be improved
upon in the upcoming releases to support
greater Splunk query translations to XQL.

7.2.10.2.8.
Graph query
results



🔒🟣

content

Building
Cortex
Query
Language
(XQL)
queries in
the Query
Builder
requires a
Data
Collection
add-on.

📝🟢

note

To display
the result of
as a time
duration,
choose the
graph type
Single Value
and enable
Show as
Time. You
can then
select the
Time Unit
(millisecond,
second,
minute, or
hour) and
the Display
format.

💡🔵

tip

Alternatively,
you can use
the Cortex
Agentic
Assistant to
generate
custom

graphs and
charts using
natural
language
prompts. By
simply
prompting
the agent, it
will build
and execute
the query,
returning the
visual
representatio
n. For more
information,
see Use
natural
language to
query and
visualize
your data.

7.2.10.3. XQL
query entities



7.2.10.3.1.
Create
authenticatio
n query



7.2.10.3.2.
Create event
log query



7.2.10.3.3.
Create file
query



7.2.10.3.4.
Create
image load
query



7.2.10.3.5.
Create
network
connections
query



7.2.10.3.6.
Create
network
query

 📝🟢

note

When you
run the
query,

depending
on the
outcome of
the results,
the value
specified in
this field
might be
displayed in
the dst_ip
field in the
query
results. This
occurs if an
RDP event is
recorded
whereby a
user

connected
from the
source IP to
the
destination IP.

7.2.10.3.7.
Create
process
query



7.2.10.3.8.
Create
registry
query

 👀🟠

important

Ensure the
KEY NAME is
entered as a
real registry
key name,
and not as a
symbolic
link.
Otherwise,
the query
will not
retrieve
results.
Instead of
HKEY_LOCAL
_MACHINE\S
ystem\Curre
ntControlSet,
which is a
symbolic
link, use
KEY_LOCAL_
MACHINE\Sy
stem\Control
Set001.
Instead of
HKEY_CURRE
NT_USER,
use

HKEY_USERS
\<SID>,
where SID is
either a SID
of the
current...

7.2.10.3.9.
Query
across all
entities



7.2.10.4.
Overview of
the Query
Center



📝🟢

note

Very short
queries
might not be
listed.
You cannot
cancel
correlation
queries.
The default
retention

period for
historic
queries is
aligned with
issue
retention.

7.2.10.4.1. Edit
and run
queries in
Query
Center



📝🟢

note

If query
limits are
applied to
your tenant,
the number
of
concurrent
running
queries is
limited per
user. If query
usage is
reaching the
defined limit,
a system
message
warns you

that a high
query load is
impacting
performance.
If you
exceed the
limit, new
queries are
blocked
until query
usage drops.
You can
view all
active
queries
under Query
Center...

📝🟢

note

You can
cancel your
own queries.
To cancel
queries run
by other
users, you
must have
View/Edit
permissions
for
Configuratio
ns → Query
Management.
By default,
Instance
administrator
s have
View/Edit
permission.

📝🟢

note

Cancelled
queries
show a
Canceled
status. You
can see

details of all
canceled
queries in
the Query
History tab.
You cannot
cancel
correlation
rule queries.
If you cancel
a scheduled
query, only
the current
query is
cancelled.
Future
recurrences

of the
scheduled
query are
not affected.

7.2.10.4.1.1.
Query
Center
reference
information

 📝🟢

note

Certain
fields are
exposed and
hidden by
default. An
asterisk (*) is
beside every
field that is
exposed by
default.

7.2.10.5.
Manage
scheduled
queries

 7.2.10.5.1.
Scheduled
Queries
reference
information

 📝🟢

note

Certain
fields are
exposed and
hidden by
default. An
asterisk (*) is
beside every
field that is
exposed by
default.

7.2.10.6.
Manage
your query

library



📝🟢

note

The Query
to Library
option is
only
available if
your role has
the Create
Queries
capability.
For more
information,
see Manage
access to
saved
queries.

📝🟢

note

When the
tenant-level
setting
Owners and
editors can
change the
general
access is
unselected,
the drop-
down is
disabled and
only an
administrator
can

configure
this option.

👀🟠

important

The ability
to create,
edit, or
share
queries is
governed by
access

management.
If certain
options are
unavailable,
contact your
administrator
. For more
information,
see Manage
access to
saved
queries.

7.2.11. Quick
Launcher

 📝🟢

note

For hosts,
Cortex
Cloud
displays
results for
exact
matches but
supports the
use of
wildcard (*)
which
changes the
search to
return
matches
that contain
the specified
text. For
example, a
search of
compy-7*
will return
any hosts
beginning
with compy-
7 such as
compy-7000,
compy-7abc,
and so forth.

7.2.12.
Research a
known threat



7.3.
Customize
cases and
issues

 7.3.1.
Customize
cases and
issues



7.3.1.1. Set up
case scoring



📝🟢

note

For scoped
users, a
small lock
icon
indicates
that you
don't have
permissions
to edit a rule.

💡🔵

tip

Right-click
an issue field
to add it as
match
7.3.1.2. criteria.
Create a
starring
configuration

 💡🔵

tip

Right-click
an issue field
to add it as
match
criteria.

7.3.1.3.
Service
Level
Agreements
(SLAs) for
issue
resolution



📝🟢

note

These filters
support
filtering of
whole days
only, for
example
Resolution
SLA > 1
filters for
issues that
have a
Resolution
SLA of
greater than
one day.

🧭➡️

navigation

Settings →
Configuratio
ns → Object
Setup →
Issues

Settings →
Configuratio
ns → Object
Setup →
Issues and
select the
SLA Rules
tab

7.3.1.4.
Create issue
exceptions



☑️🟣

prerequisite

You must
have
Exception
Approver
Admin
permission
to add or
delete issue
exception
approvers
and to turn
off and on
approver

functionality.
You must
have
Exception
Management
Admin
permission
to create
issue
exception
rules, delete
or disable
exception
rules, and
view the All
Exception
Rules page.

📝🟢

note

Issue
exceptions
apply
specifically
to issues,
not
individual
findings.
Issue
exceptions
do not
automaticall
y update
standard
dashboards,
reporting
widgets, or
compliance
reporting
profiles. To
view the
aggregate
impact of
exceptions,
you must
manually
filter the
issues table.
Exceptions
do not alter
Attack Path
policies.
Exceptions
do not
bypass any...

7.3.1.4.1.

Configure
the issue
exception
approval
workflow



☑️🟣

prerequisite

You must
have
Exception
Approver
Admin
permission
to add or
delete
approvers.

☑️🟣

prerequisite

You must
have
Exception
Approver
Admin
permission
to add or
delete
approvers.

☑️🟣

prerequisite

You must
have the
Exception
Approver
Admin
permission
to disable or
re-enable
the issue
exception
approval
workflow.

🧭➡️

navigation

Settings → Configurations → Server Settings
and scroll to the Exception Management section

Settings →
Configuratio
n → General
→ Server
Settings and
scroll to the
Exception
Management
section

7.3.1.4.2.
Create an
issue
exception
rule

 📝🟢

note

You can
create a
maximum of
10 issue
exceptions
per day.

7.3.1.4.3.
Create an
exception
rule from an
issue

 📝🟢

note

You can
create a
maximum of
10 issue
exception
rules per day.

7.3.1.4.4.
View issue
Exception
Rules

 🧭➡️

navigation

Settings →
Exceptions
Configuratio
n →

Exception
Rules
7.3.1.4.5.
Disable issue
exception
rules



📝🟢

note

You cannot
re-enable an
issue
exception
rule after it
has been
disabled.

🧭➡️

navigation

Settings →
Exceptions
→

Configuratio
n →

Exception
Rules

7.3.1.4.6. View
excepted
issues



7.3.1.5.
Optimize
case

grouping in
correlations

 📝🟢

note

Field
mapping
does not
guarantee
grouping.
Cortex
Cloud
grouping
uses

machine
learning and
platform
logic that
evaluates
confidence,
context,
relationships,
and
additional
case signals.
Field
mapping
helps
influence
grouping by
contributing
relevant
artifacts, but
final
grouping
decisions are
determined
by the
platform's
overall
grouping
logic.
For...

8.

Agentic
Assistant

chat



8.1. Get
started with
Agentic
Assistant
chat

 📝🟢

note

The Cortex Agentic Assistant is currently
available in limited regions. For more
information see Cortex Agentic Assistant If your
tenant is not within one of those regions, you
have access to the Cortex Assistant.Cortex
Agentic Assistant
To enable the Cortex Agentic Assistant, go to
Settings → Configurations → General → Server
Settings → AI...

8.2. Choose
an Agentic
Assistant
agent



📝🟢

note

Only public
agents are
supported
via Slack.

📝🟢

note

Requires the
Exposure
Management
add-on.

8.3. Chat
with an
Agentic
Assistant
agent



📝🟢

note

Case data is
only loaded
when you
send your
first
message.

Opening the
chat
interface
without
sending a
prompt does
not provide
the agent
with the
case context.

📝🟢

note

An agent's
proposed
plans and
results may
contain
inaccuracies
or errors.

Always
review the
results
carefully to
ensure you

fully
understand
the
proposed
action
before
proceeding.

💡🔵

tip

You can
quickly jump
to different
product
pages within
Cortex
Cloud by
typing / in
the prompt
area. This
shortcut is a
built-in
navigation
feature that
is available
even if the
Cortex
Agentic
Assistant is
disabled.

8.4. Chat
with the
Agentic
Assistant
from Slack



☑️🟣

prerequisite

Before you
can interact
with the
Cortex
Agentic
Assistant in
Slack, the
Slack v3
integration
must be set
up correctly.
For more
information,
see the
Slack v3
integration
documentati

on.

To perform
actions in
Slack, your
Slack email
must match
your Cortex
Cloud user
email. This
ensures the
system can
strictly
follow your
assigned
permissions
(RBAC). If...

📝🟢

note

Only public
agents are
supported
via Slack.

📝🟢

note

If a team
member in
the channel
sees the
summary
and
attempts to
ask the
agent, "Give
me more

details on
the first SLA
issue," the
team
member
receives an
access

denied
message
because the
active
session is
only
available to
you, the
initiator.

8.5. Create
and run XQL
queries with
Agentic
Assistant
chat



📝🟢

note

The
TextToXQL
action is a
hidden
system
action and
does not
appear in
the Agents
Hub.

📝🟢

note

Running XQL
queries
manually
through an
agent does
not consume
compute
units. This
includes
scenarios
where you
prompt the
agent to
create and
execute a

multi-step
plan.

8.6. Use
natural
language to
query and
visualize
your data

 📝🟢

note

Once the
graph or
chart is
saved the
Widget
Library, the
link to the
chat artifact
is severed,
and the
agent does
not track
subsequent
changes
made to the
widget or
the
dashboard.

8.7. Manage
chat history



9. Review

and

report

your

security

posture
and

progress

 9.1. Monitor
dashboards
and reports



9.1.1. About
dashboards



📝🟢

note

Dashboards
that are
based on
custom
infrastructur
e cannot be
exported.
When
importing a
dashboard
that already
exists in the
system, the
imported
dashboard
overwrites
the existing
dashboard.
To avoid
overwriting
the existing
dashboard,
duplicate
and rename
the existing
dashboard
before
importing
the new
dashboard.

👀🟠

important

The ability
to create,
edit, or
share
custom
dashboards
is governed
by access
management.
If certain
options are
unavailable,
contact your
administrator
. For more
information,
see Manage
access to
custom
dashboards.

9.1.1.1.
Command
Center
dashboards



📝🟢

note

Access to
these
dashboards
requires
RBAC
permissions
under
Dashboards
& Reports.
The
Dashboards
component
must be
Enabled in
your role
and requires
View
permissions
for
Command
Center
Dashboards.
If certain
options are
unavailable,
contact your
administrator
. For more
information,
Manage
access to
custom
dashboards.
The
dashboards
are available
in dark...

9.1.1.1.1.
Cortex
Agentic
Assistant
dashboard



☑️🟣

prerequisite

The Cortex
Agentic
Assistant is
only
available
after it is
enabled, and
for tenants
in certain
regions. For
more

information,
see Cortex
Agentic
Assistant.Cor
tex Agentic
Assistant

📝🟢

note

Access to
this
dashboard
requires
RBAC
permissions
under
Dashboards
& Reports.
The
Dashboards
component
must be
Enabled in
your role
and requires
View
permissions
for
Command
Center
Dashboards.
If certain
options are
unavailable,
contact your
administrator
. For more
information,
see Manage
access to
custom
dashboards.

9.1.1.1.2.
Cloud
Security
Operations

 9.1.1.1.2.1.
Cloud
Security
Operations

 📝🟢

note

Command
Center data
may not
match the
counts on
the Issues
page, and
you may
observe
inconsistenci
es. This is
because
dashboard
data is a
snapshot of
issues
identified,
whereas the
Issues page
provides the
most up-to-
date view of
risks across
your cloud
assets. In
addition, the
Issues pages
do not
support all
the currently
available
filters on
the...

9.1.1.1.3.
Cortex
Cloud
Command
Center



9.1.1.2.
Predefined
dashboards



📝🟢

note

Select the star in the right corner of a widget to
filter the data for cases that match case-
starring policies.
A purple star indicates that the widget is
displaying only starred cases. The starring filter
is persistent and continues to show the filtered
results until you clear the star.

📝🟢

note

Users can
access all
information
on the
dashboard
when their
user access

is scoped to
view All
assets or

assigned to
the Instance
Administrato
r role.
Otherwise,
users with
granular
scoping set
to No assets
or Select
asset groups
will have
limited
access to
the
dashboard.
For more
information
on Scope-
Based
Access
Control
(SBAC), see
Manage
user...

9.1.1.3.
Reports

 9.1.1.3.1.
Report
templates



9.1.2. Build
custom
dashboards
and reports



9.1.2.1. Build
a custom
dashboard

 📝🟢

note

For agent-
related
widgets, you
can apply an
endpoint
scope to
refine the
displayed
data to only
show results
from
specific
endpoint
groups.
Select the
menu on the
top right
corner of
the widget,
select
Groups, and
select one

or more

endpoint
groups.
For case-
related
widgets, you
can refine
the
displayed
data to only
show results
from cases
that match
a...

9.1.2.2.
Manage
your Widget
Library



📝🟢

note

Unlike
dashboards,
individual
widgets
cannot be
shared with
other users
directly
from the
Widget
Library. Yet,
if you add a
Restricted
widget to a
dashboard
and then
share that
dashboard,
other users
who have
access to
the
dashboard
will also be
able to see
that specific
widget.

📝🟢

note

Any
dashboards
or reports
that include
the widget
are affected
by the
changes.

9.1.3. Fine-
tune
dashboards
and reports



9.1.3.1. Create
a custom

widget using
a script



📝🟢

note

If you have
added
arguments
to the script,
these
appear when
creating a
widget.

📝🟢

note

Available
options are
Pie, Column,
Line, and
Single Value.
To display
the result of
the script as
a time
duration,
choose the
graph type
Single Value
and enable
Show as
Time. You
can then
select the
Time Unit
(millisecond,
second,
minute, or
hour) and
the Display
format.

9.1.3.1.1.
Script-
based
widget
examples



📝🟢

note

Add the
widget tag
in the script
settings to
make the
script
available for
use in script-
based
widgets. For
more

information,
see Create a
script.

📝🟢

note

If your script
returns a
time
duration,
configure
the widget
with the
graph type
Single Value
and enable
Show as
Time..

9.1.3.2.
Create a
text widget

 📝🟢

note

From the
Dashboard
Manager,
you can

right-click
an existing
dashboard
and choose
Save as
report
template.
You can also
add text-
based
widgets
directly to
new or

existing
report
templates,
as described
below.

9.1.3.3.
Create
custom XQL
widgets



📝🟢

note

Cortex
Query
Language
(XQL)
queries
generated
from the
Widget
Library do
not appear
in the Query
Center. The
results are
used only
for creating
the custom
widget.

📝🟢

note

Whenever
the time
period is
changed in
the query
window, the
config
timeframe is
automaticall
y set to the
time period
defined, but
this won't be
visible as
part of the
query. Only
if you
manually
type in the
config
timeframe
will this be
seen in the
query.
These time
picker
options are
available in
XQL queries
when using
the Query
Builder, XQL
Widgets,...

📝🟢

note

To display
the result of
as a time
duration,
choose the
graph type
Single Value
and enable
Show as
Time. You
can then
select the
Time Unit
(millisecond,
second,
minute, or
hour) and
the Display
format.

💡🔵

tip

You can
create a
generic
dashboard
for multiple
views of the

same

dataset by
using an
asterisk (*)
when
defining the
dataset in
the XQL
widget as
dataset =
<dataset_na
me>*. The
placement
of the
asterisk in
the dataset
name

ensures that
any view
containing
this prefix
text is
displayed in
the results.
The dataset
in a query is
defined as:
If there are...

9.1.3.3.1.
Configure
filters and
inputs for
custom XQL
widgets



☑️🟣

prerequisite

Fixed filters
are based on
parameters
that are
defined in
custom XQL
widgets.
Before you
can

configure
fixed filters,
take the
following
steps:
Create
custom XQL
widgets with
parameters.
For more
information,
see Create
custom XQL
widgets.
Add the
widgets to a
custom
dashboard.
For more
information,
see Build a
custom
dashboard.

📝🟢

note

This option
only appears
if the
dashboard
contains
custom XQL
widgets with
defined
parameters.

📝🟢

note

You can define up to four parameter filters on a
single dashboard or report.

📝🟢

note

If you
specify
more than
one field,
only the first
field value is
used.

💡🔵

tip

After the
initial setup,
when you
access your
dashboard
the filters
and inputs
might need
further
refinement.
You can
make
changes to
the
configured
parameters
in the XQL
widgets, and
update the
Filters &
Inputs on
your
dashboard
until you are
satisfied
with the
results.

9.1.3.3.2.
Configure
dashboard
drilldowns



☑️🟣

prerequisite

To configure drilldowns your dashboard must
contain custom XQL widgets. In addition, if you
want to configure in-dashboard drilldowns your
custom XQL widget must contain one or more
parameters. For more information about
configuring parameters in custom XQL widgets,
see Create custom XQL widgets.

📝🟢

note

If the
selected
parameter is
configured
in other XQL
widgets on
the
dashboard,
these
widgets are
also
affected by
the drilldown.

9.1.3.3.2.1.
Variables in
drilldowns



9.1.4. Run or
schedule
reports



📝🟢

note

Report
templates
that are
based on
custom
infrastructur
e cannot be
exported.
If you import
a report
template
that already
exists in the
system, the
imported
template
will
overwrite
the existing
template. If
you do not
want to
overwrite
the existing
template,
duplicate
and rename
the existing
template
before
importing
the new
template.
You can
also...

📝🟢

note

The report
name and
description
will be
displayed in
the report
header and
are not
editable
during
customizatio
n.

📝🟢

note

For agent-
related
widgets, you
can apply an
endpoint
scope to
refine the
displayed
data to only
show results
from
specific
endpoint
groups.
Select the
menu on the
top right
corner of
the widget,
select
Groups, and
select one
or more

endpoint
groups.
For agent-
related
widgets, you
can apply an
endpoint
scope to
refine the
displayed
data to only
show...

📝🟢

note

To send
reports to
Slack, Slack
must be
configured
as an

external
application
in Cortex
Cloud. For
more

information,
see Integrate
Slack for
outbound
notifications

📝🟢

note

When
ownership is
transferred,
any existing
report
schedules
associated
with the
report
template are
automaticall
y removed.
The new
Owner must
manually
redefine the
schedule to
resume

automated
report
generation.

🧭➡️

navigation

Under
Settings →
Configuratio
ns →

General →
Notifications,
click + Add
Forwarding
Configuratio
n

10.

Discovery

Engine

 10.1.
Discovery
Engine



11. Cortex

Cloud AI

Security



11.1. What is
Cortex
Cloud AI
Security?



11.2.
Supported
services in
Cortex
Cloud AI
Security



11.3. Cortex
Cloud AI
Security
concepts



11.4. Cortex
Cloud AI
Security use
cases



11.5. How to

perform
advanced AI
Security
investigations
using XQL

 📝🟢

note

For more
information,
see Build
XQL queries.

12.

Cortex

Cloud

Applicati
on

Security



📝🟢

note

While some
features are
included by
default,
others
require a
dedicated
add-on
purchase.

12.1.
Onboard
data sources



📝🟢

note

Third-party
SCA
ingestion
(such as
Snyk,
Semgrep)
covers CVE
vulnerabilitie
s only.
License
Miscomplian
ce and
Package
Operational
Risk findings
are

produced
exclusively
by the native
Cortex
Cloud SCA
scanner. You
must enable
native SCA
scanning to
achieve
license and
operational-
risk coverage

12.1.1.
Onboard
version
control
systems



📝🟢

note

Cortex
Cloud
Application
Security
(which
includes IaC
and Secrets
scanning), is
an add-on
to a license
(such as
Posture
Security)
that must be
purchased
separately.

📝🟢

note

Disclaimer:
When
onboarding
with third-
party data
sources, we
outline the
required
steps for
setup, but
we do not
monitor
these
external
resources,

and they
may change
over time.

Always refer
to the
relevant
third-party
documentati
on for the
most current
integration
steps.

💡🔵

tip

Navigate to
Settings →
Data
Sources &
Integrations
→ + Add
New → and
enter your
VCS data
source in the
search bar.

12.1.1.1. AWS
CodeCommit



☑️🟣

prerequisite

Before you begin, ensure the following:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
AWS user permissions: To onboard
CodeCommit, the user logged into the AWS
Console must have permissions to deploy the
CloudFormation stack and authorize the...

💡🔵

tip

The instance
ID is
identical to
the stack ID
on the AWS
platform.

🧭➡️

navigation

Modules →
Application
Security →
Periodic
scans

12.1.1.2.
Azure
DevOps



☑️🟣

prerequisite

Before you begin:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
Azure DevOps permissions: Ensure the user
performing the integration holds one of the
following roles in Azure DevOps:
Project Administrator: This permission is
required to subscribe...

📝🟢

note

This forces
Azure to
bypass
browser
cookies and
issue a token
for the
correct

directory.

📝🟢

note

PATs are
static. To
onboard a
different
tenant, you
must log in
to that
specific
environment
to generate
a new token.

👀🟠

important

When
redirected
to the
Microsoft
login screen,
do not
immediately
enter your
email.

🧭➡️

navigation

Settings →
Data
Sources &
Integrations
→ + Add
New

User
Settings →
Personal
access

tokens → +
New Token

12.1.1.2.1.
Azure
DevOps
onboarding
system
architecture

📝🟢 

note

Microsoft
has
announced
the
deprecation
of legacy
OAuth
methods by
2026,
making Entra
ID the
required
standard for
long-term
support.

12.1.1.3.
Bitbucket
Cloud



☑️🟣

prerequisite

Before you begin:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
Bitbucket permissions: In Bitbucket, grant the
user performing the Cortex application
authorization the following permissions. The
level of access required depends on the
modules you...

📝🟢

note

Ensure that
you receive
the Instance
Successfully
Created
message on

this step,
indicating
successful
instance
creation.

12.1.1.4.
Bitbucket
Data Center



☑️🟣

prerequisite

Before you begin:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
Bitbucket authorization (PAT): You must
generate a Personal Access Token (PAT) within
your Bitbucket Server account settings to
authenticate the connection. For detailed
instruction on...

📝🟢

note

For more
information
about the
Transporter,
including
setup
instructions,
refer to
Transporter
over Broker
VM.

📝🟢

note

Ensure that
you receive
the Instance
Successfully
Created
message on

this step,
indicating
successful
instance
creation.

📝🟢

note

By default,
the
permissions
of the
access token
are set

according to
your current
access level.
It is essential
to define
two levels of
permissions,
Project and
Repository
permissions.
The
Repository
permissions
inherit from
Project
permissions,
requiring
Repository
permissions
to match or
exceed
Project
permissions
Providing
read and
write
permissions...

📝🟢

note

For
additional
security, it is
recommende
d to set an
expiry
automaticall
y. The expiry
date of a
token
cannot be
changed
after it is
created. You
can see the
expiry dates
for all your
tokens on
Profile
picture →
Manage
account →
Personal
access

tokens.

👀🟠

important

Always refer
to the
Bitbucket
documentati
on for
information
relating to
creating a
PAT.

12.1.1.5.
GitHub
Cloud



☑️🟣

prerequisite

Before you begin:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
In GitHub, grant the user performing the Cortex
application authorization the following
permissions:
Organization Owner: Only an Organization
Owner can directly authorize and install...

🧭➡️

navigation

Verify
integration:
On Data
Sources,
select Code
Providers →
GitHub SaaS
→ View
more and
confirm that
the status of
your

integrated
GitHub
instance is

'Connected 12.1.1.6.
GitHub
Enterprise
(On-Prem)



☑️🟣

prerequisite

Before you begin:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
GitHub permissions: You must have
Organization Owner permissions to install the
Cortex application. Users with only repository-
level admin permissions cannot complete the
installation...

📝🟢

note

The domain
is the
hostname
associated
with your
GitHub
Enterprise
(On-Prem)
instance.

📝🟢

note

For more
information
about the
Transporter,
including
setup
instructions,
refer to
Transporter
over Broker
VM.

📝🟢

note

Ensure that
you receive
the Instance
Successfully
Created
message on

this step,
indicating
successful
instance
creation.

12.1.1.7.
GitLab SaaS



☑️🟣

prerequisite

Before you begin:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
In Gitlab, the following permissions are required
to integrate the application:
Maintainer (Project-level). Grants sufficient
permissions to configure external integrations,
manage...

📝🟢

note

A repository can only be integrated with a
single instance. The first instance that connects
with the repository will be the one that the
repository is assigned to. This means that if
multiple integrations attempt to connect to the
same repository, only the first integration to
establish the connection will be associated with
that repository.

12.1.1.8.
GitLab Self
Managed
(On-Prem)



☑️🟣

prerequisite

Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
Gitlab permissions: Authorize the user
integrating Cortex Cloud Application Security
with your GitLab Self Managed (On-Prem)
instances with the following permissions:
Maintainer permissions. Grants...

📝🟢

note

The domain
is the
hostname
associated
with your
GitLab Self
Managed
(On-Prem)
instance.

📝🟢

note

For more
information
about the
Transporter,
including
setup
instructions,
refer to
Transporter
over Broker
VM.

📝🟢

note

Ensure that
you receive
the Instance
Successfully
Created
message on

this step,
indicating
successful
instance
creation.

12.1.2.
Onboard
CI/CD
systems



12.1.2.1.
CircleCI for
CI/CD
pipeline
scans



☑️🟣

prerequisite

Before you begin:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
CircleCI user requirements:
Permissions: To enable Cortex Cloud visibility
for all CircleCI projects, a version control
system (VCS) user with integration permissions
must be...

📝🟢

note

This
integration
utilizes a
Personal
Access
Token (PAT)
for
authenticatio

n

CircleCI
onboarding
offers both
code and
CI/CD
scanning. A
single
integrated
instance
supports
either code
or CI
scanning,
but not
both. If you
require both
code and CI
scanning for
your
CircleCi
environment,
you must
create two
separate
integrations,
selecting the
appropriate...

12.1.2.2.
Jenkins for
CI/CD
pipeline
scans



☑️🟣

prerequisite

Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
In Jenkins:
To install and configure the Cortex plugin in
Jenkins, you must be a Jenkins Administrator
with Overall/Administer permissions
Ensure the build server allows outbound HTTPS
(Port 443...

📝🟢

note

Jenkins
onboarding
offers both
code and
CI/CD
scanning. A
single
integrated
instance
supports
either code
or CI
scanning,
but not
both. If you
require both
code and CI
scanning for
your Jenkins
servers, you
must create
two
separate
integrations,
selecting the
appropriate
scanning
type for
each. To
onboard
Jenkins for
code scans,
refer to
Jenkins for...

📝🟢

note

The
integration is
added on
the console
but
integration is
pending, and
will only be
completed
after
completing
step 5
below. You
can view the
pending
integration
on the
Jenkins
Instances
page: Select
Data
Sources →
Jenkins →
View Details.
The type of
integration is
Pipeline Risks

📝🟢

note

Always refer
to the
official
Jenkins
documentati
on when
installing
plugins on
Jenkins
servers.

12.1.3.
Integrate CI
tools



🧭➡️

navigation

Settings →
Data
Sources &
Integrations
→ hover
over a CI
tool → View
Details

Delete an
instance:
Right-click
on an

instance of
the CI tool
→ Delete
instance →
Delete

12.1.3.1. AWS
CodeBuild



☑️🟣

prerequisite

Before you
begin:
User
permissions:
Ensure the
user

performing
the
integration
has
permissions
to edit
pipeline
configuration
s (such as
YAML files)
and manage
secrets/cred
entials within
the CI
platform to
store the
Cortex
Cloud API
key securely

📝🟢

note

Do not
change the
names of the
environment
variables
provided by
Cortex
Cloud. They
are required
for proper
integration
and
functionality.

📝🟢

note

The code is
only a
reference.
Replace the
placeholder
values with
your build-
specific
values.

12.1.3.2.
CircleCI for
code scans



☑️🟣

prerequisite

Before you
begin:
User
permissions:
Ensure the
user

performing
the
integration
has
permissions
to edit
pipeline
configuration
s (such as
YAML files)
and manage
secrets/cred
entials within
the CI
platform to
store the
Cortex
Cloud API
key securely

📝🟢

note

CircleCI onboarding offers both code and
CI/CD scanning. A single integrated instance
supports either code or CI scanning, but not
both. If you require both code and CI scanning
for your CircleCi environment, you must create
two separate integrations, selecting the
appropriate scanning type for each. To onboard
CircleCI for CI/CD scans, refer to...

📝🟢

note

Do not
change the
names of the
environment
variables
provided by
Cortex
Cloud. They
are required
for proper
integration
and
functionality.

👀🟠

important

The cortex-
secrets
naming
convention
for the
context is
mandatory
to ensure

functionality
and must
not be
changed.

12.1.3.3.
Connect
Cortex CLI



☑️🟣

prerequisite

System
requirements:
macOS (Intel
Core i7, such
as Sequoia):
To ensure all
functionalitie
s work
correctly,
you must
install the
vectorscan

dependency
via
Homebrew,
using this
command:
brew install
vectorscan
RHEL 8.10
and Red Hat
UBI9. The
following
prerequisites
must be met:
Install
patchelf
Install zstd
Ubuntu 20
requires the
prefetch...

📝🟢

note

The Cortex
CLI requires
an API key
with the
Standard
security level.

📝🟢

note

On macOS
arm 64
architecture
you must
unpack the
downloaded
file to
retrieve the
executable. 12.1.3.4.
GitHub
Actions



☑️🟣

prerequisite

Before you
begin:
User
permissions:
Ensure the
user

performing
the
integration
has
permissions
to edit
pipeline
configuration
s (such as
YAML files)
and manage
secrets/cred
entials within
the CI
platform to
store the
Cortex
Cloud API
key securely

📝🟢

note

Do not
change the
names of the
environment
variables
provided by
Cortex
Cloud. They
are required
for proper
integration
and
functionality.

📝🟢

note

The code is
only a
reference.
Replace the
placeholder
values with
your build-
specific
values.

12.1.3.5.
Jenkins for
code scans



☑️🟣

prerequisite

Grant
Administrato
r permissions
to the user
integrating
Cortex
Cloud
Application
Security
with Jenkins
Egress path:
Create an
egress path
to establish
the
designated
route for
outbound
data
transmission
from Cortex
Cloud to
third party
services. For
more

information
about
configuring
egress

paths, refer
to Egress
configuration
sEgress
configuration
s

☑️🟣

prerequisite

For Cortex
Cloud
Application
Security CI
tools, you
must store
secrets in
Jenkins
Credentials
for use in
your Jenkins
pipelines
using either
of these
methods:
Plain text
storage:
Store
secrets

directly as
plain text in
Jenkins
Credentials.
Access them
in your
pipeline
using the
credentials
function,
which
retrieves the
secret

directly as
plain...

📝🟢

note

Jenkins
onboarding
offers both
code and
CI/CD
scanning. A
single
integrated
instance
supports
either code
or CI
scanning,
but not
both. If you
require both
code and CI
scanning for
your Jenkins
servers, you
must create
two
separate
integrations,
selecting the
appropriate
scanning
type for
each. To
onboard
Jenkins for
CI/CD scans,
refer to
Jenkins for...

📝🟢

note

This step is
only
required for
new pipelines
For private
repositories,
ensure the
necessary
credentials
are

configured
in Jenkins
Credentials

📝🟢

note

This ensures
your build
runs within a
Docker
environment.
If a node
without
Docker is
used, the
build will fail.

12.1.3.6.
Terraform
Cloud (Run
Tasks)



☑️🟣

prerequisite

Before you
begin:
User
permissions:
Ensure the
user

performing
the
integration
has
permissions
to edit
pipeline
configuration
s (such as
YAML files)
and manage
secrets/cred
entials within
the CI
platform to
store the
Cortex
Cloud API
key securely
Procure a
Terraform
cloud
license that
is either a
trial license
or a TF
Cloud
license at
the TEAM &...

📝🟢

note

Skip this
step if you
plan on
using an
existing
token.

📝🟢

note

Cortex
Cloud
Application
Security
performs a
scan of
Terraform
templates
on selected
workspaces
based on the
Run Stage.

12.1.3.7.
Terraform
Enterprise
(Run Tasks)



☑️🟣

prerequisite

Before you
begin:
Ensure
access to a
Terraform
Enterprise
console to
enable you
to provide a
user or team
token that
authorizes
Cortex
Cloud
Application
Security to
access

workspaces
and helps
regulate run
configuration
s

Terraform
Enterprise
version
compatibility
: Ensure Run
Tasks for
workspaces
on is
compatible
with version
1.1.9 and
above
Terraform...

📝🟢

note

For
container
image
vulnerabilitie
s, Cortex
Cloud
Application
Security
performs
'Image
Referencer'
scans within
Terraform
Enterprise
(Run Tasks),
as full SCA

scans are

not

currently
supported.

📝🟢

note

Skip this
step if you
plan on
using an
existing
token.

📝🟢

note

Cortex
Cloud
Application
Security
performs a
scan of
Terraform
templates
on selected
workspaces
based on the
Run Stage.

12.1.4. CLI
pipeline
code
snippets

☑️🟣 

prerequisite

User
permissions:
Ensure the
user

performing
the
integration
has
permissions
to edit
pipeline
configuration
s (such as
YAML files).

12.1.5.
Onboard
private
package
registries



12.1.5.1. JFrog
Artifactory



12.1.5.2.
Onboard
JFrog
Artifactory



☑️🟣

prerequisite

Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
JFrog permissions:
The permissions associated with the user
configured during the onboarding process
determine the scope of scan results. Only
repositories and artifacts the user can access
are...

📝🟢

note

To authorize
the scanner
to connect

through your
firewall,
select
Enable
access by
IPs, and
copy the
displayed
source IPs to
your
organization'
s allowlist.

📝🟢

note

For Maven:
Select Mirror
Registry if
this
repository
mirrors an
external
repository
Use the
Mirror Of
value to
define the
duplication
scope:
* mirrors all
requests
Type a
request
[value]:
Mirrors only
specific
requests
(such as
central).
For Maven:
Select Mirror
Registry if
this
repository
mirrors an
external
repository
Use the
Mirror Of...

12.1.6. Ingest
third-party
data sources



📝🟢

note

Only
onboarded
and scanned
repositories
can be
mapped.

12.1.6.1.
Semgrep



☑️🟣

prerequisite

Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
Ensure that you have a connected version
control system (VCS) and repositories
Semgrep: A valid Semgrep API token
Note
To create a Semgrep API token, in Semgrep,
navigate to Settings → Tokens →...

📝🟢

note

Mapping
establishes
relationships
between
Semgrep
projects and
Cortex
Cloud code
repositories,
simplifying
access

management
and enabling
risk analysis
at the
repository
level,
including
displaying
findings on
the tenant
Only
mapped
projects are
ingested

📝🟢

note

Requisite
RBAC
permissions
are required
for edit and
delete
operations.

👀🟠

important

Deletion is
irreversible.
To restore
Semgrep
findings
after
deletion,
reconfigure
the
integration
and wait for
the next
polling cycle
to re-ingest
findings
from
Semgrep.

12.1.6.1.1.
Semgrep
Software
Composition
Analysis
(SCA) data
ingestion

🧭➡️ 

navigation

Application
Security →
Issues →
Vulnerabilitie
s 12.1.6.1.2.
Semgrep
Static
Application
Security
Testing
(SAST) data
ingestion

🧭➡️ 

navigation

Application
Security →
Issues →
Code
Weaknesses

12.1.6.2. Snyk 

☑️🟣

prerequisite

Permissions: The following user permissions are
required:
Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
Ensure that you have a connected version
control system (VCS) and repositories
Snyk permissions and requirements:
Permissions: The...

📝🟢

note

Select Test
Connection
to verify
that Cortex
Cloud can
connect to

your Snyk
organization.

📝🟢

note

SCA requires
Snyk Open
Source or
Snyk
Container
projects
configured
in the
organization
SAST
requires
Snyk Code
enabled and
projects
configured
in the
organization

📝🟢

note

Mapping
establishes
relationships
between
Snyk
applications
and Cortex
Cloud code
repositories,
simplifying
access

management
and enabling
risk analysis
at the
repository
level,
including
displaying
findings on
the tenant
Only
mapped
applications
are ingested

📝🟢

note

Requisite
RBAC
permissions
are required
for edit and
delete
operations.

👀🟠

important

Snyk Free
tier is not
supported. A
paid Snyk
plan (Team,
Business, or
Enterprise) is
required to
ingest
findings
from Snyk.

12.1.6.2.1.
Snyk
Software
Composition
Analysis
(SCA)
ingestion

🧭➡️ 

navigation

Modules →
Application
Security →
Vulnerabilitie
s (under
Issues 12.1.6.2.2.
Snyk Static
Application
Security
Testing
(SAST) data
ingestion

🧭➡️ 

navigation

Modules →
Application
Security →
Code
Weaknesses

Modules →
Application
Security →
AppSec
Coverage

12.1.6.3.
SonarQube



☑️🟣

prerequisite

Cortex Cloud user permissions: Ensure you have
View/Edit permissions for Data Sources and
Integrations (RBAC: AppSec Admin or Instance
Administrator)
Ensure that you have onboarded a version
control system (such as GitHub, GitLab) and
have connected repositories
SonarQube and SonarCloud setup:
Token requirements: The API token for both
SonarCloud...

📝🟢

note

Mapping
establishes
relationships
between
SonarQube
Applications
and Cortex
Cloud
Application
Security
code
repositories,
simplifying
access

management
and enabling
risk analysis
at the
repository
level,
including
displaying
findings on
the tenant
Only
mapped
projects will
be ingested

📝🟢

note

Verify that the Connector Created Successfully
message is displayed on the page.

12.1.6.4.
Veracode



☑️🟣

prerequisite

Permissions:
The
following
user

permissions
are required:
Cortex
Cloud:
Instance
Admin,
AppSec
Admin or
GRBAC
permissions.
For more
information
on AppSec
Admin
permissions,
refer to
Code
Security
user roles
and
permissionsC
ode Security
user roles
and
permissions
Veracode:
At minimum,
Reviewer
permissions
are required
Permissions:
The
following...

📝🟢

note

Only
mapped
applications
will be

ingested. 📝🟢

note

This is the
recommende
d option to
ensure

complete
coverage
and
successful
operation of
all features.

📝🟢

note

Mapping
establishes
relationships
between
Veracode
projects and
Cortex
Cloud
Application
Security
code
repositories,
simplifying
access

management
and enabling
risk analysis
at the
repository
level,
including
displaying
findings on
the tenant
Only
mapped
projects will
be ingested

📝🟢

note

Verify that the Connector Created Successfully
message is displayed on the page.

12.1.6.5.
Generic 3rd
Party
AppSec
Collector



12.1.6.5.1.
Tenant
(console)
workflow



📝🟢

note

Editing a
collector
does not
regenerate
the API
credentials.
The existing
credentials
remain valid
after editing
the
collector
name.

👀🟠

important

The Token ID
and API
Token are
displayed
only once.
Copy and
store the
credentials
securely
before
proceeding.
If the
credentials
are lost, edit
the
collector to
regenerate
the
credentials.

⚠️🟡

caution

Deleting a
collector
permanently
removes the
collector
instance and
invalidates
the
associated
API
credentials.
Existing
findings that
were

previously
ingested
through the
deleted
collector
remain in the
platform
and are not
affected by
the deletion.

🧭➡️

navigation

Settings →
Data
Sources &
Integrations
→ 3rd Party
AppSec
Collector

Settings →
Data
Sources →
3rd Party
AppSec
Collector

12.1.6.5.2. API
workflow

📝🟢 

note

The apiUrl
returned
during
collector
creation
resolves to
the public
upload
endpoint at
/public_api/a
ppsec/v1/coll
ectors/{colle
ctorId}. Use
the apiUrl
value
directly in
upload
requests.

12.1.6.5.3.
Upload
findings
from CI/CD
pipelines



📝🟢

note

Store all
collector
credentials
(Token ID,
API Token,
API URL) as
encrypted
secrets or
credentials
in the CI/CD
platform. Do
not
hardcode
credentials
in pipeline
configuration
files.

👀🟠

important

Verify that
the upload
request
returns a
2xx HTTP
status code.
A non-2xx
response
indicates
that the
upload
failed and
findings
were not

ingested.
Configure
the CI/CD
pipeline step
to fail on
non-2xx
responses to
prevent
silent
ingestion
failures. In
the cURL
examples,
add the --
fail flag or
check the
exit code to
detect
upload
failures.

12.1.6.5.4.
Technical
requirements
and SARIF
specification
s



12.1.6.5.5.
Troubleshoot
ing



12.1.7.
Manage
data source
integrations



12.1.8.
Transporter
over Broker
VM



☑️🟣

prerequisite

Ensure you
have the
necessary
permissions
and have
already set
up your
Broker VM
Confirm that
your Broker
is v 28 or
above
Whitelist IP
addresses to
enable
access to
Cortex
Cloud
resources.

The IP
addresses
for the
Transporter
are in the
Broker VM
Resources
section of
the Enable
access to

required
PANW
resources

document
Open port
4052
(inbound),
which...

📝🟢

note

The broker enables multiple connections,
allowing a single Transporter applet to connect
to various VCS instances
Multiple Transporter applets cannot be created
on a single Broker VM. Each Broker VM is
limited to a single Transporter applet instance

⛔🔴

warning

The
Transporter
applet is not
supported
for
FedRAMP
customers.

12.1.8.1. Set
up a
Transporter
applet on
Broker VM



☑️🟣

prerequisite

Permissions:

To configure
and manage
Transporter
applet
settings, you
must have
permissions
to manage
Broker
Service
configuration
s (such as an
Instance
Administrato
r)

📝🟢

note

You cannot
add a
Transporter
applet
through
Clusters.

🧭➡️

navigation

your AppSec
Transporter
under the
App column
→ Configure
→ Add
Connection
(on the
redirected
AppSec
Transporter
setup page

12.1.8.2. Set
up a
Transporter
on your VCS



☑️🟣

prerequisite

Ensure you
have user
permissions
to onboard
data sources.
Note
The
dedicated
AppSec
Admin role
includes the
required
permissions.
Ensure you
have user
permissions
to onboard
data sources.
Before you
begin you
must setup a
Transporter
applet on
your Broker
VM. Refer to
Set up a
Transporter
applet on
Broker VM
for more
information

📝🟢

note

GitLab
Enterprise
webhook
internal IP
restriction:
GitLab
Enterprise's
security
policies
prevent
webhook
subscriptions
to internal
IP addresses
(such as
broker
addresses).
These
subscriptions
can only be
enabled
through
specific
instance

configuration
. For more
information,
refer to
https://docs.
gitlab.com/se
curity/webho
oks/.

🧭➡️

navigation

Settings →
Data
Sources &
Integrations
→ + Add
New → , and
search for
your VCS on-
premises
data source

12.1.9.
Manage
integrations
via data
source APIs

🧭➡️ 

navigation

API key pair:
Generate an
API key (Key
ID + Key
Secret) from
Settings →
Configuratio
ns → API
Keys in the
Cortex
Cloud tenant

12.2. Cortex
Cloud
Application
Security
dashboard

🧭➡️ 

navigation

Modules →
Application
Security →
AppSec
Dashboard

12.3.
Application
Security
Posture
Management
(ASPM)



📝🟢

note

ASPM does
not execute
scans.

Detection

and finding
generation
belong to
Code
Security
(native
scanners)
and the
Supply
Chain
Security
pillar (VCS
repository
and CI/CD
pipeline
analysis).
ASPM
evaluates
the findings
against
unified
policies and
orchestrates
the resulting
issues
through to
remediation.

🧭➡️

navigation

You can
view AppSec
Admin
permissions
in the tenant
by
navigating
to Settings
→

Configuratio
ns → Roles
→ AppSec
Admin

12.3.1. Code
to Cloud



🧭➡️

navigation

Full Code to
Cloud (C2C)
coverage is
achieved
when Cortex
Cloud can
resolve the
following
chain:
Repository
→ Pipeline
→ Image →
(optional
Registry) →
Runtime
resources

(including
VMs, VM
images, and
IaC-defined
infrastructur

e

12.3.1.1.
Supported
integrations



12.3.1.2.
Code to
Cloud
context and
visibility

🧭➡️ 

navigation

Topology
graph:
Located in
the Business
Applications
side card,
this tab
visualizes
the entire
path to
production
(Code →
Build →
Deploy →
Run),
allowing you

12.3.1.3.
Code to
Cloud
troubleshooti
ng

📝🟢 

note

Empty state
messages
and
troubleshooti
ng guidance
for missing
lineage are
only visible
to users with
an active
license.

12.3.2. ASPM
Command
Center

12.3.2.1. 
Operational
workflows



12.3.3.
Applications



12.3.3.1.
Defining
Business
Applications



12.3.3.1.1.
Defining
business
applications
by Criteria



12.3.3.1.2.
Manage
Criteria via
the tenant UI

🧭➡️ 

navigation

Modules →
Application
Security →
Application
Management
→ Criteria
12.3.3.1.3.
Define
applications
by VCS
Criteria



☑️🟣

prerequisite

Data source:
Your Version
Control
System
(such as
GitHub,
GitLab) must
already be
onboarded
as a Data
Source
Permissions:
You must
have
View/Edit
permissions
for Access
Management
SBAC: You
can only
create

applications
from VCS
entities
(Organization
s, Projects,
or

Repositories)
that are
already
included in
your SBAC
Asset Groups

📝🟢

note

This
workflow
allows you
to unify
assets
across

disparate
providers
(such as
grouping a
GitHub
repository
and a
Bitbucket
repository
into one
application)
if they share
naming
conventions

12.3.3.1.4.
Define
applications
by cloud
tag-based
Criteria



☑️🟣

prerequisite

Permissions:
You must
have
View/Edit
permissions
for Access
Management,
or a role
that includes
these
permissions.

📝🟢

note

You can only
create

applications
based on the
entities from
onboarded
Cloud
accounts
listed in the
Cloud card.

📝🟢

note

Kubernetes (K8S) labels: K8S labels are
supported as tags for asset grouping only when
they originate from a supported cloud provider;
AWS, GCP, Azure, or OCI.

12.3.3.1.5.
Manage
criteria via
the public
API



12.3.3.1.6.
How to
manually
build an
application



☑️🟣

prerequisite

Before you
begin,
ensure you
have
connected
the
necessary
data
sources.

Refer to
How to
onboard
data sources
for more
information.H
ow to
onboard
data sources

📝🟢

note

These filters

are

represented
by icons
displayed on
the Code
pane after
selecting a
VCS.

📝🟢

note

To edit
application
assets, click
the Clear All
icon before
clicking
Finish. This
clears all
application
data,
allowing you
to restart
the
application
building
process
from the
beginning.

12.3.3.2.
Application
management
and visibility

🧭➡️ 

navigation

Inventory →
All Assets →
Business
Applications
(under
Application

12.3.3.3.
Manage
applications
via public
APIs

📝🟢 

note

The
application
public API
does not
execute
scans,
enforce
policies, or
create
issues.
Scanning,
policy
enforcement,
and issue
creation are

performed
by their
respective
subsystems
(Unified
Application
Security
policies,
scanner

orchestration
).

12.3.3.4.
Business
application
assets



12.3.3.4.1.
Business
application
expanded
asset details

📝🟢 

note

The
topology
graph is
available
only when
all
application
components
(code,
pipeline,
build and
deploy), are
configured.

12.3.3.4.2.
Export
business
application
data



12.3.3.5.
Scope user
access to

applications
(Application
SBAC)



🧭➡️

navigation

Configure
user scopes
in Cortex
Cloud by
navigating
to Settings
→

Configuratio
ns → Access
Management

12.3.3.5.1.
Enable
SBAC in the
Cortex
Cloud tenant



☑️🟣

prerequisite

RBAC
permissions:
To configure
user scopes
you must
have
Administrato
r or

View/Edit
RBAC
permissions
for Access
Management
(under
Configuratio
ns).

📝🟢

note

Exclusions
(roles not
governed by
SBAC):
Certain roles
cannot have
SBAC
applied. For
these roles,
access and
permissions
are

managed
through
Role-Based
Access
Control
(RBAC). You
must

manually
ensure that
these roles
have all
necessary
base
permissions
(for example
Edit/View
permissions
to assets),
because
SBAC is
bypassed
and does
not impose
its usual...

🧭➡️

navigation

Settings →
Configuratio
ns →

General →
Server
Settings →
Enable
Scope
Based
Access
Control

12.3.3.5.2.
Create an
application-
based Asset
Group

📝🟢 

note

You cannot
create SBAC
based on
static groups.

12.3.3.5.3.
Scope user
access to an

application



📝🟢

note

For more
information
about User
Groups,
refer to User
group
management.

🧭➡️

navigation

Settings →
Configuratio
ns → User
Groups
(under
Access
Management

Right-click
on a group
in the table
→ Edit
Group →
select the
Scope tab

Assets →
Select asset
groups →
select an
Asset Group
associated
with
applications

Settings →
Configuratio
ns → Users
(under
Access
Management

the chevron
icon (>) in
the Assets
field →
Select Asset
groups →
select the
user group

scoped to
the
application
(see above

12.3.3.5.4.
Create
application-
scoped
policies



📝🟢

note

Application-
scoped
policies
apply to
both code
and CI/CD
configuration
policies.

🧭➡️

navigation

Modules →
Application
Security →
AppSec
Polices
(under
Policy
Management
) → Add
Policy

Add Filter →
Business
Application
Names →
enter the
required
application
name

12.3.3.5.5.
Terraform
workflow for
Asset Groups



12.3.4.
Repository
as an asset



12.3.4.1.
Understandin
g repository
assets via
the UI

🧭➡️ 

navigation

To access
repository
assets,
under
Inventory,
select All
Assets →
Code →
Repositories

12.3.4.2.
Investigate
repository
assets

🧭➡️ 

navigation

From the
Repositories
table, right-
click a
repository
→ Open in
Agentic
Assistant →
select
Application
Security
from the
agents
menu, and
query
repository-
specific
insights (for
example,
scan

coverage,

risk posture,
or gaps

12.3.4.3.
Manage
repositories
via API



📝🟢

note

The public
API base
path is
/public_api/a
ppsec/v1/rep
ositories.

📝🟢

note

The
excludedPat
hs field
replaces the
entire
excluded
paths list; to
add a new
path,
retrieve the
current

configuration
first,
append the
new path,
and send the
complete
array
SAST: SAST
scanner

configuration
is not
available
through the
public API
and must be
managed
through the
console UI

⛔🔴

warning

Always retrieve the current branch list before
updating. The PUT operation replaces the entire
branch selection, meaning any omitted
branches will be unintentionally removed from
scanning.

12.3.5.
Coverage

12.3.5.1. 
Coverage in
the user
interface

🧭➡️ 

navigation

Modules, →
Application
Security →
AppSec
Coverage

12.3.6.
Urgency



12.3.6.1.
Urgency and
code to
cloud
traceability

👀🟠 

important

Without
code to
cloud
traceability,
deployment-
related
urgency

signals
cannot be
computed.
Scanner-
specific
signals
(EPSS Score,
KEV Status,
Reachability,
Secret
Validation,
CWE Top 25,
OWASP Top
10) are
always
available
regardless
of code to
cloud
traceability.

12.3.6.2.
Prioritize
issues by
Urgency
levels



12.3.6.3.
Urgency
metrics



12.3.6.4. View
Urgency in
the tenant



📝🟢

note

EPSS Score,
CVSS Score,
Exploit
Maturity,
Exploit
Availability,
Fixable, Is
Deployed,
Visibility,
Validation
metrics are
displayed
even when
the value is
false or zero.
Other
metrics are
displayed
only when
the value is
present and
non-null
For more
about
Urgency
metrics refer
to Urgency
metrics

💡🔵

tip

Combine
Urgency
filters with
scanner type
filters to
focus on the
highest-risk
issues for a
specific
scanner

category
(for
example,
Top Urgent
vulnerability
issues or
Urgent
secrets
issues).

12.3.7.
Backlog
baseline



12.3.7.1.
Backlog use
cases



12.3.7.2.
Issue/Finding
classification
by scanner



📝🟢

note

In some
cases/vendor
s, this is not
accurate as

findings are
deleted
every time
new findings
are

uploaded. In
such cases,
the feature
may not be
accurate or

supported
For SAST,
the vendor
does not
support
policy in pull
requests

📝🟢

note

Scanner
updates and
new

detections:
When a
security
scanner is
updated to
support new
languages,
detection
rules, or
capabilities,
any issues
discovered
by these
new features
for existing
code are
classified as
part of the
backlog.

12.3.7.3.
Using
Backlog

🧭➡️ 

navigation

Settings →
Configuratio
n →

Application
Security →
AppSec
Issues

Configuratio
ns and
enable Are
SBOM issues
considered
new

Permissions
for all other
capabilities,
such as
viewing
issues or
applying
policies, are
defined by
the existing
RBAC/SBAC
policies and
the user's
specific
issue
management
capabilities

Settings →
Configuratio
n →

Application
Security →
AppSec
Issues

Configuratio
ns and
enable Are
SBOM issues
considered
new

12.3.8.
Service Lead
Agreements
(SLA)

12.3.8.1. 
Configure
and monitor
Cortex
Cloud
Application
Security
SLAs

🧭➡️ 

navigation

Settings →
Configuratio
ns →

Application
Configuratio
n (under
Application
Security

12.3.9.
Compliance
for Cortex
Cloud
Application
Security



12.3.9.1.
Infrastructur
e-as-Code
(IaC)
compliance



12.3.9.2.
Manage IaC
compliance



☑️🟣

prerequisite

Add these
properties
to the IaC
Rules table
through the
Table
Settings
Menu, as
they are not
exposed by
default.

🧭➡️

navigation

Modules →
Application
Security →
AppSec
Polices →
Add Policy

12.3.9.3.
CI/CD
Compliance

12.3.9.3.1. 
Create
CI/CD
compliance
reports



📝🟢

note

The CI/CD
module
supports
GitHub and
GitLab
provider
types.

📝🟢

note

For more
information
about about
Asset
Groups,
refer to
Asset groups.

📝🟢

note

CIS GitLab
Benchmark,
CIS GitHub
Benchmark,
and the
OWASP Top
10 CI/CD
Risks
standards
are

supported.

📝🟢

note

For more
information
about
assessment

profiles,
refer to Use

an

assessment

profile to
run

compliance
checks on
your assets.

🧭➡️

navigation

Posture
Management
→

Compliance
→ Reports 12.3.9.4.
Terraform
workflow for
Compliance
assessments



12.3.10.
Unified
Application
Security
policies



12.3.10.1.
Core
concepts



📝🟢

note

The Policy
Type
selection
determines
the available
configuration
options for
conditions,
filters, and
triggers.

📝🟢

note

For a full
description
of each
finding type
and its
applicability
(Code vs.
Image), refer
to Reference
A: Finding
type details.

📝🟢

note

For a
complete
list of all
available
filters by
category, as
well as
logical
groups, refer
to Reference
B: Condition
filters and
logic.

📝🟢

note

To understand the specific calculation logic
and how grace periods align with your
organization's SLAs, refer to Reference E: Grace
period logic and configuration.

📝🟢

note

Application
scoping is
currently
not

supported
for Build
Images

📝🟢

note

Images do
not have a
default
severity. You
must select
a severity to
continue
when any
image
trigger is
selected.

12.3.10.2.
Tenant (UI)
workflow



📝🟢

note

SBAC
restricts
table
visibility. The
policy table
displays only
policies
scoped to
assets within
the user's
assigned
application
boundary.
Policies
scoped to
assets
outside the
user's SBAC
scope are
not visible

🧭➡️

navigation

Posture
Management
→ Rules &
Policies →
Policy →
Application
Security The
policy table
displays all
policies (out-
of-the-box
(OOTB) and
custom

Posture
Management
→ Rules &
Policies →
Policy →
Application
Security

12.3.10.3.
Create a
policy

📝🟢 

note

To create a
new asset
group or

application,
select the
correspondin
g link in the
scope step.
The link
opens the
asset group
or

application
configuration
in a new tab

12.3.10.4.
View and
manage

policy details



⚠️🟡

caution

Deleting a
policy
closes all
issues
created by
that policy
during the
next

periodic
scan. This
action
cannot be
undone.
Consider
disabling the
policy
instead of
deleting the
policy to
preserve

issue history.

🧭➡️

navigation

Modules →
Application
Security →
Issues → All Modules →

Application
Security →
Scans

12.3.10.5. API
workflow

📝🟢 

note

The public API remains backward compatible
with existing API consumers. The API returns
additional fields for the unified policy structure,
including the new policy type, trigger
configuration, and Override Severity settings.

12.3.10.6.
Cortex CLI
workflow



12.3.10.7. IDE
workflow



12.3.10.8.
Terraform
workflow for
policies



12.3.10.9.
Reference A:
Finding type
details



12.3.10.10.
Reference B:
Condition
filters and
logic

📝🟢 

note

When
multiple
finding
types are
selected
within a
single
condition
group, only
the Common
filters (such
as Severity
or Backlog
Status) are
available for
selection.

12.3.10.11.
Reference C:
Scope
mapping
details



12.3.10.12.
Reference D:
Trigger and
actions
mapping



12.3.10.13.
Reference E:
Grace
period logic
and
configuration



12.3.10.14.
Reference F:
Engine
evaluation
and Urgency
logic

📝🟢 

note

If code to
cloud
traceability
is absent
(the
repository is
not linked to
a relevant
application),
the engine
cannot
compute
deployment
signals, and
Urgency will
display as
Not
Applicable

12.3.10.15.
Reference G:
Finding type
to trigger
mapping



📝🟢

note

Malware
findings are
applicable
to image
scans only
When a CI
Scan
includes
image
scanning,
Malware
findings are
evaluated
PR Scan and
Periodic
Scan do not
evaluate
Malware
findings

📝🟢

note

CI/CD
Configuratio
n Scanners
policies
support only
the Periodic
Scan trigger
PR Scan and
CI Scan
triggers are
not available
for the
CI/CD
Configuratio
n Scanners
policy type

📝🟢

note

Drift Detection Scanner policies support only
the Periodic Scan trigger PR Scan and CI Scan
triggers are not available for the Drift Detection
Scanner policy type

12.3.10.16.
Reference H:
Action
availability
by trigger



12.3.10.17.
Reference I:
Audit
logging

🧭➡️ 

navigation

Settings →
Configuratio
ns →

Management
→ Audit
Logs to
review
policy audit
entries

12.3.11.
Application
Security
Rules



📝🟢

note

The AppSec
rules API
manages
rule
definitions
and
metadata. It
does not
execute
scans,
create
issues, or
enforce
policies.

12.3.11.1.
Roles and
permissions

📝🟢 

note

Custom
roles can be
configured
with either
View or
View/Edit
permissions
for AppSec
rules
API keys
inherit the
permissions
of the role
assigned
during key
generation.
To perform
write
operations
(create,
modify,
delete)
through the
AppSec
rules API,
the API key
must have a
role that
includes
View/Edit
permissions

12.3.11.2.
Rules
inventory

🧭➡️ 

navigation

To filter
rules
relating to
Secrets,
select filter
icon →
Scanner
(from the
Select field)
→ Secrets
(from the
Value field

12.3.11.3.
Create
custom
Cortex
Cloud
Application
Security
rules

📝🟢 

note

See
Configure
YAML file
properties
below for
more details.

12.3.11.4.
Configure
YAML file
properties



12.3.11.5.
Terraform
workflows
for rules



12.3.11.6.
Manage
custom rules



12.3.12.
Manage
code
weakness
issues



12.3.12.1.
Navigate to
SAST code
weakness
issues



📝🟢

note

The default
sort order is
by Severity
(descending).
The default
filter shows
issues with a
status of
New and In
Progress.
Resolved
issues are
hidden by
default.

📝🟢

note

Findings in the Findings tab are raw scanner
output and do not have resolution statuses,
SLA tracking, or assignees. To track remediation
for a specific finding, create or update a unified
policy that matches the finding pattern to
generate an actionable issue in the Issues tab.

🧭➡️

navigation

To access
SAST code
violation
issues, under
Modules,
select
Application
Security →
Issues →
Code
Weaknesses

12.3.12.2.
Understand
the Code
Weaknesses
table



12.3.12.3.
Investigate
and
remediate
code
weakness
issues



👀🟠

important

Important:
Urgency
signals are
populated
only when
the SAST
asset is
traced to
deployed
cloud
resources

through the
Code-to-
Cloud
mapping. If
no Code-to-
Cloud trace
exists,
urgency

signals
display as
Not
Applicable

🧭➡️

navigation

From the
main issues
table: Right-
click on an
issue in the
table →
Change
Status →
[Select a
status

From the main issues table: Right-click on an
issue in the table → Change Assignee → [Select
a user

From the
main issues
table: Right-
click on an
issue in the
table →
Change
Severity →
[Select a
severity

12.3.12.4.
Code
weakness
findings

📝🟢 

note

Findings in the Findings tab are raw scanner
output. They do not have resolution statuses,
SLA tracking, or assignees. To track remediation
for a specific finding, create or update a unified
policy that matches the finding pattern to
generate an actionable issue in the Issues tab.

12.4.
Software
supply chain
security



12.4.1.
Supply
Chain assets



12.4.1.1. VCS
organization
assets



12.4.1.1.1.
Understandin
g VCS
organization
assets

🧭➡️ 

navigation

To access
VCS
organization
assets,
under
Inventory,
select All
Assets →
Code →
VCS
Organization
s

12.4.1.1.2.
Investigate
VCS
organization
assets



12.4.1.1.3.
Manage VCS
organization
assets



12.4.1.2. VCS
collaborator
s as assets



12.4.1.2.1. In-
depth
Collaborator
asset
information



12.4.1.2.2.
Manage
Collaborator
assets



12.4.1.3.
Repository
as an asset



12.4.1.3.1.
Understandin
g repository
assets via
the UI



12.4.1.3.2.
Investigate
repository
assets



12.4.1.3.3.
Manage
Repository
assets



12.4.1.3.4.
Export
Software Bill
of Materials
(SBOM)



12.4.1.3.5.
Manage
issues
detected in
repositories

📝🟢 

note

You can also
find the
repository
issues in the
general
issue
inventory
table, and in
the
dedicated
inventory of
issues for
each
scanner type
(see step 4
above for
details).

12.4.1.4.
Technologies
as assets



📝🟢

note

The filter
supports
wildcard
filtering and
is case-
insensitive.

12.4.1.4.1.
Manage
repository
technologies

📝🟢 

note

The filter
supports
wildcard
filtering and
is case-
insensitive.

12.4.1.4.2.
Troubleshoot
ing and FAQs



12.4.1.5.
Software
packages as
assets



12.4.1.5.1.
Understandin
g software
package
assets



12.4.1.5.2.
Investigate
software
package
assets and
security
issues



12.4.1.6.
CI/CD
instance as
an asset



12.4.1.6.1.
Understandin
g CI/CD
instance
assets

🧭➡️ 

navigation

To access
CI/CD
instance
assets,
under
Inventory,
select All
Assets →
Code →
CI/CD
Instances

12.4.1.6.2.
Investigate
and manage
CI/CD
instance
assets

📝🟢 

note

Navigate to
the
dedicated
Application
Security →
Issues →
CI/CD Risks
page to
manage the
CI/CD risks
remediation
lifecycle at
scale
through bulk
status

updates,
team
assignments,
and SLA
tracking for
compliance
monitoring.

12.4.1.6.3.
Manage
CI/CD
pipeline
instances



12.4.1.7.
CI/CD
pipeline as
an asset



12.4.1.7.1.
Understandin
g CI/CD
pipeline
assets

🧭➡️ 

navigation

To access
CI/CD
pipeline
assets,
under
Inventory,
select All
Assets →
Code →
CI/CD
Pipelines

12.4.1.7.2.
Investigate
and manage
CI/CD
pipeline
assets



📝🟢

note

This requires
active CI/CD
integrations
and
successful
build log
analysis.
Pipelines
without
successful
build log
analysis
display only
the
repository
and pipeline
nodes

📝🟢

note

Navigate to
the
dedicated
Application
Security →
Issues →
CI/CD Risks
page to
manage the
remediation
lifecycle at
scale
through bulk
status

updates,
team
assignments,
and SLA
tracking for
compliance
monitoring.

🧭➡️

navigation

From the
CI/CD
Pipelines
table, right-
click a
pipeline row
→ Open in
Agentic
Assistant →
Application
Security
from the
agents menu

12.4.1.7.3.
Manage
CI/CD
pipeline
assets



12.4.1.8.
Tools as an
asset



📝🟢

note

Although attributes are identical across
inventories, their values for the same tool can
differ, most commonly in Risk Factors and Type.
This is because the inventory reflects your live
environment, which may include different
versions or configurations than the catalog—for
example, a package may not have been
upgraded or may be deployed differently.

🧭➡️

navigation

In the
inventory
table, right-
click on a
tool →
Change
Status →
select a
status

12.4.1.8.1.
Active tools



12.4.1.8.1.1.
Supply
Chain Tools

🧭➡️ 

navigation

To access
Supply
Chain Tools,
select
Modules →
Application
Security →
Supply
Chain Tools
(under 3rd
Party Tools

12.4.1.8.1.2.
Expanded
Supply
Chain tool
information

📝🟢 

note

AppSec
Admin user
permissions
are required
to perform
these
actions.

12.4.1.8.2.
Supply chain
catalog

🧭➡️ 

navigation

To access
the Supply
Chain
Catalog,
select
Modules →
Application
Security →
Supply
Chain
Catalog
(under 3rd
party tools

12.4.2.
Software
Composition
Analysis
(SCA)
scanners



👀🟠

important

Utilizing
native SCA
scanner

capabilities
requires an
AppSec add-
on license
(Posture
Management,
Runtime, or
Premium
Cortex).

12.4.2.1.
Supported
Software
Composition
Analysis
(SCA)
frameworks
and
languages



12.4.2.2.
Software
Composition
Analysis
(SCA)
vulnerability
issues



12.4.2.2.1.
Understand
the
Vulnerabilitie
s table



12.4.2.2.2.
Investigate
and
remediate
CVE
vulnerability
issues



📝🟢

note

The Open
Fix PR action
is available
only for CVE
vulnerability
issues
detected
through
periodic
scans where
an

automated
fix exists.
The action is
not available
for issues
detected
through PR
scans. Verify
that the Is
Fixable
indicator is
True before
attempting
to open a fix
pull request.

👀🟠

important

Important:
Urgency
signals
related to
deployment
context are

populated
only when
the SCA
asset is
traced to
deployed
cloud
resources

through the
Code-to-
Cloud
mapping. If
no Code-to-
Cloud trace
exists,
deployment-
related
urgency

signals
display as
Not
Applicable.
SCA-
specific
signals
(EPSS Score,
KEV Status,
Reachability,
Has Fix) are
always
available...

12.4.2.2.3.
Investigate
CVE
vulnerabilitie
s findings



📝🟢

note

Findings in the Findings tab are raw scanner
output and do not have resolution statuses,
SLA tracking, or assignees. To track remediation
for a specific finding, create or update a unified
policy that matches the finding pattern to
generate an actionable issue in the Issues tab.

🧭➡️

navigation

To access
CVE
findings,
under
Modules,
select
Application
Security →
Issues →
Vulnerabilitie
s → click the
Findings tab

12.4.2.2.4.
Manage SCA
CVE
vulnerability
issues



12.4.2.3.
License
miscomplian
ce issues



12.4.2.3.1.
Understand
the Licenses
table

📝🟢 

note

Important:
License
miscomplian
ce issues do
not have
urgency

signals.
Unlike CVE
vulnerability
issues,
license
issues are
legal
compliance
risks — not
security
vulnerabilitie
s. Urgency
signals such
as EPSS
score, KEV
status,
reachability
analysis, and
Code-to-
Cloud
deployment
context do
not apply to
license
compliance
evaluation.

12.4.2.3.2.
Take action
on license
miscomplian
ce issues

👀🟠 

important

Automated fix pull requests are not available
for license miscompliance issues. Remediation
requires human judgment to evaluate license
compatibility, negotiate licensing terms, identify
alternative packages, or accept the license with
documented justification.

12.4.2.3.3.
License
miscomplian
ce findings

🧭➡️ 

navigation

To access
license
miscomplian
ce findings,
under
Modules,
select
Application
Security →
Issues →
Licenses →
click the
Findings tab

12.4.2.3.4.
Open-
source

software
license
categories



12.4.2.3.5.
Manage
license
miscomplian
ce issues



12.4.2.4.
Ingest third-
party SCA
data

🧭➡️ 

navigation

Modules →
Application
Security →
Issues →
Vulnerabilitie
s

12.4.3. CI/CD
Risks



📝🟢

note

Cortex
Cloud
Application
Security
CI/CD
pipeline
scans create
a

comprehensi
ve inventory
of all CI/CD
pipelines in
your
environment.
For more
information
refer to
CI/CD
pipeline as
an asset.

12.4.3.1.
CI/CD
pipeline
issues

🧭➡️ 

navigation

To access
CI/CD
pipeline risks
issues, under
Modules,
select
Application
Security →
Issues →
CI/CD Risks

12.4.3.2.
Expanded
CI/CD risks
issue
information

📝🟢 

note

Different
issue types
include
different
properties;
therefore,
not all
properties
are available
for every
issue.

12.4.3.3. VCS
and CI/CD
pipeline risk
findings



12.4.4. CI/CD
Policies



💡🔵

tip

For Cortex
Cloud Code
policies,
refer to
Unified
Application
Security
policies.

12.4.4.1.
CI/CD
policies user
roles and
permissions



12.4.4.2.
CI/CD
policies
inventory

📝🟢 

note

To view all
out-of-the-
box (OOTB)
policies,
filter by
Policy
Owner =
System.

12.4.4.3.
Create
CI/CD
configuration
policies



📝🟢

note

When you
select a
category
(such as
Application),
a list of
available
asset values
for that type
is displayed.
The policy
will
automaticall
y apply to
all assets in
the
displayed
list unless
you apply
additional
filters using
specific
asset
attributes
(such as the
Business
Application
Names filter).

📝🟢

note

SBAC scope-
based
limitations
do not apply
to Asset
Types

📝🟢

note

Periodic
scan is the
only trigger
that is
supported
for CI/CD
policies.

12.4.4.3.1.
Cortex
Cloud
Application
Security
CI/CD policy
Condition
attributes



12.4.4.4.
Manage
CI/CD
policies



📝🟢

note

You cannot
edit out-of-
the-box
(OOTB)
policies. 📝🟢

note

The
duplicated
policy will
include the
word "clone"
in its name
and must be
renamed.
12.4.5. CI/CD
Rules



📝🟢

note

Out-of-the-
box rules
cannot be
modified
Custom
CI/CD rules
are not

supported

12.4.5.1.
CI/CD rules
roles and
permissions



12.4.5.2.
CI/CD rules
inventory



12.5. Code
Security



📝🟢

note

Code
Security
covers

detection
and analysis
engines. For
governance
and
orchestration
layer, see
Application
Security
Posture
Management
(ASPM).

🧭➡️

navigation

You can
view the
granular
permissions
assigned to
each role in
the tenant
by
navigating
to Settings
→

Configuratio
ns → Roles

12.5.1. Code
Security
assets



12.5.2.
Software
packages as
assets



12.5.2.1.
Supported
Software
Composition
Analysis
(SCA)
frameworks
and
languages



12.5.2.2.
Understandin
g software
package
assets

🧭➡️ 

navigation

To access
software
package
assets,
under
Inventory,
select All
Assets →
Code →
Software
Packages

12.5.2.3.
Investigate
software
package
assets and
security
issues



12.5.3.
Infrastructur
e-as-Code
(IaC)
resources as

assets



12.5.3.1.
Supported
frameworks
and
languages



12.5.3.2.
Access and
filter IaC
assets

🧭➡️ 

navigation

Inventory,
select All
Assets →
Code → IaC
Resources 12.5.3.3.
Investigate
IaC assets



12.5.4. Code
Security
scanners



12.5.5.
Secrets
scans



12.5.5.1.
Navigate to
secrets
issues



📝🟢

note

The default
sort order is
by Severity
(descending).
The default
filter shows
issues with a
status of
New and In
Progress.
Resolved
issues are
hidden by
default.

🧭➡️

navigation

Under
Modules,
select
Application
Security →
Issues →
Secrets

12.5.5.2.
Understand
the secrets
issues table



12.5.5.3.
Investigate
and
remediate
secrets
issues

👀🟠 

important

Urgency
signals
related to
deployment
context are

populated
only when
the secrets
asset is
traced to
deployed
cloud
resources

through the
Code-to-
Cloud
mapping.

12.5.5.4.
Secrets
findings

📝🟢 

note

Review
excluded
findings:
Investigate
findings that
were

excluded by
policy filters
to confirm
that
exclusions
are

intentional
and do not
suppress
critical IaC
misconfigura
tions.

12.5.6.
Infrastructur
e as Code
(IaC)
misconfigura
tion scanner



12.5.6.1.
Navigate to
IaC
misconfigura
tion issues



📝🟢

note

The default
sort order is
by Severity
(descending).
The default
filter shows
issues with a
status of
New and In
Progress.
Resolved
issues are
hidden by
default.

🧭➡️

navigation

To access
IaC
misconfigura
tion issues,
under
Modules,
select
Application
Security →
Issues → IaC
misconfigura
tion

12.5.6.2.
Understand
the IaC
misconfigura
tions table



12.5.6.3.
Investigate
and
remediate
IaC
misconfigura
tion issues

📝🟢 

note

The Open
Fix PR action
is available
only for
issues
detected
through
periodic
scans where
an

automated
fix exists.
The action is
not available
for issues
detected
through PR
scans.

12.5.6.4. IaC
misconfigura
tionfindings



📝🟢

note

Findings in the Findings tab are raw scanner
output and do not have resolution statuses,
SLA tracking, or assignees. To track remediation
for a specific finding, create or update a unified
policy that matches the finding pattern to
generate an actionable issue in the Issues tab.

🧭➡️

navigation

To access
IaC
misconfigura
tion findings,
under
Modules,
select
Application
Security →
Issues → IaC
Misconfigura
tions → click
the Findings
tab

The Cortex
Cloud
Application
Security IaC
misconfigura
tions
Findings
table is a
filtered view
of the
comprehensi
ve Findings
inventory,
located at
Cases &
Issues →
Issues →

Findings tab 12.5.7. IaC
Drift
Detection
scans



12.5.7.1.
Navigate to
IaC drift
detection
issues



📝🟢

note

The default
sort order is
by Severity
(descending).
The default
filter shows
drift findings
with a status
of New and
In Progress.
Resolved
drift findings
are hidden
by default.

🧭➡️

navigation

Modules →
Application
Security →
IaC Drifts
12.5.7.2.
Understand
the IaC drift
detection
table



12.5.7.3.
Investigate
and
remediate
IaC drift
issues



📝🟢

note

The Revert
to IaC State
action
requires
write
permissions
on the
target cloud
account.

Verify that
the Cortex
Cloud
service
account has
the
necessary
IAM
permissions
before
initiating a
revert
operation.

📝🟢

note

Updating
the IaC
template
codifies the
current
cloud state
as the new
desired
state. Verify
that the
cloud
resource

configuration
complies
with security
policies
before
merging the
generated
pull request.

🧭➡️

navigation

From the
main drift
table:Right-
click on a
drift issue in
the table →
Change
Status →
Change
Status

From the
main drift
table:Right-
click on a
drift issue in
the table →
Change
Assignee →
Change
Status

12.5.7.4.
Investigate
IaC drift
detection
findings



📝🟢

note

Findings in
the Findings
tab are raw
scanner

output and
do not have
resolution
statuses,
SLA
tracking, or
assignees. To
track
remediation
for a
specific drift
finding,
create or

update a
unified
policy that
matches the
finding
pattern to
generate an
actionable
issue in the
Issues tab.

🧭➡️

navigation

Modules →
Application
Security →
IaC Drifts
(under
Issues) →
click the
Findings tab

12.5.8.
Software
Composition
Analysis
(SCA)
scanners



12.5.8.1.
Software
Composition
Analysis
(SCA)
vulnerability
issues



12.5.8.1.1.
Navigate to
CVE
vulnerability
issues



📝🟢

note

The default
sort order is
by Severity
(descending).
The default
filter shows
issues with a
status of
New and In
Progress.
Resolved
issues are
hidden by
default.

📝🟢

note

Findings in
the Findings
tab are raw
scanner

output and
do not have
resolution
statuses,
SLA
tracking, or
assignees. To
track
remediation
for a
specific
finding,
create or

update a
policy that
matches the
finding
pattern to
generate an
actionable
issue in the
Issues tab.

12.5.8.1.2.
Understand
the
Vulnerabilitie
s table



12.5.8.1.3.
Investigate
and
remediate
CVE
vulnerability
issues



📝🟢

note

The Open
Fix PR action
is available
only for CVE
vulnerability
issues
detected
through
periodic
scans where
an

automated
fix exists.
The action is
not available
for issues
detected
through PR
scans. Verify
that the Is
Fixable
indicator is
True before
attempting
to open a fix
pull request.

👀🟠

important

Important:
Urgency
signals
related to
deployment
context are

populated
only when
the SCA
asset is
traced to
deployed
cloud
resources

through the
Code-to-
Cloud
mapping. If
no Code-to-
Cloud trace
exists,
deployment-
related
urgency

signals
display as
Not
Applicable.
SCA-
specific
signals
(EPSS Score,
KEV Status,
Reachability,
Has Fix) are
always
available...

12.5.8.1.4.
Investigate
CVE
vulnerabilitie
s findings

📝🟢 

note

Findings in the Findings tab are raw scanner
output and do not have resolution statuses,
SLA tracking, or assignees. To track remediation
for a specific finding, create or update a unified
policy that matches the finding pattern to
generate an actionable issue in the Issues tab.

12.5.8.2.
License
miscomplian
ce issues



12.5.8.2.1.
Navigate to
license
miscomplian
ce issues



📝🟢

note

The default
sort order is
by Severity
(descending).
The default
filter shows
issues with a
status of
New and In
Progress.
Resolved
issues are
hidden by
default.

📝🟢

note

Findings in the Findings tab are raw scanner
output and do not have resolution statuses,
SLA tracking, or assignees. To track remediation
for a specific finding, create or update a unified
policy that matches the finding pattern to
generate an actionable issue in the Issues tab.

🧭➡️

navigation

Application
Security →
Issues →
Licenses
12.5.8.2.2.
Understand
the Licenses
table

📝🟢 

note

Important:
License
miscomplian
ce issues do
not have
urgency

signals.
Unlike CVE
vulnerability
issues,
license
issues are
legal
compliance
risks — not
security
vulnerabilitie
s. Urgency
signals such
as EPSS
score, KEV
status,
reachability
analysis, and
Code-to-
Cloud
deployment
context do
not apply to
license
compliance
evaluation.

12.5.8.2.3.
Take action
on license
miscomplian
ce issues

👀🟠 

important

Automated fix pull requests are not available
for license miscompliance issues. Remediation
requires human judgment to evaluate license
compatibility, negotiate licensing terms, identify
alternative packages, or accept the license with
documented justification.

12.5.8.2.4.
How license
miscomplian
ce issues fit
in the
Application
Security
ecosystem

🧭➡️ 

navigation

Configure
unified
policies for
license
compliance
at Modules

→

Application
Security →
AppSec
Policies

12.5.8.2.5.
Open-
source

software
license
categories



12.5.8.3.
Package
operational
risk scanner



12.5.8.3.1.
Understand
the package
operational
risk table



12.5.8.3.2.
Investigate
and
remediate
package
operational
risk issues



📝🟢

note

Automated fix pull requests are not available
for package operational risk issues. Remediation
requires human judgment to evaluate
alternative packages, assess API compatibility,
and perform the migration.

👀🟠

important

Package operational risk issues do not have
urgency signals based on exploit intelligence.
Unlike CVE vulnerability issues, operational risk
issues assess the health and sustainability of a
package, not active security vulnerabilities.
Urgency signals such as EPSS score, KEV status,
and reachability analysis do not apply to
operational risk evaluation....

12.5.8.3.3.
How

package
operational
risk issues fit
in the
Application
Security
ecosystem

🧭➡️ 

navigation

Configure unified policies for operational risk at
Modules → Application Security → AppSec
Policies

12.5.9. API
workflows
for Code
Security
issues



📝🟢

note

When
blockingPolic
iesIds is non-
empty, the
PR was
blocked,
preventing
vulnerable
code from
merging into
the
protected
branch.

📝🟢

note

The severity query parameter is the only server-
side filter available on the issue endpoint.
Filtering by detection method, resolution status,
or assignee requires client-side processing.

📝🟢

note

License
compliance
(License
Security)
and package
operational
risk
(Operational
Risk
Security) are
not

represented
as separate
keys in
issuesTypesB
reakdown.
To retrieve
these issue
types, filter
client-side
by
detection_m
ethod from
the full issue
set.

📝🟢

note

Application
issue
summaries
return

counts only,
not
individual
issue
records. To
retrieve
individual
records, list
periodic
scans for
the
application's
repositories,
then use
GET
/scans/{scanI
d}/issues.

💡🔵

tip

Use the issuesTypesBreakdown and issuesFound
fields in the scan list response to identify scans
with Critical or High issues before drilling into
individual records, thereby reducing API calls.

🧭➡️

navigation

Generate an
API key pair
from
Settings →
Configuratio
ns → API
Keys in the
Cortex
Cloud
console

12.5.10.
Application
Security
scans

management



12.5.10.1.
Manage
scans

through the
tenant (UI)

📝🟢 

note

While scans
provide a
comprehensi
ve inventory
of all issues
detected
during a
scan,
dedicated
inventories
are also
maintained
for specific
scan types
for more
granular
management.
For more
information,
refer to
Code
Security
scanners.

12.5.10.2.
Branch
periodic
scans

📝🟢 

note

The
inventory
table
displays
scan issues
for visibility
only;
remediation
is not
available
here. To
resolve
issues,
navigate to
the
dedicated
issue type
inventory,
where you
can manage
and
remediate
them.

12.5.10.3. Pull
Request
scans

📝🟢 

note

The
inventory
table
displays
scan issues
for visibility
only;
remediation
is not
available
here. To
resolve
issues,
navigate to
the
dedicated
issue type
inventory,
where you
can manage
and
remediate
them.

12.5.10.4. CI
scans

📝🟢 

note

The
inventory
table
displays
scan issues
for visibility
only;
remediation
is not
available in
scan

management.
To resolve
issues,
navigate to
the
dedicated
issue type
inventory,
where you
can manage
and
remediate
them.

12.5.10.5.
Scan health
and status
reference

📝🟢 

note

Periodic
scans do not
have a scan
status.
Periodic
scans

produce
issues and
findings but
do not
evaluate
blocking
policies. The
scan health
status is the
only health
indicator for
periodic
scans.

12.5.10.6.
Manage
repository
scan

configuration
s



12.5.10.7.
Monitor data
source

instances
health



12.5.10.8.
Manage
scans

through
public APIs



12.5.11.
Application
Security CLI



12.5.12. IDE 

📝🟢

note

Not all
remediation
options are
available for
all findings
or all type
of scan
category.

12.5.12.1.
System
requirements



12.5.12.2.
Visual
Studio (VS)
Code and
VS Code
compatible
IDEs



☑️🟣

prerequisite

Before you
begin (These
prerequisites
apply to
both VS
Code and
VS Code
compatibles):
Permissions:
CLI Read
only
permissions.
Refer to
Cortex CLI
for more
information
about
permissions
Environment
setup
macOS and
Windows:
Install
Python 3.9.x
to 3.12.x
Install
Node.js
version 22
and above
for SCA
scans (such
as

vulnerabilitie
s scans)
Environment...

☑️🟣

prerequisite

You must
insert your
API key and
API ID values
into the
Settings
before
providing
the tenant
URL.

📝🟢

note

Enforcement
rules and CA
certificates
are not

applicable
to the open-
source

project.

👀🟠

important

When
generating
an API key,
ensure you
select the
Standard
security
level. IDE
and CLI
scans will
fail if the
security
level of the
API key is set
to Advanced.

💡🔵

tip

You can
access

Extensions
using the
keyboard
shortcuts
Ctrl+Shift+X
(Windows)
or

Cmd+Shift+
X (macOS).

🧭➡️

navigation

Settings →
Configuratio
ns → API
Keys (under
Integration)
→ click
Copy API
URL

Install
through VS
Code IDE
Access the
Activity bar
→

Extensions
→ Cortex
Cloud →
Install

Access the
Activity bar
→

Extensions
→ Cortex
Cloud →
Install

12.5.12.2.1.
How to use
the Cortex
Cloud
extension in
VS Code



📝🟢

note

Not all
remediation
options are
available for

all issues. 📝🟢

note

The
suppression
is scoped to
the file.

12.5.12.3.
JetBrains



☑️🟣

prerequisite

Permissions:
CLI Read
only
permissions.
Refer to
Cortex CLI
for more
information
about
permissions
Environment
setup
macOS and
Windows:
Install
Python 3.9.x
to 3.12.x
Install
Node.js
version 22
and above
for SCA
scans (such
as

vulnerabilitie
s scans)
Environment
setup
macOS and
Windows:
Install
Python 3.9.x
to 3.12.x
Install
Node.js
version 22
and...

☑️🟣

prerequisite

You must
insert your
API key and
API ID values
into the
Settings
before
providing
the tenant
URL.

📝🟢

note

The Cortex
AppSec
code
security
plugin
supports all
JetBrains
products.

👀🟠

important

When
generating
an API key,
ensure you
select the
Standard
security
level. IDE
and CLI
scans will
fail if the
security
level of the
API key is set
to Advanced.

🧭➡️

navigation

Settings →
Plugins →
select the
Marketplace
tab →
search for
Prisma
Cloud →
Install → OK

Settings →
Tools →
Cortex
Cloud

12.5.12.3.1.
How to use
the JetBrains
Cortex
Cloud
extension



📝🟢

note

After
selecting Fix
Available,
the number
of issues
displayed in
the issue
categories
(such as IaC)
reflect the
number of
fixable
issues for
that type.

📝🟢

note

The
correspondin
g file
simultaneous
ly opens in
the Code
editor,
highlighting
the issue
within its
exact code
context. See
below for

more

information.

📝🟢

note

Not all types of remediation are available for all
issue categories. For example, fixes are not
available for License issues.

📝🟢

note

Secrets and
Licenses
category
issues are
typically
mitigated by
following
the
guidance in
the
Documentati
on.

12.5.13.
Developer
Suppressions

📝🟢 

note

The check
IDs and rule
names are

placeholders
for
illustration
purposes
and do not
correspond
to actual
checks.

12.6. API
endpoints



12.7.
Terraform
workflows



🧭➡️

navigation

To
authenticate,
you must
generate a
dedicated
API key in
the Cortex
Cloud
console
under
S tti →

12.7.1.
Manage
resources

📝🟢 

note

For a
complete
list of
supported
IaC and
CI/CD
li

13. Cloud

ASM



13.1. What is
Cloud ASM?



13.2. Cloud
ASM
concepts



13.2.1.
Scanning



13.2.1.1.
Scanning
cadences



13.2.1.2.
Scanning
ports and
protocols

📝🟢 

note

The
following
lists are not
exhaustive.
For current
and
complete
lists, contact
your
customer
success

team.

13.2.1.3.
Scanning
activity



13.2.2.
Network
mapping



13.3. Enable
Cloud ASM



📝🟢

note

Discovered
services will
increase
workload
consumption.
ASM data
will begin to
appear in
your tenant
and be
included in
your

licensing
workload
meter within
72 hours.

🧭➡️

navigation

Settings → Configurations → Attack Surface →
Data Management

13.4. Attack
Surface
Management
detections



13.4.1. Attack
surface rules

🧭➡️ 

navigation

Modules →
Attack
Surface →
Policies →
Attack
Surface
Rules

13.4.2.
Externally
inferred
CVEs

📝🟢 

note

An
externally
inferred CVE
might
impact your
service or
asset, but
additional
investigation
is required
to confirm
that the CVE
is actually
present.

13.5. Attack
surface
assets

🧭➡️ 

navigation

The internet-
facing
assets that
were

discovered
in an attack
surface
management
(ASM) scan
and
attributed to
your
organization
are available
in the
inventory on
the Inventory
→ Assets →
All Assets →
External
Surface
pages

13.6. Review
your

unmanaged
cloud
services

🧭➡️ 

navigation

Inventory →
Assets → All
Assets →
External
Surface →
Services

13.7. Review
unmanaged
cloud issues



14.

Network

exposure
detection



14.1. What is
Cloud
Network
Analyzer?



14.2. Internet
exposure
detection

📝🟢 

note

Currently,
CNA
supports
only an
isolated
architecture
with an AWS
Gateway
Load
Balancer
(GWLB)
within a
single VPC.
Other more
centralized
topologies
including
one security
VPC that
forwards
traffic to
other
workload
VPCs are
currently
not

supported.

14.3.
Outbound
exposure
detection



14.4. East-
west
exposure
detection



14.5.
Investigate
an internet
exposure



📝🟢

note

The Network
tab is
currently
only
available for
virtual
machines.

📝🟢

note

The Network tab is only displayed when you
have access to the main asset and associated
ones, such as security groups, VPCs and
subnets. For more information on Scope-Based
Access Control (SBAC) for configuring granular
scoping, see Manage user scope.Manage user
scope

📝🟢

note

The Network
tab is
currently
only
available for
virtual
machines.

📝🟢

note

The Network tab is only displayed when you
have access to the main asset and associated
ones, such as security groups, VPCs and
subnets. For more information on Scope-Based
Access Control (SBAC) for configuring granular
scoping, see Manage user scope.Manage user
scope

14.6.

Configure
trusted IPs

🧭➡️ 

navigation

Inventory →
Network
Configuratio
n → Trusted
Networks

15.

Cortex

Cloud

Data

Classifica

tion



15.1. What is
Cortex
Cloud Data
Classification
?

🧭➡️ 

navigation

To access
Data
Classification

management,
click
Settings →
Configuratio
ns → Data
Classification

15.2. How to
create and
validate a
custom data
pattern



📝🟢

note

Once a
custom data
pattern is
saved, it runs
on all data in
the same
way as any
out-of-the-
box (OOTB)
pattern,
becoming
globally
applicable
for all
modules
using Cortex
Cloud Data
Classification

.

📝🟢

note

For more
information,
see How to
disable and
enable data
patterns in
Data
Classification
.How to
disable and
enable data
patterns in
Data
Classification

15.2.1.
Custom
data
patterns:
Guardrails
and syntax
guide



15.3. How to
disable and
enable data
patterns in
Data
Classification



📝🟢

note

Disabling
data
patterns can
cause

changes in
your data
profile
results and
stop
detection of
these data
patterns.

📝🟢

note

For more
information
about data
patterns in
data
classification
, see What is
Cortex
Cloud Data
Classification
?.

🧭➡️

navigation

In the lower
left part of
the screen,
click
Settings →
Configuratio
ns → Data
Classification
→ Data
Patterns
15.4. How to
create and
validate a
custom data
profile



📝🟢

note

If you
choose two
data
locations,
only the
filters they
have in
common will
be included
in the
possible
filter options.

📝🟢

note

For more
information,
see How to
disable and
enable data
profiles in
Cortex
Cloud Data
Classification

.

👀🟠

important

Deleting a
data profile
deletes all
past data
associated
with it in all
modules
using Cortex
Cloud Data
Classification
after a
warning
notification
is displayed.

👀🟠

important

Enabling and
disabling a
data profile
removes or

re-adds the
data profile
results to
the data
objects; that
is, files and
tables.

15.5. How to
disable and
enable data
profiles in
Cortex
Cloud Data
Classification



📝🟢

note

Existing
results on
past scans
are updated
in the Asset
and Object
inventories
within two
hours after
being
disabled or
enabled.

📝🟢

note

For more
information
about data
profiles in
Cortex
Cloud Data
Classification
, see What is
Cortex
Cloud Data
Classification
?.What is
Cortex
Cloud Data
Classification
?

🧭➡️

navigation

In the lower
left part of
the screen,
click
Settings →
Configuratio
ns → Data
Classification
→ Data
Profiles

15.6. How to
report a
false
positive in
Cortex
Cloud Data
Classification

📝🟢 

note

Alternatively,
in Cortex
Command
Center, click
Help →
Submit a
Support
Case.

16.

Cortex

Cloud

Data

Security



16.1. What is
Cortex
Cloud Data
Security?



16.2.
Supported
assets in
Cortex
Cloud Data
Security



🔒🟣

content

This feature
is included
with a
Cortex
XSIAM
Premium
license. It is
also
included
with any
other Cortex
XSIAM
license that
has the
Cloud
Posture
Security or
Cloud
Runtime
Security
add-on.

📝🟢

note

CMK
(customer-
managed
key) in SQL
Server is
only
supported in
outpost
mode.

📝🟢

note

The list
above refers
to only data
classification
; however,
Cortex
Cloud Data
Security
discovers
and
monitors all
cloud assets
and services
for usage
and
misconfigura
tions.

📝🟢

note

For
information
about
configuring
data
classification
asset types,
see How to

configure
the scanning
settings for
supported
services.

16.3. Cortex
Cloud Data
Security
concepts



16.4. Cortex
Cloud Data
Security use
cases



16.5. Data
Inventory



📝🟢

note

To clear the
filter in the
filter panel,
click Delete.

🧭➡️

navigation

When you
select
Inventory →
All Assets →
Data → All
Data Assets,
a list of all
data assets
appears

16.6. How to
review errors
in Cortex
Cloud Data
Security



16.7. How to

configure
the scanning
settings for
supported
services

📝🟢 

note

If you do not
select a
scanning
cadence, the
default
setting is
applied. For
more

information,
contact your
Customer
Support
team.

16.8. How to
perform
advanced
Data
Security
investigations
using XQL

📝🟢 

note

For more
information,
see Build
XQL queries.
16.9. How to
onboard
Databricks

📝🟢 

note

If you turn
on this
feature,
both the
cloud and
region will
be used for
scanning,
possibly
incurring
cost and
requiring
adherence
to certain
compliance
policies.

16.10. How to
onboard
Microsoft
365

☑️🟣 

prerequisite

You have
generated a
Globally
Unique
Identifier
(GUID), also
known as a
Universally
Unique
Identifier
(UUID). You
will need
this ID for
the tenant
you want to
use for the
Microsoft
365 instance.
In order to
use

Microsoft
365, you
must be
registered
with
Microsoft
Azure.

16.11. How to
onboard on-
premise
assets to
Cortex
Cloud Data
Security



📝🟢

note

Your data is
scanned on
the Broker
VM itself,
and only the
metadata
and
classification
results are
transmitted
from the on-
premise
environment
to Cortex
Cloud.

📝🟢

note

For
information
about
activating
the DSPM
Fileshare
applet, see
Activate
DSPM
Fileshare.

📝🟢

note

For
information
about
activating
the DSPM
Database
applet, see
Activate
DSPM
Database.

16.12. How to
onboard
Snowflake



☑️🟣

prerequisite

In order to
use

Snowflake,
you must be
registered
with one of
these cloud
providers:
Amazon
AWS,
Microsoft
Azure, or
Google
Cloud
Platform
(GCP).
Ensure you
have the
necessary
account
permissions
to onboard.
It is
recommende
d to use
Account
Admin as the
role for the
onboarding.

📝🟢

note

The account
identifier
can be
found using
the user
information
at the
bottom left.
Hover over
the account
you wish to
onboard and
select the
copy option
at the top
right. The
account
identifier is
usually of
the format:
(organization
).[account]

16.13. How to
use

information
protection
labels in
Cortex
Cloud Data
Security

📝🟢 

note

Only If you
have
connected a
Microsoft
365 instance
to Cortex
Cloud, the
information
protection
labels
appear. See
How to
onboard
Microsoft
365.

17.

Cortex

Cloud

Identity
Security



17.1. What is
Cortex
Cloud
Identity
Security?



17.2. Review
and improve
your Identity
Security
posture



📝🟢

note

Least
Privilege
Access
(LPA) is not
supported in
OCI.

💡🔵

tip

Reducing
the number
of non-
human
entities that
are either
dormant or
have
administrativ
e privileges
is the most
effective
way to
improve your
security
score.

17.3. How
does
Effective
Permission
Calculation
work?

📝🟢 

note

Unlike GCP
(organization
s/folders/pro
jects) or
Azure
(managemen
t

groups/subs
criptions),
OCI uses a
nested
compartmen
t structure.

17.4. Cortex Cloud Identity Security functionality 

📝🟢

note

Access to
the Weak
Password
widget and
column
requires the
ITDR add-on.

📝🟢

note

Once a user
updates
their
password to
a strong
password,
they will no
longer
appear in
the filtered
weak
password
results.

📝🟢

note

For more
information
about
Effective
Permission
Calculation,
see How
does
Effective
Permission
Calculation
work?.

📝🟢

note

When a
policy is
analyzed
and
categorized
as excessive,
a relevant
finding and
highlight is
attached to
that policy
and the
various
identities
being
granted
excessive
permissions.

📝🟢

note

Each action
must be
defined as
either being
administrativ
e or not.
The
administrativ
e tag is an
additional
tag for
administrativ
e actions,
along with
one of the
other access
level types.
For example,
the AWS
action
iam:CreateGr
oup is
categorized
as config
and has the
administrativ
e tag as well.

👀🟠

important

If you turn
off the audit
logs, even
briefly, this
temporarily
impacts the
accuracy of
the Last
Access data,
potentially
showing
permissions
as unused
when
actually
they were
active. Full
accuracy is
restored 90
days after
you re-
enable the
audit logs.

17.5. Unified
Human
Identities



17.6. Achieve
the principle
of least
privilege
access



📝🟢

note

In the case
of Amazon
AWS, Cortex
Cloud
Identity
Security also
uses AWS
Identity and
Access
Management
(IAM)
Access
Advisor
insights to
expand the
coverage of
supported
actions.

👀🟠

important

If you turn
off the audit
logs, even
briefly, this
temporarily
impacts the
accuracy of
the Last
Access data,
potentially
showing
permissions
as unused
when
actually
they were
active. Full
accuracy is
restored 90
days after
you re-
enable the
audit logs.

17.7. Explore
permissions
using the
simple and
advanced
access

tables



17.8. Create a
custom
detection
rule in
Cortex
Cloud
Identity
Security

🧭➡️ 

navigation

Posture
Management
→ Rules &
Policies →
Rules →
Cloud
Security

17.9. Perform
advanced
Identity
Security
investigations
using XQL

📝🟢 

note

For more
information,
see Build
XQL queries.
17.10. Ingest
logs and
data from
Okta

☑️🟣 

prerequisite

Administrato
r privileges:
Your Okta
user must
have a role
capable of
creating API
tokens, such
as Read-
only
Administrato
r, Super
Administrato
r, or

Organization
Administrato
r. For more
information,
see the Okta
Administrato
rs

Documentati
on.

17.11. Enable
inactive
human
identity logs
on Azure in
Cortex
Cloud
Identity
Security

📝🟢 

note

For more
information,
see Ingest
logs from
Microsoft
Azure Event
Hub.

17.12.
Manage
RBAC and
SBAC in
Cortex
Cloud
Identity
Security

👀🟠 

important

You can use
the Cortex
Cloud
Identity
Security
RBAC roles
to define
access to
the various
sections and
functionalitie
s of Cortex
Cloud
Identity
Security, but
these roles
do not
directly
control the
specific
data a user
sees within
those
sections.
Data
visibility is
further
refined and
limited by
scope-
based
access

control
(SBAC)
capabilities.

18.

Vulnerabi

lity

managem
ent



18.1.
Vulnerability
management
in Cortex
Cloud



🔒🟣

content

Requires the
Cortex
Cloud
Posture
Security,
Cortex
Cloud
Runtime
Security,
Exposure
Management,
Cortex
XSIAM
Premium or
ASM add-on.

📝🟢

note

Cortex
Cloud
vulnerability
management
provides the
ability to
identify and
assess

runtime
vulnerabilitie
s in every
asset across
traditional IT
and cloud
environments
. For
vulnerabilitie
s detected
in your
software
development
lifecycle
through
application
security
scans, refer
to the
Cortex
Cloud
Application
Security
documentati
on.

18.1.1. Cortex
Cloud
vulnerability
concepts

📝🟢 

note

CVE is an
acronym for
Common
Vulnerabilitie
s and
Exposures,
which is a
list of
publicly
disclosed
security
threats. We
often use
the term
"CVE" to
refer to a
vulnerability
that has
been a
assigned a
CVE ID.
Cortex
Cloud
identifies
CVEs and
non-CVE
vulnerabilitie
s.

18.1.2.
Vulnerability
Management
dashboard



18.2. Cortex
Vulnerability
Risk Score

🧭➡️ 

navigation

Posture
Management
→

Vulnerability
Management
→

Vulnerability
Issues

18.3.
Vulnerability
policies



18.3.1. Create
a

vulnerability
policy

🧭➡️ 

navigation

Posture
Management
→ Rules &
Policies →
Policies →
Vulnerability
Management

18.3.2.
Update the
Ignored
CVEs, Asset
Groups, and
Assets policy



18.3.3.
Modify a
vulnerability
policy

🧭➡️ 

navigation

Posture
Management
→ Rules &
Policies →
Vulnerability
Management

18.3.4.
Configure a
block grace
period

🧭➡️ 

navigation

Posture
Management
→ Rules &
Policies →
Vulnerability
Management
→

Vulnerability
Policies -
Prevention

18.3.5. Enable
or disable a
vulnerability
policy



18.4.
Investigate
and
remediate
vulnerabilitie
s



18.4.1. View
all
Vulnerabilitie
s

🧭➡️ 

navigation

Posture
Management
→

Vulnerability
Management
→

Vulnerabilitie
s

18.4.2. View
vulnerability
issues



18.4.3. View
All
Vulnerability
Findings

🧭➡️ 

navigation

Posture
Management
→

Vulnerability
Management
→

Vulnerability
Issues and
click the All
Vulnerability
Findings
button

18.4.4. View
vulnerable
assets

🧭➡️ 

navigation

Posture
Management
→

Vulnerability
Management
→

Vulnerable
Assets

18.5.
Vulnerability
Intelligence

🧭➡️ 

navigation

Posture
Management
→

Vulnerability
Management
→

Vulnerability
Intelligence

18.6. Recast
CVSS scores
and CVSS
severities



19. Cloud

Security
Rules

and

Policies



19.1. Cloud
security
rules and
policies



19.2. Cloud
security rules

📝🟢 

note

Findings are
only
generated
for OOTB
rules. 19.3. Cloud
security
policies

📝🟢 

note

If you have
the
following
Scope
Based
Access
Control
(SBAC)
settings in
place, User
Settings →
Cases and
Issues Scope
→ Select
domains →
Posture, you
may
encounter a
Case
mismatch in
Issues/Cases/
Findings
counts. This
is because
the Case
count on the
Rules page
captures
Cases
belonging to
the Posture
domain.
Whereas
Platform
pages,
capture
Issues...

19.4. Create
and manage
cloud
security rules



🧭➡️

navigation

You can
view and
manage
cloud
posture
security
rules from
the Posture
Management
→ Rules &
Policies →
Rules →
Cloud
Security
page

19.4.1. Create
an attack
path rule



19.4.2.
Create a
configuration
rule



19.4.2.1.
Guidelines
for creating
cloud
security rules



19.4.2.2.
Cloud
security rule
status for
custom

configuration
rules



19.4.3.
Create a
data rule



19.4.4.
Create an
identity rule



19.4.5.
Create a
network
exposure
rule



19.4.6.
Create an AI
rule



19.4.7. View
cloud
security rule
status



19.4.8. Edit a
cloud
security rule



19.4.9. Enable
or disable a
rule



19.4.10. Use
an existing
rule to
create a new
one



19.4.11.
Delete a
custom
cloud
security rule



19.5. Create
and manage
cloud
security
policies



19.5.1. Create
a cloud
security
policy

🧭➡️ 

navigation

Posture
Management
→ Rules &
Policies →
Policies →
Cloud
Security

19.5.2. Edit a
cloud
security
policy



19.5.3. Enable
or disable a
policy



19.5.4. Use an
existing
policy to
create a new
one



19.5.5. Delete
a custom
cloud
security
policy



20. Cloud

Workload

Policies

and Rules



☑️🟣

prerequisite

Users need
View/Edit
RBAC
permissions
(under
Policies →
Compute
Policies) or
the Instance
Administrato
r role to
view, edit,
and modify
Cloud
Workload
Policies
policies.

👀🟠

important

Users with
SBAC
granular
scoping (in
addition to
the RBAC
permissions
required for
Cloud
Workload
Policies) can
only view
Cloud
Workload
Policies,
when their
access is
scoped to
any of the
available
options: All
assets, No
assets, or
Select asset
groups. For
more

information
on granular
scoping, see
Manage user
scope. When
no SBAC
restriction is...

20.1. How
policies and
rules work
together



20.2. Cloud
Workload
Policies



📝🟢

note

Issues are
automaticall
y resolved
when the
finding is no
longer
applicable
to the asset
or when the
affected
asset is
removed
from the
inventory.
For more
details on
Prevent and
create
issues, see
Cloud
Workload
Preventive
Action.

20.2.1. Types
of Cloud
Workload
Policies

20.2.1.1. 
Trusted
image cloud
workload
policies



📝🟢

note

Registry
scanning is
for finding
problems
that are
objectively
issues
regardless
of context,
organization,
or scope.
Trust,
however, is
subjective.
Depending
on the scope
and other
factors, an
issue may or
may not be
problematic,
and can
change over
time
depending
on the
context.

📝🟢

note

Trusted
image
policies do
not override
other
policies and
their rules,
such as
malware
policies,
misconfigura
tion policies,
and secrets
policies.

⚠️🟡

caution

The default
behavior for
the prevent
action when
trust criteria
are

unavailable
is Create an
Issue. Be
aware that
changing
this default
will mean an
incomplete
deployment
file can
block a
workload.

20.2.2. Cloud
Workload
Policies page



📝🟢

note

Keep the
following
caveats in
my mind
when
working with
Policies:
Instance
Administrato
rs are able
to view all
facets of
policies
without
restrictions,
even if
Scope
Based
Access
Control
(SBAC) roles
are in effect.
Learn more
about
SBAC.Manag
e user scope

If you've
been
assigned a
custom role
with
View/Edit
permissions
limited by
SBAC, you
may not be...

20.2.2.1.
Widgets
panel

20.2.2.1.1. 
Show or hide
the widget
panel



20.2.2.2.
Change the
layout of the
policies table



20.2.2.3.
Policy
Details Panel



20.2.3.
Enable or
disable a
Cloud
Workload
Policy

🧭➡️ 

navigation

Posture
Management
→ Rules &
Policies →
Policies →
Cloud
Workload

20.2.4.
Create a
Cloud
Workload
Policy



📝🟢

note

The
Evaluation
stage for
Misconfigura
tion policies
is supported
only in the
Runtime
SDLC stage
and is
enforceable
through the
Kubernetes
Admission
Controller
for clusters
on-boarded
using the
Posture
Management
(KSPM)
Connector.

📝🟢

note

Use Create a
new Custom
Detection
Rule to
define and
add new
custom rules
as required.

📝🟢

note

Each rule
may support
different
actions.
While some
include both
Create an
Issue and
Prevent and
Create an
Issue, others
provide only
the Create
an Issue
option.

📝🟢

note

All non-relevant (non-compute) assets are
automatically excluded from the included asset
list.

📝🟢

note

Currently, dynamic asset groups for Kubernetes
Prevent Policies support only the following
attributes:
Kubernetes Resource Cluster: The cloud
identifier for the cluster.
Kubernetes Resource Namespace: The
namespace where the resource is located.
Kubernetes Resource Labels: The labels
assigned to the resource.
Kubernetes Resource Category: The category...

📝🟢

note

See Cloud
Workload
Preventive
Action, to
learn more
about the
Prevent
action
behavior and
prerequisites.
If the
Prevent and
create an
issue action
is selected,
the
preventive
actions in
the CI
pipeline will
trigger a Fail
Pipeline by
returning an
exit code of
2 in the CI
tool.

📝🟢

note

Choosing
this option
results in the
automatic
selection of
all other
asset scopes
in the list.

📝🟢

note

See Cloud
Workload
Preventive
Action, to
learn more
about the
Prevent
action
behavior and
prerequisites.

📝🟢

note

See Cloud
Workload
Preventive
Action, to
learn more
about the
Prevent
action
behavior and
prerequisites.

📝🟢

note

Choosing
this option
results in the
automatic
selection of
all other
asset scopes
in the list.

📝🟢

note

See Cloud
Workload
Preventive
Action, to
learn more
about the
Prevent
action
behavior and
prerequisites.

20.2.5. Use
an existing
policy to
create a new
Cloud
Workload
policy



20.2.6. Edit a
Cloud
Workload
Policy



20.2.7. Delete
a Cloud
Workload
Policy



20.2.8. Cloud
Workload
Preventive
Action



20.3. Cloud
Workload
Rules



20.3.1.
Default (pre-
defined)
Rules



20.3.2.
Custom
(user-
defined)
Rules



20.3.3. Cloud
Workload
Rules page



📝🟢

note

Keep the
following
caveats in
my mind
when
working with
Rules:
Instance
Administrato
rs are able
to view all
facets of
Rules
without
restrictions,
even if
Scope
Based
Access
Control
(SBAC) roles
are in effect.
Learn more
about
SBAC.Manag
e user scope

If you've
been
assigned a
custom role
with
View/Edit
permissions
limited by
SBAC, you
may not be
able...

20.3.3.1.
Filter page
results



20.3.3.2.
Change the
layout of the
rules table



20.3.3.3. Rule
details panel



20.3.4.
Create a
new Custom
Detection
Rule



📝🟢

note

The custom
rule must
use the
match term
(not deny or
others) to
function
properly.

📝🟢

note

Only users
with the
following
roles can
enable or
disable
Custom
Code
Execution:
Account
Admin
Instance
Administrato

r

Deployment
Admin
Privileged
Security
Admin

📝🟢

note

Only
Custom
Detection
Rules (not
built-in
rules) can be
assigned to
custom
controls.

👀🟠

important

The custom
Python
scripts are
intended to
be executed
exclusively
for
compliance
checks and
validations.
To ensure
the scripts
are used
properly and
no security
risks or
unintended
changes
occur, the
system
implements
the
following
restrictions
and
safeguards:
Only a
predefined
set of
Python
libraries and
functions
required for
compliance
checks are...

20.3.5. Use
an existing
rule to
create a new
Custom
Detection
Rule

🧭➡️ 

navigation

Posture
Management
→ Rules &
Policies →
Rules →
Cloud
Workload

20.3.6. Edit a
Custom
Detection
Rule



20.3.7. Delete
a Custom
Detection
Rule



21. Base

Images
Rule

🧭➡️ 

navigation

Posture
Management
→ Rules &
Policies →
Rules →
Base Images

Inventory →
Assets → All
Assets →
Compute →
Container
Images

22.

Monitor

and track

complian
ce

adherenc

e



22.1. Cortex
compliance
flow



22.2. Choose
compliance
standards
from the
compliance
catalog



🧭➡️

navigation

Cortex
provides
lists of
available
standards
and controls
in the
Standards
and
Controls
catalogs
under
Posture
Management
→

Compliance
→ Catalogs

22.2.1.
Standards
Catalog

22.2.1.1. Use 
a built-in or
custom
standard



22.2.2.
Controls
catalog

22.2.2.1. Use 
a built-in or
custom
control



22.2.2.1.1.
Associate a
custom
control to a
detection
rule



📝🟢

note

Custom
rules can
only be
associated
with custom
compliance
controls.

📝🟢

note

You can only
associate
custom

compliance
controls to
the
following
cloud
security rule
types:
ConfigIdentit
yAI

🧭➡️

navigation

Posture
Management
→ Rules &
Policies →
Rules →
Cloud
Workload or
Cloud
Security

22.2.2.1.2.
Create a
new Custom
Detection
Rule



📝🟢

note

The custom
rule must
use the
match term
(not deny or
others) to
function
properly.

📝🟢

note

Only users
with the
following
roles can
enable or
disable
Custom
Code
Execution:
Account
Admin
Instance
Administrato

r

Deployment
Admin
Privileged
Security
Admin

📝🟢

note

Only
Custom
Detection
Rules (not
built-in
rules) can be
assigned to
custom
controls.

👀🟠

important

The custom
Python
scripts are
intended to
be executed
exclusively
for
compliance
checks and
validations.
To ensure
the scripts
are used
properly and
no security
risks or
unintended
changes
occur, the
system
implements
the
following
restrictions
and
safeguards:
Only a
predefined
set of
Python
libraries and
functions
required for
compliance
checks are...

22.3. Use an
assessment

profile to
run

compliance
checks on
your assets



🧭➡️

navigation

Under
Posture
Management
→

Compliance
→

Assessment
Profiles,
click Create
New
Assessment

22.3.1.
Configuring
assessments
for custom
compliance
standards
based on
custom
cloud
security rules



22.4. View
and manage
compliance
assessments

and reports



📝🟢

note

Compliance
assessment

results may
take up to
six hours to
be
generated.

22.4.1.
Compliance
score

📝🟢 

note

The status
of controls
is
determined
by the
evaluation
of the
associated
rules. If an
asset fails a
check
against any
rule
associated
with a
control, that
control is
considered
failed for
that asset.

22.4.2.
Assessments

📝🟢 

note

If there are no rules associated with the control,
the control will be assigned a severity of low.

22.4.3.
Reports

🧭➡️ 

navigation

The Posture
Management
→

Compliance
→ Results →
Reports
page shows
a table
listing
compliance
assessment

report files

22.5. View
the
compliance
assessment
of an
individual
asset



22.6.
Compliance
Overview
Dashboard

📝🟢 

note

If you just
created an
assessment

profile, it
may take up
to 10
minutes for
it to appear
in the
Assessment
filter. As a
workaround,
you can
refresh the
URL in the
browser.

23. Data

managem
ent



23.1. Broker
VM



23.1.1. What
is the Broker
VM?

📝🟢 

note

The Broker
VM is a
closed,
hardened
appliance.
To maintain
its security
integrity and
performance
standards,
third-party
agents
cannot be
installed on
the Broker
VM.

23.1.2. Set up
and
configure
Broker VM



📝🟢

note

The Broker VM comes with a 512 GB disk.
Therefore, deploy the Broker VM with thin
provisioning, meaning the hard disk can grow up
to 512 GB but will do so only if needed.

📝🟢

note

When DHCP
is not
enabled in
your
network and
there isn't an
IP address
for your
Broker VM,
configure
the Broker
VM with a
static IP
using the
serial
console
menu.

📝🟢

note

When
configuring
a Local
Agent
Settings
applet with
installer and
content

caching, you
need to
configure an
SSL
certificate
for the
Broker VM
as explained
in the task
below. For

more

information
on specific
requirements
for the
Local Agent
Settings
applet, see
Activate
Local Agent
Settings.
Keep in mind
that several
Broker VM
applets,
such as the...

📝🟢

note

When a
Broker VM is
disconnected
for more
than 30
days, it will
have to go
through a re-
registration
process.

📝🟢

note

When DHCP
is not
enabled in
your
network and
there isn't an
IP address
for your
Broker VM,
configure
the Broker
VM with a
static IP
using the
serial
console
menu.

📝🟢

note

You can

configure
another
Broker VM
as a proxy
server for
this Broker
VM by
selecting the
HTTP type.
When
selecting
HTTP to
route Broker
VM
communicati
on, you need
to add the IP
Address and
Port number
(set when
activating
the Agent
Proxy) for
another
Broker VM
registered in
your tenant.
This
designates
the other
Broker VM
as a proxy
for this
Broker VM.

📝🟢

note

The Palo
Alto
Networks
Broker VM
supports
only strong
cipher
SHA256-
based
certificates.
MD5/SHA1-
based
certificates
are not

supported.

📝🟢

note

If adding a
CA
certificate
to the Broker
VM is not
possible,
ensure that
you've
added the
Broker
Service
FQDNs to
the SSL
Decryption
Exclusion list
on your
firewalls. See
Enable
Access to
Cortex XDR.

👀🟠

important

There can
be instances
in which the
Broker VM
requires up
to 50% of
the incoming
bandwidth
as outgoing.
Such
instances
can be,
network
instability
between the
Broker VM
and Cortex
Cloud, or
data that is
being
collected,
but not well
compressed.

👀🟠

important

The internal
network for
the Broker
VM must be
unique and
reserved.
Other
devices
should not
use the
same IP as
the Broker
VM internal
network as it
can lead to
communicati
on issues
with the
Broker VM.

👀🟠

important

If you use
SSL
decryption
in your
firewalls and
proxies, see
the
Understandin
g CA
certificate
functionality
in Broker VM
deployments
section
below. In
addition,
verify that
the proxies
used
support
HTTP/2,
gRPC-
specific
headers, and
HTTP/2
trailers, and
the
inspection
policies
support
gRPC traffic.
Any devices
that you use
with this
configuration
...

👀🟠

important

When
configuring
more than
one network
interface,
ensure that
only one
Default
Gateway is
defined. The
rest must be
set to 0.0.0.0,
which
configures
them as
undefined. In
addition, we
recommend
assigning
each
network
interface to
a different
subnet, as
oppose to
configuring
two
interfaces
on the same
subnet
which can
potentially
cause

unexpected...

👀🟠

important

Internal IP
must be:
Formatted
as

prefix/mask,
for example
192.0.2.1/24.
Must be
within /8 to
/24 range.
Cannot be
configured
to end with
a zero.

For Broker
VM version
9.0 and
earlier,
Cortex
Cloud will
only accept
172.17.0.0/16.

👀🟠

important

We strongly recommend disabling SSH
connectivity when it's not being used.
Therefore, activate SSH connectivity when it's
needed and disable it right afterwards.
When generating a new SSH key ensure to avoid
embedding the domain-style username, by not
using any backslashes (\) in the comment field,
to ensure the SSH key passes validation.

23.1.2.1.
Broker VM
image
installations



23.1.2.1.1. Set
up Broker
VM on
Alibaba
Cloud



☑️🟣

prerequisite

Download a
Cortex
Cloud
Broker VM
QCOW2
image. For
more

information,
see the
virtual
machine
compatibility

requirements
in Set up
and
configure
Broker VM.

📝🟢

note

For Linux
and
Windows
uploads, you
can use

Alibaba
Cloud's
graphical
management
tool called
ossbrowser.

📝🟢

note

The Object
Storage
Service must
be created
in the same
Region as
the image of
the virtual
machine.

🧭➡️

navigation

Hamburger
menu →

Elastic
Compute
Service →
Instances &
Images →
Instances

23.1.2.1.2. Set
up Broker
VM on
Amazon
Web
Services



☑️🟣

prerequisite

Download a
Cortex
Cloud
Broker VM
VMDK
image. For
more

information,
see the
virtual
machine
compatibility

requirements
in Set up
and
configure
Broker VM.
You need to
set up an
AWS VM
Import role
(vmimport)
before
running the
import-
snapshot CLI
command. If
the role
vmimport
does not
exist or does
not have the
required
permissions,
you can
create it...

📝🟢

note

Assigning
security
groups can

take up to 15
minutes. 👀🟠

important

A t3.xlarge
(16 GB RAM)
is the lowest
machine
type that
can be used
as an

instance
type to meet
the
mandatory 4
vCPU
requirement.

🧭➡️

navigation

In the Cortex
Cloud Web
Console ,
select
Settings →
Configuratio
ns → Data
Broker →
Broker VMs
→ Add
Broker →
VMDK

In the AWS
Console,
select All
services →
Storage →
S3

All services
→ EC2 →
IMAGES →
AMIs and
locate your
AMI image
using the
task ID
23.1.2.1.3. Set
up Broker
VM on
Google
Cloud
Platform
(GCP)



☑️🟣

prerequisite

Download a
Cortex
Cloud
Broker VM
VMDK
image. For
more

information,
see the
virtual
machine
compatibility

requirements
in Set up
and
configure
Broker VM.
To complete
the set up,
you must
have G
Cloud
installed and
have an
authenticate
d user
account.

☑️🟣

prerequisite

Before importing a GCP image using the gcloud
CLI, ensure that you update the Google Cloud
components to version 371.0.0 and above using
the following command:

📝🟢

note

The import tool uses Cloud Build API, which
must be enabled in your project. For the import
to work, Cloud Build service account must have
compute.admin and iam.serviceAccountUser
roles. When using the Google Cloud console to
import the image, you will be prompted to add
these permissions automatically.

23.1.2.1.4. Set
up Broker
VM on KVM
using Ubuntu

☑️🟣 

prerequisite

Download a
Cortex
Cloud
Broker VM
QCOW2
image. For
more

information,
see the
virtual
machine
compatibility

requirements
in Set up
and
configure
Broker VM.

23.1.2.1.5. Set
up Broker
VM on
Microsoft
Azure



☑️🟣

prerequisite

Download a
Cortex
Cloud
Broker VM
VHD (Azure)
image. For
more

information,
see the
virtual
machine
compatibility

requirements
in Set up
and
configure
Broker VM.

📝🟢

note

Extraction
can take up
to a few
hours.

📝🟢

note

Upload can
take up to a
few hours.

📝🟢

note

For more
information,
see the
Azure
Documentati
on.

📝🟢

note

Creating the
VM can take
up to 15
minutes. The
Broker VM
Web UI is
not
accessible
during this
time.

📝🟢

note

For more
information
on creating
a rule in an
Azure VM,
see Create a
Security
Rule in the
Azure
Documentati
on.

23.1.2.1.6. Set
up Broker
VM on
Microsoft
Hyper-V

☑️🟣 

prerequisite

Download a
Cortex
Cloud
Broker VM
VHD image.
For more
information,
see the
virtual
machine
compatibility

requirements
in Set up
and
configure
Broker VM.

23.1.2.1.7. Set
up Broker
VM on
Nutanix
Hypervisor



☑️🟣

prerequisite

Download a
Cortex
Cloud
Broker VM
QCOW2
image. For
more

information,
see the
virtual
machine
compatibility

requirements
in Set up
and
configure
Broker VM.

📝🟢

note

Saving the
image to
Nutanix
hypervisor
can take
time as it's a
large file.
We
recommend
verifying
periodically
that the
connection
is alive for
the upload
process to
finish
successfully.

📝🟢

note

Creating the
VM can take
up to 15
minutes. The
Broker VM
Web user
interface is
not
accessible
during this
time.

23.1.2.1.8. Set
up Broker
VM on
VMware
ESXi using
vSphere
Client

☑️🟣 

prerequisite

Ensure you
have a
virtualization
platform
installed
that is
compatible
with an OVA
image, and
have an
authenticate
d user
account.
Download a
Cortex
Cloud
Broker VM
OVA image.
For more
information,
see the
virtual
machine
compatibility

requirements
in Set up
and
configure
Broker VM.

23.1.2.2.
Broker VM
data
collector
applets

🔒🟣 

content

Some data
collector
applets
require the
Data
Collection
add-on.

23.1.3.
Manage
Broker VM



📝🟢

note

For more
information

on

troubleshooti
ng errors
and
warnings for
these broker
applets, see
Troubleshoot
Broker VM
applet errors.

📝🟢

note

Certain
fields are
also
exposed in
the Clusters
tab, when a
Broker VM
node is
added to a
High
Availability
(HA) cluster,
and each
cluster node
is expanded
to view the
Broker VM
nodes table.
An asterisk
(*) is beside
every field
that is also
included in
the Broker
VM nodes
table for
each HA
cluster.

🧭➡️

navigation

Settings →
Configuratio
ns → Data
Broker →
Broker VMs

23.1.3.1. Edit
Broker VM
Configuratio
n



📝🟢

note

For all
Broker VM
nodes
added to a
HA cluster,
you can also
Configure
the Broker
VM nodes
from the
Clusters tab.

📝🟢

note

Internal IP
must be:
Formatted
as

prefix/mask,
for example
192.0.2.1/24.
Must be
within /8 to
/24 range.
Cannot be
configured
to end with
a zero.

For Broker
VM version
9.0 and
lower,
Cortex
Cloud
accepts only
172.17.0.0/16.

📝🟢

note

Make sure
you save the
password
before
closing the
window. The
only way to
re-generate
a password
is to disable
ssh and re-
enable.

23.1.3.2.
Increase
Broker VM
storage
allocated
for data
caching



📝🟢

note

Follow your
hypervisor
documentati
on to
understand
how to add
a persistent
disk storage
to your VM.

📝🟢

note

If your disk is
not listed
and you
didn't
shutdown
your Broker
VM in your
hypervisor
before
manually
adding a
disk to the
VM, you'll
need to
reboot the
Broker VM
before the
disk details
are detected
by the
Broker VM.
This can be
performed
either in the
hypervisor
or directly in
the Broker
VMs page.

⛔🔴

warning

This
operation is
irreversible,
and will
make the
disk become
an integral
part of the
broker,
where
disconnectin
g the disk
will result in
errors and
data loss.

23.1.3.3.
Monitor
Broker VM
using
Prometheus



📝🟢

note

For all
Broker VM
nodes
added to a
HA cluster,
you can also
Configure
the Broker
VM nodes
from the
Clusters tab.

📝🟢

note

For more
information
on Docker,
see the
Docker
website.

23.1.3.4.
Collect
Broker VM
Logs

📝🟢 

note

The
Download
Logs
(<TIMESTAM
P>) is only
displayed
when you've
downloaded
your logs
previously
using
Generate
New Logs.

23.1.3.5.
Upgrade
Broker VM

👀🟠 

important

After a
Broker VM
upgrade,
your broker
may require
a reboot to
finish
installing
important
updates. A
notification
about this
will be sent
to your
Cortex
Cloud
console
Notification
Center.

23.1.3.6.
Import
Broker VM
Configuratio
n



📝🟢

note

If this option
is selected,
the
shutdown
process can

take up to 10
minutes to
allow the
broker to
finish
offloading
the data
cache.

👀🟠

important

This option
can only be
used on
Broker VMs
with version
20.0 and
later, and is
only suitable
for
importing a
configuration
of brokers
in the same
version, or
from a
broker in an
older version
to a broker
in a newer
version.

👀🟠

important

If your source Broker VM configuration includes
a WEC applet, you'll need to ensure that you
update the DNS record of this Broker VM's
FQDN to point to the target Broker VM IP
address.

23.1.3.7.
Open Live
Terminal

📝🟢 

note

You can
either
restart_route
s or reboot
the Broker
VM for the
changes in
the
/etc/network
/routes file
to take
affect.

23.1.3.8. Add
Broker VM
to cluster



23.1.3.9.
Switchover
Primary
Node in
Cluster



23.1.3.10.
Remove
from Cluster

📝🟢 

note

If you've
enabled a
Load
Balancer
Health-
Check on
the cluster,
you need to
exclude this
Broker VM
from your
Load
Balancer
settings.

23.1.4.
Manage
Broker VM
data
collector
applets



📝🟢

note

For more
information
on the
Broker VM
applet
connectivity
status, see
Manage
Broker VM.

23.1.4.1.
Activate
DSPM
Fileshare



☑️🟣

prerequisite

Set up and
configure
Broker VM
Know the
complete
path to the
files and
folders that
you want
Cortex
Cloud to
monitor.
Necessary
user

permissions
to access
the network
shares. For
the SMB
connection
type, you
need the
user name

and
password.

📝🟢

note

The applet
list displays
only the
applets for
which you
have
permissions.

📝🟢

note

By default,
all
configured
connections

are saved. 📝🟢

note

For details
regarding
the
connection
fields, see
the table
above under
File Share
Connection.

📝🟢

note

You can add
multiple
connections
under a
single
instance of
the DSPM
Fileshare
applet by
returning to
the File
Share
Connection
screen and
clicking Add
Connection.
Each new
connection
can be of
either the
NFS or SMB
connection
type.

23.1.5. Broker
VM High
Availability
Cluster



📝🟢

note

For Load
Balancing,
you must
install a
Load
Balancer in
your
network,
which will
distribute
the incoming
data
between the
nodes.

📝🟢

note

The
following
applets
aren't
supported
when
configuring
Broker VMs
in HA
clusters:
DSPM
Fileshare,
Registry
Scanner, and
Transporter.

23.1.5.1.
Configure
High
Availability
Cluster



☑️🟣

prerequisite

Be sure you
do the
following
task before
creating a
cluster from
an existing
Broker VM:
If the Broker
VM is
explicitly
specified in
some Agent
Settings
profile,
which means
Cortex
Cloud
agents
retrieve
release
upgrades
and content
updates
from this
Broker VM,
you must
change the
Broker VM's
current

designated
role. To do
this, you
need to
modify the...

📝🟢

note

The Broker
VM Load
Balancer
Health
Check
requires
HTTP/1.1 or
higher.
Legacy
HTTP/1.0 is
no longer
supported.
Ensure your
external
load
balancer is
configured
to use
HTTP/1.1 or
above when
performing
health
checks.

👀🟠

important

You can only
create a new
cluster from
an existing
Broker VM,
when the
Broker VM
version is
19.0 and
later, and
the STATUS
is Connected.
The Create a
Cluster from
this Broker
option is
only listed if
the Broker
VM is not
already
added to a
cluster.

👀🟠

important

When the
Protocol is
set to
HTTPS, you
may need to
perform a
few follow-
up steps to
establish a
validated
secure SSL
connection
with the
Broker VM.
If you're
using your
own

Certificate
Authority
(CA) to sign
the
certificates,
you'll need
to place the
CA in the
client, such
as the Load
Balancer,
and upload
the
certificates
to the Broker
VM.

If you're...

23.1.5.2.
Manage
Broker VM
clusters



🧭➡️

navigation

After you've configured a cluster, you can
manage all your Broker VM clusters from the
Clusters tab on the Broker VMs page (Settings
→ Configurations → Data Broker → Broker VMs
→ Clusters

23.1.5.2.1.
View cluster
details



📝🟢

note

For more
information

on

troubleshooti
ng errors
and
warnings for
these
applets, see
Troubleshoot
Broker VM
applet errors.

🧭➡️

navigation

The Clusters
tab of the
Broker VMs
page
(Settings →
Configuratio
ns → Data
Broker →
Broker VMs)
enables you

23.1.5.2.2.
Edit cluster

🧭➡️ 

navigation

Settings →
Configuratio
ns → Data
Broker →
Broker VMs,
and select
the Clusters
tab

23.1.5.2.3.
Add applet
to cluster



📝🟢

note

With Cortex
XDR Prevent,
it's only
relevant to
configure a
HA cluster
with a Local
Agent
Settings
applet as
this is the
only applet
supported
for this
product
license. The
other
applets are
collector
applets,
which are
only
available in
Cortex XDR
Pro or
Cortex
XSIAM.

📝🟢

note

For more
information

on

troubleshooti
ng errors
and
warnings for
these
applets, see
Troubleshoot
Broker VM
applet errors.

23.1.5.2.4.
Add Broker
VM to
cluster



23.1.5.2.5.
Remove
cluster



23.1.6. Broker
VM
notifications



23.1.7.
Monitor
Broker VM
activity

📝🟢 

note

Certain
fields are
exposed and
hidden by
default. An
asterisk (*) is
beside every
field that is
exposed by
default.

23.1.8.
Troubleshoot
Broker VM
applet errors



23.2. Dataset
management



🔒🟣

content

Managing
Roles
requires an
Account
Admin or
Instance
Administrato
r role. For

more

information,
see

Predefined
user roles.

☑️🟣

prerequisite

Dataset
Management
requires
View/Edit
RBAC
permissions
for Data
Management
(under
Configuratio
ns → Data
Management
), which are
the same
permissions
required for
Parsing
Rules, Data
Model Rules,
and Event
Forwarding.

📝🟢

note

Cold storage, in addition to a cold storage
license, requires compute units (CU) to run cold
storage queries. For more information on CU,
see Manage compute units.

📝🟢

note

Only
datasets
that are
already
handled as
part of the
GB license
are

supported
for this
license. In
addition, the
retention
configuration
is only
available in
Cortex
Cloud, as
opposed to
the public
APIs.

📝🟢

note

Certain
fields are
exposed and
hidden by
default. An
asterisk (*) is
beside every
field that is
exposed by
default.
Datasets
include
dataset
permission
enforcement
s in the
Cortex
Query
Language(X
QL), Query
Center, and
XQL
Widgets. For
example, to
view or
access any
of the
endpoints
and
host_inventor
y datasets,
you need
role-based
access

control
(RBAC)...

📝🟢

note

You'll only
be able to
save the
dataset view
if the query
contains no
errors;

otherwise,
the Save
button is
disabled.

📝🟢

note

For more
information

on user

roles, see
Manage user
roles.

👀🟠

important

Cortex
Cloud
enforces
retention on
all log-type
datasets
excluding
Host
Inventory,
Vulnerability
Assessment,
Metrics, and
Users.

👀🟠

important

This column
is updated
once a day.
Therefore, if
the dataset
was created
or updated
by the
target or
lookup
flows, it's
possible that
the Last
Updated
value is a
day behind
when the
queries or
reports were
run as it was
before this
column was
updated.

23.2.1. What
are datasets?



☑️🟣

prerequisite

Dataset
Management
requires
View/Edit
RBAC
permissions
for Data
Management
(under
Configuratio
ns → Data
Management
), which are
the same
permissions
required for
Parsing
Rules, Data
Model Rules,
and Event
Forwarding.

📝🟢

note

CEF and
LEEF logs
are often
wrapped in a
Syslog
envelope
(RFC
3164/5424).
Cortex
Cloud
automaticall
y detects
and strips
the Syslog
header to
extract the
reporting
device IP
and
hostname
before
parsing the
inner CEEF
or LEEF
payload.

📝🟢

note

For more
information
on tasks
specific to
lookup
datasets,
see Lookup
datasets.

📝🟢

note

Schema
changes to
datasets
may not be
reflected in
the
autocomplet
e

suggestions
and
definitions
as you type
in real time
the XQL
query and
can appear

with a slight
delay.

👀🟠

important

By default,
forensic
datasets are
not included
in XQL query
results,
unless the
dataset
query is
explicitly
defined to

use a

forensic
dataset.

23.2.2.
Lookup
datasets



☑️🟣

prerequisite

Dataset
Management
requires
View/Edit
RBAC
permissions
for Data
Management
(under
Configuratio
ns → Data
Management
), which are
the same
permissions
required for
Parsing
Rules, Data
Model Rules,
and Event
Forwarding.

📝🟢

note

A lookup
dataset can
only be
deleted if
there are no
other
dependencie
s. For
example, if a
Correlation
Rule is
based on a
lookup
dataset, you
wouldn't be
able to
delete the
lookup
dataset until
you removed
the dataset
from the
XQL query
of the
Correlation
Rule.

👀🟠

important

When you
create or
add data to
a lookup
dataset
using the
target stage,
the _time
field won't
be included
by default
unless you
explicitly
add it with
the fields
stage.

23.2.2.1.
Import a
lookup
dataset



☑️🟣

prerequisite

Dataset
Management
requires
View/Edit
RBAC
permissions
for Data
Management
(under
Configuratio
ns → Data
Management
), which are
the same
permissions
required for
Parsing
Rules, Data
Model Rules,
and Event
Forwarding.

☑️🟣

prerequisite

When
uploading a
CSV, TSV, or
JSON file,
ensure that
the file
meets the
following
requirements:
The
maximum
size for the
total data to
be imported
into a
lookup
dataset is 30
MB from the
Dataset
Management
page.
Otherwise,
the limit is
50 MB using
Cortex
Query
Language
(XQL) or
APIs.
Field names
can contain
characters
from
different
languages,
special...

👀🟠

important

The name of
a dataset
created
from a TSV
file must
always
include the
extension.
For example,
if the
original file
name is
mrkdptusrsn
ov23.tsv, you
can save the
dataset with
the name
marketing_de
pt_users_Nov
_2023.tsv.

23.2.2.2.
Download
JSON file of
lookup
dataset



☑️🟣

prerequisite

Dataset
Management
requires
View/Edit
RBAC
permissions
for Data
Management
(under
Configuratio
ns → Data
Management
), which are
the same
permissions
required for
Parsing
Rules, Data
Model Rules,
and Event
Forwarding.

🧭➡️

navigation

Open the
Settings →
Configuratio
ns → Data
Management
→ Dataset
Management
page

23.2.2.3. Set
time to live
for lookup
datasets

☑️🟣 

prerequisite

Dataset
Management
requires
View/Edit
RBAC
permissions
for Data
Management
(under
Configuratio
ns → Data
Management
), which are
the same
permissions
required for
Parsing
Rules, Data
Model Rules,
and Event
Forwarding.

23.2.3.
Monitor
datasets and
dataset
views activity



☑️🟣

prerequisite

Dataset
Management
requires
View/Edit
RBAC
permissions
for Data
Management
(under
Configuratio
ns → Data
Management
), which are
the same
permissions
required for
Parsing
Rules, Data
Model Rules,
and Event
Forwarding.

📝🟢

note

Certain
fields are
exposed and
hidden by
default. An
asterisk (*) is
beside every
field that is
exposed by
default.

23.3.
Manage
Event
Forwarding



☑️🟣

prerequisite

Event
Forwarding
requires
View/Edit
RBAC
permissions
for Data
Management
(under
Configuratio
ns → Data
Management
), which are
the same
permissions
required for
Parsing
Rules, Data
Model Rules,
and Dataset
Management.

📝🟢

note

After
exporting
logs, you
can

download
them from
GCP for up
to 14 days.
The Pub/Sub
subscription
messages
are available
for 7 days.

📝🟢

note

Requires the
Endpoints
Event
Forwarding
add-on. 📝🟢

note

Requires the
GB Event
Forwarding
add-on

👀🟠

important

Since data is
aggregated
and
compressed,
it may take
up to two
hours until
the data is
available in
the
forwarding
bucket.

🧭➡️

navigation

Under
Settings →
Configuratio
ns → Data
Management
→ Event
Forwarding,
activate one
or more of
the
following:
Enable GB
Event
Forwarding
Enable
Endpoints
Event
Forwarding

Under
Settings →
Configuratio
ns → Data
Management
→ Event
Forwarding,
activate one
or more of
the following

23.3.1.
Endpoints
Event
Forwarding -
included/excl
uded fields
by event type

☑️🟣 

prerequisite

Event
Forwarding
requires
View/Edit
RBAC
permissions
for Data
Management
(under
Configuratio
ns → Data
Management
), which are
the same
permissions
required for
Parsing
Rules, Data
Model Rules,
and Dataset
Management.

23.4.
Manage
compute
units



📝🟢

note

For more
information,
see Import
historical
data into
cold
storage.Impo
rt historical
data into
cold storage

📝🟢

note

The CU
consumption
of cold
storage
queries is
based on the
number of
days in the
query time
frame. For
example,
when
querying 1
hour of a
specific day,
the CU of
querying this
entire day is
consumed.
When
querying 1
hour that
extends past
2 days, such
as from
23�50 to
00�50 of the
following
day, the CU
of querying
these two
days is
consumed.

23.4.1.
Compute
units usage



📝🟢

note

To enable
the add-on,
select
Settings →
Cortex
Cloud
LicenseAddo
ns tile, and
select the
Compute
Unit tile and
Enable.

🧭➡️

navigation

Settings →
Configuratio
ns → Data
Management
→ Compute
Unit Usage

24.

Cortex

Cloud

Data

Sources



24.1. What
are Cortex
Cloud data
sources?



📝🟢

note

Some
content

packs
provide
parsing rules
and data
model rules
for data

sources

ingested
using a
Syslog
Collector
applet of
the Broker
VM or for
standard
data
sources, and
won't be
listed in the
Data
Sources &
Integrations
page.

👀🟠

important

We are
actively
working to
evolve the UI
to create a

single,
unified point
of
configuration
for all data
ingestion
and
integration
points. This
documentati
on is
structured
to help you
navigate the
current

segmented
process until
that UI
update is
available.

24.2.
Complete
data source
catalog



24.3. Vendor-
specific
data sources



👀🟠

important

In some
cases, the
same vendor
is available
through one
or more of
these
options:
standard
data source,
CSP
onboarding
data source,
or content

pack
integration.
Check the
available
descriptions
for each
data source
in both the
user

interface
and
documentati
on to decide
which option
is more
suitable for
your needs.

24.3.1.
Amazon S3



24.3.1.1.
Ingest audit
logs from
AWS Cloud
Trail



📝🟢

note

For more
information

on

configuring
data
collection
from
Amazon S3
using AWS
CloudTrail,
see the AWS
CloudTrail
Documentati
on.

📝🟢

note

For more
information
on creating
an AWS
CloudTrail
trail, see
Create a trail.
If you
already have
an Amazon
S3 bucket
configured
with AWS
CloudTrail
audit logs,
skip this step
and go to
Configure
an Amazon
Simple
Queue
Service
(SQS).

📝🟢

note

It is your
organization'
s

responsibility
to define a
retention

policy for
your
Amazon S3
bucket by
creating a
Lifecycle
rule in the
Management
tab. We
recommend
setting the
retention
policy to at
least 7 days
to ensure
that the
data is
retrieved
under all
circumstance
s.

📝🟢

note

The
CloudWatch
Logs -
optional
settings are
not

supported
and should
be left
disabled.

📝🟢

note

Ensure that
you create
your
Amazon S3
bucket and
Amazon
SQS queue
in the same
region.

📝🟢

note

For more
information
on granting
permissions
to publish
messages to
an SQS
queue, see
Granting
permissions
to publish
event
notification
messages to
a destination.

📝🟢

note

If your
receive an
error when
trying to
save your

changes,
you should
ensure that
the
permissions
are set up
correctly.

📝🟢

note

It your
organization'
s

responsibility
to ensure
that the user
who
performs
this task of
creating the
access key is
designated
with the
relevant
permissions.
Otherwise,
this can
cause the
process to
fail with

errors.

Skip this
step if you
are using an
Assumed
Role for
Cortex
Cloud.

📝🟢

note

For more
information,
see

Managing
access keys
for IAM users.

📝🟢

note

Skip this
step if you
are using an
Assumed
Role for
Cortex
Cloud.

📝🟢

note

For more
information
on granting
permissions
to publish
messages to
an SQS
queue, see

Granting
permissions
to publish
event
notification
messages to

a destination. 24.3.1.2.
Ingest
network
flow logs
from
Amazon S3



📝🟢

note

For more
information

on

configuring
data
collection
from
Amazon S3,
see the
Amazon S3
Documentati
on.

📝🟢

note

For the
Amazon S3
bucket
created
using
CloudFormat
ion, it is the
customer's
responsibility
to define a
retention
policy by
creating a
Lifecycle
rule in the
Management
tab. We
recommend
setting the
retention
policy to at
least 7 days
to ensure
that the
data is
retrieved
under all
circumstance
s.

📝🟢

note

To create a new VPC, click Launch VPC Wizard.
For more information, see AWS VPC Flow Logs.

📝🟢

note

It is the
responsibility
of the
customer's
organization
to ensure
that the user
who
performs
this task of
creating the
access key is
designated
with the
relevant
permissions.
Otherwise,
this can
cause the
process to
fail with

errors.

Skip this
step if you
are using an
Assumed
Role for
Cortex
Cloud.

📝🟢

note

For more
information,
see

Managing
access keys
for IAM users.

📝🟢

note

Skip this
step if you
are using an
Access Key
to provide
access to
Cortex
Cloud.

24.3.1.3.
Ingest
generic logs
from
Amazon S3



📝🟢

note

For more
information

on

configuring
data
collection
from
Amazon S3,
see the
Amazon S3
Documentati
on.

📝🟢

note

You need to
set up an
Amazon S3
data
collector to
receive
generic logs
when
collecting
logs from
BeyondTrust
Privilege
Management
Cloud. For
more

information,
see Ingest
logs from
BeyondTrust
Privilege
Management
Cloud.

📝🟢

note

It is the
customer's
responsibility
to define a
retention

policy for
your
Amazon S3
bucket by
creating a
Lifecycle
rule in the
Management
tab. We
recommend
setting the
retention
policy to at
least 7 days
to ensure
that the
data is
retrieved
under all
circumstance
s.

📝🟢

note

Ensure that
you create
your
Amazon S3
bucket and
Amazon
SQS queue
in the same
region.

📝🟢

note

For more
information
on granting
permissions
to publish
messages to
an SQS
queue, see
Granting
permissions
to publish
event
notification
messages to
a destination.

📝🟢

note

If your
receive an
error when
trying to
save your

changes,
you should
ensure that
the
permissions
are set up
correctly.

📝🟢

note

It is the
responsibility
of your
organization
to ensure
that the user
who
performs
this task of
creating the
access key is
assigned the
relevant
permissions.
Otherwise,
this can
cause the
process to
fail with

errors.

Skip this
step if you
are using an
Assumed
Role for
Cortex
Cloud.

📝🟢

note

For more
information,
see

Managing
access keys
for IAM users.

📝🟢

note

Skip this
step if you
are using an
Assumed
Role for
Cortex
Cloud.

📝🟢

note

For more
information
on granting
permissions
to publish
messages to
an SQS
queue, see

Granting
permissions
to publish
event
notification
messages to
a destination.

📝🟢

note

-The Vendor
and Product
defaults to
Auto-Detect
when the
Log Format
is set to CEF
or LEEF.
-For a Log
Format set
to CEF or
LEEF, Cortex
Cloud reads
events row

by row to
look for the
Vendor and
Product
configured
in the logs.
When the
values are
populated in
the event
log row,
Cortex
Cloud uses
these values
even if you
specified a
value in the
Vendor...

24.3.1.4.
Ingest
network
Route 53

logs from
Amazon S3



📝🟢

note

For more
information

on

configuring
data
collection
from
Amazon S3
for Route 53
DNS logs,
see the AWS
Documentati
on.

📝🟢

note

For more
information
on creating
a Stack, see
Creating a
stack on the
AWS
CloudFormat
ion console.

📝🟢

note

For the
Amazon S3
bucket
created
using
CloudFormat
ion, it is the
customer's
responsibility
to define a
retention
policy by
creating a
Lifecycle
rule in the
Management
tab. We
recommend
setting the
retention
policy to at
least 7 days
to ensure
that the
data is
retrieved
under all
circumstance
s.

📝🟢

note

It is the
responsibility
of the
customer's
organization
to ensure
that the user
who
performs
this task of
creating the
access key is
designated
with the
relevant
permissions.
Otherwise,
this can
cause the
process to
fail with

errors.

Skip this
step if you
are using an
Assumed
Role for
Cortex
Cloud.

📝🟢

note

For more
information,
see

Managing
access keys
for IAM users.

📝🟢

note

Skip this
step if you
are using an
Access Key
to provide
access to
Cortex
Cloud.
24.3.1.5.
Create an
assumed role



☑️🟣

prerequisite

You need
ensure you
have an
Amazon S3
bucket and
Amazon
Simple
Queue
Service
(SQS)
already
configured
as it's
needed to
configure an
IAM policy.
The S3
bucket and
SQS
required
depends on
how you
plan to
configure
your
Amazon S3
data source:
When using
a

CloudFormat
ion script
provided by
Cortex
Cloud to
configure
Amazon S3
with SQS
notifications,
you'll...

📝🟢

note

You can
retrieve your
bucket's
ARN by
opening the
Amazon S3
Console in a
browser
window. In
the Buckets
section,
select the
bucket, click
Copy ARN,
and paste
the ARN in
the field.
You can
retrieve the
SQS queue
ARN by
opening
another
instance of
the AWS
Management
Console in a
browser
window, and
opening the
Amazon
SQS
Console,
and
selecting
the...

📝🟢

note

For more
information,
see the AWS
instructions.

24.3.1.6.
Configure
data
collection
from
Amazon S3
manually



📝🟢

note

If you
already have
an Amazon
S3 bucket
configured
with VPC
flow logs
that you
want to use
for this
configuration
, you do not
need to
perform the
prerequisite
steps
detailed in
the first two
bullets.

📝🟢

note

It is your
responsibility
to define a
retention
policy for
your
Amazon S3
bucket by
creating a
Lifecycle
rule in the
Management
tab. We
recommend
setting the
retention
policy to at
least 7 days
to ensure
that the
data is
retrieved
under all
circumstance
s.

📝🟢

note

For
resources

such as
Amazon S3,
SQS, VPC
flow logs,
and event
notifications,
you may use
the default
configuration
s; yet, these
settings
remain fully
customizable
to meet

your specific
environment
requirements.

📝🟢

note

If you
already have
an Amazon
S3 bucket
configured
with VPC
flow logs,
skip this step
and go to
Configure
an Amazon
Simple
Queue
Service
(SQS).

📝🟢

note

Ensure that
you create
your
Amazon S3
bucket and
Amazon
SQS queue
in the same
region.

📝🟢

note

It is the
responsibility
of the
customer's
organization
to ensure
that the user
who
performs
this task of
creating the
access key is
designated
with the
relevant
permissions.
Otherwise,
this can
cause the
process to
fail with

errors.

Skip this
step if you
are using an
Assumed
Role for
Cortex
Cloud.

📝🟢

note

Skip this
step if you
are using an
Assumed
Role for
Cortex
Cloud.

24.3.2.
Amazon
Web
Services



24.3.3. API
Security

24.3.3.1. 
Ingest data
for API
security



24.3.3.1.1.
Ingest AWS
API Gateway



📝🟢

note

You must
generate a
new key if
you did not
save. 📝🟢

note

If no logs are
showing,
send some
API requests
on Postman
or CURL.

👀🟠

important

Leave
'–'filter-
pattern
empty as
displayed
above.

24.3.3.1.2.
Ingest Azure
APIM



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

Microsoft
Azure APIM
service must
be running
before
starting to
configure
the
integration.

📝🟢

note

From the
URL, save
the UUID
and the
resource

group -
/resource/su
bscriptions/<
UUID>/resour
ceGroups/<R
esourceGrou
p>.
The UUID is
the Azure
account/sub
scription ID
and the
resource

group, which
is the group
where the
APIM
Service is
defined.

📝🟢

note

Use the
search to
navigate to
the specific
section.

📝🟢

note

131072 bytes = 128 KB. This value determines
the size (in bytes) of request and response
bodies to send to Cortex. Any bytes beyond this
limit are truncated.

📝🟢

note

If any other
inbound
policies
should be
added, they
must be
added after
these
elements.

👀🟠

important

If you want
to add
additional
data to the
<outbound>,
add it at the
start of the
<outbound>
code.

24.3.3.1.3.
Ingest
Apigee Proxy



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

The steps
include the
prerequisites
that run the
automated
script that
deploys files
and
configuration
s to Apigee.
For manual
configuration
, refer to the
section
Manual
deployment.

📝🟢

note

If you have
an existing
hookand
would like to
integrate
with the
shared flow,
run the
deploy.sh
script, and
select n' and
exit at the
prompt to
create a new
hook. Refer
to the
section
Connect to
existing hook.

📝🟢

note

Start with
the hook in
pre-proxy.

📝🟢

note

Archive
deployments
are currently
in preview
and are
subject to
change.

👀🟠

important

This should
only be
done when
there are
already
existing
hooks, and
API security
shared flows
can't be
hooked as a
standalone.
Run the
deployment
script, but
skip step 9
by passing n.
This step
publishes
API security
shared flows
to the
desired
Apigee
environment
without
setting them
to flow
hooks.

24.3.3.1.4.
Ingest Kong



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

Contact
support to
obtain the
custom

plugin file. 📝🟢

note

The queue
parameters
can be
updated to
change
when the
plugin
mirrors data
to Cortex.

📝🟢

note

Use commas
to separate
tags.

👀🟠

important

The size of
the buffer
must be
equal or
larger than
the max
body size
setting in the
plugin
configuration
, on every

data plane
node.

24.3.3.1.5.
Ingest-F5



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

Extract the
files from
the F5
plugin to a
folder
before
selecting to
upload to F5.

📝🟢

note

The provider
for
"context_pro
vider"
should
always be
uppercase.

Supported
providers:
AWS, GCP,
Azure, On-
prem.

📝🟢

note

Select
panw_apisec
_set_ssl_data
only if your
client SSL
profile is
enabled.

24.3.4. Azure
Event Hub

24.3.4.1. 
Ingest logs
from
Microsoft
Azure Event
Hub



🔒🟣

content

Requires the
Data
Collection
add-on.

☑️🟣

prerequisite

Ensure that
you do the
following
tasks before
you begin
configuring
data
collection
from Azure
Event Hub.
Before you
set up an
Azure Event
Hub,
calculate
the quantity
of data that
you expect
to send to
Cortex
Cloud,
taking into
account

potential
data spikes
and
potential
increases in
data
ingestion,
because
partitions
cannot be
modified
after
creation....

📝🟢

note

For more
information
on Azure
Event Hub
audit logs,
see

Overview of
Azure
platform
logs.

📝🟢

note

These logs
are from the
management
plane.

📝🟢

note

Even though you can collect Microsoft Entra ID
Activity logs and Microsoft Entra ID Sign-in logs
using the Azure Event Hub data collector, we
recommend using the Microsoft Office 365
data collector, because it is easier to configure.
In addition, ensure that you do not configure
both collectors to collect the same types of
logs, because if you do so,...

📝🟢

note

These logs
are from the
data plane.

📝🟢

note

For every
resource

that you
want to
confiure,
you'll have
to repeat
this step, or
use Azure

policy for a
general
configuration
.

📝🟢

note

There are
additional
log
categories
displayed.
We
recommend
selecting all
the available
options.

📝🟢

note

When you
Normalize
and enrich
audit logs,
the log
format is
automaticall
y

configured.
As a result,
the Log
Format
option is
removed and
is no longer
available to
configure
(default).

📝🟢

note

When you
Normalize
and enrich
audit logs,
the Vendor
and Product
fields are
automaticall
y

configured,
so these
fields are
removed as
available
options
(default).

⛔🔴

warning

Misconfigura
tion of Event
Hub
resources

could cause
ingestion
delays.
In an existing
Event Hub
integration,
do not
change the
mapping to
a different
Event Hub.
Do not use
the same
Event Hub
for more
than two
purposes.

24.3.5.
BeyondTrust
Privilege
Management
Cloud

24.3.5.1. 
Ingest logs
from
BeyondTrust
Privilege
Management
Cloud

📝🟢 

note

For a Log
Format set
to

Beyondtrust
Cloud ECS,
the
following
fields are
automaticall
y set and
not

configurable.
Vendor:
Beyondtrust
Product:
Privilege
Management
Compression:

Uncompresse
d

24.3.6.1. 24.3.6. Box 
Ingest logs
and data
from Box



🔒🟣

content

Requires the
Data
Collection
and Identity
Threat
Detection &
Response
add-ons.

☑️🟣

prerequisite

Set up an Enterprise Box plan.
Important
To collect Box Shield Alerts, you must purchase
Box Shield and it must be enabled on Box
enterprise.
Set up an Enterprise Box plan.
Create a valid Box account that is assigned to a
role with sufficient permissions for the data you
want to collect. For example, create an account
assigned to an Admin role to...

📝🟢

note

The Fetch
Intervals are
non-

configurable.
📝🟢

note

Collecting
Box Shield
Alerts
requires
implementing
Box Shield,

📝🟢

note

There is a current bug with the Groups API from
Box. If you don't configure the Box app with the
proper permissions for managing groups data,
the Groups API from Box won't return an error
message to Cortex Cloud indicating that the
API failed to receive the data, and the Groups
data will not be collected.

📝🟢

note

For any
future
change that
you make to
your Box
app, ensure

that you
send the
changes for
approval to
the
administrator
, who will
need to
approve
them as
explained
above.

📝🟢

note

Inventory
data
snapshots
are

collected
every 10
minutes.

24.3.7. Check
Point
FW1/VPN1



24.3.8. Cisco
ASA
firewalls and
AnyConnect



24.3.9.
Corelight
Zeek



24.3.10.
Databricks

24.3.10.1. 
How to
onboard
Databricks

📝🟢 

note

If you turn
on this
feature,
both the
cloud and
region will
be used for
scanning,
possibly
incurring
cost and
requiring
adherence
to certain
compliance
policies.
24.3.11.
Dropbox

24.3.11.1. 
Ingest logs
and data
from
Dropbox



🔒🟣

content

Requires the
ITDR Module
add-on and
Data
Collection
add-on.

☑️🟣

prerequisite

Set up an
Advanced
Dropbox
plan.
Create a
Dropbox
Business
admin
account
with
Security
admin
permissions,
which is
required to
authorize
Cortex
Cloud to
access the
Dropbox
Business
account and
generate the
OAuth 2.0
access token.

📝🟢

note

The Fetch
Interval is

non-

configurable.
📝🟢

note

When the
App Key
field is
empty, the
here link in
the tooltip is
disabled.
When an
incorrect
App Key is
entered,
clicking the
link results in
a 404 error.

📝🟢

note

Whenever
you change
the
permissions
of the
Dropbox
app, we
recommend
that you
generate a
new Access
Code for the
Dropbox
data
collector
instance so
that the
permissions
match the
updates.

📝🟢

note

Event data is
collected
every 60
seconds
with a 10
minute lag
time.

📝🟢

note

Inventory
data
snapshots
are

collected
every 10
minutes. 24.3.12.
Elasticsearch
Filebeat



📝🟢

note

You can

configure
collecting
container

logs from
Google
Kubernetes
Engine using
Elasticsearch
Filebeat
with a
Custom -
Filebeat
based
Collector or
with a
content

pack
Integration.
For more
information,
see Google
Kubernetes
Engine.

24.3.12.1.
Ingest logs
from
Elasticsearch
Filebeat

🔒🟣 

content

Requires the
Data
Collection
add-on.
24.3.13.
Forcepoint
DLP



24.3.14.
Fortinet
Fortigate



24.3.15.
Google
Cloud
Platform

24.3.15.1. 
Ingest logs
and data
from a GCP
Pub/Sub



📝🟢

note

When collecting flow logs, we recommend that
you include GKE annotations in your logs, which
enable you to view the names of the containers
that communicated with each other. GKE
annotations are only included in logs if
appended manually using the custom metadata
configuration in GCP. For more information, see
VPC Flow Logs Overview. In addition, to...

📝🟢

note

For a Log
Format set
to CEF or
LEEF, Cortex
Cloud reads
events row

by row to
look for the
Vendor and
Product
configured
in the logs.
When the
values are
populated in
the event
log row,
Cortex
Cloud uses
these values
even if you
specified a
value in the
Vendor and
Product
fields in the
GCP data
collector
settings. Yet,
when the
values are
blank in the...

📝🟢

note

For a Log
Format set
to CEF or
LEEF, Cortex
Cloud reads
events row

by row to
look for the
Vendor and
Product
configured
in the logs.
When the
values are
populated in
the event
log row,
Cortex
Cloud uses
these values
even if you
specified a
value in the
Vendor and
Product
fields in the
GCP data
collector
settings. Yet,
when the
values are
blank in the...

24.3.16.
Google
Workspace

24.3.16.1. 
Ingest logs
and data
from Google
Workspace



🔒🟣

content

Requires the
ITDR Module
add-on and
Data
Collection
add-on.

📝🟢

note

For all types
of data
collected
via the
Admin
Reports API,
except
alerts and
emails, the
log events
are

collected
with a preset
lag time as
reported by
Google
Workspace.
For more
information
on these lag
times for the
different
types of
data, see
Google
Workspace
Data
retention
and lag
times.

📝🟢

note

Spam emails
from the
compliance
email
account,
and from all
other
monitored
email
accounts,
are not
collected.
Any draft
emails
written in
the
compliance
email
account are

collected by
the Google
Workspace
data
collector,
and are then
deleted even
if the email

was never

sent.

📝🟢

note

For more
information
on the entire
process, see
Perform
Google
Workspace
Domain-
Wide
Delegation
of Authority.

📝🟢

note

You don't
need to add
permissions
to the GCP
Project-org
service
account.

📝🟢

note

When
collecting
Google
Workspace
alerts with
other types
of data,
except
emails, you
need to
configure a
service
account in
Google with
the
applicable
permissions
to collect
events from
the Google
Reports API
and alerts
from the
Alert Center
API. If you
prefer to use
different
service
accounts to
collect
events and
alerts
separately,
you'll need
to...

📝🟢

note

For more
information
on the
Admin
Reports API
scopes, see
OAuth 2.0
Scopes for
Google APIs.

📝🟢

note

For more
information
on the Gmail
API scopes,
see OAuth
2.0 Scopes
for Google
APIs.

📝🟢

note

For more
information
about the
events
collected
from the
various
Google
Reports, see
Google
Workspace
Reports API
Documentati
on.

24.3.17.
Google
Kubernetes
Engine



📝🟢

note

It's also
possible to
use a

Custom -
Filebeat
based
Collector to
ingest logs
related to
file activity
on your

endpoints
and servers
without
using the
Cortex XDR
agent. For
more

information,
see

Elasticsearch
Filebeat.

24.3.17.1.
Ingest logs
from Google
Kubernetes
Engine



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

Cortex
Cloud
supports
logs in single
line format
or multiline
format. For

more

information
on handling
messages

that span
multiple
lines of text
in
Elasticsearch
Filebeat,
see Manage
Multiline
Messages.

24.3.18.
HTTP log
collector

24.3.18.1. Set 
up an HTTP
log
collector to
receive logs



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

-The Vendor
and Product
defaults to
Auto-Detect
when the
Log Format
is set to CEF
or LEEF.
-For a Log
Format set
to CEF or
LEEF, Cortex
Cloud reads
events row

by row to
look for the
Vendor and
Product
configured
in the logs.
When the
values are
populated in
the event
log row,
Cortex
Cloud uses
these values
even if you
specified a
value in the
Vendor...

📝🟢

note

Each record
cannot
exceed 5 MB
in size.

24.3.19.
Kubernetes



24.3.19.1.
Onboard the
Kubernetes
Connector



📝🟢

note

This option
is not
supported
for Fargate.
📝🟢

note

Enabling
Realtime
Protection
installs the
agent on
your
Kubernetes
clusters as a
DaemonSet.

📝🟢

note

For Fargate,
you must
provide the
cluster
resource

identifier.
The format
of the
identifier is
arn:aws:eks:
<region>:
<account-
id>:cluster/<
cluster-
name>.

📝🟢

note

Basic
authenticatio
n is only
supported in
Posture
Management.
If deploying
Realtime
Protection,
select None .

📝🟢

note

If you install
the latest
version but
select the N-
1 strategy,
this policy
will take
effect
starting
from the
next

upgrade
cycle (it will
not

immediately
downgrade
your current
installation).
If you
choose an
older version
and keep the
latest
strategy, the
latest
version will
be installed.

24.3.19.2.
What's new
in
Kubernetes
Connector?



24.3.19.3.
Supported
Kubernetes
distributions



📝🟢

note

Does not
include
Autopilot.

📝🟢

note

In Google
Container-
Optimized
OS release
100 and
earlier,
where the
FANOTIFY
EXEC flag is
not

supported,
the Kernel
configuration
may be
partial for
the user
mode agent
to properly
function. In
such cases,
the agent
will fallback
to

asynchronou
s mode.
In RHCOS
version 4.12
and earlier,
the Kernel
configuration
may be
partial for
the user
mode agent
to...

24.3.20.
Microsoft
Azure



24.3.21.
Microsoft
Office 365

24.3.21.1. 
Ingest logs
from
Microsoft
Office 365



🔒🟣

content

Requires the ITDR Module add-on and Data
Collection add-on. For viewing alerts and Azure
AD Activity Logs only the Data Collection add-
on is required.

📝🟢

note

Ingesting
Microsoft
Entra ID
(formerly
known as
Azure AD)
authenticatio
n and audit
events from
Microsoft
Graph API
requires a
Microsoft
Azure
Premium 1 or
Premium 2
license.
Alternatively,
if the
directory
type is
Azure AD
B2C, the
sign-in
reports are
accessible
through the
API without
any
additional
license
requirement.
To ingest
email logs
and data...

📝🟢

note

When
auditing is
turned off
from the
default
setting, you
need to first
turn on

auditing for
your

organization
to collect
Microsoft
Office 365
audit events
from the
Management
Activity API.
Log
duplication
of up to 5%
in Microsoft
products is
considered
normal. In
some cases,
such as
login to a
portal using
MFA, two
log entries
are recorded
by design.

📝🟢

note

To address
Azure
reporting
latency,
there is a 10-
minute
latency
period for
Cortex
Cloud to
receive
Azure AD
logs.

📝🟢

note

For more
information,
see the
Office 365
Management
Activity API
schema.

📝🟢

note

In FedRAMP
environments
, Azure sign-
in logs are
not

supported,
due to
vendor
technical
constraints.

📝🟢

note

Use this
option when
you don't
want to
grant
permissions
for Azure
AD
Authenticati
on and
Azure AD
Audit.

👀🟠

important

Migration
Advisory:
Microsoft
Graph
Security API
v1 to v2
Microsoft
will retire
the Legacy
Alerts (v1)
API endpoint
in April
2026. To
avoid
service
interruption,
instances
currently
using the
legacy
Alerts (v1)
option must
manually
transition to
the v2 GA
endpoint by
selecting the
Alerts →
Security
Alerts V2
option.
Schema
impact:
Moving from
v1 to...

👀🟠

important

Microsoft
has
deprecated
the Legacy
Alerts (v1)
API in April
2026. To
avoid
service
interruption,
you must
migrate to
the v2
endpoint
before this
date.

24.3.22.
Microsoft
Office 365
(email)

24.3.22.1. 
Ingest logs
and data
from
Microsoft
365



🔒🟣

content

Email
content

visibility and
licensing:
Email
subjects and
bodies are
stored in an
encrypted
format to
ensure data
privacy. To
view this
content or
generate
alerts for it,
an Email
Security
module
license is
required.
Without the
license:
Sensitive
email
content

(subject,
body, and
attachments)
remains
encrypted
and is not
accessible
for viewing
or...

☑️🟣

prerequisite

A user account with the Microsoft Azure
Account Administrator role is required to set up
a new Microsoft 365 email collector.
The following Microsoft Graph API permissions
are required:
Mailbox access (read-write)
Read and write mail in all mailboxes
Read contacts in all mailboxes
Read all user mailbox settings
User information, groups,...

📝🟢

note

For other
logs from
Microsoft
Office 365,
use the
Office 365
data
collector.
For more
information,
see Ingest
logs from
Microsoft
Office 365.

24.3.23.
Microsoft
365 (Posture)

24.3.23.1. 
How to
onboard
Microsoft
365

☑️🟣 

prerequisite

You have
generated a
Globally
Unique
Identifier
(GUID), also
known as a
Universally
Unique
Identifier
(UUID). You
will need
this ID for
the tenant
you want to
use for the
Microsoft
365 instance.
In order to
use

Microsoft
365, you
must be
registered
with
Microsoft
Azure.

24.3.24.1. 24.3.24. Okta 
Ingest logs
and data
from Okta

☑️🟣 

prerequisite

Administrato
r privileges:
Your Okta
user must
have a role
capable of
creating API
tokens, such
as Read-
only
Administrato
r, Super
Administrato
r, or

Organization
Administrato
r. For more
information,
see the Okta
Administrato
rs

Documentati
on.

24.3.25.
OneLogin

24.3.25.1. 
Ingest logs
and data
from
OneLogin



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

Event data is
collected
every 30
seconds.

📝🟢

note

Inventory
data
snapshots
are

collected
every 10
minutes.

24.3.26.
Oracle
Cloud
Infrastructur

e



24.3.27.
PingFederate



24.3.28.
PingOne

24.3.28.1. 
Ingest
authenticatio
n logs and
data from
PingOne



🔒🟣

content

Requires the
Data
Collection
add-on.

🧭➡️

navigation

Reporting →
Subscriptions
→ Add
Subscription

24.3.29.
Salesforce

24.3.29.1. 
Ingest and
run

Salesforce
automation
and
remediation



☑️🟣

prerequisite

RBAC
permissions:
Requires
View/Edit
permissions
for Log
Collections,
Data
Sources, and
Integrations
(under
Configuratio
ns & Data
Collections).
Content
packs:
Ensure the
Salesforce
and Base
content

packs are
installed or
updated to
the latest
version.
Gateway
permissions:
Requires
Account
Admin or
Instance
Administrato
r permissions
for
configuring...

📝🟢

note

Do not
include
https:// or
trailing
slashes. 📝🟢

note

The new
Salesforce
integration
has the
description:
Salesforce
CRM
services for
identity
management,
automation,
remediation
and SaaS
Posture
Security.

👀🟠

important

You cannot
proceed
with the
authenticatio

n

configuration
until you
provide a
domain URL
and click
Apply.When
you click
Apply, the
instance is
connected
to a unique
URL in
Salesforce,
and it
cannot be
changed. To
change it,
you need to
exit the
wizard and
start over.

👀🟠

important

If a
'''connected
application''
for Cortex
Cloud data
collection
already
exists, there
is no
obligation to
migrate to
an '''External
Client App''
- but as
Connected
Applications
are

deprecated
by
Salesforce,
it is
recommende
d to migrate.

24.3.30.
Snowflake

24.3.30.1. 
How to
onboard
Snowflake



☑️🟣

prerequisite

In order to
use

Snowflake,
you must be
registered
with one of
these cloud
providers:
Amazon
AWS,
Microsoft
Azure, or
Google
Cloud
Platform
(GCP).
Ensure you
have the
necessary
account
permissions
to onboard.
It is
recommende
d to use
Account
Admin as the
role for the
onboarding.

📝🟢

note

The account
identifier
can be
found using
the user
information
at the
bottom left.
Hover over
the account
you wish to
onboard and
select the
copy option
at the top
right. The
account
identifier is
usually of
the format:
(organization
).[account]

24.3.31.
Windows
DHCP via
Elasticsearch
Filebeat

24.3.31.1. 
Ingest logs
from
Windows
DHCP using
Elasticsearch
Filebeat



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

To avoid
formatting
issues in
filebeat.yml,
we

recommend
that you edit
the text file
inside the
user

interface,
instead of
copying it
and editing
it elsewhere.
Validate the
syntax of
the YML file
before you
finish
creating the
profile.

📝🟢

note

For more
information

on

configuring
the
filebeat.yml
file, see the
Elastic
Filebeat
Documentati
on.

📝🟢

note

To avoid
formatting
issues in
your

filebeat.yml,
we

recommend
that you use
the
download
example file
to make your
customizatio
ns. Do not
copy and
paste the
code syntax
examples
provided
later in this
procedure
into your file.

📝🟢

note

The
tokenizer
definition is
dependent
on the
Windows
server

version that
you are

using as the
log format
differs.
-For

platforms
earlier than
Windows
Server 2008,
use "%{id},%
{date},%
{time},%
{description},
%
{ipAddress},
%
{hostName},
%
{macAddress
}"
-For
Windows
Server 2008
and 2008
R2, use "%
{id},%
{date},%
{time},%
{description},
%
{ipAddress},
%
{hostName},
%{mac...

24.3.32.
Workday

24.3.32.1. 
Ingest report
data from
Workday



🔒🟣

content

Requires the
Data
Collection
and Identity
Threat
Detection &
Response
add-ons.

☑️🟣

prerequisite

Create an
Integration
System User
that is
designated
to access
the custom
report from
Workday for
data
collection in
Cortex
Cloud.
Create an
Integration
System
Security
Group for
the
Integration
System User
created in
Step 1 for
accessing
the report.
When
setting this
group
ensure to
define the
following:
Type of
Tenanted
Security
Group:
Select...

📝🟢

note

Any user
with
permissions
to view
issues and
cases can

view the
Workday
data.

📝🟢

note

For more
information

on

completing
any of the
prerequisite
steps, see
the Workday
documentati

on.

📝🟢

note

The Business
Object
changes
depending
on the Data
Source
selected.

📝🟢

note

Cortex
Cloud uses
a structured
schema
when
integrating
Workday
data. To get
the best
Analytics
results,
specify all
the fields
marked with
an asterisk
from the
recommende
d schema.
workday_use
r_id*
full_name*
workday_ma
nager_user_i
d*
manager*
worker_type*
position_title
*

department*
private_email
_address*
business_em
ail_address*
employment_
start_date*
em...

24.3.33.
Zscaler
Internet
Access



24.3.34.
Zscaler
Private
Access



24.4. Cloud
service
provider
(CSP)
onboarding



24.4.1. Cloud
service
provider
onboarding



24.4.2.
Understand
CSP
onboarding
tiers and
licensing



24.4.3.
Amazon
Web
Services
cloud
onboarding



24.4.3.1.
Prerequisites

📝🟢 

note

The
permissions
included in
Sid:
CortexBaseM
oduleDeploy
ment are

always
required for
the
discovery
module and
stack
lifecycle
management.
The
permissions
included in
Sid:
CortexAuditL
ogsModule
are required
only when
deploying
the audit log
collection
module. It
enables
creating S3,
SQS, SNS,
CloudTrail,
and KMS
resources.

24.4.3.2.
Onboard
Amazon
Web
Services



24.4.3.3. How
to onboard
Amazon
Web
Services



📝🟢

note

Scanning
with an
outpost may
require
additional
AWS
permissions
and may
incur
additional
CSP costs.

👀🟠

important

It is critical
to ensure

that your
KMS key
region and
SNS topic
region are
the exact
same as the
AWS region
where you
are

deploying
the
CloudFormat
ion stack.

24.4.3.4. How to onboard Amazon Web Services

with foundational configuration 

🔒🟣

content

Onboarding
Amazon
Web
Services
(AWS) using
the
foundational
configuration
is included
with Cortex
Cloud NG
SIEM, Cortex
Cloud
Enterprise,
and Cortex
Cloud
Enterprise+
licenses. For
more details
on the CSP
onboarding
tiers and
licensing,
see

Understand
CSP
onboarding
tiers and
licensing.

👀🟠

important

It is critical
to ensure

that your
KMS key
region and
SNS topic
region are
the exact
same as the
AWS region
where you
are

deploying
the
CloudFormat
ion stack.

24.4.3.5.
Deploy the
CloudFormat
ion template
in AWS



24.4.3.6.
AWS post-
deployment
verification



24.4.4.
Microsoft
Azure cloud
onboarding



24.4.4.1.
Onboard
Microsoft
Azure



24.4.4.2.
Prerequisites



24.4.4.3.
How to
onboard
Microsoft
Azure



📝🟢

note

Scanning
with an
outpost may
require
additional
GCP
permissions
and may
incur
additional
CSP costs.

📝🟢

note

Cortex
Cloud
performs a
live
verification
of each
tenant's
approval
status
against
Azure each
time the list
is displayed.
If a
previously
approved
tenant no

longer
shows a
green
checkmark,
the Cortex
Cloud
application
may have
been
removed
from the
Azure
tenant. Click
Approve in
Azure to re-
authorize.

👀🟠

important

It is critical
to ensure

that your
namespace
and Event
Hub belong
to the
specific
Azure
subscription
being
onboarded.
Cross-
subscription
or

centralized
logging is
not

currently
supported.

24.4.4.4.
How to
onboard
Microsoft
Azure with
foundational
configuration



🔒🟣

content

Onboarding
Microsoft
Azure using
the
foundational
configuration
is included
with Cortex
Cloud NG
SIEM, Cortex
Cloud
Enterprise,
and Cortex
Cloud
Enterprise+
licenses. For
more details
on the CSP
onboarding
tiers and
licensing,
see

Understand
CSP
onboarding
tiers and
licensing.

📝🟢

note

Cortex
Cloud
performs a
live
verification
of each
tenant's
approval
status
against
Azure each
time the list
is displayed.
If a
previously
approved
tenant no

longer
shows a
green
checkmark,
the Cortex
Cloud
application
may have
been
removed
from the
Azure
tenant. Click
Approve in
Azure to re-
authorize.

👀🟠

important

It is critical
to ensure

that your
namespace
and Event
Hub belong
to the
specific
Azure
subscription
being
onboarded.
Cross-
subscription
or

centralized
logging is
not

currently
supported.

24.4.4.5. Finalize Microsoft Azure onboarding by
executing the authentication template 

☑️🟣

prerequisite

Before you
begin,
ensure you
have:
An Azure
subscription.
A user with
the required
permissions
for the
relevant
scope

(subscription,

management
group,
tenant). We
recommend
you create a
dedicated
role.
Tenant ID
and
subscription
ID. You can
view these in
Microsoft
Azure Portal
in
Management
groups.
Installed
Terraform on
your local
machine.
You can...

☑️🟣

prerequisite

Before you
begin,
ensure you
have:
An Azure
subscription.
A user with
the required
permissions
for the
relevant
scope

(subscription,

management
group,
tenant). We
recommend
you create a
dedicated
role.
Tenant ID
and
subscription
ID. You can
view these in
Microsoft
Azure Portal
in
Management
groups.
Installed the
Azure CLI
tool.
Authorization
to create...

☑️🟣

prerequisite

Before you
begin,
ensure you
have:
An Azure
subscription.
A user with
the required
permissions
for the
relevant
scope

(subscription,

management
group,
tenant). We
recommend
you create a
dedicated
role.
Tenant ID
and
subscription
ID. You can
view these in
Microsoft
Azure Portal
in
Management
groups.
Installed the
Azure CLI
tool.
Authorization
to create...

👀🟠

important

Do not
delete or
move the
Terraform
files from
this folder. It
will prevent
you from
being able
to edit your
cloud
instance in
the future.

24.4.5. GCP
cloud
onboarding



24.4.5.1.
Onboard
Google
Cloud
Platform



24.4.5.2.
Prerequisites



24.4.5.3. How
to onboard
Google
Cloud
Platform



📝🟢

note

Scanning
with an
outpost may
require
additional
GCP
permissions
and may
incur
additional
CSP costs.

📝🟢

note

If you want to connect to your GCP Workspace,
you must first complete onboarding with the
option disabled. Once the GCP cloud instance
is created, perform the steps detailed in
Connect Google Workspace with your GCP
cloud instance.

24.4.5.4.
How to
onboard
GCP with
foundational
configuration

🔒🟣 

content

Onboarding Google Cloud Platform (GCP)
using the foundational configuration is included
with Cortex Cloud NG SIEM, Cortex Cloud
Enterprise, and Cortex Cloud Enterprise+
licenses. For more details on the CSP
onboarding tiers and licensing, see Understand
CSP onboarding tiers and licensing.

24.4.5.5.
Deploy the
Terraform
authenticatio
n template
in GCP



☑️🟣

prerequisite

Before you
begin,
ensure you
have:
A GCP
account.
Permission
to create the
required
resources in
Google
Cloud
Deployment
Manager.
Installed
Terraform on
your local
machine.
You can
download
Terraform
from the
official
Terraform
website and
follow the
installation
instructions
for your
operating
system.
Installed the
GCP gcloud
CLI tool.
Reviewed
the...

📝🟢

note

The
directory
you create
must be a
subdirectory
of the home
directory.

👀🟠

important

You must not delete or move the Terraform files
from this folder. It will prevent you from being
able to edit your cloud instance in the future.

24.4.5.6.
Connect
Google
Workspace
with your
GCP cloud
instance



☑️🟣

prerequisite

Ensure you
have the
Super Admin
role in
Google
Workspace.

☑️🟣

prerequisite

Ensure you
have the
organization
ID of the
Google
Workspace
you want to
connect:
Log in to
your Google
Admin
Console. and
navigate to
Account →
Account
settings →
Profile. Next
to Customer
ID is your
organization
ID.
Ensure you
have the
organization
ID of the
Google
Workspace
you want to
connect:
Log in to
your Google
Admin
Console. and
navigate...

24.4.6.
Oracle
Cloud
Infrastructur
e cloud
onboarding



24.4.6.1.
Onboard
Oracle
Cloud
Infrastructur

e



24.4.6.2.
Prerequisites



24.4.6.3. How
to onboard
Oracle
Cloud
Infrastructur

e



24.4.6.4.
How to
onboard
Oracle
Cloud
Infrastructur
e with
foundational
configuration

🔒🟣 

content

Onboarding Oracle Cloud Infrastructure (OCI)
using the foundational configuration is included
with Cortex Cloud NG SIEM, Cortex Cloud
Enterprise, and Cortex Cloud Enterprise+
licenses. For more details on the CSP
onboarding tiers and licensing, see Understand
CSP onboarding tiers and licensing.

24.4.6.5.
Deploy the
Terraform
authenticatio
n template
in OCI

☑️🟣 

prerequisite

Before you
begin,
ensure you
have:
An Oracle
Cloud
Infrastructur
e account
and the
tenancy
OCID.
Permission
to deploy a
custom

template
and create
its resources
in OCI.
Installed
Terraform on
your local
machine.
You can
download
Terraform
from the
official
Terraform
website and
follow the
installation
instructions
for your
operating
system.
Installed...

24.4.7.
Alibaba
Cloud cloud
onboarding



24.4.7.1.
Security
capabilities
and
deployment
planning



24.4.7.2.
Resource
inventory



24.4.7.3.
Security
model and
authenticatio

n



24.4.7.4.
Onboard
Alibaba
Cloud



24.4.7.5.
Prerequisites



24.4.7.6. How
to onboard
Alibaba
Cloud



☑️🟣

prerequisite

Before you
begin,
ensure you
have:
Installed
Terraform on
your local
machine.
You can
download
Terraform
from the
official
Terraform
website and
follow the
installation
instructions
for your
operating
system.
Installed the
Alibaba
Cloud CLI
tool.

📝🟢

note

If the
following
error

appears,
contact
support:
"Validation
failed: No
valid
managed
outpost
found for
cloud
provider
ALIBABA_CL
OUD".

👀🟠

important

Do not
delete or
move the
Terraform
files from
this folder. It
will prevent
you from
being able
to edit your
cloud
instance in
the future.

24.4.7.7.
Alibaba
Cloud post-
deployment
verification



24.4.8.
Manually
connect a
cloud
instance



24.4.9.
Manage
cloud
instances

📝🟢 

note

Currently,
automation
permission
errors or

missing
automation
permissions
do not
affect the
Automation
health
status. You
can view any
permission
errors or

missing
permissions
in the the
Permissions
health status.

24.4.10.
Pending
cloud
instances



24.4.11. Edit
your
onboarded
CSP
configuration

👀🟠 

important

When using Terraform authentication
templates, you must execute the updated
Terraform template from the same folder where
the original Terraform template was executed.

24.4.12.
Update
cloud
permissions
after Cortex
release
updates



☑️🟣

prerequisite

Ensure that
the user
account
used to
modify
permissions
has the
necessary

privileges
within both
the Cortex
platform
and your
cloud
environment,
for example,
AWS or
Azure.
You received
a

notification
regarding a
new version
available
that requires
permission
updates, or
viewed a
Needs
Update
status in the
Data
Sources &
Integrations
page.

📝🟢

note

Instances
requiring
updates will
not change
their
connection
status, for
example,
Connected,
Warning,
Error,
Disabled,
due to the
pending
permission
update.

24.4.13.
Troubleshoot
errors on

cloud
instances

📝🟢 

note

Errors
related to
collection of
audit logs in
the cloud
instance are
recorded in
the
collection_au
diting
dataset. For
more

information,
see Audit
logs fields
and query
examples.

24.4.14.
Outposts



👀🟠

important

Outpost scan is an alternative to the
recommended standard cloud scan. Cloud scan
is recommended because it is fully managed by
Palo Alto Networks and incurs no compute
costs for your organization. Outpost scan is an
advanced deployment model reserved for
specific data residency or architectural
requirements.

24.4.14.1.
Outpost
fundamental
s and
planning



📝🟢

note

Notes:
Before you
create your
outpost,
verify that
your internet
connection
is active. An
active
internet
connection
is necessary
for the
notification
to be sent to
Cortex
Cloud to
create the
new outpost.
(Azure) Due
to limitations
in Terraform,
the Azure
subscription
name cannot
contain
blanks. Take
this into
account
while
onboarding.

👀🟠

important

While
outposts
provide
maximum
control over
the scanning
environment,
cloud scan
mode is the
recommende
d default for
most
organizations
.

🧭➡️

navigation

Before
onboarding
your Cortex
Cloud with
the cloud
service
provider
(CSP)
onboarding
wizard,
create an

outpost by
navigating
to Settings
→ Data
Sources &
Integrations
→ Outposts

Settings →
Data
Sources &
Integrations
→ Add New

View and
manage
existing
outposts by
navigating
to Settings
→ Data
Sources &
Integrations
→ Outposts

24.4.14.2.
Create an
outpost



☑️🟣

prerequisite

Before you begin, ensure you have:
An AWS account
Permission to create a stack and its resources in
AWS
Installed Terraform on your local machine. You
can download Terraform from the official
Terraform website and follow the installation
instructions for your operating system.
Installed the AWS CLI tool and configured your
profile with the aws...

☑️🟣

prerequisite

Before you
begin,
ensure you
have:
A GCP
account
Permission
to create the
required
resources in
Google
Cloud
Deployment
Manager
Installed
Terraform on
your local
machine.
You can
download
Terraform
from the
official
Terraform
website and
follow the
installation
instructions
for your
operating
system.
Installed the
GCP gcloud
CLI tool

☑️🟣

prerequisite

Before you
begin,
ensure you
have:
An active
Azure
subscription.
Installed the
Azure CLI
tool.
Permission
to deploy a
custom

template
and create
its resources
in Microsoft
Azure
("Owner" or
"Contributor"
on the
designated
outpost
subscription
scope, and
Active
Directory
"Cloud
Application
Administrato
r" or
"Application
Administrato
r" privileged...

📝🟢

note

Verify that
your internet
connection
is active. An
active
internet
connection
is necessary
for
notifications
to be sent to
Cortex
Cloud for
creating the
new outpost.
If you are
unable to
establish an
internet
connection,
contact
customer

support for
a manual
workaround.

📝🟢

note

When
creating an
outpost for
a specific
Azure
subscription,
the outpost
account
must be in
the same
Azure
organization
as the
monitored
subscriptions
.

📝🟢

note

Due to
limitations in
Terraform,
the Azure
subscription
name cannot
contain
blanks.

👀🟠

important

While
outposts
provide
maximum
control over
the scanning
environment,
cloud scan
mode is the
recommende
d default for
most
organizations
. For details,
see When to
choose
outpost scan.

🧭➡️

navigation

Start the
outpost
creation
wizard by
navigating
to Settings
→ Data
Sources &
Integrations
→ Outposts
and clicking
New Outpost

A new
outpost is
created in
pending
status and
can be
viewed in the
Outpost
page at
Settings →
Data
Sources &
Integrations
→ Outposts

24.4.15.
Introduction
to Terraform
for Cloud
service
provider
(CSP)
onboarding



👀🟠

important

Always run
these
commands
in the same
folder where
the original
.tf files and
.terraform
folder live—
this is where
the state is
stored.

👀🟠

important

This .tfstate
state file is
critical
because it is
needed by
the
terraform
destroy
command to
clean up
created
resources. It
is critical
that you
never delete
this file.

24.4.16.
Cloud
service
provider
permissions



⚠️🟡

caution

Reducing
permissions
below the
recommende
d level may
cause

specific
commands
to fail or
limit
functionality
in future
pack
updates.

24.4.16.1.
Amazon
Web
Services
provider
permissions



📝🟢

note

Unified Cortex platform cloud content packs
require a specific set of automation permissions
to enable full integration with your cloud
environment. Before configuring access for
these packs, review the automation permission
scope guidelines.

👀🟠

important

All
conditional
capabilities
documented
below
require the
mandatory
Base and
Discovery
Engine
permissions
to be
deployed
alongside
them. Base
provides the
foundational
CortexPlatfo
rmRole and
AWS-
managed
read-only
baseline.
Discovery
Engine
extends that
baseline
with the
asset-
inventory
coverage

that every
other
capability
assumes.

24.4.16.2.
Microsoft
Azure
provider
permissions

📝🟢 

note

Unified Cortex platform cloud content packs
require a specific set of automation permissions
to enable full integration with your cloud
environment. Before configuring access for
these packs, review the automation permission
scope guidelines.

24.4.16.3.
Google
Cloud
Platform
provider
permissions

📝🟢 

note

Unified Cortex platform cloud content packs
require a specific set of automation permissions
to enable full integration with your cloud
environment. Before configuring access for
these packs, review the automation permission
scope guidelines.

24.4.16.4.
Oracle
Cloud
Infrastructur
e provider
permissions



24.5.
Container
Registry
Scanning



24.5.1.
Overview of
container
registry
scanning



24.5.1.1.
Registry
Components



24.5.1.2. How
Container
Registry
Scanning
Works



24.5.2.
Configure
registry
scanning for
cloud
accounts

☑️🟣 

prerequisite

Ensure that
you have
performed
the all steps
till
Additional
Security
Capabilities
as listed in
the
onboarding
wizard for
the required
CSP:
Onboard
Amazon
Web
ServicesOnb
oard
Amazon
Web
Services
Onboard
Google
Cloud
PlatformOnb
oard Google
Cloud
Platform
Onboard
Microsoft
AzureOnboar
d Microsoft
Azure
Onboard
Oracle
Cloud
Infrastructur
eOnboard
Oracle...

24.5.3.
Modify the
container
registry
scanning
scope



24.5.4. Scan
re-

evaluation
process



24.5.5.
Connect
Docker Hub
registry



☑️🟣

prerequisite

Ensure an
Outpost is
connected
to your
tenant. ☑️🟣

prerequisite

Set up and
configure
Broker VM
Configure
High
Availability
Cluster

📝🟢

note

If you
choose
Azure as the
Cloud
Provider,
you must
also select
the Tenant
Id. The
Tenant Id is
required to
approve
Cortex as an
enterprise
application
in your
Azure tenant.

📝🟢

note

If you
choose
Azure as the
cloud
provider,
only
Outposts
associated
with the
selected
tenant ID are
displayed.

📝🟢

note

The list of
Broker VMs
displays only
VMs that
support
registry
scanning.
The list of
high-
availability
Clusters
displays only
clusters that
contain at
least one
VM
supporting
registry
scanning.
The registry
scanning
status for
each VM
appears in
brackets if it

was

previously
activated for
that specific
VM.

24.5.5.1.
Manage a
Docker Hub
connector

📝🟢 

note

If you
selected
Scan with
Broker VM
mode, you
can't change
to a
different
scan mode
(such as
Cloud
Discovery or
Scan with
Outpost)
when you
edit the
instance.
When
editing an
instance

configured
for Scan
with Broker
VM, you
must re-
enter your
authenticatio

n

credentials,
including
Username,
Password,
and CA
certificate.

24.5.6.
Connect
Docker V2
compliant
container
registry



☑️🟣

prerequisite

Ensure an
Outpost is
connected
to your
tenant.
Outposts

☑️🟣

prerequisite

Ensure one
of the
following is
configured:
Set up and
configure
Broker VM.
Configure
High
Availability
Cluster.

📝🟢

note

If you
choose
Azure as the
Cloud
Provider,
you must
also select
the Tenant
Id. The
Tenant Id is
required to
approve
Cortex as an
enterprise
application
in your
Azure tenant.

📝🟢

note

If you
choose
Azure as the
cloud
provider,
only
Outposts
associated
with the
selected
tenant ID are
displayed.

📝🟢

note

The list of
Broker VMs
displays only
VMs that
support
registry
scanning.
The list of
high-
availability
Clusters
displays only
clusters that
contain at
least one
VM
supporting
registry
scanning.
The registry
scanning
status for
each VM
appears in
brackets if it

was

previously
activated for
that specific
VM.

24.5.6.1.
Manage a
Docker V2
connector

📝🟢 

note

If you
selected
Scan with
Broker VM
mode, you
can't change
to a
different
scan mode
(such as
Cloud
Discovery or
Scan with
Outpost)
when you
edit the
instance.
When
editing an
instance

configured
for Scan
with Broker
VM, you
must re-
enter your
authenticatio

n

credentials,
including
Username,
Password,
and CA
certificate.

24.5.7.
Connect
GitLab
container
registry



☑️🟣

prerequisite

Ensure an
Outpost is
connected
to your
tenant. ☑️🟣

prerequisite

Set up and
configure
Broker VM
Configure
High
Availability
Cluster

📝🟢

note

When both
the group ID
and project
ID are
provided,
the system
retrieves
container
images from
all projects
within the
specified
group as
well as from
the specified
project.
If neither the
group ID nor
the project
ID is
provided,
the system
retrieves
container
images from
all registries
(across all
groups and
projects)
accessible
to the
authenticate
d...

📝🟢

note

When both
the group ID
and project
ID are
provided,
the system
retrieves
container

images from
all projects
within the
specified
group as
well as from
the specified
project.
If neither the
group ID nor
the project
ID is
provided,
the system
retrieves
container

images from
all registries
(across all
groups and
projects)
accessible
to the
authenticate
d...

📝🟢

note

If you
choose
Azure as the
Cloud
Provider,
you must
also select
the Tenant
Id. The
Tenant Id is
required to
approve
Cortex as an
enterprise
application
in your
Azure tenant.

📝🟢

note

If you
choose
Azure as the
cloud
provider,
only
Outposts
associated
with the
selected
tenant ID are
displayed.

📝🟢

note

When both
the group ID
and project
ID are
provided,
the system
retrieves
container
images from
all projects
within the
specified
group as
well as from
the specified
project.
If neither the
group ID nor
the project
ID is
provided,
the system
retrieves
container
images from
all registries
(across all
groups and
projects)
accessible
to the
authenticate
d...

📝🟢

note

When both
the group ID
and project
ID are
provided,
the system
retrieves
container

images from
all projects
within the
specified
group as
well as from
the specified
project.
If neither the
group ID nor
the project
ID is
provided,
the system
retrieves
container

images from
all registries
(across all
groups and
projects)
accessible
to the
authenticate
d...

📝🟢

note

The list of
Broker VMs
displays only
VMs that
support
registry
scanning.
The list of
high-
availability
Clusters
displays only
clusters that
contain at
least one
VM
supporting
registry
scanning.
The registry
scanning
status for
each VM
appears in
brackets if it

was

previously
activated for
that specific
VM.

📝🟢

note

When both
the group ID
and project
ID are
provided,
the system
retrieves
container

images from
all projects
within the
specified
group as
well as from
the specified
project.
If neither the
group ID nor
the project
ID is
provided,
the system
retrieves
container

images from
all registries
(across all
groups and
projects)
accessible
to the
authenticate
d...

📝🟢

note

When both
the group ID
and project
ID are
provided,
the system
retrieves
container
images from
all projects
within the
specified
group as
well as from
the specified
project.
If neither the
group ID nor
the project
ID is
provided,
the system
retrieves
container
images from
all registries
(across all
groups and
projects)
accessible
to the
authenticate
d...

24.5.7.1.
Manage a
Gitlab
Container
Registry
connector

📝🟢 

note

If you
selected
Scan with
Broker VM
mode, you
can't change
to a
different
scan mode
(such as
Cloud
Discovery or
Scan with
Outpost)
when you
edit the
instance.
When
editing an
instance

configured
for Scan
with Broker
VM, you
must re-
enter your
authenticatio

n

credentials,
including
Username,
Password,
and CA
certificate.

24.5.8.
Connect
Harbor
registry



☑️🟣

prerequisite

Ensure an
Outpost is
connected
to your
tenant. ☑️🟣

prerequisite

Ensure one
of the
following is
configured:
Set up and
configure
Broker VM.
Configure
High
Availability
Cluster.

📝🟢

note

If you
choose
Azure as the
Cloud
Provider,
you must
also select
the Tenant
Id. The
Tenant Id is
required to
approve
Cortex as an
enterprise
application
in your
Azure tenant.

📝🟢

note

If you
choose
Azure as the
cloud
provider,
only
Outposts
associated
with the
selected
tenant ID are
displayed.

📝🟢

note

The list of
Broker VMs
displays only
VMs that
support
registry
scanning.
The list of
high-
availability
Clusters
displays only
clusters that
contain at
least one
VM
supporting
registry
scanning.
The registry
scanning
status for
each VM
appears in
brackets if it

was

previously
activated for
that specific
VM.

24.5.8.1.
Manage a
Harbor
connector

📝🟢 

note

If you
selected
Scan with
Broker VM
mode, you
can't change
to a
different
scan mode
(such as
Cloud
Discovery or
Scan with
Outpost)
when you
edit the
instance.
When
editing an
instance

configured
for Scan
with Broker
VM, you
must re-
enter your
authenticatio

n

credentials,
including
Username,
Password,
and CA
certificate.

24.5.9.
Connect
JFrog
container
registry



☑️🟣

prerequisite

Ensure an
Outpost is
connected
to your
tenant. ☑️🟣

prerequisite

Set up and
configure
Broker VM
Configure
High
Availability
Cluster

📝🟢

note

If you
choose
Azure as the
Cloud
Provider,
you must
also select
the Tenant
Id. The
Tenant Id is
required to
approve
Cortex as an
enterprise
application
in your
Azure tenant.

📝🟢

note

If you
choose
Azure as the
cloud
provider,
only
Outposts
associated
with the
selected
tenant ID are
displayed.

📝🟢

note

The list of
Broker VMs
displays only
VMs that
support
registry
scanning.
The list of
high-
availability
Clusters
displays only
clusters that
contain at
least one
VM
supporting
registry
scanning.
The registry
scanning
status for
each VM
appears in
brackets if it

was

previously
activated for
that specific
VM.

24.5.9.1.
Manage a
JFrog
connector

📝🟢 

note

If you
selected
Scan with
Broker VM
mode, you
can't change
to a
different
scan mode
(such as
Cloud
Discovery or
Scan with
Outpost)
when you
edit the
instance.
When
editing an
instance

configured
for Scan
with Broker
VM, you
must re-
enter your
authenticatio

n

credentials,
including
Username,
Password,
and CA
certificate.

24.5.10.
Connect
Sonatype
Nexus
registry



☑️🟣

prerequisite

Ensure an
Outpost is
connected
to your
tenant.
Outposts

☑️🟣

prerequisite

Ensure one
of the
following is
configured:
Set up and
configure
Broker VM.
Configure
High
Availability
Cluster.

📝🟢

note

If you are
using a CA
certificate,
enter the
server IP
address
instead of
the registry
url.

📝🟢

note

If you
choose
Azure as the
Cloud
Provider,
you must
also select
the Tenant
Id. The
Tenant Id is
required to
approve
Cortex as an
enterprise
application
in your
Azure tenant.

📝🟢

note

If you
choose
Azure as the
cloud
provider,
only
Outposts
associated
with the
selected
tenant ID are
displayed.

📝🟢

note

If you are
using a CA
certificate,
enter the
server IP
address
instead of
the registry
URL.

📝🟢

note

The list of
Broker VMs
displays only
VMs that
support
registry
scanning.
The list of
high-
availability
Clusters
displays only
clusters that
contain at
least one
VM
supporting
registry
scanning.
The registry
scanning
status for
each VM
appears in
brackets if it

was

previously
activated for
that specific
VM.

📝🟢

note

If you are
using a CA
certificate,
enter the
server IP
address
instead of
the registry
URL.

24.5.10.1.
Manage a
Sonatype
connector

📝🟢 

note

If you had
selected
Scan with
Broker VM
mode, you
can't change
to a
different
scan mode
(such as
Cloud
Discovery or
Scan with
Outpost)
when you
edit the
instance.
When
editing an
instance
configured
for Scan
with Broker
VM, you
must re-
enter your
authenticatio

n

credentials,
including
Username,
Password,
and CA
certificate.

24.6. Generic
on-premise
data
collectors



24.6.1. Broker
VM data
collector
applets



24.6.1.1.
Activate
Apache
Kafka
Collector



🔒🟣

content

Requires the
Data
Collection
add-on.
This data
source is
only
available in
your tenant
if the tenant

was

activated
before
October 1,
2025 with an
active Data
Collection
add-on.

☑️🟣

prerequisite

Apache
Kafka
version 2.5.1
and above.
Kafka
cluster set
up on
premises,
from which
the data will
be ingested.
Privileges to
manage
Broker
Service
configuration
, such as
Instance
Administrato
r privileges.
Create a
user in the
Kafka
cluster with
the
necessary
permissions
and the
following
authenticatio
n details:
Broker
Certificate
and Private
Key for...

📝🟢

note

For CEF and
LEEF logs,
Cortex
Cloud takes
the vendor
and product
names from
the log
itself,
regardless
of what you
configure on
this page.

24.6.1.2.
Activate
CSV
Collector



🔒🟣

content

Requires the
Data
Collection
add-on.
This data
source is
only
available in
your tenant
if the tenant

was

activated
before
October 1,
2025 with an
active Data
Collection
add-on.

☑️🟣

prerequisite

Set up and
configure
Broker VM.
Ensure that
you share
the
applicable
CSV files.
Know the
complete
file path for
the Windows
directory.

📝🟢

note

When you
implement a
wildcard file
search,
ensure that
the CSV
files share
the same
columns and
header rows
as all other
logs that are
collected
from the
CSV files to
create a

single
dataset.

📝🟢

note

The CSV
Collector
checks for
new CSV
files every 10
minutes.

24.6.1.3.
Activate
Database
Collector



🔒🟣

content

Requires the
Data
Collection
add-on.
This data
source is
only
available in
your tenant
if the tenant

was

activated
before
October 1,
2025 with an
active Data
Collection
add-on.

☑️🟣

prerequisite

Set up and
configure
Broker VM

📝🟢

note

The DB
Collector
does not
support
passwords
containing
semicolons
(;).

📝🟢

note

The
reference
data
ingested
using the DB
Collector is
counted
towards
license
utilization.

👀🟠

important

When
configuring
the DB
collector to
work with an
Oracle
database
and enabling
this option,
ensure the
following
steps are
completed
for a
successful
connection:
Ensure you
have the
certificate
of the
Database
server.

Upload
certificate
to Broker CA
Trust Store
(Conditional):
This step is
only
required if
the Oracle
server's
certificate is
self-signed...

👀🟠

important

The
Database
Collector
applet
supports a
single row
size of up to
65 KB.
Ensure your
source data
does not
exceed this
limit per
record to
avoid
ingestion
errors.

24.6.1.4.
Activate
DSPM
Fileshare



☑️🟣

prerequisite

Set up and
configure
Broker VM
Know the
complete
path to the
files and
folders that
you want
Cortex
Cloud to
monitor.
Necessary
user

permissions
to access
the network
shares. For
the SMB
connection
type, you
need the
user name

and
password.

📝🟢

note

The applet
list displays
only the
applets for
which you
have
permissions.

📝🟢

note

By default,
all
configured
connections

are saved. 📝🟢

note

For details
regarding
the
connection
fields, see
the table
above under
File Share
Connection.

📝🟢

note

You can add
multiple
connections
under a
single
instance of
the DSPM
Fileshare
applet by
returning to
the File
Share
Connection
screen and
clicking Add
Connection.
Each new
connection
can be of
either the
NFS or SMB
connection
type.
24.6.1.5.
Activate
Files and
Folders
Collector



🔒🟣

content

Requires the
Data
Collection
add-on.
This data
source is
only
available in
your tenant
if the tenant

was

activated
before
October 1,
2025 with an
active Data
Collection
add-on.

☑️🟣

prerequisite

Set up and
configure
Broker VM.
Know the
complete
path to the
files and
folders that
you want
Cortex
Cloud to
monitor.
Ensure that
the user
permissions
for the
network
share
include the
ability to
rename and
delete files
in the folder
that you
want to

configure
collection.

📝🟢

note

Cortex
Cloud only
supports
ingestion of
files
encoded in
UTF-8
format.

📝🟢

note

When using the Linux file share, including the
Linux share with NFS, a Username and Password
are not required, so these fields are grayed out
in the screen.

📝🟢

note

In Batch
mode, the
Files and
Folders
Collector
supports
collecting
logs from a
network
share for a
maximum
file size of
500 MB.

📝🟢

note

When
uploading
JSON files,
Cortex
Cloud only
parses the
first level of
nesting and
only
supports
single line
JSON
format, such
that every
new line
means a

separate
entry.

📝🟢

note

Use this
option only
in cases
where your
files contain
some sort of
"header"
lines, such
as a general
description,
an

introduction,
a disclaimer,
or similar,
and you
want to skip
ingesting
them. The
Lines to Skip
are not part
of the file
format. For
example, in
CSV files,
there is no
need to skip
lines.

📝🟢

note

When the
storing
method is
Replace, the
maximum
size for the
total data to
be imported
into a
lookup
dataset is 30
MB each
time the
data is
fetched.
The
inventory
data
ingested
using the
Files and
Folders
collector is
counted
towards
license
utilization.
When you
use a JOINT
function
with a
lookup table
in a query or
correlation
rule, make
sure you...

📝🟢

note

You can't
specify a file
name that's
the same as
a system file
name.

The name of
a dataset
created
from a tsv
file must
always
include the
extension. If
the original
file name is
mrkdptusrsn
ov23.tsv, you
can name

save the
dataset with
the name
marketing_de
pt_users_Nov
_2023.tsv.

📝🟢

note

The Vendor
and Product
defaults to
Auto-Detect
when the
Log Format
is set to CEF
or LEEF.

24.6.1.6.
Activate FTP
Collector



🔒🟣

content

Requires the
Data
Collection
add-on.
This data
source is
only
available in
your tenant
if the tenant

was

activated
before
October 1,
2025 with an
active Data
Collection
add-on.

☑️🟣

prerequisite

Set up and
configure
Broker VM.
Ensure that
the user
permissions
for the FTP,
SFTP, or
FTPS
include the
ability to
rename and
delete files
in the folder
that you
want to

configure
collection.
When
setting up an
FTPS
Collector
with a server
using a Self-
signed
certificate,
you must
upload the
certificate
first to the
Broker VM
as a Trusted
CA...

📝🟢

note

When
configuring
an SFTP
connection,
Cortex
Cloud
expects the
private key
to be in the
RSA format
that is
included in
the -----
BEGIN RSA
PRIVATE
KEY-----
tag. Cortex
Cloud does
not support
providing
the private
key in the
OpenSSH
format from
the -----
BEGIN
OPENSSH
PRIVATE
KEY----- tag.
When using
ssh-keygen
using a Mac,
you get the
OpenSSH
format by...

📝🟢

note

When
uploading
JSON files,
Cortex
Cloud only
parses the
first level of
nesting and
only
supports
single line
JSON
format, such
that every
new line
means a

separate
entry.

📝🟢

note

Use this
option only
in cases
where your
files contain
some sort of
"header"
lines, such
as a general
description,
an

introduction,
a disclaimer,
or similar,
and you
want to skip
ingesting
them. The
Lines to Skip
are not part
of the file
format. For
example, in
CSV files,
there is no
need to skip
lines.

📝🟢

note

The Vendor
and Product
defaults to
Auto-Detect
when the
Log Format
is set to CEF
or LEEF.

24.6.1.7.
Activate
Local Agent
Settings



📝🟢

note

When
routing
traffic
through a
Broker VM
proxy that
sits behind a
firewall, you
may
experience
intermittent
agent
disconnectio
ns if the
firewall is
configured
to drop
challenge
ACK reset
(RST)
packets. For
environments
using a Palo
Alto
Networks
Next-
Generation
Firewall
(NGFW), see
this
Knowledge
Base article
for details
on managing
this behavior
via...

📝🟢

note

This is
assuming a
standard
hardware
setup with
2vCPU 8 GB
memory.

📝🟢

note

Broker VMs
can act as
as a proxy

for routing
XDR
Collector
traffic to the
Cortex
Cloud
tenant. The
Broker VM
does not
cache XDR
Collector
installers.

📝🟢

note

When you
install your
XDR agents,
you need to
configure
the IP
address of
the Broker
VM and a
port number
during the
installation.
You can use
the default
8888 port or
set a custom
port. You are
not

permitted to
configure
port
numbers
between 0-
1024 and
63000-
65000, or
port
numbers
4369, 5671,
5672, 5986,
6379, 8000,
9100, 15672,
25672.
Additionally,..
.

👀🟠

important

A FQDN
must be
configured
for Agent
Installer and
Content
Caching to
function
properly.

👀🟠

important

You can only
enable
Agent
Installer and
Content
Caching,
when in the
Broker VM
Configuratio
n, you've
uploaded
your signed
SSL Server
Certificate
and key and
set the
FQDN. For
more

information,
see the
Agent
Installer and
Content
Caching
requirements
explained
above.

24.6.1.8.
Activate
NetFlow
Collector



🔒🟣

content

Requires the
Data
Collection
add-on.

☑️🟣

prerequisite

Set up and
configure
Broker VM

📝🟢

note

Since
multiple
network
devices can
send data to
a single
NetFlow
Collector,
we

recommend
that you
configure a
maximum of
50 NetFlow
Collectors
per Broker
VM applet,
with a
maximum
aggregated
rate of
approximatel
y 50K flows
per second
(FPS) to
maintain
system
performance.

📝🟢

note

Since Cortex
Cloud
reserves

some port
numbers, it
is best to
select a port
number that
is not in the
range of 0-
1024 (except
for 514), in
the range of
63000-
65000 or has
one of the
following
values: 4369,
5671, 5672,
5986, 6379,
8000, 8888,
9100, 15672,
or 28672.

📝🟢

note

The options
available
change,
according to
the set
Source
Network
value.

24.6.1.9.
Activate
Network
Mapper



☑️🟣

prerequisite

After you
have
configured
and
registered
your Broker
VM, you can
choose to
activate the
Network
Mapper
application.

📝🟢

note

Each IP address range can receive multiple scan
requests based on it's availability.

📝🟢

note

IP address
ranges are

displayed
according to
what you
defined as
your
Network
Parameters.

24.6.1.10.
Activate
Registry
Scanner



☑️🟣

prerequisite

To initialize
registry
scanning on
your Broker
VM, you
must first
add the
necessary
data
connectors.
For details,
see:

Connect
Docker Hub
registry
Connect
Docker V2
compliant
container
registry
Connect
GitLab
container
registry
Connect
Harbor
registry
Connect
JFrog
container
registry
Connect
Sonatype
Nexus
registry
To initialize
registry...

📝🟢

note

You cannot
activate the
Registry
Scanner
directly on a
new or

existing
Broker VM.
You can only
activate or
deactivate
existing
Registry
Scanner
applets. To
activate or
deactivate
existing
applets, see
Step 4 under
Verify
Registry
Scanner
connection
section.

24.6.1.11.
Syslog
Collector
applet



24.6.1.11.1.
Activate
Syslog
Collector



🔒🟣

content

Requires the
Data
Collection
add-on.

☑️🟣

prerequisite

Set up and
configure
Broker VM

📝🟢

note

Once
configured,
you cannot
change the
Port/PROTO
COL. If you
don't want
to use a
data source,
ensure to
remove the
data source
from the list
as explained
in Task 5.

📝🟢

note

The Vendor
and Product
defaults to
Auto-Detect
when the
Log Format
is set to CEF
or LEEF.
For a Log
Format set
to CEF or
LEEF, Cortex
Cloud reads
events row

by row to
look for the
Vendor and
Product
configured
in the logs.
When the
values are
populated in
the event
log row,
Cortex
Cloud uses
these values
even if you
specified a
value in the
Vendor and...

📝🟢

note

The server
certificate
and private
key pair is
expected in
a PEM
format.
Cortex
Cloud will
notify you
when your
certificates
are about to
expire.

📝🟢

note

Because
some port
numbers are
reserved by
Cortex
Cloud , you
must choose
a port
number that
is not:
In the range
of 0-1024
(except for
514)
In the range
of 63000-
65000
Values of
4052, 4369,
5671, 5672,
5986, 6379,
8000, 8888,
9100, 15672,
or 28672

📝🟢

note

CORELIGHT
is not
available for
a UDP
protocol. 24.6.1.11.2.
Ingest logs
from a
Syslog
receiver

🔒🟣 

content

Requires the
Data
Collection
add-on.

24.6.1.11.3.
Check Point
FW1/VPN1

24.6.1.11.3.1. 
Ingest logs
from Check
Point
firewalls



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

Logs with
sessionid = 0
are dropped.
Destination
Port data is
available
only in the
raw logs.

24.6.1.11.4.
Cisco ASA
firewalls and
AnyConnect



24.6.1.11.4.1. Ingest logs from Cisco ASA firewalls
and AnyConnect 🔒🟣 
content

Requires the
Data
Collection
add-on.
24.6.1.11.5.
Corelight
Zeek

24.6.1.11.5.1. 
Ingest logs
from
Corelight
Zeek

🔒🟣 

content

Requires the
Data
Collection
add-on.
24.6.1.11.6.
Forcepoint
DLP

24.6.1.11.6.1. 
Ingest logs
from
Forcepoint
DLP

🔒🟣 

content

Requires the
Data
Collection
add-on.
24.6.1.11.7.
Fortinet
Fortigate

24.6.1.11.7.1. 
Ingest logs
from
Fortinet
Fortigate
firewalls

🔒🟣 

content

Requires the
Data
Collection
add-on.

24.6.1.11.8.
Next-
Generation
Firewall

24.6.1.11.8.1. 
Ingest Next-
Generation
Firewall logs
using the
Syslog
collector



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

In the
following
procedure,
general
information
is provided
for NGFW
and
Panorama.
For detailed
instructions,
consult the
documentati
on for your
specific
devices and
Panorama
version, to
ensure that
you have
configured
log
forwarding
correctly for
all the log
types that
you would
like to
forward to
Cortex
Cloud. The
following
steps only
cover...

📝🟢

note

To avoid the
possible
effects of
line
formatting,
do not
copy/paste
the message
formats
directly into
the PAN-OS
web
interface.
Instead,
paste into a
text editor,
remove any
carriage
return or line
feed
characters,
and then
copy and
paste into
the web
interface.

📝🟢

note

From version
10.0 and
later, the log
format
documented
for log types
(Traffic,
Threat, and
URL)
exceeds the
maximum
supported
2048
characters in
the Custom
Log Format
tab on the
firewall and
Panorama.
Select the
CEF keys
and values
to limit the
number of
characters
to 2048, as
per your
requirements.

24.6.1.11.9.
PingFederate

24.6.1.11.9.1. 
Ingest
authenticatio
n logs from
PingFederate

🔒🟣 

content

Requires the
Data
Collection
add-on.
24.6.1.11.10.
Zscaler
Internet
Access

24.6.1.11.10.1. 
Ingest logs
from Zscaler
Internet
Access

🔒🟣 

content

Requires the
Data
Collection
add-on.

24.6.1.11.11.
Zscaler
Private
Access

24.6.1.11.11.1. 
Ingest logs
from Zscaler
Private
Access



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

For more
information

on

configuring
the
parameters
on the
screen, see
the Zscaler
Private
Access
(ZPA)
documentati
on for
Configuring
a Log
Receiver.

📝🟢

note

You can only
configure a
ZPA log
receiver to
collect one
type of log
with your
Syslog
Collector in
Cortex
Cloud. To
configure
more that
one log
type, you'll
need to add
another log
receiver.

24.6.1.12.
Activate
Transporter



☑️🟣

prerequisite

Permissions:
To configure
and manage
Transporter
applet
settings, you
must have
permissions
to manage
Broker
Service
configuration
s (such as an
Instance
Administrato
r)
Set up and
configure
Broker
VMSet up
and
configure
Broker VM
Confirm that
your Broker
is v 28 or
above
Whitelist IP
addresses to
enable
access to
Cortex
Cloud
resources.

The IP
addresses...

⛔🔴

warning

The
Transporter
applet is not
supported
for
FedRAMP
customers.

24.6.1.13.
Activate
Windows
Event
Collector



🔒🟣

content

Requires the
Data
Collection
add-on.

☑️🟣

prerequisite

Set up and configure Broker VM
Broker VM version 8.0 and later
You have knowledge of Windows Active
Directory and Domain Controllers.
You must configure different settings related to
the FQDN where the instructions differ
depending on whether you are configuring a
standalone Broker VM or High Availability (HA)
cluster.
Standalone broker
A...

📝🟢

note

You can also
activate the
Windows
Event
Collector on
Windows
Core. For
more

information,
see Activate
Windows
Event
Collector on
Windows
Core.

📝🟢

note

By default,
Cortex
Cloud
collects
Palo Alto
Networks
predefined
Security
events that
are used by
the Cortex
Cloud
detectors.
Removing
the Security
collector
interferes
with the
Cortex
Cloud
detection
functionality.
Restore to
Default to
reinstate the
Security
event
collection.

📝🟢

note

You must
install the
WEF
certificate
on every
Windows
Server,
whether DC
or not, for
the WEFs
that are
supposed to
forward logs
to the
Windows
Event
Collector
applet on
the Broker
VM.

📝🟢

note

You must
install the
WEF
certificate
on every
Windows
Server,
whether DC
or not, for
the WEFs
that are
supposed to
forward logs
to the
Windows
Event
Collector
applet on
the Broker
VM.

📝🟢

note

This step
overrides
any local
policy
settings. 📝🟢

note

You must
type out the
name, do
not select
the name
from the
browse
button.

📝🟢

note

If Windows
Firewall is
enabled on
your event
forwarders,
you will have
to define an
outbound
rule to
enable the
WEF to
reach port
5986 on the
WEC.

24.6.1.13.1.
Activate
Windows
Event
Collector on
Windows
Core



🔒🟣

content

Requires the
Data
Collection
add-on.

☑️🟣

prerequisite

Set up and configure Broker VM
Broker VM version 8.0 and later
You have knowledge of Windows Active
Directory and Domain Controllers.
You must configure different settings related to
the FQDN where the instructions differ
depending on whether you are configuring a
standalone Broker VM or High Availability (HA)
cluster.
Standalone broker
A...

📝🟢

note

By default,
Cortex
Cloud
collects
Palo Alto
Networks
predefined
Security
events that
are used by
the Cortex
Cloud
detectors.
Removing
the Security
collector
interferes
with the
Cortex
Cloud
detection
functionality.
Restore to
Default to
reinstate the
Security
event
collection.

📝🟢

note

You must
install the
WEF
certificate
on every
Windows
Server,
whether DC
or not, for
the WEFs
that are
supposed to
forward logs
to the
Windows
Event
Collector
applet on
the Broker
VM.

📝🟢

note

This step
overrides
any local
policy
settings. 📝🟢

note

You must
type out the
name, do
not select
the name
from the
browse
button.

📝🟢

note

If Windows
Firewall is
enabled on
your event
forwarders,
you will have
to define an
outbound
rule to
enable the
WEF to
reach port
5986 on the
WEC.

24.6.1.13.2.
Renew WEC
certificates



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

You must
install the
WEF
certificate
on every
Windows
Server,
whether DC
or not, for
the WEFs
that are
supposed to
forward logs
to the
Windows
Event
Collector
applet on
the Broker
VM.

📝🟢

note

You can see more than one
ca.wec.paloaltonetworks.com and
forwarder.wec.paloaltonetworks.com file from a
previous installation in the directory, so select
the file with the most extended Expiration Date.
You can verify that you are using the correct
certificate:
To verify the client certificate in the Personal →
Certificates folder is related to the...

📝🟢

note

Only
perform this
step under
the
following
conditions:
You have
completed
the WEF
certification
renewal
process for
ALL clients
in your
environment.
Otherwise,
events from
the WEFs
that you did
not install
the new
client
certificate
will not be
collected by
the WEC.
You are
approaching
the WEC
server CA
certificate
expiration
date, which
is 2 years...

📝🟢

note

If this query
does not
display
results with
a timestamp
from after
the renewal
process, it
could
indicate that
the renewal
process is
not

complete, so
wait a few
minutes
before
running
another
query. If you
are still
having a
problem,
contact
Technical
Support.

👀🟠

important

After you
receive a
notification
for renewing
your WEC
CA
certificate,
we

recommend
that you do
not add any
new WEF
clients until
the WEC
certification
renewal
process is
complete.
Events from
these WEF
clients that
are added
afterwards
will not be
collected by
the server
until the
WEC
certificates
are renewed.

24.6.2. XDR
Collectors



📝🟢

note

Ingestion of
log events
larger than 5
MB is not
supported. 24.6.2.1. XDR
Collector
audit logs

🔒🟣 

content

Requires the
Data
Collection
add-on.
24.6.2.2. XDR
Collector
machine
requirements
and
supported
operating
systems

🔒🟣 

content

Requires the
Data
Collection
add-on.

24.6.2.3.
Resources
required to
enable
access to
XDR
Collectors



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

Some of the
IP addresses
required for
access are

registered in
the United
States. As a
result, some
GeoIP
databases
do not
correctly
pinpoint the
location in
which IP
addresses
are used. All
customer
data is
stored in
your

deployment
region,
regardless
of the IP
address
registration
and restricts
data
transmission
through any
infrastructur
e to that...

📝🟢

note

Throughout
this topic,
refers to the
chosen
subdomain
of your
Cortex
Cloud
tenant and
<region> is
the region in
which your
Strata
Logging
Service is
deployed.

24.6.2.4.
Manage XDR
Collectors



🔒🟣

content

Requires the
Data
Collection
add-on.

24.6.2.4.1.
XDR
Collectors
installation
resource for
Windows
and Linux



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

In some
circumstance
s, such as
after an XDR
Collectors
upgrade, the
configured
settings in
the XML
configuration
file can be
erased. Yet,
this won't
affect the
saved
settings in
the XDR
Collectors
database.

24.6.2.4.2.
Create an
XDR
Collector
installation
package



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

XDR
Collectors
cannot be
moved
between
Cortex
Cloud
managing
servers. In
this
situation,
you need to
uninstall the
existing
collector,
and then
install a new
collector
using an
installation
package
from the

new

managing
server. For
more

information

on

uninstalling,
see Uninstall
the XDR
Collector.

📝🟢

note

Since Cortex
Cloud relies
on the
installation
package ID
to approve
XDR
Collector
registration
during
install, it is
not
recommende
d to delete
the
installation
package for
any active
on-premise
collector
machines.
Hiding the
installation
package will
remove it
from the
default list
of available
installation
packages,
and can be
useful to
eliminate...

24.6.2.4.3.
Install the
XDR
Collector
installation
package for
Windows



🔒🟣

content

Requires the
Data
Collection
add-on.

24.6.2.4.3.1.
Install the
XDR
Collector on
Windows
using the MSI



📝🟢

note

When the package is executed using the MSI, an
installation log is generated in
%TEMP%\MSI<Random characters>.log by
default.

📝🟢

note

If the XDR
Collector
does not
connect to
Cortex
Cloud, verify
your internet
connection
on the
collector
machine. If
the XDR
Collector
still does not
connect,
verify that
the
installation
package has
not been
removed
from the
Cortex
Cloud tenant.

24.6.2.4.3.2.
Install the
XDR
Collector on
Windows
using
Msiexec



24.6.2.4.4.
Install the
XDR
Collector
installation
package for
Linux



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

We
recommend
that you
perform a
Linux RPM
or Linux DEB

installation. 📝🟢

note

If the XDR Collector does not connect to
Cortex Cloud, verify your Internet connection
on the collector machine. If the XDR Collector
still does not connect, verify the installation
package has not been removed from the Cortex
Cloud management console.

📝🟢

note

The XDR
Collector
does not
support
proxy
communicati
on in
environments
where
proxy
authenticatio
n is required.

👀🟠

important

When performing a XDR Collector installation
or upgrade in Linux using a shell installer, the
/tmp folder cannot be marked as noexec.
Otherwise, the installation or upgrade fails. As a
workaround, before the installation or upgrade,
use the following command:

24.6.2.4.5.
Configure
the XDR
Collector
upgrade
scheduler



🔒🟣

content

Requires the
Data
Collection
add-on.

🧭➡️

navigation

In Cortex
Cloud,
select
Settings →
Configuratio
ns → XDR
Collectors

→

Configuratio
n

24.6.2.4.6.
Set an
application
proxy for
XDR
Collectors



🔒🟣

content

Requires the
Data
Collection
add-on.

🧭➡️

navigation

In Cortex
Cloud,
select
Settings →
Configuratio
ns → XDR
Collectors

→

Administratio

n

24.6.2.4.7.
Set an alias
for an XDR
Collector
machine



🔒🟣

content

Requires the
Data
Collection
add-on.

🧭➡️

navigation

Settings →
Configuratio
ns → XDR
Collectors

→

Administratio
n 24.6.2.4.8.
Upgrade
XDR
Collectors



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

The XDR
Collector
keeps the
name of the
original
installation
package
after every
upgrade.

24.6.2.4.9.
Uninstall the
XDR
Collector

🔒🟣 

content

Requires the
Data
Collection
add-on.

24.6.2.5.
Define XDR
Collector
machine
groups



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

XDR
Collectors
only support
IPv4

addresses. 📝🟢

note

Disconnectin
g Directory
Sync in your
Cortex
Cloud
deployment
can affect
existing
collector
machine
groups and
policy rules
based on
Active
Directory
properties.

24.6.2.6.
About
Cortex XDR
Collector
content

updates

🔒🟣 

content

Requires the
Data
Collection
add-on.

24.6.2.7. XDR
Collector
profiles



🔒🟣

content

Requires the
Data
Collection
add-on.

24.6.2.7.1.
Add an XDR
Collector
profile for
Windows



📝🟢

note

Ingestion of
log events
larger than 5
MB is not

supported. 📝🟢

note

Fileset
validation is
enforced.
You must
enable at
least one
fileset in the
module,
because
filesets are
disabled by
default.
Cortex
Cloud
collects all
logs in either
an

uncompresse
d JSON or
text format.
Compressed
files, such as
the gzip
format, are
not

supported.
Cortex
Cloud
supports
logs in single
line format
or multiline
format. For

more...

24.6.2.7.1.1.
Ingest logs
from
Windows
DHCP using
Elasticsearch
Filebeat



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

Although
this
enrichment
is available
when
configuring
a Windows
DHCP
collector for
a cloud data
collection
integration,
we

recommend
configuring
Cortex
Cloud to
receive
Windows
DHCP logs
with an XDR
Collector
Windows
Filebeat
profile,
because it is
simpler to
set up.

📝🟢

note

Configuration activities include editing the
filebeat.yml file. To avoid formatting issues in
this file, use the template provided by Cortex
Cloud to make your customizations. We
recommend that you edit the file inside the user
interface, instead of copying it and editing it
elsewhere. Validate the syntax of the YML file
before you finish creating your...

📝🟢

note

Configuratio
n activities
include
editing the
filebeat.yml
file. To avoid
formatting
issues in this
file, use the
example file
provided by
Cortex
Cloud to
make your
customizatio
ns. Do not
copy and
paste the
code syntax
examples
provided
later in this
procedure
into your
filebeat.yml
file. Validate
the syntax
of the YML
file before
you finish
creating...

📝🟢

note

The
tokenizer
definition is
dependent
on the
Windows
server

version that
you are

using,
because the
log format
differs.
For

platforms
earlier than
Windows
Server 2008,
use "%{id},%
{date},%
{time},%
{description},
%
{ipAddress},
%
{hostName},
%
{macAddress
}"
For
Windows
Server 2008
and 2008
R2, use "%
{id},%
{date},%
{time},%
{description},
%
{ipAddress},
%
{hostName},
%...

24.6.2.7.1.2.
Ingest
Windows
DNS debug
logs using
Elasticsearch
Filebeat



📝🟢

note

Detailed
logs are
significantly
larger,
because
more

information
is added to
the logs.

📝🟢

note

To avoid
formatting
issues in
your

filebeat.yml
file, we
recommend
that you
validate the
syntax of
the file.

24.6.2.7.2.
Add an XDR
Collector
profile for
Linux



📝🟢

note

Ingestion of
log events
larger than 5
MB is not

supported. 📝🟢

note

Fileset
validation is
enforced.
You must
enable at
least one
fileset in the
module,
because
filesets are
disabled by
default.
Cortex
Cloud
collects all
logs in either
an

uncompresse
d JSON or
text format.
Compressed
files, such as
the gzip
format, are
not

supported.
Cortex
Cloud
supports
logs in single
line format
or multiline
format. For

more...

24.6.2.8.
Apply
profiles to
collection
machine
policies



🔒🟣

content

Requires the
Data
Collection
add-on.

🧭➡️

navigation

Settings →
Configuratio
ns → XDR
Collectors
→ Policies
→ +Add
Policy

24.6.2.9. XDR
Collector
datasets

🔒🟣 

content

Requires the
Data
Collection
add-on.

24.7. Palo
Alto
Networks
integrations



🔒🟣

content

Data
collection
may require
an add-on.

📝🟢

note

Roll-back of
Strata
Logging
Service
integration
migration is
not

supported.

24.7.1. About
Palo Alto
Networks
integrations



📝🟢

note

For Palo
Alto
Networks
Integrations
there is an
option to
turn on or
off the
collection of
URL and File
log types.
For more
information,
see

Collecting
URL and File
log types.

📝🟢

note

Roll-back of
Strata
Logging
Service
integration
migration is
not

supported.

24.7.2. Cloud
Next-
Generation
Firewall

24.7.2.1. 
Ingest data
from Cloud
Next-
Generation
Firewall



🔒🟣

content

Requires the
Data
Collection
add-on.

☑️🟣

prerequisite

Cortex
Cloud RBAC
permissions:
Requires
View/Edit
permissions
for Data
Sources
(under
Configuratio
ns → Data
Collections).
Cloud
Service
Provider
(CSP)
account
permissions:
Configuratio
n of data
ingestion
from
multiple
accounts

and regions
requires
Super User
permissions
on both the
Cortex
Cloud
tenant and
on the
device
accounts.
Note
Cross CSP...

📝🟢

note

If you
change a
device
region, you
must first
disconnect
the device
from Cortex
Cloud and
then
reconnect it
after the
region
change is
complete.

👀🟠

important

This cross-
account
support is
limited to
the same
SFDC
hierarchy; it
does not
extend to
MSSP
scenarios
where the
customer

and provider
own

separate
ends of the
solution.

24.7.3. Next-
Generation
Firewall



24.7.3.1.
Ingest data
from Next-
Generation
Firewall



🔒🟣

content

Requires the
Data
Collection
add-on.

☑️🟣

prerequisite

Ensure that
you have
completed
the
following on
the NGFW
or Panorama
side:
For
Panorama
only, ensure
that the
Panorama
Cloud
Services
plugin is
installed.
Enable log
forwarding
profiles on
firewall rules.
On the
Cortex
Cloud side,
ensure that
you have
user role
permissions
for Data
Collection >
Data
Sources &
Integrations.
Configuratio
n of data...

📝🟢

note

Roll-back of
Strata
Logging
Service
integration
migration is
not

supported.

📝🟢

note

If you
change a
device
region, you
must first
disconnect
the device
from Cortex
Cloud and
then
reconnect it
after the
region
change is
complete.
If your
firewalls are
located in a
different
region, or
bandwidth
issues are
encountered
due to large
log size, you
can ingest
NGFW logs
in CEF
format,
using the
Syslog
collector.
However, the
Syslog
solution...

📝🟢

note

In the
following
procedure,
general
information
is provided
for NGFW
and
Panorama.
For detailed
instructions,
consult the
documentati
on for your
specific
devices and
Panorama
version.

📝🟢

note

Cortex
Cloud does
not validate
your firewall
credentials,
you must
ensure the
certificates
or PSK
details have
been
updated in
your
firewalls in
order for
data to
stream.

📝🟢

note

It can take
an hour or
longer after
connecting
the firewall
in Cortex
Cloud until
you start
seeing
notifications
that the
certificate
has been
approved,
and that the
logging
service
license has
appeared on
the firewall.

📝🟢

note

IOC and
BIOC issues
are

applicable
on stitched
data only,
and are not
available on
raw data.

👀🟠

important

This cross-
account
support is
limited to
the same
SFDC
hierarchy; it
does not
extend to
MSSP
scenarios
where the
customer

and provider
own

separate
ends of the
solution.

💡🔵

tip

You can see an overview of ingestion status for
all log types, and a breakdown of each log type
and its daily consumption quota on the NGFW
Ingestion Dashboard.

24.7.3.2.
Ingest Next-
Generation
Firewall logs
using the
Syslog
collector



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

In the
following
procedure,
general
information
is provided
for NGFW
and
Panorama.
For detailed
instructions,
consult the
documentati
on for your
specific
devices and
Panorama
version, to
ensure that
you have
configured
log
forwarding
correctly for
all the log
types that
you would
like to
forward to
Cortex
Cloud. The
following
steps only
cover...

📝🟢

note

To avoid the
possible
effects of
line
formatting,
do not
copy/paste
the message
formats
directly into
the PAN-OS
web
interface.
Instead,
paste into a
text editor,
remove any
carriage
return or line
feed
characters,
and then
copy and
paste into
the web
interface.

📝🟢

note

From version
10.0 and
later, the log
format
documented
for log types
(Traffic,
Threat, and
URL)
exceeds the
maximum
supported
2048
characters in
the Custom
Log Format
tab on the
firewall and
Panorama.
Select the
CEF keys
and values
to limit the
number of
characters
to 2048, as
per your
requirements.

24.7.4. Ingest
data from
Prisma
Access



🔒🟣

content

Requires the
Data
Collection
add-on.

☑️🟣

prerequisite

Configuratio
n of data
ingestion
from
multiple
accounts
requires
Super User
permissions
in both
Cortex
Cloud
tenant and
Prisma
Access
accounts.

📝🟢

note

Roll-back of
Strata
Logging
Service
integration
migration is
not

supported.

📝🟢

note

If you
change a
device
region, you
must first
disconnect
the device
from Cortex
Cloud and
then
reconnect it
after the
region
change is
complete.

📝🟢

note

Cortex
Cloud does
not validate
your Prisma
Access
account
credentials.
You must
ensure the
account has
been
deployed in
order for
data to
stream.

24.7.5. Ingest
logs from
Prisma
Access
Browser



🔒🟣

content

Requires the
Data
Collection
add-on.

📝🟢

note

If you
change a
device
region, you
must first
disconnect
the device
from Cortex
Cloud and
then
reconnect it
after the
region
change is
complete.

24.7.6. Ingest
detection
data from
Strata
Logging
Service

📝🟢 

note

IOC and
BIOC issues
are

applicable
on stitched
data only
and are not
available on
raw data.

24.7.7. IoT
Security

🔒🟣 24.7.7.1. Ingest alerts and assets from IoT Security  

content

Requires the
Data
Collection
add-on.
24.7.8.
Collecting
URL and File
log types

24.7.8.1. 
Detectors
connected
to URL and
File log types



24.8. Cloud
Posture and
Runtime
Security
data sources



24.8.1. How
to onboard
Databricks

📝🟢 

note

If you turn
on this
feature,
both the
cloud and
region will
be used for
scanning,
possibly
incurring
cost and
requiring
adherence
to certain
compliance
policies.

24.8.2. How
to onboard
on-premise
assets to
Cortex
Cloud Data
Security



📝🟢

note

Your data is
scanned on
the Broker
VM itself,
and only the
metadata
and
classification
results are
transmitted
from the on-
premise
environment
to Cortex
Cloud.

📝🟢

note

For
information
about
activating
the DSPM
Fileshare
applet, see
Activate
DSPM
Fileshare.

📝🟢

note

For
information
about
activating
the DSPM
Database
applet, see
Activate
DSPM
Database.

24.8.2.1.
Activate
DSPM
Fileshare



☑️🟣

prerequisite

Set up and
configure
Broker VM
Know the
complete
path to the
files and
folders that
you want
Cortex
Cloud to
monitor.
Necessary
user

permissions
to access
the network
shares. For
the SMB
connection
type, you
need the
user name

and
password.

📝🟢

note

The applet
list displays
only the
applets for
which you
have
permissions.

📝🟢

note

By default,
all
configured
connections

are saved. 📝🟢

note

For details
regarding
the
connection
fields, see
the table
above under
File Share
Connection.

📝🟢

note

You can add
multiple
connections
under a
single
instance of
the DSPM
Fileshare
applet by
returning to
the File
Share
Connection
screen and
clicking Add
Connection.
Each new
connection
can be of
either the
NFS or SMB
connection
type.

24.8.2.2.
Activate
DSPM
Database



📝🟢

note

The applet
list displays
only the
applets for
which you
have
permissions.

📝🟢

note

By default,
all
configured
connections

are saved. 📝🟢

note

For details
regarding
the
connection
fields, see
the table
above under
Database
Connection.

📝🟢

note

If you get a
Test
connection
failed
message,

check your
parameters
and try again.

📝🟢

note

You can add
multiple
connections
under a
single
instance of
the DSPM
Database
applet by
returning to
the
Database
Connection
screen and
clicking Add
Connection.

⛔🔴

warning

Set up and
configure
Broker VM
Know the
database
engine, host,
and port for
the
database
you want to
connect.

24.8.3. How
to onboard
Microsoft
365

☑️🟣 

prerequisite

You have
generated a
Globally
Unique
Identifier
(GUID), also
known as a
Universally
Unique
Identifier
(UUID). You
will need
this ID for
the tenant
you want to
use for the
Microsoft
365 instance.
In order to
use

Microsoft
365, you
must be
registered
with
Microsoft
Azure.

24.8.4. Ingest
logs and
data from
Okta

☑️🟣 

prerequisite

Administrato
r privileges:
Your Okta
user must
have a role
capable of
creating API
tokens, such
as Read-
only
Administrato
r, Super
Administrato
r, or

Organization
Administrato
r. For more
information,
see the Okta
Administrato
rs

Documentati
on.

24.8.5.
Activate
Registry
Scanner



☑️🟣

prerequisite

To initialize
registry
scanning on
your Broker
VM, you
must first
add the
necessary
data
connectors.
For details,
see:

Connect
Docker Hub
registry
Connect
Docker V2
compliant
container
registry
Connect
GitLab
container
registry
Connect
Harbor
registry
Connect
JFrog
container
registry
Connect
Sonatype
Nexus
registry
To initialize
registry...

📝🟢

note

You cannot
activate the
Registry
Scanner
directly on a
new or

existing
Broker VM.
You can only
activate or
deactivate
existing
Registry
Scanner
applets. To
activate or
deactivate
existing
applets, see
Step 4 under
Verify
Registry
Scanner
connection
section.

24.8.6. How
to onboard
Snowflake



☑️🟣

prerequisite

In order to
use

Snowflake,
you must be
registered
with one of
these cloud
providers:
Amazon
AWS,
Microsoft
Azure, or
Google
Cloud
Platform
(GCP).
Ensure you
have the
necessary
account
permissions
to onboard.
It is
recommende
d to use
Account
Admin as the
role for the
onboarding.

📝🟢

note

The account
identifier
can be
found using
the user
information
at the
bottom left.
Hover over
the account
you wish to
onboard and
select the
copy option
at the top
right. The
account
identifier is
usually of
the format:
(organization
).[account]

24.8.7.
Activate
Transporter



☑️🟣

prerequisite

Permissions:

To configure
and manage
Transporter
applet
settings, you
must have
permissions
to manage
Broker
Service
configuration
s (such as an
Instance
Administrato
r)
Set up and
configure
Broker
VMSet up
and
configure
Broker VM
Confirm that
your Broker
is v 28 or
above
Whitelist IP
addresses to
enable
access to
Cortex
Cloud
resources.

The IP
addresses...

⛔🔴

warning

The
Transporter
applet is not
supported
for
FedRAMP
customers.

🧭➡️

navigation

Settings →
Configuratio
ns → Broker
VMs (under
Data Broker

the Brokers
tab →
locate your
Broker VM
→ hover and
click + Add
under the
Apps
column →
AppSec
Transporter

the Appsec
Transporter
under the
App column
→

Deactivate
→ Confirm
when
prompted
All existing
connections
are deleted
but their
configuration
s are saved
in the
database

the Appsec
Transporter
under the
App column
→

Deactivate
→ Confirm
when
prompted

24.9.
Administratio
n and
troubleshooti
ng



24.9.1.
Manage
instances



24.9.1.1. Add
a new data
source or

instance



📝🟢

note

If a data

source

contains
multiple
integrations,
the
integration
configured
as the
default
integration
will used by
the Data
Onboarder.
The default
integration
of the
content

pack is
indicated in
each
content

pack's
documentati
on. The
other
integrations
are available
for
configuration
in the Data
Sources &
Integrations
page after
installing the
content...

📝🟢

note

If you are
adding a
new instance
to an

existing data
source,
these
options are
View only.
You can
adjust the
view-only
options on
the relevant
page in the
system, for
example
Correlations,
Playbooks,
or Scripts.
If you are
adding a
new instance
to an

existing data
source,
these
options are
View only.
You can
adjust the
view-only
options on
the relevant...

24.9.1.2. How
to configure
the scanning
settings for
supported
services

📝🟢 

note

If you do not
select a
scanning
cadence, the
default
setting is
applied. For
more

information,
contact your
Customer
Support
team.

24.9.1.3.
Manage
cloud
instances

📝🟢 

note

Currently,
automation
permission
errors or

missing
automation
permissions
do not
affect the
Automation
health
status. You
can view any
permission
errors or

missing
permissions
in the the
Permissions
health status.

24.9.1.4.
Update
cloud
permissions
after Cortex
release
updates



☑️🟣

prerequisite

Ensure that
the user
account
used to
modify
permissions
has the
necessary

privileges
within both
the Cortex
platform
and your
cloud
environment,
for example,
AWS or
Azure.
You received
a

notification
regarding a
new version
available
that requires
permission
updates, or
viewed a
Needs
Update
status in the
Data
Sources &
Integrations
page.

📝🟢

note

Instances
requiring
updates will
not change
their
connection
status, for
example,
Connected,
Warning,
Error,
Disabled,
due to the
pending
permission
update.

24.9.1.5.
Pending
cloud
instances



24.9.1.6.
Troubleshoot
errors on

cloud
instances

📝🟢 

note

Errors
related to
collection of
audit logs in
the cloud
instance are
recorded in
the
collection_au
diting
dataset. For
more

information,
see Audit
logs fields
and query
examples.

24.9.1.7.
Manage
Kubernetes
Connector
instances

📝🟢 

note

After uninstalling the Kubernetes connector,
the connector status updates to Not connected
48 hours after the uninstall process is initiated.

24.9.2.
Integrations



📝🟢

note

In addition
to content

packs that
you install
from
Marketplace,
related
content

packs are
automaticall
y
downloaded
when you
adopt
playbooks or
edit tasks
that require
content
items such
as scripts or
integrations.

📝🟢

note

If the
integration
was installed
from a
content

pack, you
need to
duplicate
the
integration
before
editing.

📝🟢

note

Integration
commands
are only
available
when the
integration
instance is
enabled.
Some
commands
depend on a
successful
connection
between
Cortex
Cloud and
third-party
integrations.

💡🔵

tip

In the
Playground,
you can
clear the
context
data, if
needed,
which
deletes
everything in
the
Playground
context
data, but
does not
affect the
actual issue
or case. To
clear the
context, run
!DeleteConte
xt all=yes'
from the CLI
or click
Clear
Context
Data while
viewing the
context data.

🧭➡️

navigation

Settings →
Configuratio
ns →

Marketplace
24.9.2.1. Add
an

integration
instance



24.9.2.2.

Configure
integration
permissions

🧭➡️ 

navigation

Settings →
Configuratio
ns → Data
Collection

→

Integration
Permissions

24.9.2.3.
Troubleshoot
Integrations

📝🟢 

note

If an
integration
instance
consistently
encounters
API rate
limits when
running on
the tenant,
consider
configuring
the instance
to run on an
engine to
change the
source of
the
outbound
traffic.

24.9.3. Verify
collector
connectivity

📝🟢 

note

For more
information

on

troubleshooti
ng data
collector
applet
errors, see
Troubleshoot
Broker VM
applet
connectivity.

24.9.4.
Overview of
data
ingestion
metrics



☑️🟣

prerequisite

For Cortex Cloud to monitor data ingestion
health and create health issues, you must
enable Cortex - Analytics. Go to Configurations
→ Cortex - Analytics. For more information, see
Enable the Analytics Engine and Identity
Analytics.Enable the Analytics Engine and
Identity Analytics

24.9.4.1.
Creating
correlation
rules to
monitor data
ingestion
health



24.9.4.2.
Measuring
data
freshness

📝🟢 

note

The max_delay metric is taken from the
maximum bucket value with a restricted limit;
therefore, metrics show whole numbers.
The median and ninetieth_percentile metrics are
statistical calculations that give an
approximation of the real value; therefore,
metrics show decimal numbers.
Time slots with a zero log count or zero byte
count display records...

24.9.5. About
health issues



☑️🟣

prerequisite

For Cortex
Cloud to
monitor data
ingestion
health and
create
health
issues, you
must enable
the
following
settings
under
Configuratio
ns:

Cortex -
Analytics:
Go to
Configuratio
ns → Cortex
- Analytics.
For more
information,
see Enable
the
Analytics
Engine and
Identity
Analytics.Ena
ble the
Analytics
Engine and
Identity
Analytics

📝🟢

note

The Health
Issues page
displays
issues that

were

triggered
after July
2024. To see
health issues
that were
triggered
before this
date, click
Legacy
Health Issues.

📝🟢

note

Cortex
Cloud
enforces the
dedup logic
to health
issues. This
logic
reduces the
likelihood of
identical
health issues
from
flooding the
issues
dataset.

24.9.5.1.
Investigate
and resolve
health issues



24.9.5.2.
Monitor data
ingestion
health



24.9.5.3.
Monitor
correlation
rules

📝🟢 

note

In the
dataset, the
Query start
time and
Query end
time indicate
the
timeframe of
the data
that was
queried. The
actual start
and end
times of the
correlation
rule
execution
are recorded
in the _time
field for the
Initiated and
Completed
entries.

25.

Marketpla
ce



25.1. Cortex
Marketplace

🧭➡️ 

navigation

Settings →
Configuratio
ns →

Marketplace)
or at Cortex
Developer
Docs
Marketplace

25.2.
Content
Pack
Support
Types

📝🟢 

note

Palo Alto
Networks is
not liable for
and does
not warrant
or support
any content
pack
produced by
a third-party
publisher.
Palo Alto
Networks
does not
support
content

packs that
do not have
official
available
documentati
on.

25.3. Cortex
Cloud
content



25.4.
Manage
content

packs



📝🟢

note

Optional
content

packs that
are already
installed are
treated like
they are
required
content

packs to
preserve
content
integrity.

📝🟢

note

Cortex
Cloud
includes a
built-in
default mail
sender. You
also have
the option
of installing
a different
mail sender
content

pack, such
as Microsoft
Exchange
Online.

📝🟢

note

In addition
to content

packs that
you install
from
Marketplace,
related
content

packs are
automaticall
y
downloaded
when you
adopt
playbooks or
edit tasks
that require
content
items such
as scripts or
integrations.

📝🟢

note

Third-party
product
integrations
are

developed
and tested
against a
specific
product
version. For
products
that are on-
prem or
cloud-based
with specific
API versions,
the version
developed
and tested
against will
be included
in the
integration's
documentati
on. Newer
versions of
the product
are not

always
immediately
tested, and
it is
expected
that...

⚠️🟡

caution

If you want
to

downgrade,
any content
that
depends on
the content
pack
including
any
customizatio
ns may be
deleted if it
does not
exist in the
target
content

pack version.

⚠️🟡

caution

If another
content

pack is
dependent
on the
content

pack you
want to
delete, it
may break
the other
content

pack. You
can reinstall
the content
pack, but
you cannot
restore
detached
and
customized
content.

💡🔵

tip

You can also find content packs that require
updates by going to Settings → Data Sources &
Integrations and filtering by Pack Version =
Update Available. If you click on an integration
in the filtered list, there is a link to the content
pack in Marketplace for updates.

25.5.
Marketplace
FAQs



25.6.
Content
changes
when
upgrading
Cortex
Cloud
versions



26.

Serverless

function

posture

security



26.1.
Onboard
cloud
providers for
serverless
functions



📝🟢

note

When
scanning
serverless
functions
with layers,
those layers
need to be
from the
same cloud
account.

📝🟢

note

Only
functions
containing
zip files are
supported. 👀🟠

important

Cortex
supports
Google
Cloud
Functions:
1st gen and
2nd gen
Cloud
Functions
API.
26.2.
Serverless
function
posture
policies



26.2.1.
Manage
serverless
function
policies



📝🟢

note

You can
select
multiple
types to
view all your
serverless
function
rules across
your cloud
providers.

📝🟢

note

Refer to
Create
serverless
function
policies for
more

information
on how to
define the
steps of a
policy in the
wizard.

26.2.2.
Create
serverless
function
policies



📝🟢

note

You can
select
multiple
options.
📝🟢

note

For more
information
about rules,
refer to
Manage
serverless
function
rules.

26.3.
Serverless
function
posture rules



26.3.1.
Manage
serverless
function
rules



📝🟢

note

You can
select
multiple
types to
view all your
serverless
function
policies
across your
cloud
providers.

📝🟢

note

Refer to
Create
serverless
function
rules for

more

information
on how to
define the
steps of a
rule in the
wizard.

26.3.2.
Create
serverless
function
rules



26.3.3.
Create an
attack path
rule for
serverless
functions

🧭➡️ 

navigation

Under
Posture
Management,
select Rules
& Policies →
Cloud
Security
(under
Rules) →
click Create
Rule

in as the
operator →
enter the
CVE ID →
Search

> or >= as

the operator
→ Severity
level (such
as High,
Low) →
Search

> or >= as

the operator
→ enter a
score →

Search

26.3.4.
Create a
configuration
rule for
serverless
functions



26.3.5.
Create a
network
exposure
rule for
serverless
functions



26.4.
Serverless
function
usage

🧭➡️ 

navigation

To access
serverless
function
assets,
under
Inventory,
select All
Assets →
Compute →
Serverless
Functions

Posture
Management
→

Vulnerability
Management
) →
Vulnerability
Issues

Add Filters
→ Asset
Category →
Serverless
Function

Posture
Management
→

Vulnerability
Management
) →
Vulnerable
Assets

Add Filters
→ Asset
Category →
Serverless
Functions

27.

Serverless

function

runtime

security



27.1.
Overview



27.2. Set up
serverless
function
protection



27.3.
Serverless
runtime
issues

📝🟢 

note

Issues
triggered
within 24
hours,
sharing the
same name

and
description,
will be
aggregated
into cases
along with
issues from
the same
function per
execution.

28.

Cortex

CLI



📝🟢

note

This role is
not

supported
for CWP, as
the CWP
system does
not support
offline mode.

📝🟢

note

There are no
preconfigure
d CLI-
specific
roles. Add
the CLI Tools
permission
to an

existing role
or create a
dedicated
custom role.

28.1.
Connect
Cortex CLI



☑️🟣

prerequisite

System
requirements:
macOS (Intel
Core i7, such
as Sequoia):
To ensure all
functionalitie
s work
correctly,
you must
install the
vectorscan

dependency
via
Homebrew,
using this
command:
brew install
vectorscan
RHEL 8.10
and Red Hat
UBI9. The
following
prerequisites
must be met:
Install
patchelf
Install zstd
Ubuntu 20
requires the
prefetch...

📝🟢

note

The Cortex
CLI requires
an API key
with the
Standard
security level.

📝🟢

note

On macOS
arm 64
architecture
you must
unpack the
downloaded
file to
retrieve the
executable.

🧭➡️

navigation

If needed,
retrieve your
public API
URL by
navigating
to Settings
→

Configuratio
ns → API
Keys and
clicking
Copy API
URL, then
paste it into
the code

Windows
installer →
uninstall via
Settings →
Apps →
Installed
apps, or
delete the

28.2.
Authenticate
credentials



28.3. Cortex
CLI usage

📝🟢 

note

For more
information
about CLI
usage for
CWP, refer
to Cortex
CLI for
Cloud
Workload
Protection
For more
information
about CLI
usage for
API Security,
refer to
Cortex CLI
for API
Security
For more
information
about CLI
usage for
Cortex
Cloud
Application
Security,
refer to
Cortex CLI
usage for
Cortex
Cloud
Application
Security

28.4. Self-
service API

keys for CLI
scans



28.5. Cortex
CLI common
command
line
reference
guide



28.6. Cortex
CLI for API
Security



☑️🟣

prerequisite

Ensure you
have the
required
user

permissions.
Refer to
Cortex CLI
for more
information
Onboard
and install
the Cortex
CLI. Refer to
Connect
Cortex CLI
for more
information
Ensure your
application
exposes
APIs and
provides a
correspondin
g OpenAPI
Specification
file
Ensure that
you have
installed
Java v 11 and
above

28.6.1.
Cortex CLI
API Security
command
line
reference
guide



28.7. Cortex
CLI for
Cloud
Workload
Protection



☑️🟣

prerequisite

Ensure you
have the
required
user

permissions.
Refer to
Cortex CLI
for more
information
Onboard
and install
the Cortex
CLI. Refer to
Connect
Cortex CLI
for more
information
Verify that
Java version
11 and above
is installed:
Run java -
version in
your
terminal. If
not, refer to
Java SE
Development
Kit 11.0.25
for
information
about
installing
Java

☑️🟣

prerequisite

Before you
begin,
ensure you
have sudo
privileges to
execute the
image scan.

📝🟢

note

CWP does
not support
container
image secret
scanning for
systems
running on
ARM
architecture.

📝🟢

note

For available
CWP
commands,
refer to
Cloud
Workload
Protection
command
line
reference.

📝🟢

note

--archive:
When used
with image
scan, sets
the scan
source to an
archive file.
When used
with image
sbom,
indicates the
SBOM
should be
exported
from an
archive file
The --
archive flag
can also be
explicitly set
as --

archive=true
--archive-
format
<value>: The
image
archive
format (such
as docker-
archive or
oci-archive).
Default:
docker-
archive.

28.7.1. Cloud
Workload
Protection
command
line
reference



28.8. Cortex
CLI for Code
Security



☑️🟣

prerequisite

For the
Cortex CLI
binary:
Ensure you
have Node.js
v22 installed
on your host
machine
before
running any
scans with
the Cortex
CLI. This is
crucial to
prevent
runtime
errors, as the
CLI depends
on Node.js
for
executing
JavaScript
analysis
Note
To check
your version
of Node.js,
run node -v
To download
Node.js,
refer to the
official
Node.js...

28.8.1.
Cortex CLI
usage for
Cortex
Cloud
Application
Security



28.8.2.
Cortex CLI
Cortex
Cloud
Application
Security
command
line
reference



📝🟢

note

The repo-id
flag must
not end with
.config, .log
or .ini. -

config is
acceptable.
--repo-id
foo.config
will be
blocked
--repo-id
foo-config
will pass

👀🟠

important

The Cortex CLI Cortex Cloud Application
Security only supports single occurrences of
each flag. If the same flag is passed multiple
times, only the last provided value will be used.
For example, in the following command, only TF
CloudFormation will be the scanned framework.
./cortexcli --api-base-url <YOUR_API_URL> --
api-key <YOUR_API_KEY> --auth-id...

28.8.3.
Cortex CLI
pre-commit
hooks



☑️🟣

prerequisite

These
common

prerequisites
are required
for all types
of
installation
(both local
and global)
of the
Cortex CLI
pre-commit
hook.
Ensure you
have a
license for
Cortex
Cloud
Application
Security
Install the
Cortex
Cloud CLI
binary
locally.
Refer to
Connect
Cortex CLI
for
information
about
onboarding
the
CLIConnect
Cortex CLI
Obtain
Cortex
Cloud API...

☑️🟣

prerequisite

For local
installation:
Install the
pre-commit
framework
version 3.2.0
or greater.
Refer to
https://pre-
commit.com/
for
installation
instructions.

📝🟢

note

It is
recommende
d you
configure
credentials
for the
Cortex CLI
using a
configuration
file.

📝🟢

note

The pre-
commit
framework is
not required
for global
mode.

28.8.4. Pre-
commit
hook usage



28.8.5.
Cortex CLI
pre-receive
hooks



☑️🟣

prerequisite

Before you
begin,
ensure you
have:
Administrato
r access to
the VCS
server and
console
A valid
license for
Cortex
Cloud
Application
Security
The Cortex
Cloud CLI
binary or
Docker
image
installed on
the server
(requires
GLIBC (GNU
C library)
version 2.35
or greater).
Refer to
Connect
Cortex CLI
for
information
about
onboarding
the
CLIConnect
Cortex...

📝🟢

note

Make sure to create the directory under the
home directory of the Linux user that runs the
Git hooks. This user is typically not the root user.

28.8.6. Pre-
receive hook
usage



29.

Cortex

Cloud

XQL



29.1. Get
started with
XQL



29.1.1. XQL
language
features



💡🔵

tip

When
creating
XQL queries,
you can:

Use the up
and down
arrow keys
to navigate
through the
auto-
suggestion
commands
and
definitions.
Select an
auto-
suggestion
command by
pressing
either the
Enter or Tab
key.
Press
Shift+Enter
to add a
new line, and
easily ignore
the auto-
suggestion
output.
Close the
auto-
suggestion
output by
pressing the
Esc key.

🧭➡️

navigation

You submit
XQL queries
to Cortex
Cloud using
the
Investigation
& Response
→ Search →
Query
Builder user
interface

29.1.2. XQL
Language
Structure



📝🟢

note

Specifying a
dataset is
not required
because
Cortex
Cloud uses
xdr_data as
the default
dataset. If
you have
more than
one dataset
or lookup,
you can

change your
default
dataset by
navigating
to Settings
→

Configuratio
ns → Data
Management
→ Dataset
Management,
right-click
on the
appropriate
dataset, and
select Set as
default.

📝🟢

note

You can also
build a
query that
investigates
data in both
a cold
dataset and
hot dataset
in the same
query. In
addition, as
the hot
storage
dataset
format is the
default
option and
represents
the fully
searchable
storage, this
format is
used
throughout
this guide.
For more
information
on hot and
cold
storage, see
Dataset
management.

29.1.2.1.
Adding
comments in
queries



29.1.3.
Supported
operators

📝🟢 

note

In some
cases, using
an IN or NOT
IN operator
combined
with a
dataset and
filterfilter
stage can be
a better
alternative
to using a
join stage.join

29.1.4.
Datasets
and presets



📝🟢

note

To set up this Cloud Identity Engine (previously
called Directory Sync Service (DSS)) dataset,
you need to set up a Cloud Identity Engine.
Otherwise, you will not have a pan_dss_raw
dataset. For more information, see Set up Cloud
Identity Engine.Set up Cloud Identity Engine

📝🟢

note

INFO issues
are not
included in
this dataset.
The issue
fields
included in
this dataset
are limited
to certain
fields
available in
the API. For
the full list,
see Get
Alerts Multi-
Events v2
API.

📝🟢

note

The fields
contained in
this dataset
are a subset
of the fields
in the
xdr_data
dataset.

📝🟢

note

If the vendor
and product
are not

specified in
the
Winlogbeat
profile's
configuration
file, Cortex
Cloud
creates a
default
dataset
called
microsoft_wi
ndows_raw.

📝🟢

note

To ensure
GlobalProtec
t access
authenticatio
n logs are
sent to
Cortex
Cloud, verify
that your
PANW
firewall's
Log Settings
for
GlobalProtec
t has the
Cortex Data
Lake
checkbox
selected.

📝🟢

note

The fields
contained in
this dataset
are a subset
of the fields
in the
xdr_data
dataset.

📝🟢

note

Prisma
Access
firewalls do
not send
configuration
logs to the
Structured
Log Storage
(SLS).

📝🟢

note

You can
query in XQL
for this data
and build
widgets
based on the
xdr_data
dataset or
using the
preset
device_contr
ol.
To view in an
XQL query
these events,
the Device
Configuratio
n of the
endpoint
profile must
be set to
Block.
Otherwise,
the USB
events are
not

captured.
The events
are also
captured
when a
group of
device types
are blocked
on the...

📝🟢

note

The fields
contained in
this dataset
are a subset
of the fields
in the
xdr_data
dataset.

29.1.5. About
examples

🧭➡️ 

navigation

Investigation
& Response
→ Search →
Query
Builder →
XQL →
Query
Library

29.1.6. JSON
functions

👀🟠 

important

JSON field

names are
case

sensitive, so
the key to
field pairing
must be
identical in
an XQL
query for
results to be
found. For
example, if a
field value is
"TIMESTAMP"
and your
query is
defined to
look for
"timestamp",
no results
will be found.

29.1.7. How
to filter for
empty
values in the
results table



29.1.8.
Understandin
g string
manipulation
in XQL



29.2. Build
XQL queries



📝🟢

note

If you have
the Cortex
Agentic
Assistant,
you can use
natural
language
prompts to
create and
run XQL
queries
within the
chat
interface.
For more
information,
see Create
and run XQL
queries with
Agentic
Assistant
chat.Cortex
Agentic
Assistant

29.2.1. About
the Query
Builder



📝🟢

note

Schema
changes to
datasets
may not be
reflected in
the
autocomplet
e

suggestions
and
definitions
as you type
in real time
the XQL
query, and
can appear

with a slight
delay.

💡🔵

tip

When
creating
XQL queries,
you can:

Use the up
and down
arrow keys
to navigate
through the
auto-
suggestion
commands
and
definitions.
Select an
auto-
suggestion
command by
pressing
either the
Enter or Tab
key.
Press
Shift+Enter
to add a
new line, and
easily ignore
the auto-
suggestion
output.
Close the
auto-
suggestion
output by
pressing the
Esc key.

29.2.2. How
to build XQL
queries



📝🟢

note

Users with different dataset permissions can
receive different results for the same XQL query.
An administrator or a user with a predefined
user role can create and view queries built with
an unknown dataset that currently does not
exist in Cortex Cloud. All other users can only
create and view queries built with an existing
dataset.
When you have more...

📝🟢

note

You can
build a
query that
investigates
data in both
a cold
dataset and
a hot
dataset in
the same
query. In
addition, as
the hot
storage
dataset
format is the
default
option and
represents
the fully
searchable
storage, this
format is
used
throughout
this guide
for
investigation
and threat
hunting. For
more

information
on hot and
cold
storage, see...

👀🟠

important

Forensic
datasets are
not inlcuded
by default in
XQL query
results,
unless the
dataset
query is
explicitly
defined to

use a

forensic
dataset.

29.2.2.1. Get
started with
XQL queries



29.2.2.2.
Useful XQL

user

interface
features



📝🟢

note

Schema
changes to
datasets
may not be
reflected in
the
autocomplet
e

suggestions
and
definitions
as you type
in real time
the XQL
query and
can appear

with a slight
delay.

💡🔵

tip

When
creating
XQL queries,
you can:

Use the up
and down
arrow keys
to navigate
through the
auto-
suggestion
command
suggestions
and
definitions.
Select an
auto-
suggestion
command by
pressing
either the
Enter or Tab
key.
Press
Shift+Enter
to add a
new line, and
easily ignore
the auto-
suggestion
output.
Close the
auto-
suggestion
output by
pressing the
Esc...

29.2.2.3. XQL
Query best
practices



29.2.2.4.
Expected
results when
querying
fields



29.2.2.5.
Create XQL
query



📝🟢

note

Whenever
the time
period is
changed in
the query
window, the
config
timeframe is
automaticall
y set to the
time period
defined for
the entire
query,

including
queries that
are part of
the join
stage. Yet,
this won't be
visible as
part of the
query. Only
if you
manually
type in the
config
timeframe
will this be
seen in the
query.
These time
picker...

💡🔵

tip

When
creating
XQL queries,
you can:

Use the up
and down
arrow keys
to navigate
through the
auto-
suggestion
command
suggestions
and
definitions.
Select an
auto-
suggestion
command by
pressing
either the
Enter or Tab
key.
Press
Shift+Enter
to add a
new line, and
easily ignore
the auto-
suggestion
output.
Close the
auto-
suggestion
output by
pressing the
Esc...

💡🔵

tip

While the
query is
running, you
can navigate
away from
the page. A
notification
is sent when
the query
has finished.
You can also
Cancel the
query or run
a new query,

where you
have the
option to
Run only
new query

(cancel
previous) or
Run both
queries.

29.2.2.6.
Review XQL
query results



📝🟢

note

It's also
possible to
graph the
results
displayed.
For more
information,
see Graph
query results.

📝🟢

note

Real time
query
results are
available
only in the
Query
Builder and
in free text
query.
Real time
results are
displayed
only for
queries run
on hot
datasets.
The Sort
option is
available
only after all
the data is
retrieved.
When you
formulate
complex
queries, the
results will
be displayed
when the
query has
finished
running
completely,
and not in
real...

📝🟢

note

Results are
received
incrementally
for the first
100K
records, or
up to 100MB
worth of
records,
whichever
comes first.
After that,
the next
update is
when the
query has
finished
running
completely.

📝🟢

note

In order for
Cortex
Cloud to
provide a
histogram
for a field,
the field
must not
contain an
array or a

JSON object.

29.2.2.7.
Translate to
XQL

👀🟠 

important

This feature is still in a Beta state and you will
find that not all Splunk queries can be
converted to XQL. This feature will be improved
upon in the upcoming releases to support
greater Splunk query translations to XQL.

29.2.2.8.
Graph query
results



🔒🟣

content

Building
Cortex
Query
Language
(XQL)
queries in
the Query
Builder
requires a
Data
Collection
add-on.

📝🟢

note

To display
the result of
as a time
duration,
choose the
graph type
Single Value
and enable
Show as
Time. You
can then
select the
Time Unit
(millisecond,
second,
minute, or
hour) and
the Display
format.

💡🔵

tip

Alternatively,
you can use
the Cortex
Agentic
Assistant to
generate
custom

graphs and
charts using
natural
language
prompts. By
simply
prompting
the agent, it
will build
and execute
the query,
returning the
visual
representatio
n. For more
information,
see Use
natural
language to
query and
visualize
your data.

29.2.3. XQL
query entities



29.2.3.1.
Create
authenticatio
n query

🧭➡️ 

navigation

Investigation
& Response
→ Search →
Query
Builder 29.2.3.2.
Create event
log query



29.2.3.3.
Create file
query



29.2.3.4.
Create
image load
query



29.2.3.5.
Create
network
connections
query



29.2.3.6.
Create
network
query

📝🟢 

note

When you
run the
query,

depending
on the
outcome of
the results,
the value
specified in
this field
might be
displayed in
the dst_ip
field in the
query
results. This
occurs if an
RDP event is
recorded
whereby a
user

connected
from the
source IP to
the
destination IP.

29.2.3.7.
Create
process
query



29.2.3.8.
Create
registry
query

👀🟠 

important

Ensure the
KEY NAME is
entered as a
real registry
key name,
and not as a
symbolic
link.
Otherwise,
the query
will not
retrieve
results.
Instead of
HKEY_LOCAL
_MACHINE\S
ystem\Curre
ntControlSet,
which is a
symbolic
link, use
KEY_LOCAL_
MACHINE\Sy
stem\Control
Set001.
Instead of
HKEY_CURRE
NT_USER,
use

HKEY_USERS
\<SID>,
where SID is
either a SID
of the
current...

29.2.3.9.
Query
across all
entities



29.2.4.
Overview of
the Query
Center



📝🟢

note

Very short
queries
might not be
listed.
You cannot
cancel
correlation
queries.
The default
retention

period for
historic
queries is
aligned with
issue
retention.

29.2.4.1. Edit
and run
queries in
Query
Center



📝🟢

note

If query
limits are
applied to
your tenant,
the number
of
concurrent
running
queries is
limited per
user. If query
usage is
reaching the
defined limit,
a system
message
warns you

that a high
query load is
impacting
performance.
If you
exceed the
limit, new
queries are
blocked
until query
usage drops.
You can
view all
active
queries
under Query
Center...

📝🟢

note

You can
cancel your
own queries.
To cancel
queries run
by other
users, you
must have
View/Edit
permissions
for
Configuratio
ns → Query
Management.
By default,
Instance
administrator
s have
View/Edit
permission.

📝🟢

note

Cancelled
queries
show a
Canceled
status. You
can see

details of all
canceled
queries in
the Query
History tab.
You cannot
cancel
correlation
rule queries.
If you cancel
a scheduled
query, only
the current
query is
cancelled.
Future
recurrences

of the
scheduled
query are
not affected.

29.2.4.1.1.
Query
Center
reference
information

📝🟢 

note

Certain
fields are
exposed and
hidden by
default. An
asterisk (*) is
beside every
field that is
exposed by
default.

29.2.5.
Manage
scheduled
queries



🧭➡️

navigation

Investigation
& Response
→ Search →
Scheduled
Queries 29.2.5.1.
Scheduled
Queries
reference
information

📝🟢 

note

Certain
fields are
exposed and
hidden by
default. An
asterisk (*) is
beside every
field that is
exposed by
default.

29.2.6.
Manage
your query

library



📝🟢

note

The Query
to Library
option is
only
available if
your role has
the Create
Queries
capability.
For more
information,
see Manage
access to
saved
queries.

📝🟢

note

When the
tenant-level
setting
Owners and
editors can
change the
general
access is
unselected,
the drop-
down is
disabled and
only an
administrator
can

configure
this option.

👀🟠

important

The ability
to create,
edit, or
share
queries is
governed by
access

management.
If certain
options are
unavailable,
contact your
administrator
. For more
information,
see Manage
access to
saved
queries.

29.3. Cortex
XQL syntax,
parameters,
and
examples



30. Graph
Search



30.1. What is
Graph
Search?

☑️🟣 

prerequisite

Graph
Search
requires
View or
View/Edit
RBAC
permissions
for Graph
Search
under
Investigation
& Response
→ Search.

30.2. Get
started with
Graph
Search
queries



🔒🟣

content

This feature
is included
with a
Cortex
XSIAM
Premium
license. It is
also
included
with any
other Cortex
XSIAM
license that
has the
Cloud
Posture
Security or
Cloud
Runtime
Security
add-on.

☑️🟣

prerequisite

Graph
Search
requires
View or
View/Edit
RBAC
permissions
for Graph
Search
under
Investigation
& Response
→ Search.

🧭➡️

navigation

Understand
your assets
and findings
data: Graph
Search
queries are
based on the
current data
that has
been
collected
for assets
and findings
from the
data sources
configured
and then
sent to the
Unified
Asset
Inventory
(UAI), which
is displayed
in the All
Assets page
(Inventory →
Assets → All
Assets

30.3. How to
build Graph
Search
queries?



☑️🟣

prerequisite

Graph
Search
requires
View or
View/Edit
RBAC
permissions
for Graph
Search
under
Investigation
& Response
→ Search.

🧭➡️

navigation

You submit
Graph
Search
queries
using the
Investigation
& Response
→ Search →
Query
Builder →
Graph
Search built-
in query
interface

30.4.
Understand
Graph
Search
query results

☑️🟣 

prerequisite

Graph
Search
requires
View or
View/Edit
RBAC
permissions
for Graph
Search
under
Investigation
& Response
→ Search.

30.5. Create
Graph
Search query



☑️🟣

prerequisite

Graph
Search
requires
View or
View/Edit
RBAC
permissions
for Graph
Search
under
Investigation
& Response
→ Search.

📝🟢

note

For more
information
about the
Query
Library, see
Manage the
Graph
Search
Query
Library.

💡🔵

tip

After
running the
query, you
can view the
complete
query by
hovering
over the last
THAT... in the
Graph
Search
query

interface,
and the
query is
displayed in
a tooltip.
If your query
doesn't find
any results
or you want
to change
your query

for any
reason, you

can always
click
anywhere in
the Graph
Search
query

interface,
where your
existing
query is...

30.6. Graph
Search
examples

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🔒🟣

content

This feature
is included
with a
Cortex
XSIAM
Premium
license. It is
also
included
with any
other Cortex
XSIAM
license that
has the
Cloud
Posture
Security or
Cloud
Runtime
Security
add-on.

☑️🟣

prerequisite

Graph
Search
requires
View or
View/Edit
RBAC
permissions
for Graph
Search
under
Investigation
& Response
→ Search.

30.7. Manage
the Graph
Search
Query
Library



☑️🟣

prerequisite

Graph
Search
requires
View or
View/Edit
RBAC
permissions
for Graph
Search
under
Investigation
& Response
→ Search.

🧭➡️

navigation

From Graph
Search in the
Query
Builder
Select
Investigation
& Response
→ Search →
Query
Builder →
Graph
Search

Investigation
& Response
→ Search →
Query
Builder →
Graph
Search

Investigation
& Response
→ Search →
Query
Center

30.8. Edit
and run
queries in
Query
Center

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📝🟢

note

If query
limits are
applied to
your tenant,
the number
of
concurrent
running
queries is
limited per
user. If query
usage is
reaching the
defined limit,
a system
message
warns you

that a high
query load is
impacting
performance.
If you
exceed the
limit, new
queries are
blocked
until query
usage drops.
You can
view all
active
queries
under Query
Center...

📝🟢

note

You can
cancel your
own queries.
To cancel
queries run
by other
users, you
must have
View/Edit
permissions
for
Configuratio
ns → Query
Management.
By default,
Instance
administrator
s have
View/Edit
permission.

📝🟢

note

Cancelled
queries
show a
Canceled
status. You
can see

details of all
canceled
queries in
the Query
History tab.
You cannot
cancel
correlation
rule queries.
If you cancel
a scheduled
query, only
the current
query is
cancelled.
Future
recurrences

of the
scheduled
query are
not affected.

🧭➡️

navigation

Investigation
& Response
→ Search →
Query
Center →
Query
History

30.8.1. Query
Center
reference
information

📝🟢 

note

Certain
fields are
exposed and
hidden by
default. An
asterisk (*) is
beside every
field that is
exposed by
default.

30.9.
Supported
assets and
findings

☑️🟣 

prerequisite

Graph
Search
requires
View or
View/Edit
RBAC
permissions
for Graph
Search
under
Investigation
& Response
→ Search.

30.10. FAQ
on Graph
Search

☑️🟣 

prerequisite

Graph
Search
requires
View or
View/Edit
RBAC
permissions
for Graph
Search
under
Investigation
& Response
→ Search.